Millikelvin Confocal Microscopy of Electrostatic Exciton Traps and Filter-Function Description of Noisy Quantum Dynamics

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Colophon
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Tobias Hangleiter
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Summary

Quantum technology promises to be a paradigm shift in the precision, speed, and breadth of application of technology. Recent advances in the fields of quantum computing, quantum networks, and quantum sensing have been rapid, but large-scale utilization is still out of reach. This is largely due to the fragility of the individual quantum systems that this technology seeks to control and manipulate. While small demonstrators have by now shown these novel machines to work in principle, the task in the coming years will be to make them reproducible at scale. Leveraging the advanced semiconductor fabrication technology of today's classical computers, spin qubits in semiconductors such as Si/SiGe or GaAs/AlGaAs are hoped to meet this challenge. In this thesis, I present contributions in four different areas towards this overarching goal. I present an open-source software tool for Fourier-transform noise spectroscopy, improvements and characterization of a Millikelvin confocal microscope, optical measurements of electrostatic exciton traps, and a general filter-function formalism for the description of noisy quantum dynamics. I review the theory of noise spectroscopy based on Welch's method of overlapping periodograms and lay out the design of the hardware-agnostic, open-source `python_spectrometer` software package that facilitates interactive noise spectroscopy in everyday use. In a didactic fashion, I explore the features and capabilities of the tool and guide the reader through its operation. I then characterize a confocal microscope incorporated in a cryogen-free dilution refrigerator, applying the aforementioned noise spectroscopy tool to evaluate the displacement noise power spectral density of the system. I develop an optical technique based on a knife-edge reflectance contrast to measure the displacement noise reaching a shot-noise floor of $1\,\mathrm{nm}/\sqrt{\mathrm{Hz}}$. Using this technique, I determine a displacement noise RMS of $100\,\mathrm{nm}$. I measure the electron temperature of a GaAs/AlGaAs quantum dot in the microscope, finding $76\,\mathrm{mK}$, and demonstrate at hand of a second-order correlation measurement of a self-assembled quantum dot the setup's optical capabilities. In the experimental part of this thesis, I first provide an introduction to photoluminescence and the quantum-confined Stark effect in semiconductor quantum wells, appealing to simple analytical theory to develop an intuition for the relevant effects. I then describe the `mjolnir` measurement framework developed to conduct optical measurements of electrostatic exciton traps in semiconductor membranes, and present extensive measurements in search of signatures of single-photon emitters. Finally, I develop a comprehensive theoretical formalism for describing and analyzing the noisy dynamics of quantum systems based on filter functions. There, I combine the Magnus and cumulant expansions to derive a quantum-operations formulation of the dynamics under arbitrary classical noise that can be expressed in terms of filter functions capturing the susceptibility of a quantum system in response to noise at a given frequency. I present the `filter_functions` software package that implements the formalism and demonstrate its efficacy using several examples.

Zusammenfassung

Quantentechnologie verspricht einen Paradigmenwechsel in Bezug auf Präzision, Geschwindigkeit und Anwendungsvielfalt von Technologie. Die jüngsten Fortschritte in Quantencomputing, Quantennetzwerken und Quantensensorik waren rasant, doch eine großflächige Nutzung ist noch nicht in Sicht. Dies liegt hauptsächlich an der Fragilität der einzelnen Quantensysteme, die diese Technologie zu steuern versucht. Während kleine Demonstratoren gezeigt haben, dass diese Maschinen im Prinzip funktionieren, besteht die Aufgabe nun darin, sie in großem Maßstab reproduzierbar zu machen. Unter Nutzung fortgeschrittener Halbleiterfertigungstechnologie heutiger klassischer Computer soll diese Herausforderung mit Spin-Qubits in Halbleitern wie Si/SiGe oder GaAs/AlGaAs bewältigt werden. In dieser Arbeit stelle ich Beiträge in vier Bereichen in Richtung dieses Ziels vor. Ich präsentiere ein Open-Source-Softwaretool für die Fourier-Transformations-Rauschspektroskopie, Verbesserungen eines Millikelvin-Konfokalmikroskops, optische Messungen elektrostatischer Exzitonenfallen und einen allgemeinen Filterfunktionsformalismus zur Beschreibung rauschbehafteter Quantendynamik. Ich gebe einen Überblick über die Theorie der Rauschspektroskopie auf der Grundlage von Welchs Methode und stelle das Design des hardwareunabhängigen Open-Source-Pakets `python_spectrometer` vor, das interaktive Rauschspektroskopie im täglichen Gebrauch erleichtert. Auf didaktische Weise untersuche ich die Funktionen des Tools und führe den Leser durch dessen Bedienung. Anschließend charakterisiere ich ein konfokales Mikroskop, das in einen kryogenfreien Mischkryostaten integriert ist, und wende das oben genannte Tool an, um die spektrale Leistungsdichte des Positionsrauschens zu messen. Ich entwickle eine Technik auf der Grundlage eines Messerkanten-Reflexionskontrasts, um das Positionsrauschen zu messen, das einen Schrotrauschuntergrund von $1\,\mathrm{nm}/\sqrt{\mathrm{Hz}}$ erreicht. Mit dieser Technik bestimme ich ein Positionsrauschen (RMS) von $100\,\mathrm{nm}$. Weiterhin bestimme ich die Elektronentemperatur eines GaAs/AlGaAs-Quantenpunkts im Mikroskop zu $76\,\mathrm{mK}$. Anhand einer Messung der Korrelationsfunktion zweiter Ordnung eines selbstorganisierten Quantenpunkts demonstriere ich die optischen Fähigkeiten des Aufbaus. Im experimentellen Teil gebe ich zunächst eine Einführung in Photolumineszenz und den quantenbegrenzten Stark-Effekt in Halbleiter-Quantentöpfen und stütze mich dabei auf einfache analytische Theorie. Anschließend beschreibe ich das `mjolnir` Softwarepaket für optische Messungen von elektrostatischen Exzitonfallen in Halbleitermembranen und präsentiere umfangreiche Messungen auf der Suche nach Signaturen von Einzelphotonenemittern. Schließlich entwickle ich einen theoretischen Formalismus zur Beschreibung der verrauschten Dynamik von Quantensystemen auf der Grundlage von Filterfunktionen. Hierbei kombiniere ich Magnus- und Kumulanten-Entwicklungen zur Herleitung einer Quantenoperationsformulierung der Dynamik unter klassischem Rauschen, ausgedrückt durch Filterfunktionen. Ich präsentiere schließlich das `filter_functions` Softwarepaket, das die einfache Berechnung der Filterfunktionen ermöglicht, und diskutiere ausgewählte Anwendungsbeispiele.

Acknowledgements

It’s been a journey. In a way, this endeavor started nine years ago, when I took up a HiWi job for a few months before the start of my Bachelor’s project. Hendrik, it is a testament to you, the group you’ve built, and to the research you pursue that I have kept coming back for my Master’s and finally for my PhD projects despite external factors in my life pulling me away from Aachen. Thank you for the opportunity, and for giving me the freedom, to grow, learn, and explore. Quantum technology is a hugely interdisciplinary field of research, and your vision of the grand scheme of things as well as your appreciation of not only the experimental physics but also the theory, software, and engineering that come along with it made this group an immensely rewarding place to be.

Having been part of the group for so long, there are many people that I am grateful to have shared my time with at the institute. Simon, who ad-hoc introduced me to the Bloch sphere when I stumbled into his office looking for a Bachelor’s thesis, and whose curiosity and eagerness to learn about just about anything are inspiring. Who could have ever imagined that the world of Glam Metal is so big and yet so small?[*] Pascal, who supervised both my Bachelor’s and Master’s projects, and whose influence on both the group and my own personal growth cannot be overstated. Julian, for sticking to theory amidst so many experimentalists and teaching them lessons in diplomacy. René, with whom I shared the central office for the longest time, and who was my partner in solitude manning the office during the Covid years. Alex, who took his place – literally speaking – and continues to teach everyone sour lessons in roasting. Sebastian, who labored and toiled away in the clean room, and who I shared pains and successes with on the lone isle of optics land—it’s been a joy collaborating with you! Thomas, whose work ethic enabled him to do Sebastian’s and my job simultaneously, and whose shoulders most of my work stands on.

I have appreciated discussions in the optics team, in particular with Beata Kardynał, who was always available to share insight into the field of semiconductor optics that was new to me, and Maxim, who is wise beyond his years. Thank you also to the mechanical workshop, Frank and Helmut in particular, who always find a solution to whichever problem you come to them with. I am grateful to all my proof readers – Paul, Max, Alex, Sebastian, Evelyn, Maxim, and Dominik – that have had to cope with my complex thesis repository layout and knack for convoluted sentences.

Pursuing a PhD is more than just a job, and so I am also thankful to the people who have made it possible to reach this point or whose paths have touched mine during the course of my time here. Constanze and Ulrich, who took me in and gave me a roof over my head 13 years ago when I came to Aachen with a backpack and a duffle bag, surprised at the fact I had not found a place to live yet despite having looked for all but a few weeks

[*] This was before they were famous.

before the start of the semester. The anonymous gentleman walking his dogs in Melaten every morning who was a regular friendly face during the lockdown weeks and beyond. "Einen erfolgreichen Tag und bleiben Sie gesund!" Stefan Schael, for being a demanding but understanding neighbor.

Crucially, and emphatically unironically, my parents, who have placed me in the privileged position of being able to pursue my studies – both of them! – without financial worries or lofty expectations looming over my head. Papa, without your enduring and patient help and support I would have never made it this far.

It is perhaps fitting that as this chapter of my life closes, another, one that also started nine years ago, merely changes shape. Idske, you have been with me through all the highs and lows, you have helped me make decisions, lifted my spirits, shared my joy, made me laugh—in short, you have made my life the life it is, and I will not imagine it any other way anymore. I cannot wait to start the next part of that chapter with you.

Tobias Hangleiter
Aachen, September 2025

Preface

The present thesis consists of four parts. Each part is a largely self-contained accounting of one aspect of my work during the last years. Unfortunately, not everything always goes according to plan, and so there is not the *one* overarching theme that connects all parts. Yet it is in the nature of physics that there are always points of contact and connections.

One of these is *noise*, which I have dealt with extensively from various different points of view. In Part I, I present a software package to facilitate noise spectroscopy as an everyday tool in physics labs. The experiments that we conduct as condensed-matter physicists suffer from many kinds of noise, and this tool is intended to help analyze it. That is, given the *system*, what is the noise and how can we quantify it? In Part II, I then analyze, characterize, and improve upon an experimental setup designed to perform confocal optical spectroscopy of semiconductor nanostructures at Millikelvin temperatures. There, the topic of noise enters as an undesired but unavoidable external influence, and I apply the tools developed in Part I to characterize and mitigate the noise in order to improve the experimental capabilities of the setup. That is, given the *noise in the current state of the system*, how can we modify the system in order to ameliorate it? In Part IV, I develop a theoretical formalism to describe noise and its influence on the coherent evolution of single quantum systems. Also there, of course, the aim is to reduce the influence of noise, but in quite a different way. There, I deal with modeling the noise and the way it interacts with the intended dynamics of a controlled quantum system so as to understand and come up with ways of avoiding perturbations induced by the noise. That is, given the *noise*, how can we predict how the system reacts to it, and how might we be able to control the system in a way that is robust to noise?

Another recurring theme of the present thesis is *research software*. As physicists, much of our work – be it experimental or theoretical – relies on computers doing some aspect of it for us. Developing, and sharing, this software has thus become an integral part of physics research, and in the present thesis I contribute to the open-source research software ecosystem. In Part I, I introduce the `python_spectrometer` software package already mentioned above, which implements hardware-agnostic data acquisition, processing, and visualization for Fourier-transform noise spectroscopy. In Part III, I present optical measurements of electrostatic exciton traps in the Millikelvin confocal microscope discussed in Part II. These measurements are facilitated by the `mjolnir` measurement framework, which abstracts the complex hardware stack required to perform the measurements, allowing for minimal coding overhead going from the concept of a measurement to its execution. In Part IV, I introduce the `filter_functions` software package, which implements the formalism developed in the same part, enabling easy adoption of the techniques for the computation of noisy quantum processes.

The present thesis has been typeset with LuaLaTeX and is optimized for digital reading. The document source code is available on Github. Throughout the document, I make liberal use of cross-references and acronyms, the latter of which link to and are defined in the Glossary on page 223. As such, I recommend a PDF viewer that can preview the targets of internal hyperlinks for reading the present thesis in order to avoid having to jump back and forth within the document.

The various figures included in this document are either generated by LaTeX/TikZ or by a Python script included in the source repository in the `img/py` directory. Figure captions contain a hyperlink to the list of figure source files and parameters on page 226 whose entries point to the source file that generates the figure. For figures that contain experimental data, the entry in the list of figure source files and parameters on page 226 furthermore typically contains values of external parameters that were fixed during the measurement. In the spirit of open data, all data discussed in the present thesis are included in the `data` directory at the top level of the source repository.

Finally, I frequently employ an asinh-scale for data that spans several orders of magnitude. This scaling is asymptotically equivalent to a symmetric logarithmic scale for large $|x|$, $\operatorname{asinh}(\pm x) \sim \pm \log |x|$, but avoids the divergence close to zero, instead being linear, $\operatorname{asinh}(x) \sim x$. The crossover from linear to logarithmic behavior is governed by a parameter x_0 such that $x \to x_0 \operatorname{asinh}(x/x_0)$ and that can be freely chosen, thus influencing the representation of the data. The particular value of x_0 (corresponding to the `linear_width` parameter in `matplotlib`) in a given figure can be looked up in the script file that generates it.

Tobias Hangleiter
Aachen, September 2025

Contents

Part I

A Flexible Python Tool For Fourier-Transform Noise Spectroscopy

Introduction 1

Author contributions — *This part of the present thesis has benefited from discussions with several people as part of a course taught during the winter semester of 2022/23 [1]. The software package presented here was developed by me and has received contributions from Simon Humpohl,[a] Max Beer,[a] Paul Surrey,[a] and René Otten.[b]*

[1]: Bluhm et al. (2022), *Research Techniques for Advanced Solid State Device Experiments*

a: RWTH Aachen University and Forschungszentrum Jülich GmbH.

b: Then at RWTH Aachen University and Forschungszentrum Jülich GmbH.

In solid-state quantum technology research, devices with features on the order of the Fermi wavelength – say tens of nanometers – are embedded in a crystal containing $\sim 10^{23}$ atoms. These devices host single quantum objects such as electrons or quasiparticles—collective many-body excitations. They are controlled and measured by classical signals routed to the device through macroscopic connections like cables and fibers. The signals, in turn, are generated and analyzed by electronic (*e.g.*, transistors) or optical instruments (*e.g.*, lasers) that, in all likelihood, are again based on solid-state technology driven by the first quantum revolution [2, 3]. Ultimately, then, the experiments take place in an environment full of external influences from trams passing by, shaking the ground, to cosmic rays, creating electron-hole or breaking up Cooper pairs.

All of these different layers to such experiments are inherently – and in fact often fundamentally[1] – *noisy*, and it is the physicist's challenge to measure their sought-after effects in spite of this noise. A well-thought-out experimental setup is hence one that has been designed taking the various noise sources into consideration, and state-of-the-art experiments often need to push the frontier in order to be successful. From material choice in the sample to the signal path and the specs of the measurement equipment, many aspects need to be optimized and, in particular, characterized in order to assess the noise. Indeed, the assessment of noise might even be the entire *goal* of the experiment, for instance to evaluate material properties. In this case, the quantity being measured might not be the same quantity whose noise one is interested in but rather some function of it, and the resulting data still needs to be transformed before one is able to make any practical statements about those properties.

1: Consider Johnson-Nyquist noise in a resistor, for example.

Noise, in the sense that we are concerned with in the present thesis,[2] is a stochastic process, meaning that we cannot predict with certainty a dynamical system's time evolution; it *fluctuates* randomly around its noise-free value.[3] Only by obtaining statistics, *i.e.*, repeated observations, either by preparing the system in the same initial state or measuring for a certain amount of time, can we make any statements about the underlying stochastic process. Two questions are key to assessing these statistics: first, what is the amplitude of the fluctuations? And second, at which frequency do the fluctuations occur? If the amplitude is small enough, we might not need to care, and if the frequency is large enough, we might be able to average out the fluctuations while if it is small enough, it might appear *quasistatic* and we might be able to subtract the constant value from our signal. Although numerous other techniques for measuring and estimating a noise's properties exist, analyzing noise in frequency space by means of the Fourier transform imposes itself when considering these two notions.

2: Quantum noise does not fall under this scope as it may – disregarding vacuum fluctuations for the sake of argument – be considered *emergent*; it arises from a system entangled with a (not necessarily large) number of quantum degrees of freedom being observed, *i.e.*, tracing out the environment's degrees of freedom. Our lack of knowledge about this environment then appears as noise in our observations. See Reference 4 for a comprehensive review of the quantum case.

3: Quantum measurements are also noisy in this sense as we cannot predict the outcome of a single-shot measurement, but here the stochasticity is in the outcomes of an ensemble rather than the sequence of values in time. The two are, however, closely related through ergodicity, which we will require in Section 2.1.

This approach is the central topic of this part of the present thesis. However, I will neither be too much concerned with the theoretical side of the subject matter nor with the details of its practical application in identifying and mitigating noise.[4] Rather, I will focus on making these techniques readily and easily accessible to experimentalists in the lab. Given the arguments laid out above, noise spectroscopy, that is, the analysis of noise in Fourier space, should be an essential item in an experimentalists toolbelt. In practice, though, we are faced with several challenges. First, different experiments require different data acquisition (DAQ) hardware, all of which have both varying capabilities and software interfaces. Hence, transferring a spectroscopy code from one device or setup to another is a non-trivial task and can inhibit adoption of the technique. Albeit some instruments come with built-in spectroscopy solutions, they vary in functionality and are not easily transferred to different systems. Second, while probably everyone finding themselves in the situation is capable of computing the noise spectrum when presented with a set of time series data, inferring the correct parameters for data acquisition given the desired parameters of the resulting spectrum can, while not difficult, be cumbersome to do.[5] Lastly, noise spectroscopy is most effective when proper visualization tools are employed. Again, this is not a difficult task per se, but such things always incur overhead costs that can deter users from employing these essential techniques.

4: I refer the interested reader to the lecture material of Refences 5 and 1 and references therein.

5: Although it is of course a good exercise, and, as a physicist, one should always strive to understand the tools one is using and the underlying principles at work. This part of the present thesis is intended to provide a starting point for that.

Here, I address these points by introducing a Python software package, `python_spectrometer` [6], that tackles the entire processing chain of practical noise spectroscopy in a physics laboratory. By abstracting DAQ hardware into a unified interface, it is portable between different setups. With the goal to make noise spectroscopy as accessible as possible and lower the entry barrier, it automatically handles parameter inference and hardware constraints. Finally, it provides a comprehensive plotting solution that allows for interactively analyzing the data using various different data visualization methods. I will employ the tool to perform displacement noise spectroscopy in a cryogenic confocal microscope in Part II using two different methods that highlight its versatility. As such, this part of the present thesis is intended to serve as an introduction to noise spectroscopy and the `python_spectrometer` software package presented here aimed at potential users.

The remainder of this part is structured as follows. In Chapter 2, I review the mathematical groundwork underpinning noise spectroscopy by means of Fourier transforms of time series and discuss parameters and properties of the central quantity of interest, the power spectral density (PSD). In Chapter 3, I then present the software package by going over its design choices and giving a walkthrough of its features using a typical workflow as an example. I conclude by giving an outlook on future directions in Chapter 4.

2 Theory of spectral noise estimation

There exist various methods for estimating the properties of noise in a classical time series signal $x(t) \in \mathbb{C}$.[1] A simple metric quantifying the average noise amplitude of the signal observed from time $t = 0$ to $t = T$ is the root mean square (RMS) [8],

$$\mathrm{RMS}_x = \sqrt{\frac{1}{T}\int_0^T \mathrm{d}t\, |x(t)|^2}, \tag{2.1}$$

for which

$$\mathrm{RMS}_x^2 = \mathrm{E}[x]^2 + \mathrm{Var}[x] \tag{2.2}$$

with $\mathrm{E}[x]$ the expectation value and $\mathrm{Var}[x]$ the variance of $x(t)$. While this already tells us *something* about the noise, it is evident that a single number does not provide many clues if we were to attempt to mitigate the noise or say something qualitative about it beyond "small" and "large". In cases such as these, physics has often turned to the *spectral* representation of the function of interest. Knowing the frequency content of a function gives access to a wealth of information about the underlying contributing processes. But how can we learn how a system behaves as a function of frequency?

1: We discuss only classical noise here, meaning $x(t)$ commutes with itself at all times. For descriptions of and spectroscopy protocols for quantum noise refer to Refences 4 and 7, for example. Furthermore, $x(t)$ will in most applications be strictly real, see Section 2.4 for an extended discussion.

Consider an electrical black box – some device under test (DUT) – with two leads connected to a lock-in amplifier (LIA) as sketched in Figure 2.1. We could then simply measure the conductance $G(t) = I(t)/V(t)$ through the device for some time T with a given lock-in modulation frequency f_i, subtract the constant offset,[2] and calculate the RMS using Equation 2.1. Repeating this procedure for different modulation frequencies $\{f_i\}_i$, we would collect a set of RMS-values that we could assign to the modulation frequencies at which they were measured and thus sample the noise amplitude spectrum[3] of the conductance, $S_G(f_i)$.[4] This method has the advantage that we can choose the frequencies at which the spectrum should be sampled—in particular they do not have to be evenly spaced. However, it has two significant shortcomings. First, it is rather inefficient and therefore time-consuming. For N frequency sample points, it takes a total time of NT to acquire all data, where T needs to be chosen such that the variance of RMS_G is sufficiently small. Second, and crucially, it is not always possible to excite the system at a certain frequency and measure its response. For example, it is generally considered hard[5] to – deliberately – excite vibrations of a specific frequency in a cryostat, yet we might still be interested in the displacement spectrum inside of it. A related method is available if one has access to a frequency-tunable probe, that is, a physical system whose behavior – typically its time evolution or some measurable property after letting the system evolve – under noise is well understood within a given noise model, and that can be controlled to be sensitive to a certain frequency. One example for such methods is dynamical decoupling (DD)-based noise spectroscopy using a qubit [9–12], a protocol based on insights gained from the filter-function formalism—see Part IV. These protocols are especially well-suited for high but less so for low frequencies as they rely on observing the coherence of a qubit and hence the visibility on long time scales.

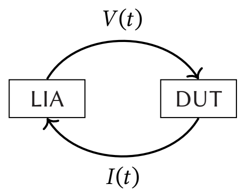

Figure 2.1: Measuring the conductance through a device under test (DUT) using a lock-in amplifier (LIA).

2: The constant part $G_0 = G(t) - \delta G(t)$ of course also holds some information about the system, for example about its bandwidth.

3: We will properly define this term below.

4: See Subsection 3.2.1 for a discussion on how the RMS at a certain frequency relates to other quantities discussed in this part.

5: And also unpopular with colleagues.

In a sense the most straight-forward way of measuring the noise spectrum presents itself as a consequence of the Wiener-Khinchin theorem [13, 14] that relates the noise spectrum to the Fourier transform of a time-dependent signal. This makes it possible to estimate the noise spectrum by simply observing a signal for a certain amount of time and performing some post-processing![6] This method will be the topic of the present chapter. It is laid out as follows. I will first derive the Wiener-Khinchin theorem for continuous stochastic processes and introduce the central quantity of interest in noise spectroscopy; the power spectral density (PSD). Then, I will discretize the method and discuss resulting side-effects before describing an efficient way of obtaining the spectrum from a finite amount of data. I will conclude by elaborating on relevant parameters and properties.

6: An interesting combination of the Fourier-transform-based and the qubit-based spectroscopy methods was recently proposed in Reference 15, where the authors compute the spectrum by Fourier-transforming the second derivative of the coherence function.

2.1 Spectrum estimation from time series

To see how spectral noise properties may be estimated from time series data, consider a continuous signal in the time domain, $x(t) \in \mathbb{C}$, that is observed for some time T. We define the windowed Fourier transform of $x(t)$ and its inverse by[7]

7: In this chapter we will always denote the Fourier transform of some quantity ψ using the same symbol with a hat, $\hat{\psi}$.

$$\hat{x}_T(\omega) = \int_0^T \mathrm{d}t\, x(t) e^{-i\omega t}, \tag{2.3}$$

$$x(t) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \hat{x}_T(\omega) e^{i\omega t}, \tag{2.4}$$

i. e., we assume that outside of the window of observation $x(t)$ is zero. The autocorrelation function of $x(t)$ is given by

$$C(t, t') = \langle x^*(t) x(t') \rangle = \lim_{T\to\infty} \frac{1}{T} \int_0^T \mathrm{d}t\, x^*(t) x(t+\tau), \tag{2.5}$$

where $\tau = t' - t$, $\langle \bullet \rangle$ is the ensemble average over multiple realizations of the process and the second equality holds true for ergodic processes. We now assume that $x(t)$ is *wide-sense stationary*. A process is wide-sense (or *weakly*) stationary if its mean is constant and $C(t, t') \equiv C(\tau)$, that is, if its autocorrelation function depends only on the time lag $\tau = t' - t$ and not on the absolute point in time.[8] Expressing $x(t)$ in terms of its Fourier representation (Equation 2.3) and reordering the integrals, we get

8: For Gaussian processes as discussed here, this also implies stationarity [16]. The property further implies that $C(\tau)$ is an even function.

$$C(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_0^T \mathrm{d}t \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \hat{x}_T^*(\omega) e^{-i\omega t} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi} \hat{x}_T(\omega') e^{i\omega'(t+\tau)} \tag{2.6}$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi} \hat{x}_T^*(\omega) \hat{x}_T(\omega') e^{i\omega'\tau} \int_0^T \mathrm{d}t\, e^{it(\omega'-\omega)} \tag{2.7}$$

The innermost integral approaches a δ-function for large T, allowing us to further simplify this under the limit as

$$C(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega'}{2\pi} \hat{x}_T^*(\omega) \hat{x}_T(\omega') e^{i\omega'\tau} 2\pi\delta(\omega' - \omega) \tag{2.8}$$

$$= \lim_{T\to\infty} \frac{1}{T} \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} |\hat{x}_T(\omega)|^2 e^{i\omega\tau} \tag{2.9}$$

$$= \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S(\omega) e^{i\omega\tau} \tag{2.10}$$

with the power spectral density (PSD)[9]

$$S(\omega) = \lim_{T\to\infty} \frac{1}{T}|\hat{x}_T(\omega)|^2 = \hat{C}(\tau) \tag{2.11}$$

Equation 2.10 is the Wiener-Khinchin theorem [13, 14] that states that the autocorrelation function $C(\tau)$ and the PSD $S(\omega)$ are Fourier-transform pairs [16]. Furthermore, defining the latter through Equation 2.11 gives us an intuitive picture of the PSD if we recall Parseval's theorem,

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi}|\hat{x}_T(\omega)|^2 = \int_0^T \mathrm{d}t\,|x(t)|^2, \tag{2.12}$$

from which it follows that

$$P = \mathrm{RMS}_x^2 = \lim_{T\to\infty}\frac{1}{T}\int_0^T \mathrm{d}t\,|x(t)|^2 = \int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi}S(\omega). \tag{2.13}$$

That is, the total power P contained in the signal $x(t)$ is given by integrating over the PSD.[10] Similarly, the power contained in a band of frequencies $[\omega_1, \omega_2]$ is given by

$$P(\omega_1,\omega_2) = \mathrm{RMS}_x(\omega_1,\omega_2)^2 = \int_{\omega_1}^{\omega_2}\frac{\mathrm{d}\omega}{2\pi}S(\omega) \tag{2.14}$$

where $\mathrm{RMS}_x(\omega_1,\omega_2)$ is the root mean square within this frequency band. These relations are helpful when analyzing noise PSDs to gauge the relative weight of contributions from different frequency bands to the total noise power.

To become familiar with the quantities $C(\tau)$ and $S(\omega)$, consider the Ornstein-Uhlenbeck process [17], the only stationary Gaussian Markovian stochastic process [18]. The autocorrelation function of the Ornstein-Uhlenbeck process is given by

$$C(\tau) = \sigma^2 e^{-\tau/\tau_c}, \tag{2.15}$$

with σ the RMS and τ_c the correlation time of the process. The PSD in turn is the Lorentzian function

$$S(\omega) = \frac{2\sigma^2\tau_c}{1+(\omega\tau_c)^2}. \tag{2.16}$$

For a given discretization time step Δt and hence bandwidth $\omega \in [0, \pi/\Delta t]$, the Ornstein-Uhlenbeck process interpolates between perfectly uncorrelated, white noise ($\Delta t/\tau_c \to \infty, S(\omega) = 2\tau_c\sigma^2$), correlated, Brownian noise ($\Delta t/\tau_c \gg 1, S(\omega) = 2\sigma^2/\tau_c\omega^2$), and perfectly correlated, quasistatic noise ($\Delta t/\tau_c \to 0, S(\omega) = \sigma^2\delta(\omega)$), although it is still Markovian in all cases. Figure 2.2 shows simulated time series of the Ornstein-Uhlenbeck process and their autocorrelation functions and PSDs for exemplary parameters: in the white noise limit ($\tau_c = 10^{-2}\Delta t$, blue), in the intermediate regime ($\tau_c = \Delta t$, magenta), and in the correlated regime ($\tau_c = 10^2\Delta t$, green). From the time series plot at the top it becomes clear that the RMS alone is insufficient to describe the properties of noisy signals as the curves differ significantly despite being normalized to their RMS. The autocorrelation functions averaged over 10^3 realizations of the noisy signals as well as their theoretical (continuous-case) value, Equation 2.15, are plotted in the middle panel, normalized to $\tau_c = \Delta t$ and $\sigma^2 = \Delta t/4$. For the white noise limit (blue), correlations are too short to be resolved with the given time discretization. The correlations decay

9: The term *power spectrum* is often used interchangeably. I will do so as well, but emphasize at this point that in digital signal processing in particular, the *spectrum* is a different quantity from the *spectral density*. See also Sidenote 19 in Chapter 3.

10: The term "power" is used loosely here; it corresponds to the physical power if $x(t)$ is a voltage drop across a $1\,\Omega$ resistor.

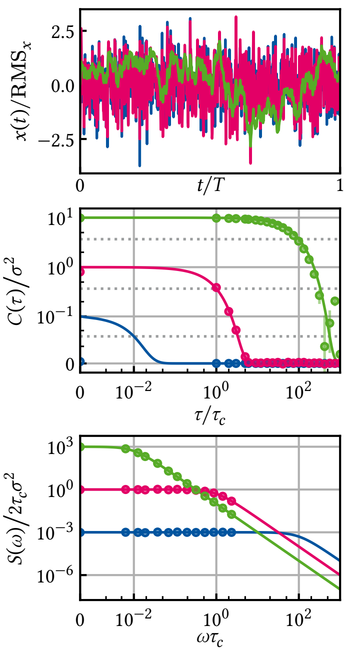

Figure 2.2: Ornstein-Uhlenbeck process. Simulated time traces, autocorrelation function, and PSD of the Ornstein-Uhlenbeck process. Top: Simulated time traces using the algorithm presented in Section C.4. The data are normalized to the computed RMS (equal to σ in the continuous case). Middle: Theoretical autocorrelation function (Equation 2.15, solid lines) computed from the simulated data averaged over 10^3 traces (circles, subset of points). Error bars indicate the standard error of the mean, axes are scaled with respect to the parameters of the magenta data, and data are plotted on an asinh-scale. Bottom: Theoretical PSD (Equation 2.16, solid lines) and periodograms computed from the simulated data using `scipy.signal.periodogram()`, *cf.* Equation 2.20, averaged over 10^3 traces (circles, subset of points). Axes are again scaled with respect to the parameters of the magenta data and plotted on an asinh-scale. Parameters are $\tau_c = \Delta t \times \{10^{-2}, 1, 10^2\}$ and $\sigma^2 = \sqrt{\tau_c}/4$ for blue, magenta, and green data, respectively.

to e^{-1} at $\tau/\tau_c = 10^{-2}, 1, 10^2$, respectively (dotted gray lines). Finally, the bottom panel shows the PSD, Equation 2.16, and its numerical estimate, again averaged over 10^3 realizations of the signal and normalized to $\tau_c = \Delta t$ and $\sigma^2 = \Delta t/4$. The cross-over from white to Brownian PSD occurs at $\omega = \tau_c$. While the simulated data for $\tau_c = 10^{-2}\Delta t$ appears perfectly white, that for $\tau_c = 10^2\Delta t$ appears perfectly $1/f$-like because the spectrum is only white below the smallest resolvable frequency $\Delta f = T^{-1}$.

Having gotten an intuition for the quantities $C(\tau)$ and $S(\omega)$, let us move on to see how the latter may be obtained from time series data with a finite time resolution such as a data acquisition instrument would produce. The fact that $S(\omega) = \hat{C}(\omega)$ (Equation 2.11) represents the starting point for the experimental spectrum estimation procedure. Instead of a continuous signal $x(t), t \in [0, T]$, consider its discretized version[11]

$$x_n, \quad n \in \{0, 1, \dots, N-1\} \tag{2.17}$$

defined at times $t_n = n\Delta t$ with $T = N\Delta t$ and where $\Delta t = f_\mathrm{s}^{-1}$ is the sampling interval (the inverse of the sampling frequency f_s), and its discrete Fourier transform

$$\hat{x}_k = \sum_{n=0}^{N-1} x_n \exp\left(-2\pi i \frac{nk}{N}\right), \quad k \in \{0, 1, \dots, N-1\}. \tag{2.18}$$

Invoking the ergodic theorem,[12] we can replace the long-term average in Equation 2.11 by the ensemble average over M realizations of the noisy signal, $\{x_n^{(m)}\}_m$, and write

$$S_n = \frac{1}{f_\mathrm{s} M} \sum_{m=0}^{M-1} \left|\hat{x}_n^{(m)}\right|^2 = \frac{1}{M} \sum_{m=0}^{M-1} S_n^{(m)} \tag{2.19}$$

where the factor f_s^{-1} is to ensure proper units, we defined the *periodogram* of $x_n^{(m)}$ by

$$S_n^{(m)} = \frac{1}{f_\mathrm{s}} \left|\hat{x}_n^{(m)}\right|^2, \tag{2.20}$$

and S_n is an *estimate* of the true PSD sampled at the discrete frequencies $\omega_n = 2\pi n/T \in 2\pi\times\{-f_\mathrm{s}/2, \dots, f_\mathrm{s}/2\}$. Equation 2.19 is known as Bartlett's method [21] for spectrum estimation.[13]

To better understand the properties of this estimate, let us take a look at the parameters Δt, N, and M. The sampling interval Δt defines the largest resolvable frequency by the Nyquist sampling theorem,

$$f_\mathrm{max} = \frac{f_\mathrm{s}}{2} = \frac{1}{2\Delta t}. \tag{2.21}$$

In turn, the number of samples N determines the frequency resolution Δf, or smallest resolvable frequency,

$$f_\mathrm{min} = \Delta f = \frac{1}{T} = \frac{1}{N\Delta t} = \frac{f_\mathrm{s}}{N}. \tag{2.22}$$

Lastly, M determines the variance of the set of periodograms $\{S_n^{(m)}\}_{i=0}^{M-1}$ and hence the accuracy of the estimate S_n.

In practice, the ensemble realizations i are of course obtained sequentially, implying that one acquires a time series of data $x_n, n \in \{0, 1, \dots, NM-$

11: We only discuss the problem of equally spaced samples here. Variants for spectral estimation of time series with unequal spacing exist [19, 20].

12: Note that the limit of perfectly correlated noise, $S(\omega) \propto \delta(\omega)$, technically does *not* correspond to an ergodic process because $C(\tau) = \text{const.}\ \forall \tau \in (-\infty, \infty)$. In practice, this is always a mathematical idealization and the spectrum is actually better described by a nascent δ-function with a small but finite width.

13: By taking the limit $M \to \infty$ one recovers the true PSD,

$$\lim_{M\to\infty} S_n = S(\omega_n).$$

The continuum limit is as always obtained by sending $\Delta t \to 0, N \to \infty, N\Delta t = \text{const.}$

1} and partitions these data into M sequences of length N. It becomes clear, then, that the Bartlett average (Equation 2.19) trades spectral resolution (larger N) for estimation accuracy (larger M) given the finite acquisition time $T = NM\Delta t$. An improvement in data efficiency can be achieved using Welch's method [22]. To see how, we first need to discuss spectral windowing.

2.2 Window functions

Partitioning a signal x_n into M sections $x_n^{(m)}, m \in \{0, 1, \dots, M-1\}, n \in \{0, 1, \dots, N-1\}$ of length N is mathematically equivalent to multiplying the signal with the rectangular *window function* given by[14]

$$w_n^{(m)} = \begin{cases} 1 & \text{if} \quad mN \le n < (m+1)N \quad \text{and} \\ 0 & \text{else} \end{cases} \tag{2.23}$$

so that $x_n^{(m)} = x_n w_n^{(m)}$. Now recall that multiplication and convolution are duals under the Fourier transform, implying that

$$\hat{x}_n^{(m)} = \hat{x}_n * \hat{w}_n^{(m)}, \tag{2.24}$$

where the Fourier representation of the rectangular window is given by[15]

$$\hat{w}_n^{(m)} = \hat{w}_n e^{-i(m+1/2)\omega_n T}, \tag{2.25}$$

$$\hat{w}_n = T \operatorname{sinc}\left(\frac{\omega_n T}{2}\right). \tag{2.26}$$

with $T = N\Delta t$.

Figure 2.3 shows the unshifted rectangular window $\hat{w}_n$ in Fourier space. We can hence understand the Fourier spectrum of $x_n^{(m)}$ as sampling $\hat{x}_n$ with the probe $\hat{w}_n^{(m)}$. However, whereas in the continuum limit (see Sidenote 13) Equation 2.26 tends towards $\delta(\omega_n)$ and thus will produce a faithful reconstruction of the true spectrum, the finite sample rate f_s of discrete signals and observation length T induce finite frequency sampling and bandwidth of the probe as well as *sidelobes*.[16] These effects incur what is known as *spectral leakage* (finite bandwidth and sidelobes) and *scalloping loss* (finite sampling), and lead to artifacts and deviations of the spectrum estimator S_n from the true spectrum $S(\omega_n)$ [16, 23].

For this reason, a plethora of *window functions* have been introduced to mitigate the effects of spectral leakage. Key properties of a window are the spectral bandwidth (center lobe width) and sidelobe amplitude between which there typically is a tradeoff.[17] A window frequently used in spectral analysis is the Hann window [25],

$$w_n^{(m)} = \begin{cases} \sin^2\left(\frac{\pi n}{N}\right) & \text{if} \quad (m-1)N \le n < mN \quad \text{and} \\ 0 & \text{else,} \end{cases} \tag{2.27}$$

with the Fourier representation of the unshifted window ($2\pi f_n = \omega_n$),

$$\hat{w}_n = \frac{T}{2} \operatorname{sinc}\left(\frac{\omega_n T}{2}\right) \frac{1}{1 - f_n^2 T^2}, \tag{2.28}$$

14: This window is also known as the boxcar or Dirichlet window.

15: $\operatorname{sinc}(x) = \sin(x)/x$.

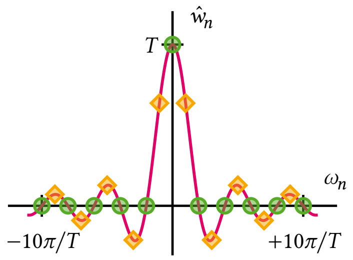

Figure 2.3: The Fourier representation of the rectangular window in continuous time (solid line) and for discrete frequencies $\omega_n = 2\pi n/T$ (circles). Introducing a phase shift, that is, shifting the window with respect to the signal in time, effectively shifts $\omega_n \to \omega_{n+\eta}$ as indicated for $\eta = 1/2$ (diamonds). This incurs scalloping loss.

16: For the optically inclined: this is akin to Fraunhofer diffraction at an aperture.

17: Wikipedia gives a good overview of existing window functions [24].

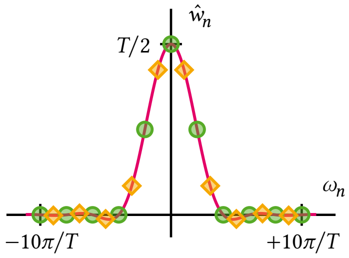

Figure 2.4: The Fourier representation of the Hann window in continuous time (solid line) and for discrete frequencies ω_n (circles). Diamonds indicate discrete sampling when the window completely out of phase with the signal (*cf.* Figure 2.3).

shown in Figure 2.4. The favorable properties of the Hann window are apparent when compared to the rectangular window in Equation 2.26 and Figure 2.3; the sidelobes are quadratically suppressed while the center lobe is broadened by a factor of two.

Another favorable property of the Hann window is that $w_0^{(0)} = w_{N-1}^{(0)} = 0$. This suppresses detrimental effects arising from a possible discontinuity ($x_0^{(0)} \neq x_{N-1}^{(0)}$) at the edge of a data segment related to the discrete Fourier transform, which assumes periodic data.[18]

18: This can usually also be achieved approximately by detrending the data before performing the Fourier transform, which is a good idea in any case.

2.3 Welch's method

Contemplating Equation 2.27, one might come to the conclusion that using a window such as this is not very data efficient in the sense that a large fraction of samples located at the edge of the window is strongly suppressed and hence does not contribute significantly to the spectrum estimate. To alleviate this lack of efficiency, one can introduce an overlap between adjacent data windows. That is, instead of partitioning the data x_n into M non-overlapping sections of length N, $x_n^{(m)}$, one shifts the mth window backward by mK with $K > 0$ the overlap. This results in M overlapping segments of the $L = N + (M-1)(N-K)$ total samples. Finally, the periodogram (Equation 2.20) is computed for each window segment and subsequently averaged to obtain the spectrum estimator (Equation 2.19).

This method of spectrum estimation is known as Welch's method [22]. One can show [22] that the correlation between the periodograms of adjacent, overlapping windows can be kept sufficiently small to avoid a biased estimate. The optimal overlap naturally depends on the choice of window; a typical value for the Hann window is $K = N/2$ with which one would obtain $M = 2L/N - 1$ windows for data of length L.

Figure 2.5 conceptually illustrates Welch's method for a trace of $1/f$ noise with $L = 300$ samples in total. Choosing the Hann window and an overlap of 50 % results in $M = 5$ segments for a window length of $N = 100$. The data in the second window is highlighted.

2.4 Parameters & Properties of the PSD

We are now in a position to discuss how the various parameters of a time series relate to both the physical parameters of the resulting spectrum es-

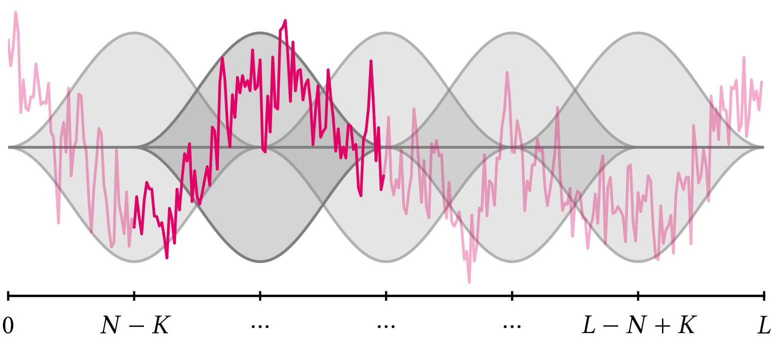

Figure 2.5: Illustration of Welch's method for spectrum estimation. The data (magenta) of length L is partitioned into $K = 2L/N - 1$ segments of length N. Each segment is multiplied with a window function (gray) which reduces spectral leakage and other artifacts. A finite overlap K between adjacent windows (gray) ensures efficient sample use.

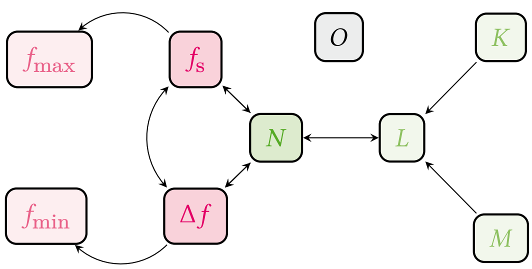

Figure 2.6: Relationships of data acquisition parameters (*cf.* Tables 2.1 and 3.1, with arrows indicating dependencies. N, f_s, and Δf are the central quantities defining the estimated spectrum's properties (darker shades). From f_s and Δf follow (bounds for) $f_{\max}$ and $f_{\min}$ (shaded red, rational numbers). From N, together with K and M, follows L, the total number of samples per data batch (shaded green, integers).

timate and to each other. To this end, we will go through the typical procedure of acquiring a spectrum estimate using Welch's method chronologically. Table 2.1 gives an overview of all relevant parameters.

To acquire data using some form of (digital) DAQ, one usually needs to specify two parameters first: the total number of samples to be acquired, L, and the sample rate f_s. This results in a measurement of duration $T = L\Delta t$ where $\Delta t = f_s^{-1}$ as previously mentioned. The choice of f_s already induces an upper bound on the first parameter characterizing the PSD estimate: the largest resolvable frequency $f_{\max} \leq f_s/2$ (*cf.* Equation 2.21, but note that we allow $f_{\max}$ to be smaller than half the sample rate in anticipation of hardware constraints). Next, we choose a number of Welch averages, M, *i. e.*, data partitions, and their overlap, K. In doing so, one fixes the number of samples per partition N and thereby induces the lower bound on the second parameter characterizing the PSD estimate: the frequency spacing $\Delta f = 1/N \leq f_{\min}$ (*cf.* Equation 2.22).[19] Finally, we can introduce a number of *outer* averages O, that is, the number of data batches that are acquired. While not directly related to Welch's method, choosing $O > 1$ can, for instance, help achieve a certain variance if the number of samples per batch, L, is limited by the data acquisition hardware, or simply allow for updating the spectrum estimate as data is being acquired. Figure 2.6 shows the relationships of the various parameters among each other. In Subsection 3.1.1, I lay out how these inter-dependencies are implemented in software.

19: Technically, the smallest resolvable frequency in a fast Fourier transform (FFT) is zero, of course. But as data is typically detrended (a constant or linear trend subtracted) before computation of the periodogram, the smallest *meaningful* frequency is given by $f_{\min}$.

Table 2.1: Overview of spectrum estimation parameters. The parameters can be assigned into four groups: 1. DAQ parameters configuring the data acquisition device, 2. Welch parameters specifying the periodogram averaging, 3. Spectrum properties induced by the above, and 4. External parameters unrelated to the others.

1. DAQ parameters	
L	Total number of samples
f_s	Sample rate
2. Welch parameters	
K	Number of overlap samples
N	Number of segment samples
M	Number of Welch segments
3. Spectrum parameters	
$f_{\min}$	Smallest resolvable frequency
$f_{\max}$	Largest resolvable frequency
4. Miscellaneous parameters	
O	Number of outer averages

To conclude this chapter, let us discuss some of the properties of stochastic processes and their autocorrelation function and PSD. Consider again the process $x(t)$. We say $x(t)$ is *Gaussian* if $x(t) \sim \mathcal{N}(\mu, \sigma^2)\,\forall t$, meaning that the value of $x(t)$ at a given point in time follows a normal distribution with some mean $\mu = \mathrm{E}[x]$ and variance $\sigma^2 = \mathrm{Var}[x]$ over multiple realizations of the process. In this case, its statistical properties are fully described by the autocorrelation function $C(\tau)$ and PSD $S(\omega)$. This is because only the first two cumulants of a Gaussian distribution are nonzero [26]. For the purpose of noise estimation, the assumption of Gaussianity is a rather weak one as the noise typically arises from a large ensemble of individual fluctuators and is therefore well approximated by a Gaussian distribution by the central limit theorem [27].[20] Even if $x(t)$ is not perfectly Gaussian, non-Gaussian contributions can be seen as higher-order contributions if viewed from the perspective of perturbation theory, and therefore the PSD still captures a significant part of the statistical properties. For this reason, the PSD is the central quantity of interest in noise spectroscopy. Let us just note at this point that techniques to estimate higher-order spectra (or *polyspectra*) exist [10, 31–33].

20: As an example, consider electronic devices, where voltage noise is thought to arise from a large number of defects and other charge traps in oxides being populated and depopulated at certain rates γ. The ensemble average over these so-called two-level fluctuators (TLFs) then yields the well-known $1/f$-like noise spectra [28, 29] (at least for a large density [30]).

For real signals $x(t) \in \mathbb{R}$, the autocorrelation function of a wide-sense stationary process $C(\tau)$ is an even function, while for $x(t) \in \mathbb{C}$ its real part is even and its complex part odd, as we can see from Equation 2.5. From this it immediately follows that for real $x(t)$ $S(\omega)$ is also an even function and one therefore distinguishes the *two-sided* PSD $S^{(2)}(\omega)$ defined over $\mathbb{R}$ from the *one-sided* PSD $S^{(1)}(\omega) = 2S^{(2)}(\omega)$ defined only over $\mathbb{R}^+$. In practice, one often works with the one-sided PSD because the negative half of the frequency domain only holds redundant information. Complex $x(t)$ such as those generated by LIAs after demodulation in turn have asymmetric, two-sided PSDs. In this chapter so far, we have implicitly employed the two-sided definition, but in the software package presented in Chapter 3, two-sided spectra are used only for complex data.

The `python_spectrometer` software package 3

In this chapter, I introduce the `python_spectrometer` Python package[1] [6] and lay out its design and functionality. This software package was developed to make it easier for experimentalists to transfer the mathematical machinery introduced in Chapter 2 to the lab. While in principle the entire process of spectrum estimation from a given array of time series data is already covered by the `welch()` routine in `SciPy` [34], obtaining the data array is not standardized. Different DAQ instruments have different capabilities, both on the hardware and the software level, and different driver interfaces to communicate with them. This implies that custom data acquisition code is required for every instrument, introducing a significant entry barrier to spectral analysis. The `python_spectrometer` package implements a simple interface to different hardware instruments that allows for changing the hardware backend without having to adapt the user-facing code and also incorporates different hardware constraints.

1: The package repository is hosted on GitLab. Its documentation is automatically generated and hosted on GitLab Pages. Releases are automatically published to PyPI and allow the package to be installed using `pip install python-spectrometer`.

What is more, noise spectroscopy tends to be a visual endeavor in practice; it is hard to compare different noise spectra based on quantitative reasoning alone. Data visualization is hence an integral part of noise spectroscopy, but plotting is not just plotting. Do we want the data to be shown on a log-log scale?[2] Do we want to show the relative magnitude of different data sets? Do we want to inspect the time traces as well? The `python_spectrometer` package addresses these questions by allowing users to interactively change features of the main plot window to adapt it to the form best suited to the situation at hand.

2: The short answer is yes, but it comes with visual side-effects that demand other ways of plotting data at times. The long answer is therefore yes, *and* …

Moreover, when concerned with noise spectrum estimation, we are typically more interested in specifying parameters of the resulting PSD rather than parameters of the underlying time series data. The package approaches data acquisition from the inverse direction: rather than inferring the spectrum properties from the time series data, users specify the properties they would like the resulting spectrum to have and the package chooses the correct parameters for data acquisition accordingly.

3.1 Package design and implementation

The `python_spectrometer` package provides a central class, `Spectrometer`, that users interact with to perform data acquisition, spectrum estimation, and plotting. It is instantiated with an instance of a child class of the `DAQ` base class that implements an interface to various DAQ hardware devices.[3] New spectra are obtained by calling the `Spectrometer.take()` method with all acquisition and metadata settings. In the following, I will go over the the design of these aspects of the package in more detail.

3: Actually *drivers* to be more precise.

```
├── doc
├── src
│   └── python_spectrometer
│       ├── daq
│       │   ├── __init__.py
│       │   ├── base.py
│       │   ├── qcodes.py
│       │   ├── settings.py
│       │   └── …
│       ├── __init__.py
│       ├── _audio_manager.py
│       ├── _plot_manager.py
│       └── core.py
├── tests
├── pyproject.toml
└── …
```

Figure 3.1: Source tree structure of the `python_spectrometer` package. Driver wrappers are placed in the `daq` subpackage. `core.py` exports the `Spectrometer` class.

3.1.1 Data acquisition

Figure 3.1 shows the directory structure of the source code. The `daq` subpackage contains on the one hand the declaration of the `DAQ` abstract base class (`base.py`) and its child class implementations (`qcodes.py`, *etc.*), and on the other the `settings.py` module, which defines the `DAQSettings` class. This class is used in the background to validate data acquisition settings both for consistency (see Section 2.4) and hardware constraints.

Table 3.1: Variable names used in Chapter 2 and their corresponding parameter names as used in `python_spectrometer` and `scipy.signal.welch()` [34].

VARIABLE	PARAMETER
L	`n_pts`
N	`nperseg`
K	`noverlap`
M	`n_seg`
O	`n_avg`
f_s	`fs`
f_{max}	`f_max`
f_{min}	`f_min`

To better understand the necessity of this functionality, consider the typical scenario of a physicist[4] in the lab. Alice has wired up her experiment, performed a first measurement, and to her dismay discovered that the data is too noisy to see the sought-after effect. She sets up the `python_spectrometer` code to investigate the noise spectrum of her measurement setup. From her noisy data she could already estimate the frequency of the most harrowing noise, so she knows the frequency band $[f_{\mathrm{min}}, f_{\mathrm{max}}]$ she is most interested in. But because she is lazy, she does not want to do the mental gymnastics to convert f_{min} to the parameter that her DAQ device understands, L (see Table 3.1), especially considering that L depends on the number of Welch averages and the overlap. Furthermore, while she could just about do the conversion from f_{max} to the other relevant DAQ parameter, f_s, in her head, her device imposes hardware constraints on the allowed sample rates she can select! The `DAQSettings` class addresses these issues. It is instantiated with any subset of the parameters listed in Table 3.1[5] and attempts to resolve the parameter interdependencies laid out in Section 2.4 and depicted in Figure 2.6 upon calling `DAQSettings.to_consistent_dict()`.[6] This either infers those parameters that were not given from those that were or, if not possible, uses a default value. Child classes of the `DAQ` class can subclass `DAQSettings` to implement hardware constraints such as a finite set of allowed sampling rates or a maximum number of samples per data buffer.

4: Let's call her Alice.

5: `DAQSettings` inherits from the builtin `dict` and as such can contain arbitrary other keys besides those listed in Table 3.1. However, automatic validation of parameter consistency is only performed for these special keys.

6: Since the graph spanned by the parameters is not acyclic, this only works *most* of the time.

For instance, Alice might want to measure the noise spectrum in the frequency band [1.5 Hz, 72 kHz]. Although she would not have to do this explicitly,[7] she could inspect the parameters after resolution using the code shown in Listing 3.1.

7: Settings are automatically parsed when passed to the `take()` method of the `Spectrometer` class.

```
>>> from python_spectrometer.daq import DAQSettings
>>> settings = DAQSettings(f_min=1.5, f_max=7.2e4)
>>> settings.to_consistent_dict()
{'f_min': 1.5,
 'f_max': 72000.0,
 'fs': 144000.0,
 'df': 1.5,
 'nperseg': 96000,
 'noverlap': 48000,
 'n_seg': 5,
 'n_pts': 288000,
 'n_avg': 1}
```

Listing 3.1: `DAQSettings` example showcasing automatic parameter resolution. `n_avg` determines the number of outer averages, *i. e.*, the number of data buffers acquired and processed individually.

```
{'f_min': 14.30511474609375,
 'f_max': 72000.0,
 'fs': 234375.0,
 'df': 14.30511474609375,
 'nperseg': 16384,
 'noverlap': 0,
 'n_seg': 1,
 'n_pts': 16384,
 'n_avg': 1}
```

Listing 3.2: Resolved settings for the same input parameters as in Listing 3.1 but for the `ZurichInstrumentsMFLIScope` backend with hardware constraints on `n_pts` and `fs`.

If the instrument she'd chosen for data acquisition had been a Zurich Instruments MFLI's Scope module,[8] the same requested settings would have resolved to those shown in Listing 3.2.[9] This is because the Scope module constrains $L \in [2^{12}, 2^{14}]$ and $f_s \in 60\,\mathrm{MHz} \times 2^{[-16,0]} \approx \{915.5\,\mathrm{Hz}, \dots, 30\,\mathrm{MHz}, 60\,\mathrm{MHz}\}$.

8: `https://docs.zhinst.com/labone_api_user_manual/modules/scope/index.html`

9: And issued a warning to inform the user their requested settings could not be matched.

As already mentioned, the `DAQ` base class implements a common interface for different hardware backends, allowing the `Spectrometer` class to be hardware-agnostic. That is, changing the instrument that is used to acquire the data does not necessitate adapting the code used to interact with the `Spectrometer`. To enable this, different instruments require small wrapper drivers that map the functionality of their actual driver onto the interface dictated by the `DAQ` class. This is achieved by subclassing `DAQ` and implementing the `DAQ.setup()` and `DAQ.acquire()` methods. Their functionality is best illustrated by the internal workflow as representatively shown in Listing 3.3.

```
daq = MyDAQ(driver_handle)

parsed_settings = daq.setup(**user_settings)
acquisition_generator = daq.acquire(**parsed_settings)

for data_buffer in acquisition_generator:
    estimate_psd(data_buffer)

# Finalize, clean up, return to user code
...
```

Listing 3.3: DAQ workflow pseudocode. A `MyDAQ` object (representing the instrument `My`) is instantiated with a driver object (for instance a `QCoDeS Instrument`). The instrument is configured with the given `user_settings`. Calling the generator function `daq.acquire()` with the actual device settings returns a generator, iterating over which yields one data buffer per iteration. The data buffers can then be passed to further processing functions (the PSD estimator in our example).

When acquiring a new spectrum, all settings supplied by the user are first fed into the `setup()` method where instrument configuration takes place. The method returns the actual device settings which are then forwarded to the `acquire()` generator function. Here, the instrument is armed (if necessary), and subsequently data is fetched from the device and yielded to the caller `n_avg` times, where `n_avg` is the number of outer averages.[10] An exemplary implementation of a `DAQ` subclass for a fictitious instrument is shown in Listing 3.4. In addition to the methods to configure the instrument and perform data acquisition, it is possible to override the `DAQSettings` property to implement instrument-specific hardware constraints such as, in this example, the number of samples per buffer being constrained to the discrete interval $[1, 2048]$. Leveraging the `qutil.domains` module [35], more complex constraints such as sample rates restricted to an internal clock rate divided by a power of two[11] can be specified.

10: *I. e.*, the number of time series data batches acquired, as opposed to the number of Welch averages `n_seg` within one batch.

11: See for example the implementation of the AlazarTech ATS9440 digitizer card.

3.1.2 Data processing

Once time series data has been acquired using a given `DAQ` backend, it could in principle immediately be used to estimate the PSD following Equation 2.19. However, it is often desirable to transform, or process, the data in some fashion. This can include simple transformations such as accounting for the gain of a transimpedance amplifier (TIA) and convert the voltage back to a current, or more complex ones such as applying calibrations. In particular, since the process of computing the PSD already involves Fourier transformation, the processing can also be performed in frequency space. In `python_spectrometer`, this can be done using a `procfn` (in the time domain) or `fourier_procfn` (in the Fourier domain). The former is specified as an argument directly to the `Spectrometer` constructor. It is a callable with signature `(x, **kwargs) -> xp`, taking as arguments the time series data and arbitrary settings passed through from the `take()` method, and returns the processed data. Listing 3.5 shows a simple function that accounts for the gain of an amplifier. The latter is specified in the `psd_estimator` argument of the `Spectrometer` constructor. This argument allows the user to specify a custom estimator for the PSD, in which case a callable is expected. Otherwise, it should be a mapping containing parameters for the default PSD estimator, `scipy.signal.welch()` [34]. Here, the keyword `fourier_procfn` should be a callable with signature `(xf, f, **kwargs) -> (xfp, fp)`.[12] That is, it should take the frequency-space data, the corresponding frequencies, and arbitrary keyword arguments and return a tuple of the processed data and the corresponding frequencies.

12: *I. e.*, the `psd_estimator` argument would be `{"fourier_procfn": fn}`.

```python
# daq/mydaq.py
import dataclasses
from qutil.domains import DiscreteInterval
from .base import DAQ
from .settings import DAQSettings

@dataclasses.dataclass
class MyDAQ(DAQ):
    handle: mydriver.DeviceHandle

    @property
    def DAQSettings(self) -> type[DAQSettings]:
        class MyDAQSettings(DAQSettings):
            ALLOWED_NPERSEG = DiscreteInterval(1, 2048)
        return MyDAQSettings

    def setup(self, **settings) -> dict:
        settings = self.DAQSettings(settings)
        parsed_settings = settings.to_consistent_dict()
        self.handle.configure(parsed_settings)
        return parsed_settings

    def acquire(self, n_avg: int, *, **settings) -> Generator:
        self.handle.arm(n_avg)
        for _ in range(n_avg):
            self.handle.wait_for_trigger()
            yield self.handle.fetch()
        return self.handle.metadata
```

Listing 3.4: Exemplary code for a `DAQ` implementation of some instrument with given driver class `DeviceHandle` in the package `mydriver`. The `MyDAQ` class is instantiated with a `DeviceHandle` instance. Optionally, the `DAQSettings` property can be overridden to implement hardware constraints or default values for data acquisition parameters. For this, the `qutil.domains` module provides several classes that represent bounded domains and sets. The `setup()` method parses the given acquisition settings and configures the instrument through the external driver interface `handle.configure()`. The `acquire()` method arms the instrument (if necessary) and loops over the number of outer averages, `n_avg`. In the body of the loop, it can wait for external triggers (or send software triggers) before yielding a batch of data fetched from the external driver interface. Once acquisition is done, the method can return arbitrary metadata to the `Spectrometer` object to attach to the stored data.

```python
def comp_gain(x, gain=1.0, **_):
    return x / gain
```

Listing 3.5: A simple `procfn`, which converts amplified data back to the level before amplification. Note the token `**_` variable keyword argument that ensures no errors arise from other parameters being passed to the function. More complex processing chains can concisely be defined with `qutil.functools.FunctionChain` that pipes the output of one function into the input of the next.

A simple example for a processing function in Fourier space is shown in Listing 3.6, which computes the (anti-)derivative of the data using the fact that

$$\frac{\partial^n}{\partial t^n} \xrightarrow{\text{F.T.}} (i\omega)^n \tag{3.1}$$

under the Fourier transform. In Section 8.3, I discuss more complex use-cases of the processing functionality included in `python_spectrometer` in the context of vibration spectroscopy.

```python
def derivative(xf, f, n=0, **_):
    return xf / (2j * pi * f)**n
```

Listing 3.6: A simple `fourier_procfn`, which calculates the (anti-)derivative.

3.2 Feature overview

Now that we have a basic understanding of the design choices underlying `python_spectrometer`, let us discuss the typical workflow of using the package. Two modes of operation are to be distinguished: first, "serial" mode, in which users record new spectra manually, and second, "live" mode, in which new data is continuously being acquired. The former is well suited to a structured approach to noise spectroscopy where data is retained persistently and discrete changes are made to the system in between subsequent data acquisitions. The latter is aimed at a more fluent workflow in which data is not retained and data acquisition runs in the background.

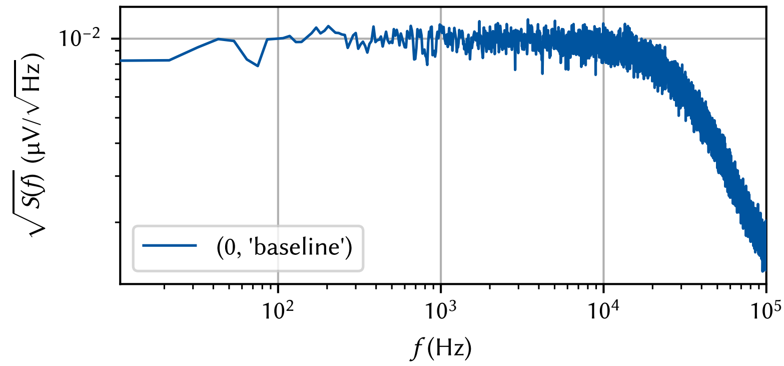

Figure 3.2: The `python_spectrometer` plot after acquiring the (here: synthetic) baseline spectrum. By default, the amplitude spectral density (ASD) = $\sqrt{\text{PSD}}$ is displayed in the main plot. Each spectrum is assigned a unique identifier key consisting of an incrementing integer and the user comment, and can be referred to by either (or both) when interacting with the object.

3.2.1 Serial spectrum acquisition

The default mode for spectrum acquisition using `python_spectrometer` revolves around the `take()` method. Key to this workflow is the idea that each acquired spectrum be assigned a comment that allows to easily identify it in the main plot. For instance, this comment could contain information about the particular settings that were active when the spectrum was recorded, or where a particular cable was placed.

Consider as an example the procedure of "noise hunting", *i. e.*, debugging a noisy experimental setup. The experimentalist,[13] having discovered that his data are noisier than expected, sets up the `Spectrometer` class with an instance of the `DAQ` subclass for the DAQ instrument connected to his sample, a Zurich Instruments MFLI.[14] Choosing to work with the demodulated data to benefit from the corresponding DAQ module's larger flexibility, he recognizes that the resulting PSD will be the two-sided version because the data returned by the LIA is complex. Since he is interested in the physical frequencies[15] he sets the lock-in's modulation frequency to 0 Hz and disables plotting the negative part of the frequency spectrum as it contains only redundant data in this case. Selecting the frequency bounds, say $f_{\text{min}} = 10\,\text{Hz}$ and $f_{\text{max}} = 100\,\text{kHz}$, and using the sensible defaults for the remaining spectrum parameters, Bob first grounds the input of his DAQ to record a *baseline* spectrum. Thus far, his code would hence look something like that shown in Listing 3.7, which produces the plot shown in Figure 3.2.

13: Let's call him Bob.

14: For the MFLI, `DAQ` subclasses for both the Scope and the DAQ module are implemented. The former gives access to the signal before and the latter to the signal after demodulation.

15: A discussion of lock-in amplification is beyond the scope of this chapter. Here I will simply note that, for finite modulation frequencies f_m, LIAs will measure the PSD in the up-converted frequency band $[-f_{\text{max}} + f_m, f_m + f_{\text{max}}]$ rather than the baseband.

```python
from python_spectrometer import Spectrometer, daq
from qutil.functools import scaled

mfli_daq = daq.ZurichInstrumentsMFLIDAQ(session, device)
spect = Spectrometer(mfli_daq, procfn=scaled(1e6),
                     plot_negative_frequencies=False,
                     processed_unit='µV')

settings = {'f_min': 1e1, 'f_max': 1e5, 'freq': 0}
spect.take('baseline', daq='grounded', **settings)
```

Listing 3.7: Setup and serial workflow using the `python_spectrometer` package. `session` and `device` are application programming interface (API) objects of the `zhinst.toolkit` driver package. It is therefore possible to simply use the driver objects that are already in use in the measurement setup. The `procfn` and `processed_unit` arguments help converting raw data into a more human-friendly unit.

The noise spectrum he obtains is white up until approximately 20 kHz where it starts falling off $\propto f^{-n}$. This is because the LIA low-pass filters the signal to suppress aliasing, and hence n is the filter order.[16] After acquiring the baseline, he next ungrounds the DAQ to obtain a representative spectrum of the noise in an actual measurement. He then proceeds by tweaking things on his setup, testing out different parameters,

16: Aliasing effects arise from finite sampling according to the Nyquist-Shannon sampling theorem [36–38]. It states that for a given physical bandwidth, the sampling rate f_s must be at least twice as large to faithfully reconstruct the signal in order to avoid aliasing, *i. e.*, the reverse of the argument we have made for the largest resolvable frequency f_{max}. Some DAQ devices perform internal aliasing rejection while others do not, and it is up to the user to ensure sufficient anti-aliasing.

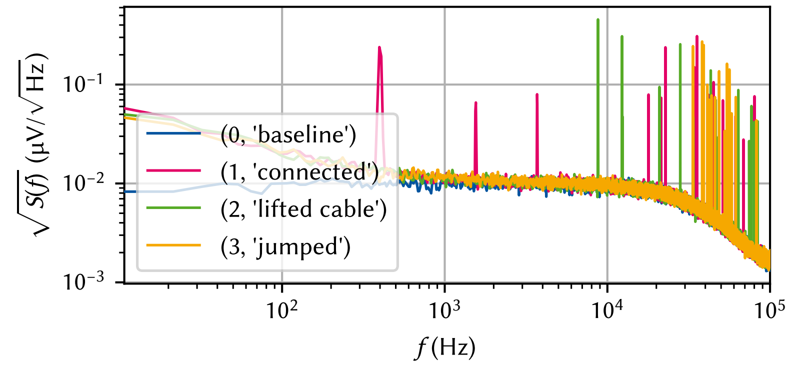

Figure 3.3: The `python_spectrometer` plot after acquiring additional (synthetic) spectra. Each spectrum is uniquely identified by a two-tuple of `(index, comment)`.

etc. Every time he changes something, he acquires another spectrum using `take()`, labeling each with a meaningful comment for identification. The code shown in Listing 3.8 would then leave him with the spectrometer plot as shown in Figure 3.3. While working, Bob realizes he'd like see the signal in the time domain as well. He easily achieves this by setting `spect.plot_timetrace = True`, which adds an oscilloscope subplot to the spectrometer figure as shown in Figure 3.4. Since his DAQ returns complex data, the absolute value $R = |X + iY|$ is plotted.

Listing 3.8: Code to acquire additional spectra. Arbitrary key-value pairs can be passed to the `take()` method, which are stored as metadata if they do not apply to any functions downstream in the data processing chain.

```
settings['daq'] = 'connected'
spect.take('connected', **settings)
spect.take('lifted cable', cable='lifted', **settings)
spect.take('jumped', **settings)
```

Bob now observes that the noise spectra he has recorded display many sharp peaks in particular at high frequencies while the $1/f$ noise floor seems pretty consistent across different measurements.[17] This makes it harder for him to evaluate whether any of his changes are actually an improvement or not. The `python_spectrometer` package allows addressing this by plotting the integrated spectra in another subplot. Bob's spectrometer figure after setting `spect.plot_cumulative = True` is shown in Figure 3.5. In the case that `spect.plot_amplitude == True`, this new subplot shows the cumulative RMS in the band $[f_{\min}, f]$,

$$\mathrm{RMS}_S(f) \equiv \mathrm{RMS}_S(f_{\min}, f), \tag{3.2}$$

and the band power (Equation 2.14) otherwise. The cumulative view helps Bob to assess the severity of individual noise peaks over broadband noise; if the cumulative RMS jumps by a large amount at a certain frequency corresponding to a peak in the spectrum, noise at this frequency contributes significantly to the overall noise power. By contrast, if the cumulative RMS increases smoothly with frequency, the system is dominated by broadband noise, which usually is limited by the instrumentation and thus less amenable to mitigation. In his case, Bob notices that the 400 Hz-peak in the `"connected"` state dominates that spectrum, while higher harmonics are less harmful. On the other hand, the `"lifted cable"` state somehow removed the peak at 400 Hz and instead introduced one at around 900 Hz!

17: I once again refer readers to the lecture materials for details on the physical origins and mitigation strategies for different types of spectral signatures [1, 5].

While the cumulative RMS plot already helps, Bob would like a more quantitative comparison of relative spectral powers. Thus, he rescales the spectra in terms of their relative powers expressed in dB[18] by applying the following settings, which changes the plot to the layout shown in Figure 3.6:

18: Recall that the decibel is defined by the ratio L_P of two powers P_1, P_2 as [39]

$$L_P = 10\log_{10}\left(\frac{P_1}{P_2}\right)\,\mathrm{dB}.$$

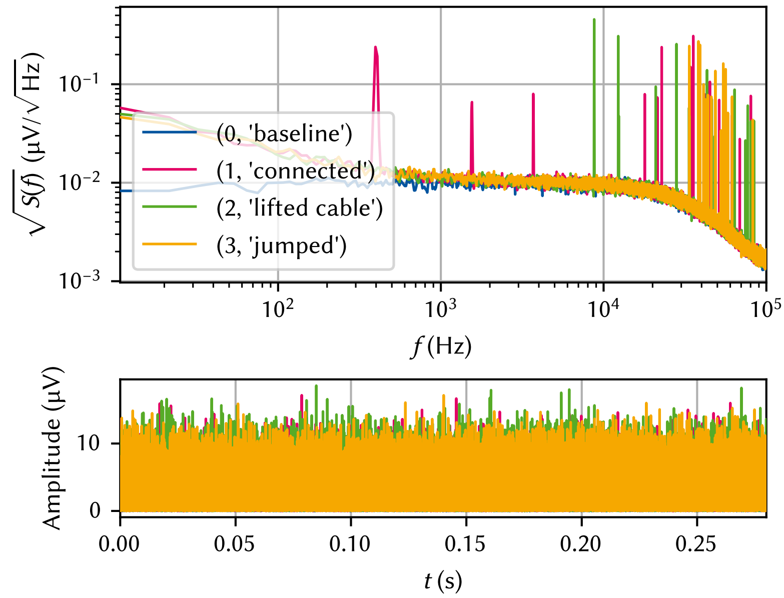

Figure 3.4: The `python_spectrometer` plot shown in Figure 3.3 when setting `spect.plot_timetrace = True`. This adds a subplot that shows the time series data from which the PSD was computed akin to what an oscilloscope would show. For complex time series, the absolute value $R = X + iY$ is plotted. Note that this is the entire time series, *i. e.*, the data of length L, which is (by default, using Welch's method) segmented for spectrum estimation.

```
spect.set_reference_spectrum('connected')
spect.plot_dB_scale = True
spect.plot_amplitude = False
spect.plot_density = False
```

The attribute `plot_density` controls whether the *power spectral density* or the *power spectrum* is plotted.[19] Scaling the data to the power spectrum instead of the density, Bob can get an estimate of the RMS at a single frequency by reading off the peak height. Additionally displaying the data in dB then gives insight into relative noise powers of different spectra. In his case, Bob can tell that the $1/f$ noise is close to 20 dB above the DAQ noise floor at 10 Hz, and that the 400 Hz peak of the `"connected"` state is more than 25 dB above the other spectra.

19: At this point, we *do* need to distinguish between the PSD and the power spectrum counter to Sidenote 9 in Chapter 2. The PSD and power spectrum are related by the equivalent noise bandwidth (ENBW),

$$\text{Spectral density} \xrightarrow{\times\text{ENBW}} \text{Spectrum},$$

which is itself a function of the sampling rate and the properties of the spectral window [23],

$$\text{ENBW} = f_s \frac{\sum_n \hat{w}_n^2}{\left[\sum_n \hat{w}_n\right]^2}. \tag{3.3}$$

Bob carries on with his enterprise and continues to acquire spectra until, finally, he finds the source of his noise! Alas, his spectrometer plot is now overflowing with plotted data when really he just wants to compare the baseline, the original, noisy state, and the final, clean spectrum. He simply calls

```
spect.hide(*range(2, 10))
```

to hide the eight spectra of unsuccessful debugging, leaving him with a plot as shown in Figure 3.7.

Finally, happy with the results, Bob serializes the state of the spectrometer to disk, allowing him to pick up where he left off at a later point in time:

```
spect.serialize_to_disk('2032-12-24_noise_hunting')
```

The next week, Bob is asked by his team about his progress on debugging the noise in their setup. Even though he is working from home that day and does not have access to the lab computer, Bob simply uses his laptop computer and pulls up the `Spectrometer` session stored on the server, allowing them to interactively discuss the spectra:

```
file = '2032-12-24_noise_hunting'
# Read-only instance because no DAQ attached
spect = Spectrometer.recall_from_disk(savepath / file)
```

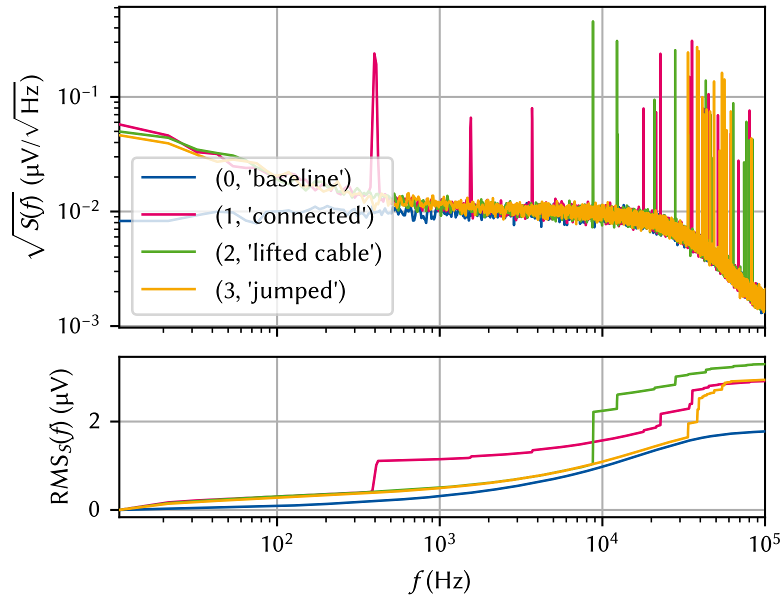

Figure 3.5: The `python_spectrometer` plot shown in Figure 3.3 when setting `spect.plot_cumulative = True`. This adds a subplot that shows the RMS (see Equation 2.14) which can be helpful in evaluating the contribution of individual peaks in the spectrum to the total noise power. Both the oscilloscope subplot (Figure 3.4) and the RMS subplot can also be shown at the same time.

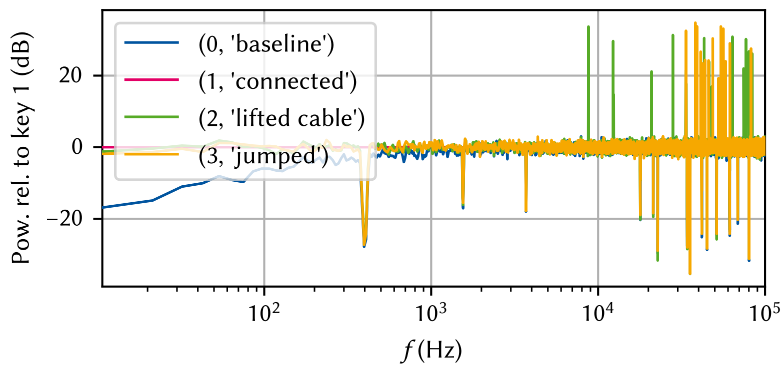

Figure 3.6: The `python_spectrometer` plot in relative mode. Starting from the state in Figure 3.3, we set `spect.plot_dB_scale = True` as well as `spect.plot_amplitude = False` and `spect.plot_density = False` to compare the relative noise powers with respect to the connected baseline.

This opens up the plot shown in Figure 3.7 again. While they cannot acquire new spectra in this state,[20] they can still use all the plotting features like showing or hiding spectra, or changing plot types as discussed above.

20: They could of course always attach a `DAQ` instance to the spectrometer and continue as they were.

3.2.2 Live spectrum acquisition

Manually recording spectra in the workflow outlined in Subsection 3.2.1 becomes tedious at some point, and experimenters tend to become negligent with keeping metadata and comments up to date as they continue to change settings. Once a certain number of spectra has been obtained, the spectrometer plot also becomes crowded, and hiding old spectra manually is cumbersome. Moreover, each time a spectrum is captured, data is saved to disk, potentially accruing large amounts of storage space. For these reasons, the `python_spectrometer` package also offers a non-persistent live mode for displaying spectra continuously.

This mode is facilitated by the `qutil.plotting.live_view` module [35] that provides asynchronous plotting functionality based on `matplotlib`. The `live_view` module supports both the multithreading and multiprocessing paradigms for concurrency in order to keep the interpreter responsive.[21] In the former case, the window hosting the figure runs in

21: Note that technically, data is also recorded in a background thread in serial mode by default.

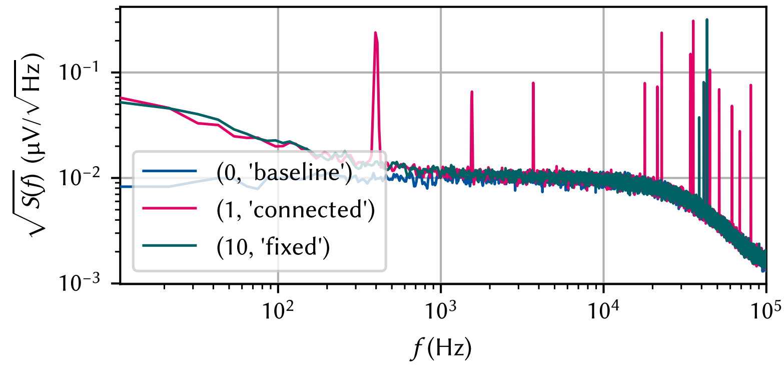

Figure 3.7: The `python_spectrometer` plot after multiple additional spectra were acquired and hidden. Hiding spectra that one is not interested in anymore is achieved through `spect.hide(*range(2, 10))`. This is reversed by `spect.show(*range(2, 10))`. Data can also be dropped (`spect.drop(key)`) or deleted (`spect.delete(key)`) from the internal cache and disk, respectively.

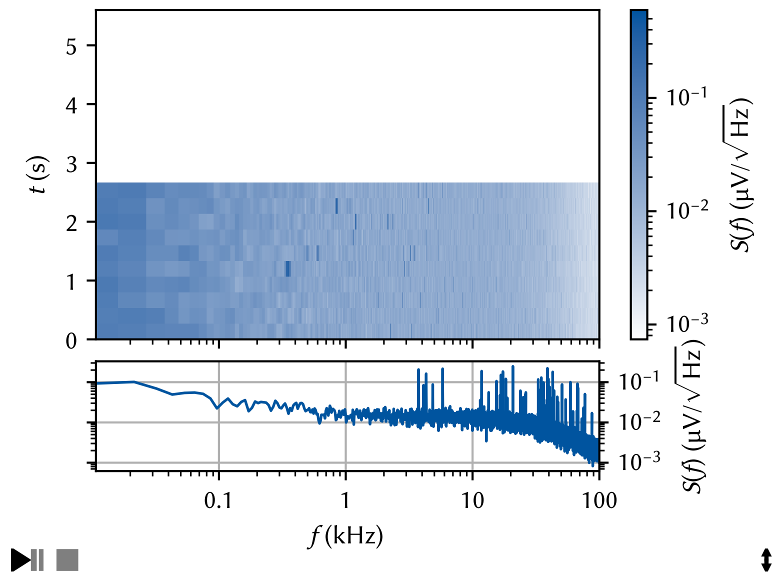

Figure 3.8: Spectrometer live view. The bottom plot shows the most recently acquired spectrum, while the top plot shows a waterfall plot of the most recent ones. Data acquisition runs in a background thread, keeping the interpreter responsive and available to interact with instruments, for example. The icons in the bottom left and right corners allow interacting with the live view.

the main thread and is kept responsive using `matplotlib`'s graphical user interface (GUI) event loop mechanisms. In the latter, the plotting takes place in a separate process, resulting in true parallelism.

The live mode is started with the `Spectrometer.live_view()` method. Data is continuously acquired in a background thread using the same `DAQ` interface as the serial mode. Instead of saving the data on disk and managing plotting from `python_spectrometer`, however, the data is fed into a queue.[22] A `live_view.IncrementalLiveView2D` object then retrieves the data from the queue and handles plotting in the GUI event loop. To start a spectroscopy session with the same parameters as in Subsection 3.2.1 with a given `Spectrometer` object (see Listing 3.7), we would call

```
view = spect.live_view(f_min=1e1, f_max=1e5, in_process=True)
```

which would open a figure window such as that shown in Figure 3.8. Similar to `take()`, the data acquisition parameters are passed to `live_view()` as keyword arguments.[23] The `in_process` argument specifies if multiprocessing (`True`) or multithreading (`False`) is used. Dictionaries with customization parameters for the `live_view` object can further be passed to the method. Functionality to take snapshots to retain live data is currently not implemented but would be a valuable addition to the noise-hunting workflow.

22: A queue is a concurrency mechanism for exchanging data between multiple threads or processes.

23: The difference is, of course, that we do not need to specify a comment since no data is retained.

Conclusion and outlook 4

In this part of the present thesis, I introduced the `python_spectrometer` software package for interactive and backend-agnostic noise spectroscopy. I first presented the theory behind spectral noise estimation based on time series analysis in Chapter 2. There, I showed how, by the Wiener-Khinchin theorem, the PSD $S(\omega)$ is related to a time-dependent signal by the Fourier transform of its autocorrelation function $C(\tau)$. After discretizing the problem, I discussed how the arising spectral windowing introduces distortions in the estimated PSD through effects such as spectral leakage and scalloping loss. The introduction of windowing naturally led to the need for a more efficient method of spectrum estimation to avoid data wastefulness. This is addressed by Welch's method, which trades spectral resolution for estimation accuracy for a given input size by dividing the input into segments and letting them overlap. Finally, I discussed the various parameters that influence the properties of the spectrum estimate produced by Welch's method.

In Chapter 3, I then introduced the `python_spectrometer` package that facilitates noise spectroscopy based on the theory developed in the previous chapter. While in principle agnostic to the specific technique (simple periodogram, Bartlett's method, *etc.*), it employs Welch's method by default to perform efficient estimation of noise power spectral densities. To make noise spectroscopy as approachable as possible, the package provides a unified interface to various hardware and software backends to manage data acquisition. Furthermore, it incorporates rich plotting features to ease data analysis and thereby expedite the feedback loop with the lab. I first outlined the design choices underlying the source code and explained the mechanism employed to infer DAQ settings from, for example, the physical parameters of the resulting spectrum estimate. I described the interface to drivers for different hardware instruments that makes it possible to use the same user code for spectrum acquisition independent of the specific instrument used for data acquisition while at the same time keeping the amount of required driver code at a minimum. Finally, I showcased the interactive features by means of a typical workflow. Here, the serial approach of recording single spectra that are saved to disk at a time stands in contrast to the non-persistent live mode, where data is continuously acquired and displayed in a separate thread or process.

There are several possible avenues for future development of the `python_spectrometer` package. An obvious case is adding support for more DAQ hardware instruments by implementing `DAQ` interfaces. Modular devices such as those offered by QBLOX[1] and Quantum Machines[2] are on track to become the new standard in quantum technology labs. Implementing drivers for these instruments would benefit the adoption of both the instruments and the `python_spectrometer` package. Another valuable addition would be supporting generic instruments abstracted by `QuMADA` [40]. `QuMADA` is a `QCoDeS`-based measurement framework that provides a unified interface to instruments in order to – in a similar spirit to `python_spectrometer`'s `DAQ` interface – abstract away internals of individual instruments and provide users with a standardized way to interact with them. This approach should naturally lend itself to a single implementation of the `DAQ` class supporting various instruments through the unified `QuMADA` interface.

1: https://www.qblox.com/research

2: https://www.quantum-machines.co/

Next, incorporating noise spectroscopy into the standard measurement workflow of quantum device experiments would allow experimentalists to quickly gauge noise levels as they are performing measurements. If for some reason the noise changed[3] the experimentalist could quickly obtain insight into the noise by analyzing the spectrum. In a client-server architecture, which is inherently asynchronous, such as Zurich Instrument's LabOne[4] software, this is already possible using the web interface. But of course the strength of the `python_spectrometer` package stems from its capacity to be utilized in conjunction with any hardware instrument.

3: As it happens often, unfortunately.

4: https://www.zhinst.com/ch/en/instruments/labone/labone-instrument-control-software

One way to implement such functionality would be to introduce a proxy `DAQ` subclass to be used together with the live mode presented in Subsection 3.2.2. This proxy class would serve as an interface to external measurement software and expose two attributes; first, a data queue, into which the external code could place arbitrary time series data that was obtained during some measurement, and second, a shared dictionary to hold acquisition parameters as these might change between measurements. Because the live view mode runs in the background, the external measurement framework could push data to the queue whenever new data was taken without obstructing the measurement workflow.

```python
# daq/tee.py
import dataclasses
import threading
import multiprocessing as mp

from .base import DAQ

@dataclasses.dataclass
class TeeDAQ(DAQ):
    settings:
     ↪ mp.managers.DictProxy
    queue: mp.JoinableQueue
    event: threading.Event

    def setup(self, **_):
        settings = self.
         ↪ DAQSettings(self.
         ↪ settings)
        return settings.
         ↪ to_consistent_dict()

    def acquire(self, **_):
        while not
         ↪ self.event.is_set():
            yield self.queue.
             ↪ get(block=True)
```

Listing 4.1: Template design for a proxy `DAQ` implementation to stream noise spectra from an external measurement framework. The `settings` attribute is a dictionary proxy shared between processes and used to pass acquisition parameters from the measurement framework to `python_spectrometer`.

Listing 4.1 shows a template design for such a `TeeDAQ` class.[5] The `setup()` method ignores the input parameters and instead obtains the current settings from the shared `settings` proxy. Similarly, instead of fetching data from an instrument itself, the `acquire()` method attempts to fetch data from the shared queue and blocks the thread if no data is present, thereby efficiently idling and consuming no resources unless triggered by the external caller. A measurement framework would then interact with the `TeeDAQ` object as exemplarized by the following code:

5: The UNIX shell command `data = $(measure(**settings) | tee daq)` should give an indication as to the naming.

```python
daq = TeeDAQ(...)
spect = Spectrometer(daq)
view = spect.live_view()
...
data = measure(fs, n_pts)
daq.settings.update(fs=fs, n_pts=n_pts)
daq.queue.put(data)
```

Measurement frameworks integrating with this interface could thus provide experimentalists live feedback on current noise levels with negligible overhead and minimal code adaptation.

Finally, it might be useful to not only allow estimating PSDs but also cross power spectral densities (CSDs) or *cross-spectra*. The cross-spectrum[6] is the Fourier transform of not the autocorrelation but the cross-correlation function $C(\tau)$ (Equation 2.5) between two random processes. Take a set of processes $\{x_1(t), x_2(t), \dots, x_n(t)\}$ that correspond to noise measured at different locations in a sample. The cross-correlation function between variables x_i and x_j is then given by[7]

6: Again, we use the two terms interchangably unless otherwise indicated, see Sidenote 9 in Chapter 2.

7: Again assuming wide-sense stationary processes.

$$C_{ij}(\tau) = \left\langle x_i(t)^* x_j(t+\tau) \right\rangle . \tag{4.1}$$

This function (and its Fourier pair the cross-spectrum $S_{ij}(\omega)$) quantifies the degree of correlation between noise at site i and noise at site j. Unlike the *auto*-spectrum (or self-spectrum), the cross-spectrum is always a complex quantity, even for real $x_i(t)$. It is not hard to see that for quantum processors, for example, these kinds of correlations could have significant impact on operation, and on error correction in particular [41–43]. To incorporate cross-spectra in the `python_spectrometer` package, only small changes should be necessary.

First, the data acquisition logic would need to be adapted. Two possible routes suggest themselves here; first, specialized `DAQ` classes could be implemented that, in place of yielding one batch of time series data, yield two batches each time they are queried. This approach first of all requires instruments with multiple channels,[8] which is not necessarily given. Furthermore, it would incur additional coding efforts by having to re-implement each `DAQ` class for cross-spectra. On the other hand, it would arguably make synchronization between channels easier to achieve.

8: If one sticks to single `DAQ` instances managing single instruments.

A less involved path would adapt the `Spectrometer` class to work with multiple `DAQ`s. This would not involve additional driver work[9] and allow the `Spectrometer` object to ensure synchronization between the different `DAQ`s. The internal workflow shown in Listing 3.3 would then need to be slightly adapted to the code shown in Listing 4.2. The downside of this approach is that synchronization of different instruments or channels would need to be taken care of externally.

9: Except possibly ensuring thread safety and timing synchronization if multiple `DAQ` objects communicate with the same physical instrument.

```
daq_1 = MyDAQ(driver_handle, channel=1)
# Or MyOtherDAQ(driver_handle_2) if another instrument
daq_2 = MyDAQ(driver_handle, channel=2)
daqs = (daq_1, daq_2)

parsed_settings = [daq.setup(**user_settings) for daq in daqs]
assert all_equal(parsed_settings), "DAQ settings do not match"

acquisition_generators = [daq.acquire(**parsed_settings[0])
                          for daq in daqs]
for data_buffers in zip(*acquisition_generators):
    estimate_csd(*data_buffers)
```

Listing 4.2: Proposed `DAQ` workflow for estimating cross-spectra. Each hardware channel (same or different instruments) is assigned to a `DAQ` object. After instrument configuration, it is asserted that the parameters match. Finally, data is fetched from both channels and fed into a CSD estimator. Note that triggering would need to be implemented externally.

Further code adaptations would involve minor changes such as replacing the spectrum estimator with `scipy.signal.csd()` [44] for CSD estimation[10] and making the plotting conform to complex data.[11]

10: In practice, working with the normalized CSD, or correlation coefficient [45, 46]

$$r_{ij}(\omega) = \frac{S_{ij}(\omega)}{\sqrt{S_{ii}(\omega)S_{jj}(\omega)}} \tag{4.2}$$

with $S_{ii}(\omega)$ the PSD of process x_i would likely be more favorable.

11: It could be worthwile to add a subplot to display both the magnitude and phase of the complex quantity $S_{ij}(\omega)$.

Part II

Characterization and Improvements of a Millikelvin Confocal Microscope

Introduction 5

Author contributions — *Parts of the results presented in this part of the present thesis have been published in Reference [47]. Thomas Descamps*[a] *and Feng Liu*[a] *originally designed the setup together with Hendrik Bluhm*[b] *and constructed it. Julian Ritzmann*[c] *and Arne Ludwig*[c] *grew the wafer on which Matthias Künne fabricated the sample used to measure the electron temperature in Section 6.2. The chip used to measure photon antibunching in Section 7.2 was grown by Xuelin Jin.*[d] *Marcus Eßer*[e] *kindly lent the accelerometer used in Section 8.2.*

[47]: Descamps et al. (2024), *Millikelvin Confocal Microscope with Free-Space Access and High-Frequency Electrical Control*

a: Then at RWTH Aachen University and Forschungszentrum Jülich GmbH.

b: RWTH Aachen University and Forschungszentrum Jülich GmbH.

c: Ruhr-Universität Bochum.

d: Forschungszentrum Jülich GmbH.

e: RWTH Aachen University.

Quantum technology is maturing and entering the realm of commercial application [48–52]. Despite this progress, its potential is arguably far from fully tapped. From the perspective of the physicist, quantum technology can not only serve the industrial sector, though, but also become a new tool in the researcher's belt to explore new physics on the nanoscale, where quantum mechanics governs the behavior of individual particles as well as many-body interactions and collective effects. Given the multidisciplinarity of the field, this presents a unique opportunity to explore in particular the intersection of different areas of physics such as optomechanics [53, 54], quantum thermodynamics [55–57] and batteries [58], quantum gravity [59, 60], and more.

In the solid state, electrical and optical properties dominate. Here, their intersection brings together two of the perhaps technologically most advanced and permeating branches of physics. The small energy scales involved in electronic *intra*-band effects or many-body correlations often necessitate extremely low temperatures in the Millikelvin range for observation. While optical effects on the other hand are fairly robust due to their large energy scales in the visible range,[1] semiconductors or semimetals such as graphene allow optical *inter*-band transitions and as such coupling between the valence and conduction band, which in themselves are predominantly governed by low-energy physics. Thus, observing many-body effects of quasiparticle excitations such as a Bose-Einstein condensate (BEC) of excitons [61–65], for example, places high demands on the experimental setup, requiring not only very low temperatures but also optical access.

1: Typically quoted as 380 nm to 750 nm corresponding to 1.65 eV to 3.26 eV.

For quantum technologies, the development of a quantum network based on optically interfaced solid-state spins is among the foremost goals [66–69]. There, single optical photons are used to transmit quantum information across long distances and qubits encoded in single spins – electronic or nuclear – are used to store and process information locally. Owing to their small magnetic moment and strong Coulomb interaction, electron spin qubits usually require temperatures well below 1 K for high-fidelity operation, and a coherent interface between these and a "flying qubit" – a photon encoding information, for example, in its helicity or time-of-arrival degree of freedom – would thus also need to be placed in an optically accessible cryostat capable of reaching such low temperatures.

Descamps et al. [47, 70] presented a free-space confocal microscope integrated into a fully wired cryogen-free dilution refrigerator (DR) (Oxford Instruments Triton 450) that facilitates such experiments. In the following, I give a brief outline of the setup described in more detail in Reference 70. The microscope is constructed with the optics fixed in place on top of the cryostat, alleviating the need to remove and re-align them when opening and closing the cryostat. Optical fibers deliver light to the cryostat from the optical table and back, simplifying the connection between the two components. On the optical table, a tunable continuous-wave (cw) Ti:sapphire laser[2] (M Squared Solstis) generates coherent radiation that is attenuated by a variable neutral-density (ND) filter mounted on an electrically controlled rotation stage (Thorlabs K10CR1/M) and can be blocked by an electrically controlled filter flipper (Thorlabs MFF101/M). The attenuated light is coupled into a single-mode fiber (SMF) patch cable (Thorlabs 780HP) and delivered to the optical head on top of the cryostat, which I describe and analyze in more detail in Chapter 7. Waveplates mounted on piezoelectric rotation stages (Attocube ECR4040) with controller Attocube AMC100 allow electromechanical control of the polarization state. The sample is mounted on three piezoelectric linear steppers (Attocube ANPx311) with controller Attocube ANC350 that allow for precise control of the sample position. Light collected from the sample is coupled into another SMF patch cable on the optical head and, on the optical table, launched into a diffraction grating spectrometer (Horiba FHR1000) with a focal length of 1 m NA-matched to the fiber by a pair of singlet lenses. The spectrometer houses two gratings (600 gr/mm and 1800 gr/mm) mounted on a rotating turret. Two exit ports selectable by an electrically controlled mirror let the dispersed light exit the spectrometer onto either a thermoelectrically cooled charge-coupled device (CCD) (Andor iDus 416) or, through an adjustable slit allowing spectral filtering, into a Hanbury Brown-Twiss (HBT) interferometer. The interferometer consists of a 50:50 beam splitter (BS) on whose exit ports the light is focused into two multi-mode fibers (MMFs) connected to an avalanche photodiode (APD)-based single-photon counting module (SPCM) (Excelitas SPCM-850-14-FC) each. A streaming time-to-digital converter (Swabian Instruments Time Tagger 20) then assigns time tags to the output pulses of the SPCMs heralding the arrival of a single photon.

2: A spectrally filtered, pulsed fs-laser (Coherent Mira 900-F) is also available but unused in the present thesis.

The entire setup thus consists of various different components of different nature – mechanical, optical, electrical, or a combination thereof – made by various different manufacturers, making the unified control of all instruments challenging. During the work on the present thesis, I extended the Python driver coverage within the `QCoDeS` framework to include all components that provide an application programming interface (API) for external control. This enables fully remote operation and allows for complete automation of the setup.[3] I furthermore implemented the bottom-loading technique for fast sample exchange, whereby the mixing chamber (MXC) is held at 10 K while the sample puck is removed or replaced using a special adapter from below. Turnaround cycles of approximately one day from base temperature to base temperature can be achieved in this way, in contrast to a full warm-up and cooldown cycle time including removal and replacement of all cryostat shields of one week. The optical alignment turns out to be remarkably stable during this procedure and it is usually not required to manually adjust it after a bottom loading cycle.

3: Remote turn-on of the laser still requires explicit user input for safety reasons.

In this part of the present thesis, I analyze, characterize, and improve upon three different parts of this measurement setup. First, I investigate the refrigeration capabilities of the DR to address the question of how the modifications for optical access impact the cooling performance (Chapter 6). There, I present measurements of various sources of heating and compare them to the available cooling power before turning attention to the achievable electron temperature in a $GaAs/Al_xGa_{1-x}As$ quantum dot. In Chapter 7, I then discuss the confocal microscope itself, that is, the optics mounted to the cryostat, including the objective and ocular lenses as well as the coupling to the SMFs. I review the fundamentals of the relevant physics in order to guide the reader through the lens selection process and subsequently compare the estimated setup efficiency to measurements, focusing on the mode profile of dipole radiation emitted from a point source inside a dielectric slab. Moreover, I characterize the laser spot on the sample using the imaging capabilities of the optical head as well as the cross-polarization extinction performance. To validate the optical performance of the microscope, I demonstrate that signatures of non-classical light, specifically photon anti-bunching from a self-assembled quantum dot (SAQD) single-photon source, can be readily observed. Lastly, in Chapter 8, I address the impact of vibrations introduced by the cryostat's operation on the microscope performance. Given that we would like to resolve features on the micrometer scale, and need to couple light emitted from the sample into a SMF with mode field diameter (MFD) of 5 µm, random displacements induced by the acoustic waves generated by the cryostat's pulse tube refrigerator (PTR) during operation can potentially limit the microscope's capabilities severely. I take a passive air spring suspension approach to decouple the cryostat from a static rigid reference frame and allow it to be displaced freely and in-phase with the external perturbation. Employing two different techniques, I characterize the vibration noise using the tools developed in Part I and show that the vibrations can indeed be attenuated to a degree sufficient for operation of the microscope.

Cryostat performance 6

An essential feature of the confocal microscope discussed in the present thesis is its capability to perform optical measurements at Millikelvin temperatures. As individual quantum systems are singled out for applications in quantum technology, thermal excitations with energy $k_B T = 86\,\mu\text{eV/K} \times T$ quickly become the dominating energy scale and overshadow the desired effects. Hence, typical energies in the solid state on the μeV scale, such as Zeeman energies of individual spins with energy $\mu_B B = 58\,\mu\text{eV/T} \times B$, require temperatures well below 1 K to suppress thermal excitations and maintain $k_B T \ll \mu_B B$.

By design the Oxford Instruments Triton 450 dry DR housing the microscope is rated for base temperatures at the mixing chamber (MXC) plate below $T_{\text{MXC}} = 10\,\text{mK}$ and a cooling power of 450 μW at $T_{\text{MXC}} = 100\,\text{mK}$. However, several factors, both passive and active, introduce additional heat loads that can potentially raise the base temperature if too large:

1. DC and RF wiring. These introduce thermal links between the sample and higher temperature stages as well as add noise that raises the electron temperature in the sample.
2. Wiring and operation of the Attocube ANPx311 nanopositioners. The nanopositioners require special low-impedance connections to ensure a large enough bandwidth for the slip-stick mode of operation. Moreover, the resistive position readout introduces additional heating.
3. Optical access. The free-space optical access requires a direct line-of-sight (LOS) port to the sample. This inevitably allows infrared radiation from room temperature into the cryostat, something that is usually painstakingly prevented when designing a cryostat. Furthermore, optical experiments involve irradiating the sample with a highly focused laser beam, some of which will be absorbed and contribute to heating.

In this chapter, I characterize the cooling performance of the dry DR housing the microscope in various ways. I first discuss the cooling power in terms of the base temperature T_{MXC} reached for different configurations in Section 6.1. This is often quoted as the bath, lattice, or *phonon* temperature,[1] and can be read out using the resistive RuO_2 thermometry setup installed in the DR. Then, in Section 6.2, I present measurements of the *electron* temperature, a quantity that is arguably more expressive of the cryostat's capability to perform precise measurements as it is more sensitive to, for example, electrical noise introduced by the wiring. In order to avoid duplication, I refer readers to References [70, Section 4.1] and [47, Section II] for more details on the cryostat layout and design.

1: This terminology can be misleading as the phonon temperature can also be measured with a quantum transport device, which will most likely see a different phonon temperature than the thermometer on the mixing chamber plate because of a temperature gradient between the heating source close to the sample and sink (the mixing chamber) close to the thermometer.

6.1 Cooling power

The base temperature (T_{MXC}) can be read out from the resistance bridge connected to the gas handling PC controlling the cryostat. Two different thermometers are installed on the mixing chamber, a Cernox® sensor that works in the range from room temperature down to around 2 K for cooldown, and a RuO_2 sensor that works from 30 K down to the Millikelvin regime for base temperature operation. The resistance bridge performs a four-terminal measurement of the sensor resistance and converts it to a temperature according to a sensor-specific calibration $T_{\mathrm{MXC}}(R_{\mathrm{RuO_2}})$. Since the particular sensor installed in the system was only calibrated down to 30 mK, I had to extrapolate the calibration curve as the system reaches lower temperatures than this, and hence the temperatures below this threshold quoted here are not guaranteed to be correct.

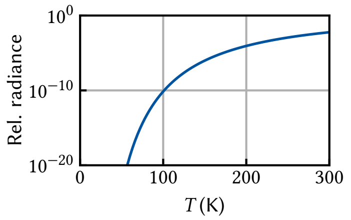

Figure 6.1: Relative black-body radiance obtained by integrating Equation 6.1 from $\lambda = 0\,\mu\mathrm{m}$ to $4.5\,\mu\mathrm{m}$ and normalizing to the total radiance. At $T = 300\,\mathrm{K}$, the fraction of total radiance residing in the high-energy part of the spectrum is still only 0.6 %.

With the configuration described in Reference 70, the DR reached a base temperature of $T_{\mathrm{MXC}} = 30\,\mathrm{mK}$. This included all DC and radio frequency (RF) wiring as well as a single anti-reflection (AR)-coated window sealing the vacuum space on top of the fridge for optical access. To investigate the influence of ambient thermal radiation entering the cryostat through the optical access port, or, more generally, radiation from the higher- to lower-temperature stages, I measured the base temperature for two additional configurations; once with an additional AR-coated window inserted into the optical path on the Cold ($\sim 80\,\mathrm{mK}$) plate, and once with three windows installed on the first pulse tube stage (PT1) ($\sim 50\,\mathrm{K}$), second pulse tube stage (PT2) ($\sim 4\,\mathrm{K}$), and Still ($\sim 800\,\mathrm{mK}$) plates. The windows (Thorlabs WW41050-B) are made from UV fused silica with AR coating.[2] The manufacturer quotes a typical reflectance of 0.25 % in the relevant wavelength range of 700 nm to 1000 nm, while the glass is largely transmissive for wavelengths below $\lambda_{\mathrm{cutoff}} \sim 4.5\,\mu\mathrm{m}$. Since the spectral radiance of thermal black-body radiation [71]

2: Depending on the manufacturer, the coating for the range 650 nm to 1050 nm is called BBAR VIS-NIR or BBAR-B.

$$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{\exp(hc/\lambda k_{\mathrm{B}} T) - 1} \tag{6.1}$$

is small up to $\lambda_{\mathrm{cutoff}}$ at low ($\leq 300\,\mathrm{K}$) temperatures as shown in Figure 6.1, we expect the windows be largely opaque to thermal radiation and hence be quite effective in reducing the radiative heat load. As further the radiative power scales with T^4 and there is at the same time more cooling power available at higher-temperature stages, installing windows there should result in an overall better performance.

Table 6.1: MXC temperature for different configurations of AR coated windows (Thorlabs WW41050-B) inside the DR.

Windows	T_{MXC} (mK)
None	30.0
Cold	11.0
PT1, PT2, Still	7.9

Table 6.1 shows the measured mixing chamber temperatures for the different window configurations. Indeed, already a single window installed on the Cold plate significantly reduces the base temperature. While the windows do introduce additional reflections of the laser spot when imaging the sample, their intensity is low enough, and their position on the camera far enough away from the main spot, to be an issue. On the contrary, the reflections can be quite useful when aligning the laser spots from the two arms of the optical head by aligning them in such a way that the reflections overlap.

Besides the ambient radiation, there is also the heat load introduced by the partial absorption of the laser excitation when conducting optical measurements. In order to both characterize the heat load and measure the absorptance $\mathscr{A}$ of our sample, I irradiated a flip-chip-bonded membrane sample (see Part III) with the laser at 815 nm and measured the increase in base temperature. The sample consists of a 220 nm thick GaAs/$\mathrm{Al}_x\mathrm{Ga}_{1-x}$As membrane glued with epoxy to a Si host substrate chip for handling. As the samples are flip-chip bonded, light that is transmitted through the chip without being absorbed or reflected will then hit the surrounding puck and scatter. Thus, to be precise, I measured $\mathscr{A} + \mathscr{T}$ this way, where $\mathscr{T}$ is the transmittance, assuming that light that exits the membrane will be absorbed *somewhere* and contribute to heating. I will nonetheless refer to the measured quantity as simply $\mathscr{A}$. In Section 13.3, I analyze the absorptance, reflectance, and transmittance of such membrane samples in more detail.

With the power meter mounted in the transmission direction of the excitation arm of the optical head (*cf.* Figure 7.1), I measured the amount of power directed towards the sample by scaling the measured power with the ratio of transmittance and reflectance of the BS, $\mathscr{T} \div \mathscr{R} \approx 15$. Assuming negligible losses on the way towards the sample, I then recorded the mixing chamber temperature as function of laser power. To relate that change in temperature to a heat load $\dot{Q}$ deposited on the sample, I calibrated the cooling power of the DR using the built-in heaters. Setting a heater power and waiting for the mixing chamber temperature to settle yields the cooling power P_{cool} as function of temperature T_{MXC}, which is expected to follow the quadratic relationship [72]

$$P_{\mathrm{cool}} = \dot{Q} = \alpha T_{\mathrm{MXC}}^2 + \beta. \tag{6.2}$$

We can then use a simple model of absorption of the laser on the sample in thermal equilibrium,

$$P_{\mathrm{cool}} = P_{\mathrm{nr}} = \mathscr{A} P_{\mathrm{laser}}, \tag{6.3}$$

where P_{nr} is the amount of power deposited into nonradiative emission channels causing heating of the lattice, to relate the incident laser power P_{laser} to the cooling power and thus obtain the absorptance $\mathscr{A}$.

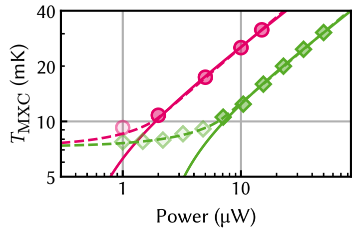

Figure 6.2: Mixing chamber temperature as function of heater (magenta) and laser (green) power. Solid lines are fits to Equation 6.2 including only the solid markers. Green dashed line is a quadratic smoothing spline fit to all laser data points. Magenta dashed line is the laser spline scaled to match the heater data with fitted factor $\mathscr{A} = 28\,\%$, corresponding to the fraction of laser power absorbed and non-radiatively emitted.

Figure 6.2 shows data sets obtained with the MXC heater (magenta) and laser (green). The solid lines are fits to Equation 6.2 including the solid data points only as at low powers there is clearly a deviation from the square-root relationship between the two. Comparing the fit results for α, we obtain $\mathscr{A} = 28.5\,\%$. Alternatively, we can fit a quadratic smoothing spline to the laser data (magenta dashed line) and fit a version scaled according to Equation 6.3 to the heater data (green dashed line). This yields $\mathscr{A} = 28.1\,\%$ in good agreement to the value obtained from fitting the theoretical model (Equation 6.2). Furthermore, the data shows that only at laser powers above 10 µW the MXC heats up appreciably.

Lastly, let us address the impact of the Attocube ANPx311 position readout. The nanopositioners are crucial in the operation of the microscope as they are used to position the sample with respect to the focal spot of the objective lens. They operate in slip-stick mode which naturally generates heat through friction when moving.[3] However, the resistive position readout also contributes to heating, so in order to assess if the readout should be switched off when it is not required to avoid unnecessarily elevated cryostat temperatures, I measured T_{MXC} as a function of readout voltage. Note that the Attocube ANC350 piezo controller offers the option of lock-in measurement for readout, which should be enabled as it improves the signal-to-noise ratio (SNR) and thus allows for a smaller readout voltage V_{AC}.

3: In physics terms, the slip-stick scheme corresponds to alternating adiabatic and diabatic ramps. Slowly ramping up a DC voltage elongates the piezoelectric element by a small amount and moves the positioner table. When quickly jumping back down to 0 V, the table slips and stays in place while the piezo contracts back to its equilibrium elongation. This way, distances much larger than the piezoelectric elongation can be travelled. Staying in the adiabatic regime results in reproducible displacements but also limits the travel.

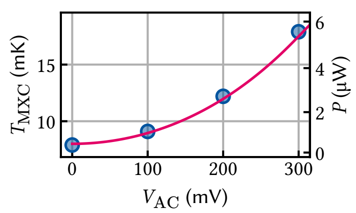

Figure 6.3: Mixing chamber temperature as function of nanopositioner AC readout voltage. The secondary axis indicates the conversion from T_{MXC} to power obtained in Figure 6.2 which is approximately linear in this regime, leading to the expected $P \sim R^{-1} V_{\mathrm{AC}}^2$ behavior. Solid line is a fit to the power with $R = 16.9\,\mathrm{k\Omega}$.

Figure 6.3 shows the temperature as function of V_{AC}, as well as the equivalent power from the calibration performed in Figure 6.2. The expected Ohmic behavior is evident, resulting in a resistance of $R = 16.9\,\mathrm{k\Omega}$. We can conclude that for $V_{\mathrm{AC}} < 100\,\mathrm{mV}$, heating from the positioner readout is negligible. In this regime, the SNR of the measurement is acceptable, and hence the readout can be safely left enabled without affecting the heat budget of the cryostat.

6.2 Electron temperature

In the previous section, I discussed various sources of heating of the cryostat temperature measured at the mixing chamber plate using a commercial resistive thermometer. An arguably more relevant metric for quantum device experiments is the electronic temperature which – depending on how it is measured in detail – corresponds to the temperature of the Fermi distribution of electrons in or coupled to a reservoir rather than the temperature of the crystal lattice. Precisely because the behavior of these quantum devices depends so sensitively on the electron temperature, one can also flip this relationship on its head and use them to *measure* the temperature, an instance of single controlled quantum systems being used as highly sensitive probes of physical quantities known as *quantum sensing* [59]. Gate-defined quantum dots (GDQDs) hosted in the two-dimensional electron gas (2DEG) of a semiconductor heterostructure offer several different ways of measuring the electron temperature that are each different in subtle ways. For example, using an adjacent charge sensor, one can measure the width of a lead[4] (charge) transition [73] or the width of the inter-dot (charge) transition in a double quantum dot (QD) [74]. Simpler still, one can measure the width of a Coulomb resonance in the conductance through a QD [73, 75]. All of these methods rely on some form of energy reference scale that relates the energy of an electron confined in a QD to some externally controlled parameter such as a voltage, however. Typically this is the so-called *lever arm* α, which is the constant of proportionality between the plunger gate voltage and the electrochemical potential μ_N for adding the Nth electron to the quantum dot [75]. It can be measured using pulsed-gate spectroscopy [76, 77], photon-assisted tunneling [78], or finite-bias spectroscopy (Coulomb diamonds) [75, 79], for example.

4: A “lead” is a reservoir of charge carriers coupled to a quantum dot.

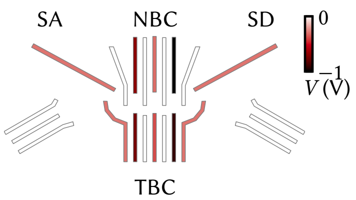

Figure 6.4: Gate layout of a quadruple quantum dot with two charge sensor QDs from Reference 80. Ohmic contacts used for transport measurements are right (left) of SA (SD). NBC and TBC are the gates in the middle of the top and bottom rows. The gates are colored according to the voltages applied with the device hosting a single large QD in the few-electron, sequential tunneling regime.

Here, I chose the simplest of these techniques that require the least tune-up to avoid unnecessarily complicating things. That is, I formed a single QD in a gated GaAs/$\mathrm{Al}_x\mathrm{Ga}_{1-x}$As heterostructure and extracted the lever arm from Coulomb diamonds and the electron temperature from the conductance trace of a Coulomb resonance at zero bias in the sequential tunneling regime. The gate layout of the device used is shown in Figure 6.4. Designed for two-qubit experiments with two-electron spin qubits by Cerfontaine [80], I tuned the device to host a single large QD in the center of the device. The gates are color-coded according to the voltages applied with the device in the few-electron regime. I applied a bias voltage at the Ohmic contact situated to the right of gate SA[5] using a home-built DecaDAC voltage source and measured the current at the Ohmic contact to the left of gate SD using a Basel Precision Instruments SP 983c I/V converter with non-inverting input shorted to ground. Gate voltages were also supplied by the DecaDAC, divided by six and low-pass filtered at room temperature in the break-out box (BOB). All but the four long gates (RFA–RFD) coming in from the top of the device were connected to the filtered DC lines as described in Reference 70.[6] The former were connected to RF lines attenuated with 20 dB, 6 dB, 0 dB, and 3 dB attenuators mounted to the PT2, Still, Cold, and MXC plates, respectively, and connected to a Zurich Instruments HDAWG but unused during these experiments.

5: I use the same naming scheme for gate electrodes as Cerfontaine [80, Chapter 7].

6: The sample is mounted on a high-frequency printed circuit board (PCB) whose on-board capacitors form the second link in a second-order *RC* low-pass filter. Between the PCB and the *z*-axis nanopositioner, there is a Cu plate connected directly to the top interface of the sample puck via flexible braids for heat sinking as the positioners have poor thermal conductivity [47]. However, this plate is also crucial for providing a ground connection for the PCB and, consequently, the functioning of the *RC* low-pass, as the positioners are also not electrically conductive.

To tune the electrochemical potential in the dot, I set up a virtual plunger gate [81] from a linear combination of the gates NBC and TBC by performing a two-dimensional sweep of TBC against NBC. From the slope m of a charge transition in the voltage space spanned by TBC and NBC,

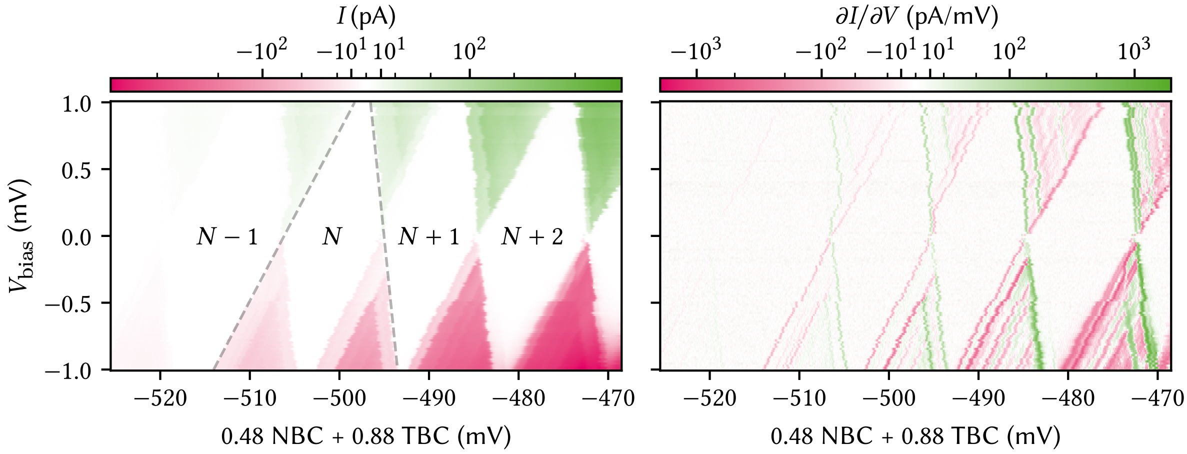

Figure 6.5: Coulomb diamond measurement of the single GDQD confined by applying static voltages to the gates shown in Figure 6.4. The left panel shows the current measured with a transimpedance amplifier (TIA) with a bias applied on one side of the device. The horizontal axis shows the virtual plunger gate swept for each bias voltage indicated on the vertical axis. The difference in slope of the dashed gray lines around the N-electron diamond yields the lever arm α converting virtual plunger gate voltage to (relative) energy inside the dot. From the fact that the slopes are not equal up to a sign we can infer that the coupling to source and drain reservoirs is unequal also, indicating that either the dot is situated closer to SA than to SD, or the topology of the tunneling contacts in the 2DEG is different due to different electrostatic potential. The right panel shows the differential conductance obtained from differentiating the data from the left panel along the plunger gate axis. Clearly visible are several additional transition lines which correspond to excited states inside the dot becoming energetically available within the bias window.

I computed the weights of the virtual gate in terms of the physical gates as

$$V_{\mathrm{p}} = \frac{1}{\sqrt{1+m^2}}\left(\mathrm{TBC} + |m| \times \mathrm{NBC}\right). \tag{6.4}$$

Following this procedure, even a dot that does not lie in the middle between the two gates, *i. e.*, couples more strongly to one than the other gate, should not move in position when changing the virtual gate voltage. Having set up the virtual plunger gate, I performed a Coulomb diamond measurement wherein the (virtual) plunger gate is swept as the source-drain bias voltage is varied.[7]

7: In this case, the source-drain bias is asymmetric as the drain is always grounded and only the electrochemical potential of the source reservoir is changed. This does not qualitatively change the physics of the measurement, though.

The left panel of Figure 6.5 shows the current of the Coulomb diamond measurement. At zero bias current can only flow through the Coulomb-blockaded QD when both source and drain are aligned with an energy level in the dot. Increasing the bias voltage opens up the *bias window*, allowing current to flow as long as the dot's energy level is inside the window. This defines areas of zero (blockaded) current within the space of plunger and bias voltage that takes on the shape of a diamond whose height (along the bias axis) is given by the energy penalty of adding another electron to the dot [75],

$$E_{\mathrm{add}} = \mu_{N+1} - \mu_N = E_c + \Delta E, \tag{6.5}$$

where μ_N is the electrochemical potential of the dot with N electrons, E_c is the charging energy due to Coulomb repulsion, and ΔE is the single-particle energy spacing. At the same time, the width of a diamond (along the gate voltage axis) is given by

$$\Delta V_{\mathrm{p}} = \alpha^{-1} E_{\mathrm{add}}, \tag{6.6}$$

implying that we can extract the lever arm α from the geometry of the diamonds. Drawing lines with slopes m_1 and m_2 along the onsets and offsets of current through the Coulomb resonance at zero bias (dashed gray lines), I obtained

$$\alpha = |\Delta m| = |m_1 - m_2| \approx 0.11\,\mathrm{eV/V}. \tag{6.7}$$

Although not of particular interest for the electron temperature, a Coulomb diamond measurement can also be used to perform excited-state spectroscopy. Computing the differential conductance $\partial I/\partial V_\mathrm{p}$ makes visible a host of additional transition lines in the conducting region of the map (right panel of Figure 6.5).[8] These correspond to excited states of the quantum dot entering the bias window. Their sign depends on the tunnel coupling of the state to source and drain [75].

8: Features inside the blockaded region can also occur, indicative of cotunneling.

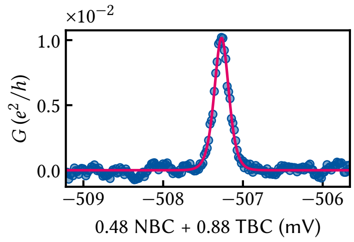

Figure 6.6: Conductance of a Coulomb resonance in the sequential tunneling regime. The magenta line is a fit to Equation 6.8 with $T = 74.9\,\mathrm{mK}$ and $\Gamma = 0.524(5)\,\mu\mathrm{eV}$.

Having measured the energy reference scale, the lever arm α, I proceeded to measure the conductance through a single Coulomb resonance as function of virtual plunger gate voltage V_p in order to measure the electron temperature. The line shape of such a resonance is qualitatively different in two limiting regimes of two competing energy scales, the tunnel coupling Γ of the quantum dot to the leads and the thermal energy $k_\mathrm{B}T$. If the former dominates, transport through the QD is said to be in the resonant (coherent) tunneling regime while if the latter dominates, it is said to be in the sequential (incoherent) tunneling regime [75, 82]. For resonant tunneling, the line shape has the form of a Lorentzian of width Γ. For sequential tunneling, the line shape of a resonance at $V_\mathrm{p}^\mathrm{res}$ is given by [82]

$$G(V_\mathrm{p}) = \frac{e^2}{2h}\frac{\Gamma}{4k_\mathrm{B}T}\cosh^{-2}\left[\frac{\alpha\left(V_\mathrm{p} - V_\mathrm{p}^\mathrm{res}\right)}{2k_\mathrm{B}T}\right] \tag{6.8}$$

in linear response (small bias). Thus, I tuned the device to small tunnel couplings (high tunnel barriers) and measured $G(V_\mathrm{p})$ with a small bias voltage. The resulting conductance trace is shown in Figure 6.6 together with a fit to Equation 6.8. From the fit we can extract the parameters $T = 74.9\,\mathrm{mK} = 6.5\,\mu\mathrm{eV}/k_\mathrm{B}$ and $\Gamma = 0.524(5)\,\mu\mathrm{eV} = 6.08(6)\,\mathrm{mK}{\times}k_\mathrm{B}$, confirming the sequential tunneling regime of $\Gamma \ll k_\mathrm{B}T$.[9] The achieved electron temperature is therefore low enough for high-fidelity qubit experiments with energy splittings $\sim 100\,\mu\mathrm{eV}$.

9: In a different tuning state of the device, the conductance would also sometimes change sign close to the resonance at small biases. The line shape in this configuration was well described by source and drain reservoirs at different temperatures,

$$G(V_\mathrm{p}) \propto f(V_\mathrm{p} - V_\mathrm{p}^\mathrm{res,S}) - f(V_\mathrm{p} - V_\mathrm{p}^\mathrm{res,D}),$$

with the Fermi-Dirac distribution

$$f(V) = \left[\exp(\alpha V/k_\mathrm{B}T) + 1\right]^{-1}.$$

The precise physical mechanism behind this behavior is not understood.

Optical path 7

THE confocal microscope integrated into a millikelvin-temperature cryogen-free DR for free-space optics experiments together with DC and AC electrical control was designed and set up by Descamps et al. [47, 70]. In this chapter, I lay out improvements to the design to improve the optical efficiency of the microscope. I review the relevant relationships between optical parameters, estimate the maximum expected efficiency, and compare it to measurements. Furthermore, I characterize the cross-polarization extinction and outline various schemes I established to automatically control the motorized stages regulating the excitation power and rejection as well as the diffraction grating spectrometer and CCD. Lastly, I demonstrate the setup's efficacy to measure photon anti-bunching in a $g^{(2)}$ measurement on self-assembled quantum dots in InGaAs.

7.1 Light coupling

While the microscope arrangement on top of and inside the cryostat is free space optics to enable imaging of the sample, illumination and collected light are routed to and from the optical table using SMFs. Convenience aside, for the illumination this is a natural choice since the guiding mode of these fibers very closely approximates the fundamental TEM_{00} laser mode [83]. For the collected light, it is less obvious that a SMF is the best choice. Coupling light – of any mode profile – in and out of fibers invariably incurs losses. Because of the small mode field diameters on the order of a few micrometers, aligning the optics for coupling is a sensitive task and subject to external disturbances such as vibrations (*cf.* Chapter 8). Moreover, even for perfect mode matching and alignment, there are reflection losses on the percent level. In Subsection 7.1.2, I discuss the coupling of collected light into the SMF in more detail. Despite these loss mechanisms, the single-mode character of the detection fiber is crucial to the microscope's operation because the cross-polarization extinction critically relies on the spatial filtering of the reflected mode by the fiber [84, 85]. I discuss the cross-polarization extinction in more detail in Subsection 7.1.4.

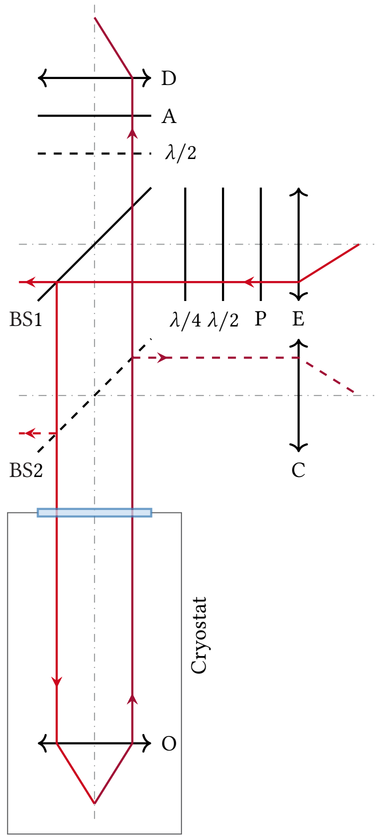

Figure 7.1: Reduced sketch of the microscope optical path. A Gaussian beam is launched from a SMF and collimated by the excitation ocular (E). It is polarized (P), passes $\lambda/2$- and $\lambda/4$-plates, and is reflected into the cryostat by a 90:10 beam splitter (BS). An objective lens (O) focuses the beam onto the sample and collects and collimates the emitted light. It exits the cryostat, is transmitted through the BS and an analyzer (A) before being focused into the SMF by the detection ocular (D). Another $\lambda/2$-plate can be inserted below the analyzer to rotate the plane of polarization, and another beam splitter can be inserted below the first to divert some of the light to a CMOS camera with ocular lens (C). Not shown is the cold mirror that deflects the collimated beam before the objective lens.

7.1.1 Choosing lenses

Figure 7.1 shows a sketch of the free space optical path. There are three lenses that need to fulfil different tasks. First, the excitation ocular (E), which collimates the Gaussian beam launched from the fiber. Next, the objective lens (O), which focuses the beam onto the sample and at the same time collects and collimates the reflected and emitted light. Finally, the collected light is focused by the detection ocular (D) into another fiber for spectral analysis. Since Gaussian beams behave fundamentally differently to geometrical optics, there are different requirements for the lens specifications. In the following, I will review the different beam behaviors and outline the rationale behind the choices made for the lenses.

The fundamental Gaussian TEM_{00} mode has the rotationally symmetric electric field profile [86]

$$E(\rho, z) = E_0 \frac{w_0}{w(z)} \exp\left\{-i\left[kz - \arctan\left(\frac{z}{z_0}\right)\right] - \rho^2\left[\frac{1}{w(z)^2} + \frac{ik}{2R(z)}\right]\right\} \quad (7.1)$$

with the beam waist radius w_0, the beam's $1/e$-radius

$$w(z)^2 = w_0^2\left(1 + \frac{z^2}{z_0^2}\right), \quad (7.2)$$

the wavefront radius of curvature

$$R(z) = z\left(1 + \frac{z_0^2}{z^2}\right), \quad (7.3)$$

the Rayleigh range

$$z_0 = \frac{\pi w_0^2}{\lambda}, \quad (7.4)$$

and where $z = 0$ at the beam waist as well as $\lambda = \lambda_0/n$ the wavelength in the propagating medium. For a SMF, the MFD is $2w_0$ and a beam launched from it expands according to Equation 7.2 with $z = 0$ in its end face.

The Rayleigh range determines the the extent of the mode's near field. At $z = z_0$, the diameter of the beam is $w(z_0) = w_0\sqrt{2}$. In the far field, the beam divergence is given by

$$\theta_{\mathrm{beam}} = \arctan\left(\frac{w_0}{z_0}\right) \approx \frac{\lambda}{\pi w_0}. \quad (7.5)$$

Collimating a Gaussian beam emerging from a SMF thus requires matching θ_{beam} with the numerical aperture (NA) of the lens such that $\mathrm{NA} \geq \sin\theta_{\mathrm{beam}}$. Conversely, coupling a beam into a SMF requires matching the fiber's MFD to the spot size, which is constrained by diffraction. From Equation 7.5 we find, by setting $w = f\tan\theta_{\mathrm{beam}}$, the rule-of-thumb

$$w_0 \approx \frac{\lambda f}{\pi w} \quad (7.6)$$

where w is the beam radius at the focusing lens.[1] For non-Gaussian beams one typically assumes a flattop profile whose diffraction pattern is given by [87],

1: Note that this disregards diffraction at the aperture and is thus only a good approximation for a clear aperture (CA) well larger than w.

$$E(\rho) = E_0 2\pi w^2 \frac{\exp(-ikf)}{f}\frac{J_1(kw\rho/f)}{kw\rho/f}, \quad (7.7)$$

where w is the radius of the lens aperture and $J_1(x)$ is the Bessel function of order one, and quotes the radius of the first Airy disk,

$$w_0 \approx 1.22\frac{\lambda f}{2w}. \quad (7.8)$$

Finally, let us note that the efficiency with which two electric field modes E_1 and E_2 can be matched, the *matching efficiency*, is given by the normalized spatial overlap integral [88],

$$\eta_{\mathrm{m}}(E_1, E_2) = \frac{\int \mathrm{d}S\,|E_1(\rho)|^2 \int \mathrm{d}S\,|E_2(\rho)|^2}{\left|\int \mathrm{d}S\,E_1(\rho)E_2(\rho)\right|^2}. \quad (7.9)$$

Excitation path Now, for as small a spot on the sample as possible, we conclude from Equation 7.6 that we should choose an objective lens with a small focal length f_{ob} (large NA) and illuminate it with a beam with a large diameter $2w$. As the lens diameter and hence the clear aperture (CA)[2] is constrained by the available space in the sample puck, the best lens was found to be Thorlabs 354330-B [89] with $f_{\mathrm{ob}} = 3.1\,\mathrm{mm}$, NA = 0.7, and infinity-side CA = 5 mm.[3] Having chosen the objective lens, we can next select the excitation ocular to match the beam diameter. For our typical excitation wavelengths around 800 nm, the best-matching SMF has MFD = $2w_0 = 5\,\mu\mathrm{m}$ [91]. Again using Equation 7.6 and solving for f, we find $f_{\mathrm{oc}} = \pi w_0 w/\lambda \approx 24.5\,\mathrm{mm}$ when setting $w = \mathrm{CA}/2$. Since w specifies the $1/e$-radius of the beam, we should choose a lens resulting in a collimated beam diameter that is smaller than the CA, *i. e.*, a shorter focal length. The lens that best matches this requirement is Thorlabs A280TM-B [92] with $f_{\mathrm{oc}} = 18.4\,\mathrm{mm}$, resulting in a collimated beam diameter of $2w \approx 3.8\,\mathrm{mm}$.[4] Collimating the Gaussian beam launched from a SMF may be viewed as transforming the beam waist $w_0 \to w$, implying that the Rayleigh range after collimation is $z_0 \approx 14\,\mathrm{m}$ (Equation 7.4), and the objective lens at a distance of $z \sim 1.5\,\mathrm{m}$ is well in the beam's near field with negligible divergence (*cf.* Equation 7.2). With the beam diameter and focal lengths set, we can compute the expected spot size to be $2w_0 \approx 0.84\,\mu\mathrm{m}$. In Subsection 7.1.3 and Section 8.3, I compare this value to measurements.

2: The CA is the diameter over which the lens specifications hold. Outside this diameter, light may still be transmitted but is not guaranteed to behave according to the lens design.

3: A lens with even higher NA exists [90] but I found it to have too short a working distance (WD) to put our flip-chipped samples into focus. Samples with a different mounting strategy might benefit from the slightly increased focusing power of that lens.

4: A lens with larger focal length and hence wider beam diameter and resulting smaller spot size exists [93] but its design wavelength is further off from our typical working wavelengths. It is also unmounted, making its integration into the optical head more cumbersome.

We have thus far addressed illumination of the sample with Gaussian laser light. What now remains to deal with is the reverse direction; that is, collection of the emitted photoluminescence and focusing it into a SMF using the detection ocular lens ("D" in Figure 7.1). Before turning our attention to that task, let us briefly compare the expected performance with the lenses chosen here to those chosen in Reference 70. There, the ocular lens had a focal length of $f_{\mathrm{oc}} = 6.2\,\mathrm{mm}$ and the objective lens $f_{\mathrm{ob}} = 4.51\,\mathrm{mm}$. With these parameters, we obtain a beam diameter of $2w \approx 1.3\,\mathrm{mm}$ just after collimation and a Rayleigh range of $z_0 \approx 1.5\,\mathrm{m}$, implying that the beam broadens by $\sim \sqrt{2}$ by the time it arrives at the objective lens $z \sim 1.5\,\mathrm{m}$ away. This would result in a spot size of $2w_0 \approx 1.3\,\mu\mathrm{m}$, roughly a factor of two larger than with the lenses we chose here.

Detection path To choose an appropriate lens for focusing light into the SMF for spectroscopic analysis, the detection ocular ("D" in Figure 7.1), let us assume that the objective lens is fully illuminated by the emitted light. This results in a beam with diameter corresponding to the infinity-side CA of the objective lens. Further neglecting beam divergence, we must then choose the focal length of the ocular such that the diffraction-limited spot size matches the MFD of the SMF. Taking the beam to have a flattop profile, an assumption that I test more closely in Subsection 7.1.2, the radius of the first Airy disk is given by Equation 7.8. For the objective lens chosen in the previous paragraph, inverting that equation leads to $f \approx 12.8\,\mathrm{mm}$. However, the first Airy disk includes slightly less of the total power than the $1/e^2$ diameter of a Gaussian beam (*cf.* Equation 7.1) corresponding to the SMF's MFD. I therefore chose a slightly larger focal length, leading to the same lens as used for the excitation ocular, Thorlabs A280TM-B, with $f_{\mathrm{oc}} = 18.4\,\mathrm{mm}$ and resulting in a mode matching efficiency for a hypothetical flattop beam of $\eta_{\mathrm{m}} = 78\,\%$.[5] Again comparing the lens chosen here with that by Reference 70 with $f_{\mathrm{oc}} = 6.2\,\mathrm{mm}$, we find that we expect only $\eta_{\mathrm{m}} = 13\,\%$ of a flattop beam's intensity focused onto the fiber end face to couple into the fiber's TEM_{00} mode.

5: Numerical optimization of the mode overlap given by Equation 7.9 for Equations 7.1 and 7.7 results in $f_{\mathrm{oc}} = 21.6\,\mathrm{mm}$ and $\eta_{\mathrm{m}} = 82\,\%$, indicating the choice fits quite well.

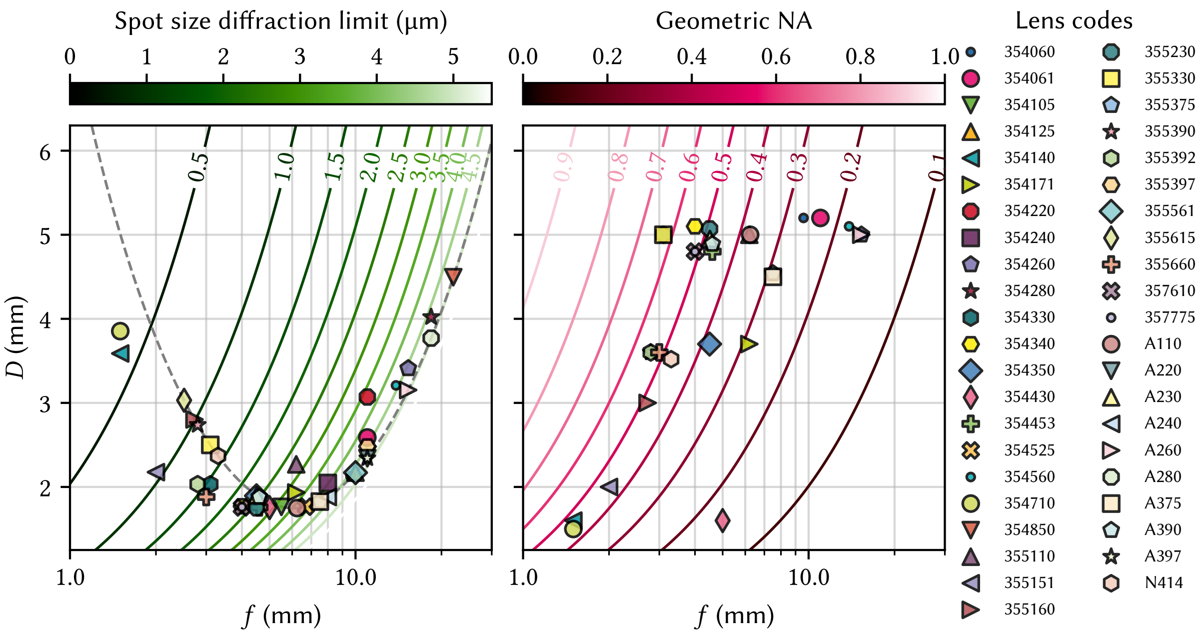

Figure 7.2: Performance of selected lenses from the Thorlabs and Edmund Optics catalogs for Gaussian (left) and geometric (right) optics. The left panel shows the diameter D of a Gaussian beam launched from a SMF with MFD = 5 μm and collimated by the given lens at the position $z = 1.5\,\text{m}$ behind the lens for various models. For lenses with different clear apertures on both sides, I assume a magnification of the beam by their ratio, resulting in the deviation from the theoretical beam diameter shown as a dashed gray line for some lenses, but note that this is a rough approximation for lens underfulling in particular. The green contours show the theoretical spot size diffraction limit, $2w_0$ (Equation 7.6), for a beam with diameter $2w = D$. To the left of the minimum of the (apparent, note the logarithmic scale) parabola, the beam diameter increases with shorter focal length because the beam divergence after the collimating lens becomes relevant as the beam's far field extends to the objective lens plane. The right panel shows the same lenses, this time plotting their (larger, if different) CA diameter D as well as the geometric NA calculated as $\text{NA} = \sin\arctan(D/2f)$ (magenta contours).

The above considerations for choosing lenses are visualized in Figure 7.2, which plots selected lenses from the Thorlabs and Edmund Optics catalog for different scenarios to help selecting models for a specific task. The left panel deals with Gaussian optics and plots the collimated beam diameter D of a Gaussian beam launched from the SMF, collimated by the given lens, and monitored at a distance of $z = 1.5\,\text{m}$ behind it. This takes into account beam expansion due to diffraction, which depends on the focal length of the collimating lens and results in the non-monotonous dependence $D(f)$. The contours indicate the expected spot size of the beam when the objective lens is illuminated by a beam of diameter D. The right panel deals with geometric optics and plots the NA computed from the CA diameter D and the focal length as $\text{NA} = \sin\arctan(D/2f)$.[6] To choose a pair of collimating and objective lenses, first pick the former from the left panel. Then choose an objective lens from the right panel to suit the focusing power needs. Identifying the objective lens in the left panel and drawing a vertical line from its focal length and a horizontal line from the collimator's beam diameter gives the expected spot size of the beam where the two lines cross.

6: Note that for high NA, this value can deviate from the value quoted by the manufacturer.

Having picked the lenses defining the characteristics of the microscope, let us now take a closer look at the expected efficiency.

7.1.2 Collection efficiency

In a confocal microscope geometry, light is collected using the same lens that is also used for illumination of the sample. For excitation with a Gaussian laser beam but non-Gaussian radiation being emitted, this means that two different beam behaviors need to be matched, a task that is likely not possible to achieve completely. In the case of photoluminescence in a pristine semiconductor quantum well (QW), the optical interband transitions are well described by in-plane dipole matrix elements [94]. If, on the other, the light emerges from a photonic crystal cavity for example, the far field pattern is close to a Gaussian mode and the considerations below need to be adjusted accordingly [95]. Here, I discuss dipole emission from the QW, which needs to be coupled into a SMF with near-Gaussian mode profile, invariably resulting in losses. A detailed analysis of the electric field profile to compute the expected coupling efficiency from the sample into the SMF is beyond our scope here as it would require taking into account the full sample and lens geometries as well as diffraction, a task only possible by employing a full-fledged numerical optics simulation suite. However, we can make some crude simplifications of the problem to estimate the order of magnitude of these effects. To this end, I model the light source as a point dipole beneath the surface of a homogeneous slab of dielectric material and the real lenses as ideal thin lenses.

Consider the situation sketched in Figure 7.3. A dipole $\mathbf{p}$ oriented along x in the plane of a GaAs QW with refractive index n buried at a depth d beneath the surface of the sample emits light into the halfspace above it. In spherical coordinates (r, ϑ, φ) oriented along x, that is, embedded in the cartesian coordinate system (y, z, x), the emitted radiation has the field components [96]

$$\mathbf{E}(r,\vartheta) = A(r)\left[E_r(r,\vartheta)\hat{\mathbf{e}}_r + E_\vartheta(r,\vartheta)\hat{\mathbf{e}}_\vartheta\right] \tag{7.10}$$

$$\mathbf{H}(r,\vartheta) = \frac{\epsilon}{\mu}A(r)H_\phi(r,\vartheta)\hat{\mathbf{e}}_\phi \tag{7.11}$$

with

$$A(r) = \frac{|\mathbf{p}|k^2}{4\pi\epsilon}\frac{\exp(ikr)}{r} \tag{7.12}$$

$$E_r(r,\vartheta) = \left(\frac{2}{k^2r^2} - \frac{2i}{kr}\right)\cos\vartheta \tag{7.13}$$

$$E_\vartheta(r,\vartheta) = \left(\frac{1}{k^2r^2} - \frac{i}{kr} - 1\right)\sin\vartheta \tag{7.14}$$

$$H_\phi(r,\vartheta) = -\left(\frac{i}{kr} + 1\right)\sin\vartheta \tag{7.15}$$

and where $\vartheta = \arctan\left(\sqrt{z^2+y^2}/x\right)$, r is the distance from the point dipole, and $k = 2\pi/\lambda = 2\pi n/\lambda_0$ and $\epsilon = \epsilon_r\epsilon_0$ are the wavenumber and the permittivity in the medium, respectively. Since kr is of order unity at $z = 0$, the semiconductor surface is in the dipole's intermediate-field regime where all terms contribute roughly equally.

The emitted light is refracted at the surface and we collect and collimate it with an objective lens (labeled "O") with $\mathrm{NA} = \sin\theta'_\mathrm{m}$ at distance f_ob above the surface of the sample, where f_ob is the focal length and θ'_m the angle of the marginal ray. The NA determines the maximum amount of light the objective lens can collect, and using Snell's law we can relate the angle of a ray outside the sample θ' to the angle inside the sample

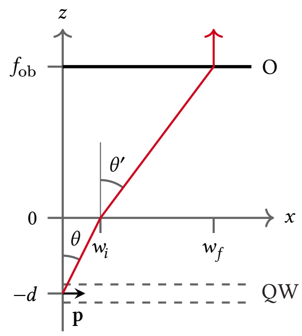

Figure 7.3: Sketch of a light source located inside a dielectric medium ($z < 0, n > 1$) emitting light in the upwards direction to collection by an objective lens in air ($z > 0, n = 1$). The red line indicates the marginal ray of the lens with focal length f_ob and CA $2w$.

$$\theta = \arctan\!\left(\sqrt{x^2+y^2}/z\right),$$

$$\sin\theta' = n\sin\theta, \tag{7.16}$$

with $n \approx 3.57$ at $\lambda_0 = 800\,\mathrm{nm}$ and $T = 0\,\mathrm{K}$. This yields for the emission angle of the marginal ray inside the semiconductor $\theta_\mathrm{m} = \arcsin(\mathrm{NA}/n) \approx 11°$, implying that only a small fraction of light escapes the sample. Indeed, according to Poynting's theorem the radiated power through the surface Σ is given by

$$\langle P\rangle = \int_\Sigma \mathrm{d}\Sigma \cdot \langle \mathbf{S}(r,\vartheta)\rangle \tag{7.17}$$

with the time-averaged Poynting vector

$$\langle \mathbf{S}(r,\vartheta)\rangle = \frac{1}{2}\,\mathrm{Re}(\mathbf{E}(r,\vartheta)\times\mathbf{H}^*(r,\vartheta)). \tag{7.18}$$

As I show in Subsection A.1.1, the fraction of the power radiated into the cone $\theta \in [0,\theta_\mathrm{m}]$ able to be collected by the objective lens, the *collection efficiency*, is thus

$$\eta_\mathrm{c} = \frac{1}{2} - \frac{1}{8n^3}\left(4n^2 - \mathrm{NA}^2\right)\sqrt{n^2-\mathrm{NA}^2} \approx 1.4\,\% \tag{7.19}$$

with $\mathrm{NA} = 0.7$ for the chosen objective lens (*cf.* Subsection 7.1.1).

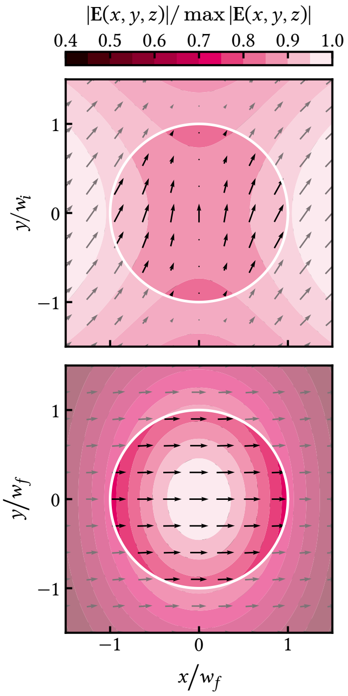

Figure 7.4: Electric field at the sample surface (top) and the objective lens plane (bottom). The white circle indicates the area from which light can be collected, corresponding to the marginal angle $w_i = d\tan\theta_\mathrm{m}$ for the upper and $w_f = \mathrm{CA}/2$ for the lower plot. The arrows represent the projection of the vector-valued electric field onto the xy-plane. At the interface, the polarization is mostly out-of-plane and along y, but in the far field, represented by the lens plane, it is almost perfectly polarized along x. The intensity profile changes from a local minimum at the center to maximal with a roughly circular dependence.

In order to estimate the fiber coupling efficiency of the light escaping the sample and collected by the objective lens focused by the ocular lens "D", we need to consider refraction and transmission of the electric field at the surface, collimation by the objective lens, as well as diffraction at the ocular lens aperture. A detailed accounting of these effects is beyond the scope of this thesis. However, let us at least gain an intuition for the magnitude of these effects. Since we observe the dipole, oriented in-plane inside the QW, from the side, it is useful to rotate the spherical coordinate system so that it is embedded in the coordinate system (x,y,z) as defined in Figure 7.3 with coordinates (r,θ,ϕ).

The magnitude of the electric field vector in the surface plane, which I derive in more detail in Subsection A.1.2, is shown in the upper panel of Figure 7.4. The circle delimits the cone of emission bounded by θ_m (radius w_i) while the arrows indicate the projection of the electric field $\mathbf{E}(x,y,z)$ onto the interface, indicating that the polarization in the plane points mostly along y at this distance from the source. Accounting for refraction and modifying the perpendicular and parallel (s and p) components of the electric field according to Fresnel's equations [87] results in the electric field in the plane of the objective lens at a distance of f_ob shown in the lower panel. Here, the circle indicates the CA of the lens with radius w_f. The picture is quite different from before. First, the field is almost exclusively polarized along x, the dipole axis, as we might have expected. Moreover, the intensity $\propto |\mathbf{E}|^2$ does not depend strongly on the azimuthal angle ϕ at this distance, allowing us to approximate the field as rotationally invariant to estimate the coupling efficiency into the SMF.[7] As shown in Subsection A.1.2, the field after collimation is thus to good approximation given by

$$\mathbf{E}(r,\theta) = \tilde{A}(r)E_x(\theta) \tag{7.20}$$

7: Note that, while a fairly good approximation for the amplitude, this is likely not a good approximation for the phase because $kw_i \sim 1$, meaning that when the light exits the sample the phase is not constant across the surface, and the approximation as a point source just below the surface emitting a spherical wave warrants further investigation, *cf.* Subsection A.1.2.

with

$$\tilde{A}(r) = \frac{|\mathbf{p}|k^2}{4\pi\epsilon}\frac{\exp(ikz)}{r} \tag{7.21}$$

$$E_x(\theta) = \frac{2n\pi\cos\theta\left[\cos\theta + n\nu(\theta) + n\nu(\theta)\cos\theta + \nu(\theta)^2\right]}{[n\cos\theta + \nu(\theta)][\cos\theta + n\nu(\theta)]} \tag{7.22}$$

and where $\nu(\theta) = \sqrt{1 - n^2\sin^2\theta}$. For $\tilde{A}(r)$, we assumed a perfect lens that transforms a spherical wave front with constant phase at constant r into a plane wave with constant phase at constant z.

The radial intensity profile ($\rho = f_{\mathrm{ob}}\tan\theta$ and $r = \sqrt{\rho^2 + f_{\mathrm{ob}}^2}$) given by the absolute value square of Equation 7.20 is shown in the upper panel of Figure 7.5 together with a flattop (magenta) and a Gaussian (green) beam profile for comparison. The intensity drops to about half its maximum at the edge of the lens aperture, $\rho = w$. We may thus expect the mode matching to be qualitatively different from the flattop behavior discussed previously when coupling this beam into a SMF with a guiding mode very closely approximating the Gaussian TEM_{00} mode.

The light that is collected and collimated by the objective lens next passes through the ocular lens in the detection arm which focuses it into the SMF. The image of the beam on the fiber end face is given by the Fraunhofer diffraction pattern generated by the wave (Equation 7.20) incident on the ocular lens aperture, which I give in Subsection A.1.3. The resulting diffraction pattern, Equation A.28, scaled with the radius ρ is plotted in the middle panel of Figure 7.5 together with the corresponding Airy disk result for a flattop beam (Equation 7.7) and the SMF's guiding Gaussian mode. The pattern of the flattop and the more accurate mode profile from Equation 7.20 are quite similar, but noticeably differ from the Gaussian mode at higher radii ρ. The lower panel shows the fraction of power included in a circle of radius ρ, $P(\rho) \propto \int_0^\rho \mathrm{d}\rho'\,\rho' I(\rho')$, demonstrating that we can expect the mode matching to be fairly good. Indeed, evaluating Equation 7.9 for the light field (E_l, Equation 7.20) and the fiber's guiding mode (E_g, Equation 7.1), results in

$$\eta_\mathrm{m}(E_\mathrm{l}, E_\mathrm{g}) \approx 83\,\% \tag{7.23}$$

for our parameters, slightly better than the naive result using the flattop beam. Together with the collection efficiency (Equation 7.19) and accounting for the transmittance of the BS, $\mathcal{T}_{\mathrm{BS}} \approx 87\,\%$,[8] the *optical efficiency* [97] from sample to fiber is thus

$$\eta_\mathrm{o} = \eta_\mathrm{c}\eta_\mathrm{m}\mathcal{T}_{\mathrm{BS}} \approx 1.0\,\%. \tag{7.24}$$

Due to the small intensities when dealing with photoluminescence (PL) emitted from a membrane sample, it is difficult to measure this efficiency directly to compare it to the theoretical expectation. What can be done rather easily is measure the reflected laser power as it is collected by the objective lens. Together with the reflectance obtained from the calibration measurement in Section 8.3, we can then estimate the various efficiencies. With the polarizer and analyzer co-polarized for maximum transmission and the laser s-polarized w.r.t. to BS1 (*cf.* Figure 7.1), I measured the efficiencies listed in Table 7.1. The "Collection + BS1" efficiency corresponds to $\mathcal{T}_{\mathrm{BS}}\mathcal{R}_\mathrm{S}$ with $\mathcal{R}_\mathrm{S}$ the reflectance of the sample and matches reasonably well the expected value of $\mathcal{R}_\mathrm{S} = |(n-1)/(n+1)|^2 \approx 32\,\%$ for GaAs. In Section 13.3, I analyze the reflectance of membrane samples in more detail using transfer-matrix method (TMM) simulations and

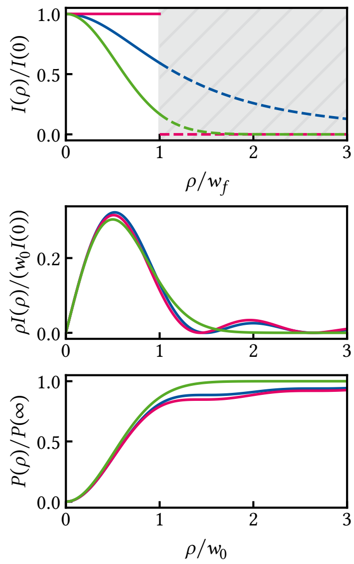

Figure 7.5: Electric field modes. Top: mode intensity of the light collected from the semiconductor at the objective lens plane (blue) in comparison to a flattop (magenta) and Gaussian TEM_{00} mode with theoretical beam diameter after collimating with the ocular lens (green). w_f is the lens CA radius. Middle: diffraction pattern of the collimated beam when focusing onto the SMF end face with the ocular lens (blue), the flattop approximation (magenta), and the fiber's guiding mode (green). The curves are scaled with the radial coordinate ρ to highlight the Airy rings. Bottom: power encased by a circle with radius ρ, $P(\rho) \propto \int_0^\rho \mathrm{d}\rho'\,\rho' I(\rho')$.

8: While the specifications of the beam-splitter are $\mathcal{T} \div \mathcal{R} = 90\,\% \div 10\,\%$, in reality $\mathcal{T} \div \mathcal{R} \approx 87\,\% \div 6\,\%$ which also varies slightly with polarization.

Table 7.1: Measured efficiencies of optical elements along the detection path for laser light reflected from the sample. All efficiencies are with respect to the previous stage, *i. e.*, the first row is the amount of power measured after BS1 divided by the power entering the cryostat

Opt. element	Efficiency (%)
Collection + BS1[a]	37.8
Analyzer[b]	67.6
Detection fiber[c]	39.1
Spectrometer[d]	22.2
Total	2.2

[a] Reflected and transmitted through windows and BS1
[b] Transmitted through the analyzer
[c] Emitted from the detection SMF
[d] Emitted from the spectrometer side exit port

find that actually thin-film interference effects are expected to decrease the bare sample reflectance to around 20 %. The transmittance of the analyzer (a nanoparticle linear film polarizer [98]) is lower than expected from the datasheet, which quotes around 80 % at 800 nm. Coupling into the SMF ("Detection fiber") is significantly worse than expected from the considerations in Subsection 7.1.1 for a Gaussian beam. Since the beam used to measure the efficiencies is launched from and collimated by the same fiber type and lens as used to collect the light, the coupling efficiency should theoretically be as large as unity. However, due to the small MFD, alignment is tricky and vibrations may be expected to significantly reduce the coupling efficiency (*cf.* Chapter 8). It is moreover unclear how well the beam retains its Gaussian TEM_{00} mode profile after traversing the entire optical path (*cf.* Figure 7.1). Lastly, the throughput efficiency of the spectrometer is also lower than expected by a factor of order two, albeit still an improvement over the 12 % reported in Reference 70. The gratings have efficiencies on the order of 50 %, and since the measurement was conducted with monochromatic light, there should be no other significant loss channels, indicating the spectrometer might not be perfectly well aligned.[9] To fully account for all losses, the photon detection efficiency (PDE) (also called quantum efficiency (QE)) of the CCD (Andor iDus 416, $\eta_{\mathrm{QE}} \sim 85\,\%$) or the SPCMs (Excelitas SPCM-850-14-FC, $\eta_{\mathrm{QE}} \sim 65\,\%$) needs to be taken into account. In summary, while certainly allowing room for improvement, the total optical efficiency of $\eta_{\mathrm{o,tot}} = 2.2\,\%$ given by the power arriving at the plane of measurement – either the CCD or the SPCMs – as fraction of the power incident on the cryostat is at an acceptable level.

9: Note that the spectrometer has an adjustable input slit that can be used to limit the spot size and thereby improve the resolution at the cost of reducing overall intensity. However, I have found this not to have a significant impact because the SMF already has a very small MFD and there is little to no improvement from further decreasing it.

7.1.3 Imaging the laser spot

As the confocal microscope is free space, we are in a position to image the sample using a white light source. We can also, though, use the imaging capabilities to inspect the laser spot focused onto the sample and compare it to the behavior expected from Subsection 7.1.1. To this end, I coupled the laser into both excitation and detection arm of the microscope and aligned their spots on top of each other on a gold gate fabricated using optical lithography – ensuring close to perfect reflectivity – by monitoring their image on the Thorlabs DCC1545M CMOS camera (*cf.* Figure 7.1). A feature of known size, for example the width of the gate, can be used to calculate the magnification of the lens system defined by O and C (*cf.* Section 8.3).[10] Once aligned and blocking the beam from each arm in turn, I recorded a picture of the spots.

10: Note that this varies slightly depending on the focal distance of sample and camera from their respective lenses. Theoretically, we expect the magnification to be given by the ratio of their focal lengths, $M = f_{\mathrm{oc}}/f_{\mathrm{ob}} \approx 32$.

From Equation 7.1, we can deduce that a perfect, aberration-free spot would have the two-dimensional intensity distribution

$$I(x, y) = I(0, 0) \exp\left\{-\frac{2x^2}{w_{0,x}^2} - \frac{2y^2}{w_{0,y}^2}\right\}, \tag{7.25}$$

where $w_{0,x(y)}$ are the beam waist radii in x and y-direction, respectively, which are equal in the perfect case and introduce ellipticity else. Allowing for ellipticity as well as a rotation of the coordinate system with respect to the axes of the CMOS camera, we can fit Equation 7.25 to the pictures obtained previously. The result is shown in Figure 7.6 for the excitation path in the upper and the detection path in the lower row. The first column shows a cropped section of the recorded images, showing good alignment (large overlap) between both spots but also some sidelobes along perpendicular axes rotated by around 45° to the camera axes.

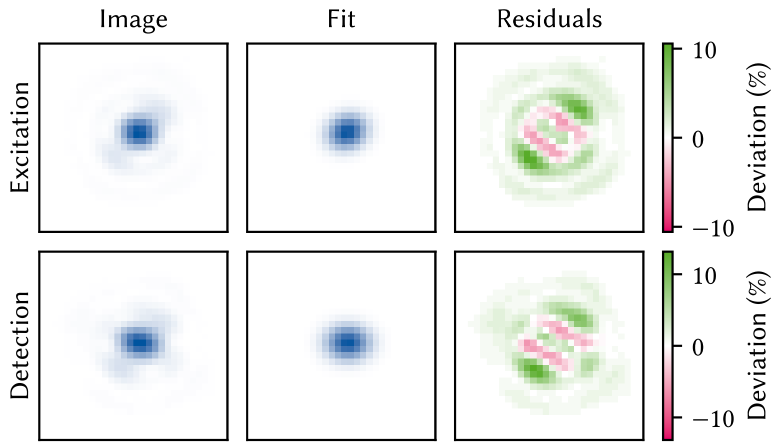

Figure 7.6: Imaging of the laser spots using the free-space imaging capabilities of the confocal microscope. Left column shows cropped images of the spot on an optical gate taken with the CMOS camera, while the middle column shows the fit to Equation 7.25 rotated by a variable amount. Right column shows the residuals of the fit, highlighting non-Gaussian components in the beam. Top row is the excitation, bottom the detection path.

This is also reflected in the relative residuals of the fitted spot that clearly shows that the spot is not described only by a Gaussian TEM_{00} mode. However, extracting the beam waist radii from the fits and scaling with the magnification factors, which display a slight asymmetry between the two axes, 30 to 27.4, we obtain reasonable results between 1.4 and 2 times larger than the diffraction limit given by the lens geometries, *cf.* Subsection 7.1.1, as shown in Table 7.2. For a more faithful measurement of the spot size one typically performs a knife-edge measurement. This has the advantage that it does not depend on additional components of the optical path such as the beam splitter BS2 and focusing lens. In Section 8.3 I perform such a measurement in the context of vibration spectroscopy.

Table 7.2: Beam waist radii extracted from the fits of Equation 7.25 to the images recorded using the imaging path of the confocal microscope.

Beam waist (µm)	$w_{0,x}$	$w_{0,y}$
Detection path	0.58	0.75
Excitation path	0.60	0.84
Diffraction limit	0.42	0.42

From experience, the sidelobes present in Figure 7.6 can be suppressed with better alignment and are most likely due to an imperfect focal distance of the fiber collimating lenses D and E resulting in a secondary (back) focal plane before the objective lens. The asymmetry resulting in elliptical spot cross sections, on the other hand, might be due to several factors including a tilt of the sample or alignment of the imaging arm of the microscope.[11] Finally, note that the excitation spot typically looks worse than the detection spot, likely because of the additional optical elements introducing beam distortions.

11: Notice that the asymmetry is due solely to the magnification factor rather than the image on the camera,

7.1.4 Cross-polarization extinction

An essential part of the confocal microscope's design is the ability to reject the excitation laser from coupling into the detection fiber. In a conventional microscope, this issue is avoided because one can arrange the illumination and detection lenses such that their optical axes are orthogonal and the light used for excitation of the sample does not scatter into the objective lens. In a confocal geometry, though, the same lens is used for excitation and detection, and unwanted excitation light dominates the response unless filtered out by means of, *e.g.*, a notch filter. This limits the optical bandwidth of the microscope and hence other techniques are desirable. Here, excitation rejection is achieved by cross-polarization of the polarizer (P, Thorlabs LPVIS050-MP2) setting the polarization axis of the beam launched from the excitation SMF and the analyzer (A, Thorlabs LPVIS050-MP2) just before coupling the light into the detection SMF. As demonstrated by Benelajla et al. [84], this technique enables extinction ratios[12] up to 10^{10} in a confocal geometry, far beyond the bare polarizer extinction ratio of up to $\sim 10^8$ [98]. In our case, vibrations and rough

12: Defined as incident power over transmitted power.

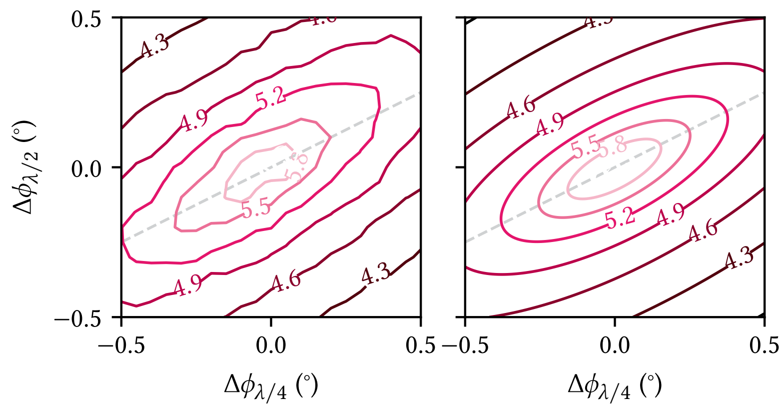

Figure 7.7: OD measured (left) and fitted (right) as function of relative $\lambda/4$ and $\lambda/2$ angles. The dashed gray line indicates the axis along which there should be no dispersion if all optics between polarizer and analyzer were polarization-independent.

sample surfaces off which the light is reflected limit the extinction ratio to $\sim 10^6$.

As the laser beam launched from the excitation fiber is already polarized, rather than rotate the polarizer, the plane of polarization is rotated using a $\lambda/2$ plate. To compensate for ellipticities introduced by the BS, windows, and sample reflection, a $\lambda/4$ plate is inserted after the $\lambda/2$ plate [99]. The waveplates are mounted on piezoelectric rotation stages (Attocube ECR4040 with Attocube AMC100 controller), allowing for computer-controlled adjustment of the angles. Previously, a third rotation stage was used to rotate the analyzer, but I found this to deteriorate the fiber coupling, most likely due to an angle-dependent beam deflection of the analyzer.[13] Furthermore, the optics behind the analyzer such as the diffraction grating have polarization-dependent properties so that it is in any case a good idea to fix the plane of polarization of the detection path after the analyzer. I therefore used only the waveplates on the excitation arm to fine-tune the excitation rejection, which is not ideal since the same argument as before naturally also applies to the optics behind these, such as the beam splitter whose ratio is polarization dependent. It is hence advisable to insert another $\lambda/2$ plate on a rotation stage before the analyzer and use it together with the $\lambda/4$ plate for rejection control as indicated in Figure 7.1.

13: This effect was less pronounced with the unmounted version of the same polarizer, Thorlabs LPVIS050. That though introduced significant etaloning with a free spectral range of $\Delta\lambda \approx \lambda^2/2nl \approx 100\,\mathrm{pm}$ matching the substrate thickness of $l = 2\,\mathrm{mm}$ [100].

To analyze the properties of the excitation rejection, I measured the optical density (OD) as a function of the relative waveplate angles $\Delta\phi_{\lambda/2(\lambda/4)}$. The OD is defined as

$$\mathrm{OD} = \log_{10}\left(\frac{P_\mathrm{i}}{P_\mathrm{t}}\right) = \log_{10}\left(\frac{1}{\mathcal{T}}\right), \tag{7.26}$$

where P_t is the transmitted power after, P_i the incident power before, and $\mathcal{T}$ the transmittance of the analyzer. To evaluate this expression, I measured the photon flux $\dot{\Phi}_\mathrm{t}$ after the spectrometer using the SPCMs when irradiating a sample with the laser and compared it to the power P measured at the optical head. Together with the optical efficiency $\eta_\mathrm{o,tot}$ measured in Subsection 7.1.2, the SPCM's PDE η_QE, and the beam splitter ratio $\mathcal{T} \div \mathcal{R}$ (*cf.* Sidenote 8), Equation 7.26 then becomes

$$\mathrm{OD} = \log_{10}\left(\frac{P \times \mathcal{R}/\mathcal{T}}{hc/\lambda \times \dot{\Phi}_\mathrm{t}/\eta_\mathrm{o,tot}\eta_\mathrm{QE}}\right). \tag{7.27}$$

Figure 7.7 shows a measurement for which the laser was focused on an exciton trap (*cf.* Part III) in the left panel. The dashed gray line indicates the axis along which the OD should theoretically be constant (rotating the $\lambda/2$ plate by $\Delta\phi$ changes the plane of polarization by $2\Delta\phi$, so the $\lambda/4$ plate should need to be rotated by the same amount $2\Delta\phi$ to retain a constant angle between plane of polarization and fast axis of the waveplate). The OD reaches maximum values around 6.4 – corresponding to an extinction ratio of 2.5×10^6 – but quickly falls off by more than an order of magnitude even for small angular displacements.[14] Qualitatively, one finds that close to the extinction maximum (maximal OD), the count rate $\dot{\Phi}_\mathrm{t}$ is a rotated paraboloid of the form

$$\dot{\Phi}_\mathrm{t} = \mathbf{a}^\top \Delta\tilde{\boldsymbol{\phi}}^2 + \dot{\Phi}_0 \tag{7.28}$$

with $\Delta\tilde{\boldsymbol{\phi}} = R(\pi/8 + \theta)\Delta\boldsymbol{\phi}$, $\Delta\boldsymbol{\phi} = (\Delta\phi_{\lambda/4}, \Delta\phi_{\lambda/2})^\top$ and $R(\theta)$ a rotation matrix. Inserting into Equation 7.27 yields

$$\mathrm{OD} = \log_{10}\left(\frac{P\mathscr{R}\lambda\eta_{\mathrm{o,tot}}\eta_{\mathrm{QE}}}{\mathscr{T}hc}\right) - \log_{10}(\mathbf{a}^\top \Delta\tilde{\boldsymbol{\phi}}^2 + \dot{\Phi}_0), \tag{7.29}$$

a fit to which is shown in the right panel of Figure 7.7, displaying good agreement with the data. From the fit, we extract $\theta = 5.6°$, indicating a non-negligible polarization dependence, as well as quadratic dispersion coefficients $\partial^2\mathrm{OD}/\Delta\tilde{\boldsymbol{\phi}}^2 = -0.048/\mathrm{deg}^2$ and $-0.007/\mathrm{deg}^2$ perpendicular (parallel) to the axis along $\pi/8 + \theta$, respectively.

14: The maximum achievable OD depends strongly on the sample surface off which the laser is reflected, see also Reference 101.

The optics between polarizer and analyzer furthermore display chromatic dispersion. When changing the excitation wavelength, the angles of $\lambda/2$ and $\lambda/4$ plate therefore need to be re-optimized. I describe the automatic calibration procedure implemented in the measurement framework to this end in Subsection 12.3.3.

7.2 Exemplary measurement of non-classical light

As a demonstration of the optical capabilities of the setup, I performed second-order coherence measurements of an InAs SAQD in GaAs. SAQDs are optically active quantum dots (OAQDs) that form at random locations during epitaxial growth. They have demonstrated excellent optical properties and show potential for technological applications such as quantum repeaters [69, 102–104]. In particular, SAQDs can be operated as single-photon sources. Because they locally deform the band structure, they can capture and confine excitons. Owing to their small spatial extents, strong Coulomb interaction shifts the energy of excitonic complexes other than the neutral exciton X^0, implying that light emitted at $h\nu = E_{X^0}$ upon recombination is certain to contain a single photon within a time window on the scale of the exciton lifetime. This so-called photon *anti-bunching* behavior[15] can be measured with a HBT interferometer and serves as a fingerprint of single-photon source behavior.

15: Also known as sub-Poissonian statistics.

The HBT interferometer consists of a symmetric ($\mathscr{T} \div \mathscr{R} = 50 \div 50$) beam splitter with SPCMs on each exit port. Photons incident on the BS will be transmitted and reflected with equal probability, and either the detectors at each output will register the photons (a “click”). If the distribution of arrival times of the incident photons is Poissonian, the times at which the two detectors will click will be uncorrelated, and hence the distribution of time differences between clicks on either detector will be uncorrelated as well. This is the classical case. In the case of sub- or super-Poissonian photon statistics, the click rate will be reduced or enhanced for a time difference at zero, implying that two photons are less or more likely to arrive at the same time, respectively. The latter case corresponds to thermal light and *photon bunching*. Here, two photons are more likely to arrive in pairs at exactly the same time. The former case is that of a single-photon source; once it has emitted a photon, a finite amount of time will pass before it can emit another, and hence the two detectors can never register a click at the same time.

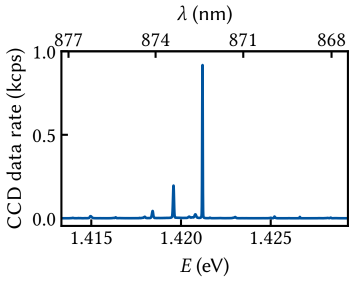

Figure 7.8: PL spectrum of a SAQD under cw excitation at 793 nm.

I loaded and cooled down an InAs/GaAs chip with SAQDs and selected a bright and isolated emission line. A typical PL spectrum under cw above-gap excitation is shown in Figure 7.8. The brightest line at 1.4212 eV = 872.39 nm has a width of around 15 µeV = 12 pm, close to the grating[16] resolution limit of around 5 pm. To perform a $g^{(2)}$ experiment, light emitted by the source is spectrally filtered using the Horiba FHR1000 diffraction grating spectrometer and sent through a 50:50 BS with an Excelitas SPCM-850-14-FC SPCM on each exit port. The clicks from the detectors are time-correlated using a counting card (Swabian Instruments Time Tagger 20), computing the second-order correlation function [105–107]

16: 1800 gr/mm, resulting in a spectral dispersion of $\mathrm{d}\lambda/\mathrm{d}x = 325$ pm/mm in the focal plane.

$$g^{(2)}(\tau) = \frac{\langle \hat{n}_1(t)\hat{n}_2(t+\tau)\rangle}{\langle \hat{n}_1(t)\rangle\,\langle \hat{n}_2(t)\rangle}, \tag{7.30}$$

where $\hat{n}_i(t) = \hat{b}_i^\dagger(t)\hat{b}_i(t), i \in \{1, 2\}$ is the photon-number operator for a single mode at detector i and averaging takes place over time. For a weakly excited artificial atom and in the absence of dephasing, one expects this function to take on the form [106, 108, 109]

$$g^{(2)}(\tau) = [1 - \exp(-\,\tau\gamma/2\,)]^2\,, \tag{7.31}$$

where γ is the Einstein A coefficient, *i. e.*, the spontaneous emission rate or inverse lifetime T_1^{-1} of the excited state. Because T_1 is on the order of 1 ns for these dots, the detector response time (dominated by the timing resolution of the SPCM, $\sigma_{\mathrm{SPCM}} = 350$ ps) plays a significant role in the measurement, and one typically fits a convolution of Equation 7.31 with the instrument-response function (IRF) of the optical setup to the data instead. Ideally, the IRF would be measured using a pulsed laser source. In the absence of this data, we instead make due with the assumption that the IRF is dominated by Gaussian timing jitter with $\sigma = \sqrt{2}\sigma_{\mathrm{SPCM}} \approx$ 500 ps, where σ_{SPCM} is the timing resolution quoted in the SPCM data sheet and the factor of $\sqrt{2}$ is due to the fact that the variance from both SPCMs adds quadratically.[17] The convolution then becomes

17: Realistically, the true IRF of the SPCMs will be some asymmetric rather than a symmetric normal distribution. This was also observed in the pulsed Stark-shift experiment in Reference 110.

$$\begin{aligned}\tilde{g}^{(2)}(\tau) &= \frac{1}{\sigma\sqrt{2\pi}}\int \mathrm{d}\tau'\, g^{(2)}(\tau')e^{-(\tau-\tau')^2/2\sigma^2} \\ &= 1 - 2K_{\gamma/2}(\tau) + K_\gamma(\tau)\end{aligned} \tag{7.32}$$

with

$$K_\alpha(\tau) = e^{\frac{\alpha^2\sigma^2}{2}}\left\{e^{-\alpha\tau}\Phi\left(\frac{\tau-\alpha\sigma^2}{\sigma}\right) + e^{\alpha\tau}\left[1-\Phi\left(\frac{\tau+\alpha\sigma^2}{\sigma}\right)\right]\right\} \tag{7.33}$$

where $\Phi(z) = (1/2)\left[1+\mathrm{erf}\left(z/\sqrt{2}\right)\right]$ is the the Gaussian cumulative distribution function.

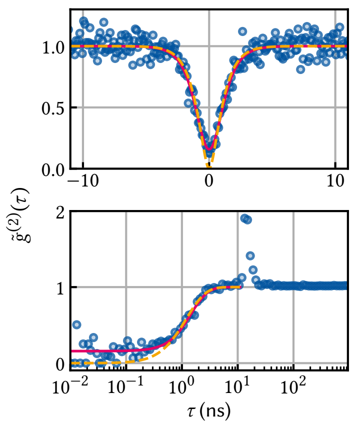

Figure 7.9: $g^{(2)}$ measurement of the emission line at 872.267 nm under cw excitation with 1 µW at 793 nm. The monochromator bandwidth was $\Delta\lambda$ = 200 pm = 325 µeV.

The upper panel of Figure 7.9 shows the measurement after a total run time of around 17 h together with a fit to Equation 7.32 (magenta line) from which we extract a lifetime of $\gamma^{-1} = 430(11)$ ps. The dashed orange line is the deconvoluted function (Equation 7.31). The lower panel shows the same measurement for logarithmically spaced time lags τ covering a wider range. A peculiar feature shows at $\tau = 13.4$ ns, where the measurement suggests a bunching of photons with $g^{(2)}(\tau) \approx 2$. The origin of this bump is not understood. One possible cause might be multiple reflections in the setup. However, in free space the delay corresponds to a time-of-flight distance of 4 m, far larger than any distances in the setup besides the path between objective and ocular lens at ~ 1.5 m. Assuming a refractive index of $n = 1.4$ for a typical SMF results in a characteristic distance of 2.8 m which also does not match any components in the setup.[18] What can be said is that it is related to the setup rather than a physical process in the sample since the feature also appeared in $g^{(2)}$ measurements on different samples. For measurements such as the one performed here, though, this effect may safely be treated as a measurement artefact and ignored. Only in case the characteristic decay time γ^{-1} approached 10 ns this would need to be investigated more carefully. Finally, I note that the dependence of the integrated peak power on excitation power for this line was superlinear, suggesting an excitonic complex rather than the neutral exciton as the source of emission. The line at 1.4196 eV did show a linear relationship and produced qualitatively the same $g^{(2)}$ results.

18: The closest match is the 10 m fiber from the optical head to the optical table.

8 Vibration noise

A microscope's performance is limited chiefly by two factors; first and foremost the resolution and imaging fidelity are limited by the systematic aberrations introduced by the optics.[1] Various types of aberrations exist, and modern microscopes usually include a complex assembly of optics to compensate for these errors. The second factor is vibration noise. This becomes more significant the higher the resolution of the microscope simply because ambient, environmental vibrations within the range of human civilization are typically on the order of 100 µm/s root mean square (RMS) [111]. Comparing that to transmission electron microscopes with atomic resolution, it is clear that these instruments require purpose-built rooms to reduce the vibration level to acceptable levels.

1: Besides the limit set by the wavelength-dependent diffraction, of course.

The demands on the microscope discussed in the present thesis are fortunately much more relaxed as the features we need to resolve are on the micrometer scale. However, we face the additional challenge of ultra-low temperatures, or rather the manner in which they are achieved. The microscope is integrated into a *dry* DR. In contrast to a *wet* DR, which uses a liquid Helium bath, these systems achieve the pre-cooling necessary for the ^{3}He/^{4}He dilution refrigeration cycle to work by adding a secondary refrigeration mechanism, usually a PTR. These are closed-cycle systems that work with ^{4}He compressed to ~21 bar on the high-pressure and ~7 bar on the low-pressure side. A rotating valve connecting high and low pressure lines to the cryostat in turn produces alternating gas flow inside a regenerator, where the gas absorbs heat at the low-temperature and and deposits heat at the high-temperature end [72, 112]. In commercial PTRs the frequency of the pulses of Helium gas, determined by the rotary valve motor, is usually fixed at values around 1.5 Hz.

Naturally, the compressor, the rotary valve motor, and the Helium pulses themselves introduce vibrations into the cryostat. While the cold foot of the PTR is not rigidly connected to the cryostat interior,[2] the entire cold head assembly rests with rubber feet on the cryostat top plate in the system's delivery status. Thus, our microscope does not only encounter passive environmental vibrations but also the active perturbation from the PTR.

2: In the Oxford Instruments Triton 450 copper braids connect the cold head to the PT1 and PT2 plates. There exist commercial systems that use gas exchange instead, for example the CryoConcept HEXA-DRY series [113].

Several authors have addressed vibration decoupling in PTRs. Caparrelli et al. [114] employed active control of a sample stage at 3 K to attenuate vibrations. Pelliccione et al. [115], Oh et al. [116], and Eßer et al. [117] constructed scanning gate microscopes in PTR-cooled systems, demonstrating excellent displacement noise. Kalra et al. [118] investigated the impact of PTR-vibrations on electrical noise while Riabzev et al. [119], Olivieri et al. [120], and Schmoranzer et al. [121] investigated the generation and decoupling of vibrations. Due to space constraints, the complexity of decoupling solutions we can apply to our microscope is limited. Nonetheless, I show in the following chapter that the rigid-body construction of the optics together with passive air spring damping reduces the vibration noise to an adequate level for our purposes.

This chapter is laid out as follows. In Section 8.1, I briefly discuss the theoretical underpinnings of vibration isolation to inform its optimization. To characterize and improve upon the isolation, I performed vibration noise spectroscopy using the techniques and tools presented in Part I. I employed two different approaches that I lay out in the following; first, using a commercial piezoelectric accelerometer (Section 8.2) and second, using the optical response of a spatial reflectance gradient (Section 8.3). As will become clear, the two approaches complement each other because they are sensitive to slightly different quantities.

8.1 Vibration isolation

A simple yet effective method of vibration isolation is to suspend the system on passive air springs. These are typically constructed with two separate air chambers, a spring and a damping chamber, connected by pneumatic tubing. The load is rigidly mounted to a plunger that rests on a diaphragm sealing the spring chamber. Excitations of the load induce oscillations in the variable spring chamber volume. The connection to the fixed-volume damping chamber provides a flow impedance[3] that manifests as a damping force to the spring chamber oscillations.

3: The speed of a fluid in laminar flow through a round pipe is proportional to the pressure gradient along the flow direction and to the square of the distance from the wall.

8.1.1 Damping theory

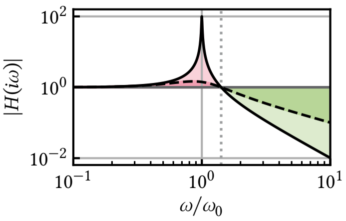

Figure 8.1: Force transmission function of a damped harmonic oscillator with $\gamma = \omega_0/200$ (solid black line) and $\gamma = \omega_0/2$ (dashed black line). Below the break frequency $\omega = \sqrt{2}\omega_0$ (dotted vertical line), external excitations are amplified (shaded red area). For larger damping γ, the amplification at resonance becomes smaller. Above $\omega = \sqrt{2}\omega_0$, excitations are attenuated (shaded green area). Both amplification below and attenuation above the break frequency become smaller as the damping rate γ is increased.

Let us adopt a simple toy model to gain an intuition for the behavior of a mass suspended on air springs as function of vibration frequency by modelling it as a damped harmonic oscillator. Consider the displacement from equilibrium $x(t)$ of the test mass m and switch on an external perturbation $u(t)$ acting on the *base* of the spring, implying that the driving force experiences both the damping rate γ and the spring stiffness $k = m\omega_0^2$ with ω_0 the resonant frequency of the undamped system. We can then compute the transfer function $H(s)$ from the Laplace transform of the Newtonian equation of motion,

$$\ddot{x}(t) + 2\gamma[\dot{x}(t) - \dot{u}(t)] + \omega_0^2[x(t) - u(t)] = 0, \tag{8.1}$$

yielding

$$H(s) = \frac{\hat{x}(s)}{\hat{u}(s)} = \frac{2\gamma s + \omega_0^2}{s^2 + 2\gamma s + \omega_0^2}. \tag{8.2}$$

The magnitude of the transfer function evaluated at $s = i\omega$ is shown in Figure 8.1 for two different dampings, $\gamma = \omega_0/200$ (solid black line) and $\gamma = \omega_0/2$ (dashed black line). Below $\omega = \sqrt{2}\omega_0$ (vertical dotted line), external impulses are in fact amplified. The maximum at the damped system's resonance $\omega_\mathrm{r} = [\omega_0^2 - \gamma^2]^{1/2}$ becomes smoothed out and smaller as the damping γ is increased but never drops below unity. This is the reason why resonance frequencies as small as possible are desirable in vibration isolation. Above this frequency, the system initially attenuates with 40 dB per decade up to $\omega = \omega_0^2/(2\gamma)$ and with 20 dB per decade beyond for $\gamma/\omega_0 \to 0$ (the underdamped case). In the strongly damped case ($\gamma/\omega_0 \to \infty$) the attenuation is only 20 dB per decade starting at $\omega = 2\gamma$.

From Equation 8.2 and Figure 8.1, we can infer two possible approaches to isolating a mass from vibrations. The first is to make the system's resonance frequency ω_0 as small as possible by resting it on a spring damping system. This maximizes the region in which external influences are attenuated. The second is to do the opposite, *i. e.*, make the entire system as stiff (large k) and thereby ω_0 as large as possible. While this minimizes the attenuation region, it also moves the amplification region close to the resonance to higher frequencies, and possibly further away from the external excitation. Consequently, this approach makes most sense if it is known that low-frequency excitations are the dominant source of vibrations.

A widely used metric for the isolation demand of vibration-sensitive equipment are the so-called vibration criterion (VC) [111, 122]. These are design standard specifications for buildings housing, for example, lithography tools. The VCs are defined in terms of band-limited RMS values similar to what we have employed in Part I (*cf.* Equation 2.14). However, instead of computing the band-limited RMS with a fixed lower band edge, one uses bands of a fixed width, typically over one-third of an octave. To be specific, the one-third octave is defined in terms of its midband frequency f_m as the interval

$$f \in f_\mathrm{m} \times \left[10^{-1/20}, 10^{1/20}\right] \approx f_\mathrm{m} \times \left[2^{-1/6}, 2^{1/6}\right] \tag{8.3}$$

whose bandwidth Δf is approximately 26 % larger than f_m and where the latter is defined referenced to 1000 Hz [123]. The 1/3 octave band RMS is then given by

$$\mathrm{RMS}_{1/3\ \mathrm{oct.}}(f_\mathrm{m}) = \sqrt{\int_{f_\mathrm{m}/10^{1/20}}^{f_\mathrm{m}\times 10^{1/20}} \mathrm{d}f\, S(f)}, \tag{8.4}$$

where $S(f)$ is the noise power spectral density (PSD) (see Chapter 2). The criteria, given as velocities rather than displacements or accelerations because it is argued that the limit to photolithography resolution is image velocity, are reproduced from Reference 111 in Table 8.1. For the typical feature sizes we would like our microscope to resolve, the VC-B criterion is a fair target. I will use them below to classify the vibration isolation of the confocal microscope.

Table 8.1: VCs and ISO guidelines [111].

1/3 octave band RMS (μm/s)	
Workshop (ISO)	800
Office (ISO)	400
Residential day (ISO)	200
Op. theater (ISO)	100
VC-A	50
VC-B	25
VC-C	12.5
VC-D	6
VC-E	3

8.1.2 Microscope isolation concept

What does this mean for our case of a dry DR? The rotary valve motor of the PTR generates pulses with frequency 1.4 Hz. Commercial damping systems that the space constraints in our lab allow to be accommodated, for example the CFM Schiller MAS 25 [124], have resonance frequencies around $f_0 = 2.5$ Hz, implying the first two harmonics of the PTR excitation fall into the amplification regime as discussed above. We are thus right in-between the two regimes and it is a-priori unclear which isolation scheme to choose without detailed mechanics simulations. Hence, the initial isolation concept for the cryostat envisaged mounting the rotary valve motor rigidly to the stiff aluminium item profile frame, which was additionally filled with sand to increase the system's resonance frequency.

However, prompted by a sudden increase in visually observed vibrations in the microscope image, I modified the cryostat frame to house three air springs [124] in the hopes of isolating the microscope from external disturbances.[4] To this end, I decoupled the frame from which the cryostat itself is suspended from the support frame resting on the lab ground as sketched in Figure 8.2. Extruding from the square footprint of the support frame at two adjacent corners and the center of the diametrically opposite side, the three air springs are mounted with the base on angle brackets connected to the support frame while their plunger is mounted to a second angle bracket connected to the cryostat frame. The springs

[124]: CFM Schiller GmbH (n.d.), *Type MAS*

4: As it turned out, the cause was a damaged nanopositioner bearing rather than environmental. Fortuitiously, the endeavor still proved successful and resulted in an improved vibration performance as I show below.

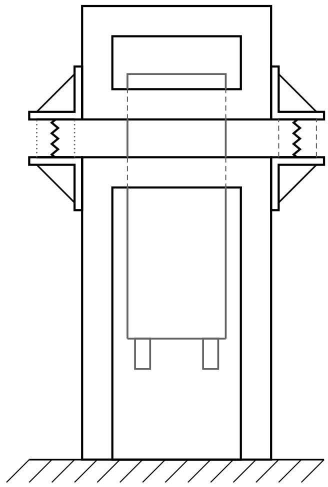

Figure 8.2: Sketch of the cryostat suspension. There are two separate frames (thick black lines), the upper of which the cryostat (grey outline) is attached to. The frames are connected by air springs mounted on angle brackets extruding from the frames.

are connected by pneumatic tubing to a central pressure regulation panel that is connected to the building's central air pressure line. The vertical placement of the springs is chosen such that when the air springs are deflated the cryostat frame rests on the support frame, establishing the same rigid connection that existed previously. This allows examining the influence of the air springs on the vibration isolation without modifying the setup by simply venting the pressurized air from the springs.

In the following, I will characterize the performance of the system with and without the air springs active using two different methods.

8.2 Accelerometric vibration spectroscopy

The most straightforward method of measuring vibration noise is an accelerometer. These are devices that convert translational forces, for example by means of a loaded spring, into electrical signals. They are mounted rigidly to the device under test (DUT) and typically connected to some sort of signal conditioner providing a constant current bias to the sensor and putting out a voltage proportional to the acceleration. The most sensitive and low-frequency designs use piezoelectric materials like Quartz crystals for sensitivities in the range of 10 V/g with a broadband noise floor of 2 µg [125].

In order to evaluate the vibration level at the sample position, I designed a small angle bracket onto which the accelerometer[5] can be screwed either in vertical or horizontal direction in the sample puck of the DR, enabling measurements of the displacement noise along the direction of gravity as well as perpendicular to it and the optical axis. The accelerometer is connected to the coaxial cables installed in the cryostat via an adapter cable from imperial 10-32 to SMA connector. Outside of the cryostat, the signal is routed to a signal conditioner that provides the necessary current bias and outputs a voltage which is digitized by a Keysight 34465A digital multimeter (DMM) connected to the measurement computer. Since the sensor's (conditioned) output is a voltage directly proportional to the acceleration, it is straightforward to compute the displacement PSD from time series data measured with the DMM using the `python_spectrometer` package presented in Chapter 3 [6]. Leveraging the `fourier_procfn` argument, we can transform the voltage data first to acceleration and then, by integration, to displacement in frequency space as indicated in Listing 8.1.

5: Wilcoxon 731–207 kindly lent by Marcus Eßer [125].

```
from qutil.signal_processing
 ↪  import fourier_space
from qutil.functools import
 ↪  chain, scaled
from qutil import const

sensitivity = scaled(1 / 9.9 /
 ↪  const.g)
fourier_procfn = chain(
    sensitivity,
    fourier_space.derivative
)
```

Listing 8.1: Functionality to transform the conditioned voltage to displacement in Fourier space. `sensitivity` is the calibration function from voltage to acceleration.

To assess the impact of the PTR and the suspension, I measured the displacement noise PSD for each combination of the two being switched on and off. The cryostat was closed, its vacuum chamber evacuated, and the magnet, a significant seismic mass, mounted as usual. The measurements are shown in Figure 8.3 together with the band-limited RMS (*cf.* Equation 2.14),

$$\mathrm{RMS}_S(f) = \sqrt{\int_{f_\mathrm{min}}^{f} \mathrm{d}f'\, S(f')} \tag{8.5}$$

with $S(f)$ the displacement noise PSD. When the PTR is switched off, the spectra with and without suspension are dominated by broadband vibration noise, although quite some structure around 15 Hz, 33 Hz, and 60 Hz can be observed.[6] When it is switched on, the PTR pulses at 1.4 Hz and a large number of its higher harmonics visually dominate the spectra.

6: Note the curious peaks slightly offset from the second and third harmonic of the PTR frequency in the spectrum with suspension enabled and PTR disabled. We may speculate that these are due to the PTRs of other cryostats in other labs in the vicinity that are transmitted through the floor. Two were running two rooms over at the time the data was acquired.

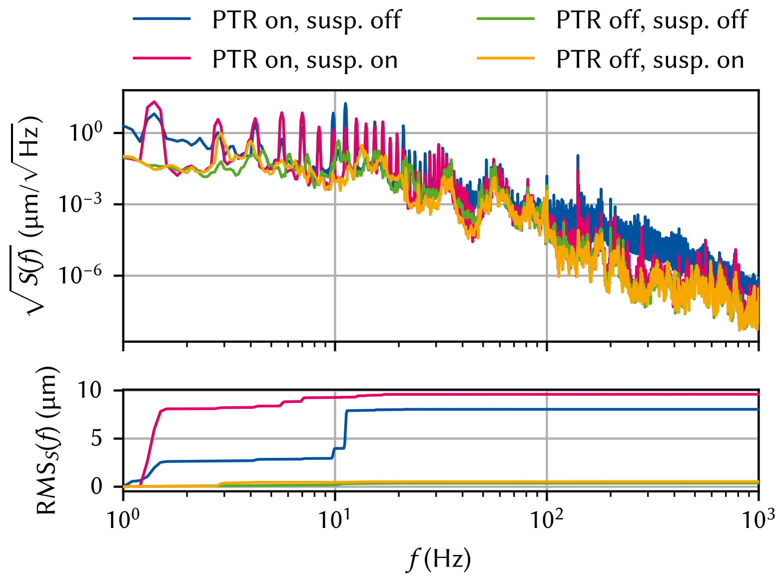

Figure 8.3: Top: displacement noise spectra acquired with the accelerometer at room temperature when the PTR is switched on (blue, magenta) or off (green and orange), and when the air suspension is switched enabled (magenta, orange) or disabled (blue, green). Bottom: band-limited RMS computed from the PSDs in the upper panel (*cf.* Equation 2.14). Turning on the PTR increases the RMS noise amplitude by more than an order of magnitude over the entire frequency spectrum. The suspension slightly worsens the total noise because the low-frequency pulses excite the system close to the air springs' resonance frequency of 2.5 Hz.

Clearly, the suspension has a larger impact in this case, matching qualitatively the behavior discussed in Section 8.1. At high frequencies, it manages to almost completely suppress the broadband excitation observed without the suspension. At low frequencies, on the other hand, the PTR harmonics are amplified to the degree that the band-limited RMS is dominated by their contribution. Only at around 10 Hz, the attenuation starts to take effect. The accelerometer noise floor can be seen to roughly fall off as $1/f^2$, compatible with white noise in the signal conditioning electronics. Overall, the PTR is found to raise the displacement noise RMS amplitude from 0.5 µm to 10 µm while the suspension, over the entire frequency range, has at best no positive influence.

This result is less than encouraging. At that level of RMS-fluctuations, we would have a slim chance of resolving micrometer-scale features using the microscope. But is the *absolute* magnitude of displacement noise at the sample position really the correct measure for the microscope performance? Indeed, if the sample oscillates in phase with the objective and ocular lenses as well as the SMF, we will still obtain a perfect imaging fidelity. So actually only the *relative* displacements of sample, lenses, and detection fiber affect the achievable resolution of the microscope. To characterize these, I developed an optical *in-situ* technique to measure the displacement noise based on knife-edge reflectance fluctuations that I will present in the following section.

8.3 Optical vibration spectroscopy

The gate electrodes on our samples are fabricated using two separate lithography processes; first, the smallest structures are written using electron-beam lithography in two steps. Then, larger structures on the order of 1 µm and above are written using optical lithography. In the region where the two overlap on the mesa to establish electrical contact, the highly reflective Ti/Au optical gates have a width of 14 µm and a height of 160 nm and lie on top of the poorly reflecting GaAs surface, resulting in a step-like reflectance profile. Scanning perpendicularly across such a straight edge between a poorly and a highly reflecting material is known as a knife-edge measurement and is frequently used to measure the spatial extent of a laser spot [126–128]. We can use the same setup to measure the displacement noise; instead of manually shifting our knife edge across the beam spot, though, we measure the reflectance fluctuations induced by the knife edge fluctuating relative to the spot due to external perturbations.

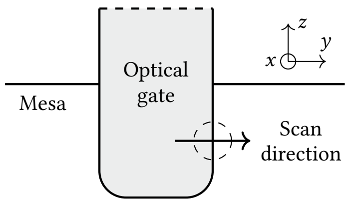

Figure 8.4: Sketch of the region of the sample used for optical vibration spectroscopy. The coordinate system follows the magnet's; z is parallel to gravity, and x is perpendicular to the QW plane. The optical gate extends further north as indicated by the dashed line.

The scenario is sketched in Figure 8.4 in the coordinate system defined by the magnet such that z is along gravity's axis and x is the out-of-plane axis. Focusing the laser (indicated by a dashed circle) onto the edge of the optical gate, we can move the sample using the y-axis nanopositioner and observe a decrease in reflected intensity if the gate is moved away from the laser and an increase if it is moved towards the laser. This gradient in reflected intensity can be inverted to obtain the vibration noise along y by monitoring the intensity as a function of time.

Let us take a closer look at the reflected intensity when the laser spot has a finite overlap with the edge of the gate. Under the simplifying assumption of a perfectly sharp drop-off and taking the reflectance of the gold gate to be unity, we can write the reflectance as function of the coordinate perpendicular to the gate edge at $y = 0$ as

$$\mathscr{R}(y) = \begin{cases} 1, & y \leq 0 \\ \mathscr{R}_0, & y > 0, \end{cases} \tag{8.6}$$

where $\mathscr{R}_0$ is the reflectance of the bare GaAs surface. Assuming a perfect Gaussian (transverse electromagnetic $(\mathrm{TEM})_{00}$ mode) beam characterized by its waist radius w_0 at which the intensity drops to $1/e^2$ of its maximum value, the laser intensity profile in 1D is given by

$$I(y) = I_0 \exp\left(-\frac{2y^2}{w_0^2}\right), \tag{8.7}$$

where $I_0 = P_0/w_0$ with P_0 the total beam power. The power reflected when the spot partially overlaps with the reflectance step can then be expressed as the convolution

$$P_\mathrm{r}(y) = \mathscr{R}(y) * I(y) = \frac{I_0 w_0}{2}\sqrt{\frac{\pi}{2}}\left[1 - (1-\mathscr{R}_0)\,\mathrm{erf}\left(\frac{y\sqrt{2}}{w_0}\right)\right] \tag{8.8}$$

in the yz focal plane, where $\mathrm{erf}(y)$ is the error function.

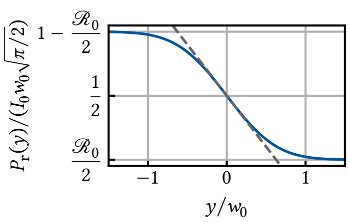

Figure 8.5: Theoretical reflected power for a Gaussian beam of width w_0 and a reflectance contrast of $1 - \mathscr{R}_0$ according to Equation 8.8. The dashed line indicates the leading order approximation at $y = 0$.

The function $P_\mathrm{r}(y)$ is plotted in Figure 8.5. The maximum contrast that can be achieved is given by $1 - \mathscr{R}_0$. Furthermore, for $y \in [-w_0/2, w_0/2]$ the function is well-approximated by

$$P_\mathrm{r}(y) \approx -I_0(1-\mathscr{R}_0)y + \frac{I_0 w_0}{2}\sqrt{\frac{\pi}{2}} \tag{8.9}$$

drawn as a dashed line. Since we measure the photon count rate rather than the power, $\Phi = P\lambda/hc$ with λ the laser wavelength, we rewrite this as

$$\Phi_\mathrm{r}(y) = -sy + \frac{\Phi_0}{2}\sqrt{\frac{\pi}{2}}. \tag{8.10}$$

where we defined the *sensitivity*

$$s = \frac{\Phi_0}{w_0}(1-\mathscr{R}_0). \tag{8.11}$$

Hence, to obtain a more sensitive probe for vibrations, meaning that small variations in y lead to large variations in Φ_r, one could either improve the reflectance contrast $1-\mathscr{R}_0$, decrease the spot size w_0, or increase the incident photon flux Φ_0.[7] In our case, the former two are fixed by the sample and the setup, respectively , whereas the latter is limited by the maximum data transfer rate of the Swabian Instruments Time Tagger 20 counting card, 9 MSa/s.

7: Note that the smaller w_0, the smaller also the maximum displacement amplitude that can be resolved as the derivative goes to zero as $y \to \pm\infty$.

Starting from Equation 8.10, it is straightforward to obtain the displacement in the vicinity of $y = 0$ as function of photon flux,

$$y(\Phi_\mathrm{r}) = \frac{w_0}{1-\mathscr{R}_0}\left[\frac{1}{2}\sqrt{\frac{\pi}{2}} - \frac{\Phi_\mathrm{r}}{\Phi_0}\right]. \tag{8.12}$$

To summarize, we can position the laser spot on the edge of an optical gate and record a time trace of the photon flux by using the Time Tagger to count the photons detected by the APDs mounted on the side exit of the spectrometer. Using Equation 8.12 we can then convert the flux into a displacement and proceed with the usual spectral noise estimation as explained in Part I.

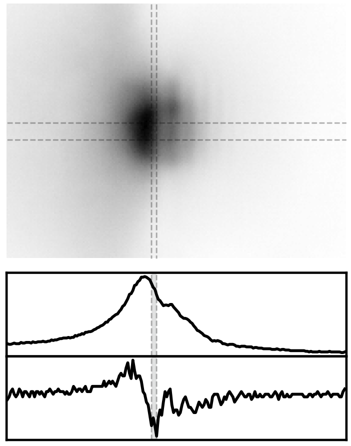

Figure 8.6: Calibration of the length reference scale. The top shows a CMOS camera image (higher intensity darker) of the white light spot on the edge of the optical gate as indicated in Figure 8.4. Several diffraction lines can be seen parallel to the edge. The vertical dashed lines indicate the region in which the intensity slope was fitted. The horizontal dashed lines indicate the extent of rows averaged over. The lower plots show a line cut along the central row of the considered region (top) and its derivative (bottom).

I will now lay out the experimental procedure of calibrating the system to (implicitly) obtain the parameters w_0, $\mathscr{R}_0$, and Φ_0. The first challenge is obtaining a proper length reference scale. While the nanopositioners on which the sample is mounted do in principle have a resistive position readout, it is extremely unreliable at small displacements. Therefore, I calibrated the relative position using the imaging arm of the optical head. Figure 8.6 depicts the procedure. Illuminating the sample with the white light, I positioned the spot on the edge of the optical gate and imaged the sample with the Thorlabs DCC1545M CMOS camera. I then extracted the position of the edge, in pixels, for several rows to obtain some statistics by fitting a linear function to the edge profile in a small region between two refraction maxima. Repeating this step for different DC voltages applied to the nanopositioner, this yields the proportionality factor between the nanopositioner DC voltage, V_DC, and the position of the gate edge on the camera. By measuring the total width of the gate on the camera image, I obtained the magnification by referencing it to the design width,

$$M = \frac{w[\mathrm{px}]}{w[\mathrm{\mu m}]} = \frac{116\,\mathrm{px}}{14\,\mathrm{\mu m}} \approx 8.3\,\mathrm{px/\mu m}. \tag{8.13}$$

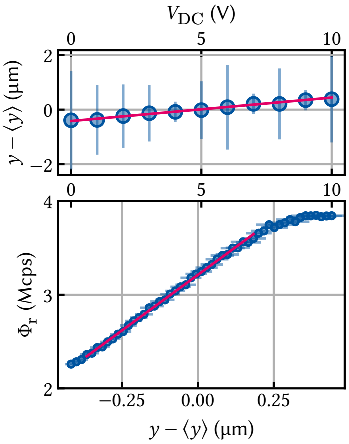

Figure 8.7: Top: linear fit of the edge positions extracted from the analysis in Figure 8.6. Error bars are propagated standard errors of the weighted average of edge positions extracted from different rows. Bottom: laser photon count rate as function of position set by the nanopositioner. Fitting the region $V_\mathrm{DC} \in [0.5, 7]\,\mathrm{V}$ yields $s \approx 2.36(2)\,\mathrm{Mcps/\mu m}$ (*cf.* Equation 8.10). Error bars on Φ_r show the standard error of the mean over multiple observations and error bars on y show the fit error from the top panel.

Again performing a linear fit to the data for different voltages then results in the linear transformation from DC voltage to position (upper panel of Figure 8.7). Of course it is also possible to fit the full knife-edge function, Equation 8.8, to the data shown in Figure 8.7. From this, the spot size w_0 and the actual bare GaAs surface reflectance $\mathscr{R}_0$ can be extracted. Here we are content with the linear approximation; refer to Subsection A.2.1 for the full fits.

Lastly, I switched from white light illumination to the laser, focused it onto the edge of a gate, and measured the photon count rate reflected off the sample as a function of V_DC, from which we can finally extract the desired sensitivity (slope) $s \approx 2.36(2)\,\mathrm{Mcps/\mu m}$ of count rate over displacement. The data and fit are shown in the bottom panel of Figure 8.7. Clearly, the count rate is linear in the displacement over a large range, indicating that for fluctuations with amplitude on the order of 100 nm RMS, the measurement sensitivity should be sufficiently robust.

We are now at last able to measure the displacement noise using `python_-spectrometer`. Setting `procfn` to the linear transformation given in

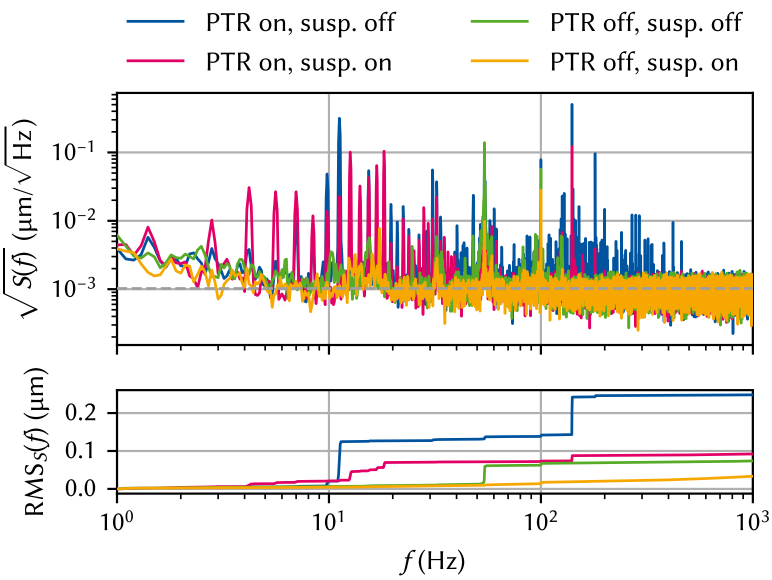

Figure 8.8: Top: displacement noise spectra acquired with the optical *in-situ* method at room temperature when the PTR is switched on (blue, magenta) or off (green and orange), and when the air suspension is switched enabled (magenta, orange) or disabled (blue, green). The dashed gray line indicates the theoretical noise floor derived in Subsection 8.3.1. Bottom: band-limited RMS computed from the PSDs in the upper panel (*cf.* Equation 2.14). The PTR has a much smaller effect than when measuring the absolute displacement noise with the accelerometer, increasing the RMS only by a factor of two. While the lowest-order PTR harmonics are amplified by up to an order of magnitude in amplitude with the suspension enabled, they contribute relatively little to the total RMS and are compensated by the superior high-frequency attenuation behavior. The total $\text{RMS}_S \approx 100\,\text{nm}$ with the cryostat in operation is below the typical µm feature size.

Equation 8.12 and measuring the counts registered by the APDs using the Time Tagger,[8] I obtained the displacement noise PSDs shown in Figure 8.8. A few things stand out. First, rather than the f^{-2} background observed with the accelerometer in Section 8.2, the noise floor is white ($S(f) = \text{const.}$) at approximately $1\,\text{nm}/\sqrt{\text{Hz}}$. To understand why, we need to take a closer look at the counting statistics, which we will postpone for a bit in order to first finish our discussion of the noise spectra. Second, the overall noise level is much – by a factor of 20 RMS – lower than with the accelerometer. We can attribute this to the fact that the optical method is sensitive to relative rather than absolute displacements. If the cryostat and the optical head were infinitely stiff we would measure no displacement noise with this method – intrinsic noise floor notwithstanding – whereas the accelerometer is still sensitive to oscillations of the cryostat on the air spring fulcrum. In that sense the optical method gives us the more pertinent results because only the displacements seen by the light travelling through the microscope ultimately matter. Third, in contrast to the accelerometer measurements, the RMS amplitude is reduced by half when the suspension is active. Although the harmonics of the PTR frequency of 1.4 Hz are again amplified by the suspension below 10 Hz, raising the band-limited RMS above that with the suspension disabled, there occurs a crossover at the eighth harmonic frequency beyond which the attenuation outweighs the amplification at low frequency.[9]

8: Since the APDs are arranged in a HBT geometry, I combined the counts of both instruments using the virtual channel functionality of the Time Tagger.

9: It furthermore appears that even a measurement whose sole electronic device is a picosecond-resolution counting card cannot escape 50 Hz power line noise (or in this case its second harmonic).

8.3.1 Noise floor

The noise floor in the optical vibration measurements shown in Figure 8.8 is qualitatively very different from that observed with the accelerometer. There, the *acceleration* noise floor was white,[10] whereas with the optical method the *displacement* noise floor is white, hinting at a different underlying mechanism.

10: Remember that as acceleration is the second time derviative of displacement, in frequency space this translates to $y \propto a/f^2$ (Equation 3.1).

To elucidate this issue, and thereby understand the limitations of the optical vibration spectroscopy technique, we model the detection event of a single photon (a "click") arriving at the detector at a random time t_i as a δ-function so that the total flux as function of time is given by

$$\Phi(t) = \sum_i \delta(t - t_i). \tag{8.14}$$

Assuming them to be uncorrelated, the time difference between subsequent clicks is exponentially distributed with average rate $\bar{\Phi}$, $|t_{i+1} - t_i| \sim \mathrm{Exp}(\bar{\Phi})$ [129]. From this it follows that the number of clicks $N(\Delta t)$ within a given time bin $t \in [s, u]$ of length $\Delta t = |u - s|$ is Poisson distributed, $N(\Delta t) \sim \mathrm{Pois}(\bar{N})$, with mean number of counts $\langle N(\Delta t)\rangle = \bar{N} = \bar{\Phi}\Delta t$ [130]. Using the formalism developed in Section 2.1, we can now compute the PSD of the stochastic process $\delta N(\Delta t) = N(\Delta t) - \bar{N}$. To this end, observe that because we assumed arrivals to be uncorrelated, $N_{u'}(\Delta t)$ for a time bin starting at $t = u'$ is independent of $N_{s'}(\Delta t)$ for a time bin starting at $t = s'$. In other words, the autocorrelation function of $\delta N(\Delta t)$ is nonzero only for the same time bins,

$$C_{\delta N(\Delta t)}(\tau) = \big\langle (N_s(\Delta t) - \bar{N})(N_u(\Delta t) - \bar{N})\big\rangle = \mathrm{Var}(N(\Delta t))\delta(\tau), \tag{8.15}$$

where $\tau = s' - u'$ and $\delta(\tau)$ is to be understood in a broad sense as zero if $|\tau| > \Delta t$ and $1/2\Delta t$ else. For the Poisson distribution the variance is equal to its mean so that we obtain

$$C_{\delta N(\Delta t)}(\tau) = \bar{N}\delta(\tau). \tag{8.16}$$

In the limit of $\Delta t \to 0$, we can then perform the Fourier transform to obtain the PSD of $\delta N = \lim_{\Delta t \to 0} \delta N(\Delta t)$,[11]

$$S_{\delta N}(\omega) = \bar{N}, \tag{8.17}$$

that is, δN is a white noise without frequency dependence.[12] $S_{\delta N}$ can be seen as the *instantaneous* number noise PSD.

As a last step, we consider once again discretely sampling the *continuous* process δN with PSD $S_{\delta N}(\omega)$ at rate $f_\mathrm{s} = \Delta t^{-1}$ in order to find an expression for the PSD of the discrete process $\delta N(\Delta t)$, $S_{\delta N(\Delta t)}(\omega)$. We know from above that $\mathrm{Var}(\delta N(\Delta t)) = \bar{N}$. On the other hand, recall from Section 2.1 that also

$$\mathrm{Var}(\delta N(\Delta t)) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S_{\delta N(\Delta t)}(\omega) = \int_{-f_\mathrm{s}/2}^{f_\mathrm{s}/2} \mathrm{d}f\, S_{\delta N(\Delta t)}(f) \tag{8.18}$$

where the last equality holds true because of the finite bandwidth of the discretely sampled signal. Since $S_{\delta N}$ is white, it follows that $S_{\delta N(\Delta t)}$ is, too, and we can directly evaluate Equation 8.18, obtaining[13]

$$S_{\delta N(\Delta t)}(\omega) = \frac{\bar{N}}{f_\mathrm{s}}. \tag{8.19}$$

To convert to the displacement noise PSD, we can simply convert units using the calibration derived above because if $N \sim \mathrm{Pois}(\bar{N})$ then so $y \sim \mathrm{Pois}(\bar{N} f_\mathrm{s}/s)$ where s is the slope of the calibration converting displacements to count rates, *i.e.*, the sensitivity (see Figure 8.7 and Equation 8.10). Hence,[14]

$$S_{\delta y(\Delta t)}(\omega) = \frac{\bar{N}}{f_\mathrm{s}} \times \left(\frac{f_\mathrm{s}}{s}\right)^2 = \frac{\bar{\Phi}}{s^2} \tag{8.20}$$

with $\delta y = y - \langle y \rangle$. This type of noise is known as *shot noise*. It was first studied in the context of electron transport by Schottky [131] and results

11: Despite appearances, $S_{\delta N}$ has units cts^2/Hz. The discrepancy stems from the difficulty in defining a continuous white noise process.

12: Note that we could have also arrived at this result directly by computing the autocorrelation function $\langle \delta N(t)\delta N(t-\tau)\rangle$ from Equation 8.14 with $N(t) = \int \mathrm{d}t\, \Phi(t)$.

13: $S_{\delta N(\Delta t)}$ also has units cts^2/Hz.

14: Note that this is the two-sided PSD; to convert to the one-sided version used in this chapter, multiply by two.

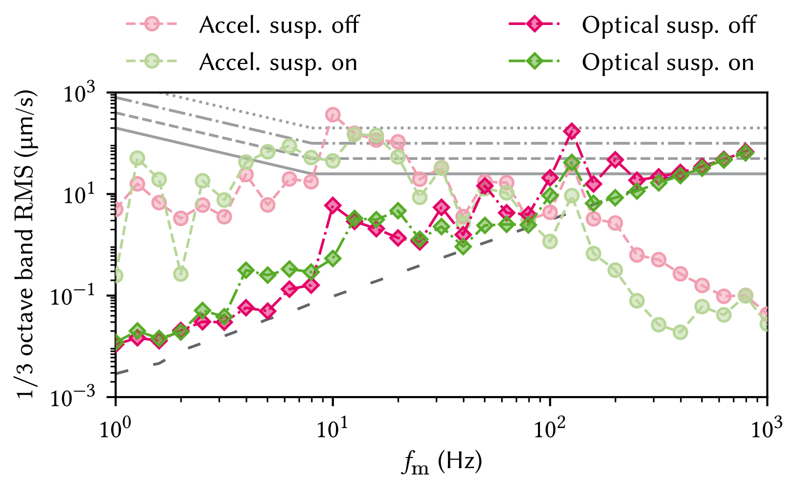

Figure 8.9: 1/3 octave band RMS velocity computed for vibration measurements with the PTR enabled and the suspension disabled (magenta) or enabled (green). Circles (diamonds) show data obtained with the accelerometer (optical method). The VCs VC-B, VC-A and first two ISO levels are indicated as gray lines (solid, dashed, dash-dotted, dotted); see Table 8.1. Accelerometer data are above the VC-B criterion for about three octaves centered around $f_m = 10\,\text{Hz}$, where even the ISO "residential day" level is breached. Optical data are more favorable, in particular with the suspension enabled (green). Towards high frequencies, the data are dominated by the wideband shot noise floor indicated by the loosely dashed dark gray line, suggesting the true displacement RMS is well below the VC-B criterion.

from the discrete nature of, in our case, photons and their stochastic emission times [132]. For the parameters in the present measurements, $\bar{\Phi} \approx 3\,\text{Mcps}$ and $s \approx 2.36\,\text{Mcps}/\mu\text{m}$ (*cf.* Figure 8.7), we obtain a shot noise floor of $S_{\delta y(\Delta t)} \approx 1\,\text{nm}/\sqrt{\text{Hz}}$ in excellent agreement with the data shown in Figure 8.8 where the theoretical value is indicated by a gray dashed line.

Inserting the theoretical expectation for the sensitivity s, Equation 8.11, into Equation 8.20, we find that

$$S_{\Delta y(\Delta t)}(\omega) = \frac{\epsilon}{\Phi_0}\left(\frac{w_0}{1-\mathscr{R}_0}\right)^2 \tag{8.21}$$

if we identify $\bar{\Phi} = \epsilon\Phi_0$ for some (fixed) setup efficiency ϵ. This shows a clear path towards improving the SNR of the method. Just as the sensitivity s is improved by increasing $\bar{\Phi}$ and $1-\mathscr{R}_0$ and by decreasing w_0, so is the shot noise floor, albeit quadratically in w_0 and $1-\mathscr{R}_0$. For example, for a reduction in spot size by a factor of two from inserting a different objective lens and a tenfold increase in maximum count rate achieved by replacing the counting card with a more powerful model,[15] our simple model predicts a noise floor of $25\,\text{pm}/\sqrt{\text{Hz}}$, a reduction by a factor of 40.

15: Swabian Instruments offers models with up to 1.2 GSa/s, although at that rate the jitter and dead time of the APDs would start to become the limiting factors [133]. Since very high frequencies are not of interest, a simpler photo-diode that supports batched measurement would also suffice.

To conclude this section, let us come back to the vibration criteria defined in Subsection 8.1.1 and evaluate the microscope based on the two different measurement methods presented in this chapter. The 1/3 octave band RMS velocities computed for the vibration spectra with PTR enabled shown in Figures 8.3 and 8.8 are plotted in Figure 8.9. Based on the data from the accelerometer (circles), the microscope does not meet the targeted level of vibration isolation (VC-B, solid gray line) over three octaves. Because this method of measuring the vibration noise is sensitive to absolute changes, we can understand qualitatively why this is the case if we view the accelerometer at the sample position as the end of a large pendulum whose fulcrum is in the center of the plane spanned by the three air springs. A rough estimate gives a resonant frequency of 0.5 Hz,[16] implying frequencies in the considered range, [4, 32] Hz, are fairly effective at exciting motion in the pendulum (*cf.* Subsection 8.1.1). By contrast, with the optical method we do not pick up on such motion because the ocular lens focusing the light into the SMF is fixed in the co-rotating frame with respect to our imagined pendulum. Indeed, the

16: The center of mass sits close to the magnet approximately $l = 1\,\text{m}$ below the springs so that we have $f = (2\pi)^{-1}\sqrt{g/l} \approx 0.5\,\text{Hz}$.

VC velocities computed for this method show that they are orders of magnitude smaller at low frequencies in particular since only deviations from the rigid body picture established above induce a change in signal. Furthermore, the RMS is dominated by the broadband shot noise floor indicated by the loosely dashed dark gray line, implying that the true vibration-induced RMS is well below our targeted VC-B criterion.

8.4 Routes for improvement

Several improvements could be made to the system if the external conditions would allow it. First, the rotary valve motor should be moved further away from the cold head.[17] As per the initial installation status, it is currently connected to the cold head with a flexible hose at a right angle and a distance of roughly 50 cm, which is below the minimum bend radius recommended by Oxford Instruments.[18] Additionally, the term "flexible" is relative here given the pressure of 20 bar. Increasing the length of the hose should reduce its relative rigidity and thereby its ability to transmit vibrations from the motor to the cold head.

17: Clearly, this will impact the performance of the PTR to some extent and should therefore be considered carefully.

18: Note that the orientation of the motor, which is horizontal with the axis, is also not the recommended configuration.

Next, the cold head should be mounted firmly to a secondary reference frame, for instance the ceiling or the lower cryostat frame on which the springs rest. An intuitively obvious step, it has also been shown in the literature that decoupling the PTR from the cryostat in this fashion leads to significant improvements in vibration isolation [120].[19] Acoustic insulation of the room and PTR flex hoses could further improve the low-frequency response of the system [116, 121]. Lastly, let me note that there also exist cryocoolers with variable operating frequency that can thus be tuned away from problematic resonances in the system [134].

19: The former option was attempted, but showed no clear improvements in the measurements for reasons unclear, see Section A.2 for additional data. It did emphatically deteriorate the interdepartmental atmosphere. Apologies to the institute on the floor above.

9 Conclusion and outlook

In this part of the present thesis I analyzed various aspects of the Millikelvin confocal microscope introduced in Refences 70 and 47. The setup offers the ability to conduct optical measurements of samples at Millikelvin temperatures while at the same time having access to state-of-the-art electrical wiring for sensitive quantum transport measurements such as those performed for semiconductor spin qubit experiments. I first covered the refrigeration aspect of the setup in Chapter 6. Housed in a commercial dry DR, the modifications made for free-space optical access introduce additional heat loads and thus impact the available cooling power budget. Specifically, I quantified the heat loads introduced by the motion stages' resistive position readout and irradiation of the sample with the laser, as well as the base temperature reached for different configurations of AR-coated windows installed inside the cryostat. With the optimal configuration of windows, I found the system to reach a base temperature below 10 mK for readout voltages below 100 mV and laser powers below 10 µW. To assess the impact on electrical performance, I measured the electron temperature in a GaAs/Al$_x$Ga$_{1-x}$As quantum dot in transport using Coulomb blockade thermometry. I tuned a device designed for spin-qubit operation into a single quantum dot in the few-electron regime and measured the width of a conductance resonance in the sequential tunneling regime in direct transport, obtaining $T = 75$ mK comparable with state-of-the-art systems.

During these experiments, the objective was not focused onto the sample while the entrance window on top of the cryostat was not covered, allowing ambient light to enter. As several studies have found, the electrical characteristics of 2DEGs in GaAs/Al$_x$Ga$_{1-x}$As heterostructures, in particular Si-doped ones, are highly sensitive to illumination, resulting in long-time transient effects and instabilities [135–138]. It would hence be interesting to study the behavior of electron temperature in relation to (wavelength-dependent) illumination, an experiment the setup described in this part of the present thesis is uniquely suited for.

In Chapter 7, I then discussed the optical characteristics of the setup, in particular the microscope and its components. I laid out the rationale behind choosing the lenses for collimating the Gaussian beam launched from the SMF, focusing it onto the sample, collecting the emitted radiation, and focusing that into another SMF. Having chosen the lenses, I analyzed the efficiency of light emitted from a point dipole located inside a dielectric slab being coupled into the SMF. From the analysis, it became clear that the precise spatial distribution of the electric field is non-trivial and warrants further attention. A more detailed numerical simulation could serve to pinpoint areas of improvement besides fairly obvious solutions such as mirrors below the sample or mode-engineering like the approach pursued in Reference 139. On the other hand, measuring the field profile experimentally using a CCD inserted into the optical path right before the detection fiber would allow better matching between theoretically expected and experimentally observed efficiencies.

A crucial point might turn out to be the mirror in front of the objective lens. As shown by Benelajla et al. [84], reflections of polarized non-plane-wave beams induce a modal transform into modes with nodes at the center that are spatially filtered by the SMF in a confocal geometry. While beneficial for cross-polarization extinction (*cf.* Subsection 7.1.4), the same effect could lead to radiation emitted from the sample not coupling into the detection fiber if it is co-polarized with the excitation beam.[1] Figure 9.1 shows a reduced sketch of the optical path. Light emitted from the sample is reflected off the mirror (M) and passes through the beam splitter BS1 and the analyzer (A) before being focused into the detection fiber by the ocular lens (D). This situation is equivalent to that described by Benelajla et al. [84] if the emitted light is cross-polarized with respect to A and has a Gaussian mode profile. As I showed in Subsection 7.1.2, the mode overlap between the dipole radiation after collimation and a Gaussian TEM_{00} mode is quite large at 83 %, implying that the effects investigated in Reference 84 for Gaussian beams will also apply to some extent to the dipole radiation considered here.

1: Of course, the cross-polarization of analyzer and polarizer will suppress emitted radiation co-polarized with the excitation in any case. The excitation with linear polarization was chosen to coherently address exciton states in a Voigt configuration. If excitation with circularly polarized light is desired, the $\lambda/4$ plate can be moved from the excitation arm to below BS1 similar to Reference 99. I tested this configuration but found the excitation rejection to be poor and reverted to the original configuration.

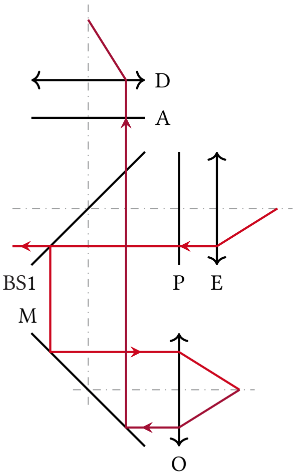

Figure 9.1: Reduced sketch of the optical path (*cf.* Figure 7.1) including the cold mirror (M). The excitation laser experiences four reflections, twice at M and once each at the sample and at BS1. The light emitted from the sample experiences a single reflection at M.

For the excitation rejection, the cold mirror potentially adds another issue. As shown by Steindl et al. [85], the giant cross-polarization extinction mechanism in a confocal geometry discovered by Benelajla et al. [84] is reduced in efficacy when additional reflections are introduced into the optical path.[2] Our situation is again slightly different to the one studied there, however, as the cold mirror is mounted perpendicular to the BS[3] and hence the effects on s- and p-polarizations upon reflection mix. A detailed analysis of the spatial mode profile would thus help elucidate the extent to which these effects matter in our experiments.

2: A halfwave plate inserted into the path was also found to reduce the rejection ratio, suggesting that the $\lambda/2$ and $\lambda/4$ plates might also impact performance. On the other hand, Kuhlmann et al. [99] found the $\lambda/4$ plate to be crucial for compensating ellipticities introduced by the setup.

3: That is, while x lies in the plane of the BS, y lies in the plane of the cold mirror if we use the coordinate system of Figure 8.4 where x is the out-of-plane sample coordinate and z is along gravity.

Lastly, in Chapter 8, I addressed the impact of vibrations, induced chiefly by the PTR, on the microscope performance. After outlining the basic principles of vibration isolation, I described the suspension scheme based on passive air springs which I installed in the setup to mitigate the vibration noise coupled into system by the PTR. I then performed measurements of the displacement noise PSD using two different methods. First, using a piezoelectric accelerometer that is directly sensitive to absolute displacements through the local acceleration experienced by the sensor, I found that during operation of the cryostat the vibrations should be considered too large for operation of the microscope, obtaining 1/3 octave band RMS velocities well into the tenths of mm/s range. However, I argued that the absolute value of displacement noise at the sample position is less indicative of the microscope performance than the relative displacement noise between sample and detection fiber, *i. e.*, the vibrations along the optical path. To quantify these, I introduced an optical *in-situ* method of measuring lateral displacement using a knife-edge measurement on a reflectance step on a sample. This technique can be employed without additional modifications or instruments besides those already present in the optical setup. It is based on the fact that the finite spot size of the Gaussian laser beam incident on a step-like reflectance profile such as that produced by a lithographic Au gate on a crystalline surface results in a linear slope in reflected intensity over the width of the laser spot. This allows measuring the reflected intensity as function of time and subsequently converting the intensity to a displacement along the gradient using a previous calibration akin to the charge sensing technique in GDQDs, following which the tools laid out in Part I can be used to obtain the noise PSD. Using this technique, I showed that the relative displacement noise is orders of magnitude smaller than the absolute, placing the octave band RMS velocities well below the targeted levels categorized by the vibration criteria. I furthermore analyzed the sensitivity of the technique, showing that the noise floor is dominated by photon shot noise, and laid out routes for enhancing the SNR.

Figure 9.2 summarizes the performance of the vibration damping for enabled PTR and the two different vibration spectroscopy techniques.

The upper panel shows the relative (instantaneous) displacement noise power for activated suspension referenced to that for deactivated suspension (*cf.* Subsection 3.2.1) on a logarithmic scale. The suspension appears to have a greater effect overall for the absolute level measured with the accelerometer. However, looks can be deceiving as evidenced by the lower panel, which shows the integrated data,

$$\text{Integrated} \equiv 10\log_{10}\left(\frac{\text{RMS}_S^2(f)}{\text{RMS}_{\text{ref}}^2(f)}\right), \tag{9.1}$$

also on a logarithmic scale (*cf.* Equation 8.5). Clearly, above 10 Hz the suspension improves the overall noise power as measured by the optical method, whereas the absolute level measured with the accelerometer does not recover from the penalty taken between 1 Hz and 10 Hz where oscillations are amplified by the air springs.

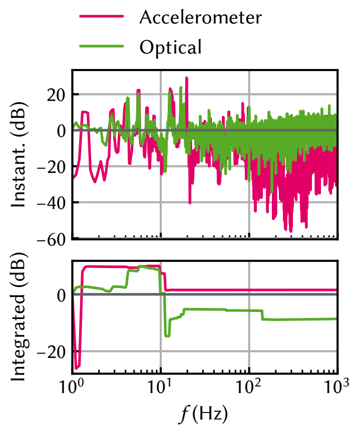

Figure 9.2: Relative displacement noise power of the setup with air spring suspension activated referenced to suspension deactivated. Lower panel shows the fractional integrated data.

The microscope characterized in this part of the present thesis has potential applications beyond the field of quantum technology. Low-temperature effects such as superconductivity and other correlated electronic phases could be studied optically [140–142]. Van der Waals heterostructures appear particularly promising as they allow to combine optically responsive direct-gap semiconductors with metallic materials whose electronic properties can be studied in transport, leading to hybridization and proximity effects due to the atomically close contact [143]. Examples of this include stacks of graphene or bilayer graphene (BLG) with transition-metal dichalcogenides (TMDs) such as MoS_2 or WSe_2 [144–147]. The latter display giant exciton binding energies that are sensitive to the dielectric screening by their close environment, thus allowing to serve as optical sensors of electronic properties [144, 148]. Recent experiments in this direction undertaken in the setup by Tebbe et al. [149] on BLG/WSe_2 show promising first results on spin-orbit interaction proximity effects and magnetic phases in BLG [150].

Part III

Optical Measurements of Electrostatic Exciton Traps in Semiconductor Membranes

Introduction 10

Author contributions — *An overview of wafers measured in this part of the present thesis is given in Table B.1. Nikolai Spitzer*[a] *grew wafer #15460 ("Honey"). Julian Ritzmann*[a] *grew wafer #15271 ("Fig"). Chao Zhao*[b] *grew wafer #M1_05_49. Thomas Descamps*[c] *fabricated the sample Doped M1_05_49-2. Sebastian Kindel*[d] *fabricated the samples Honey H13 and Fig F10.*

a: Ruhr-Universität Bochum.

b: Then at Forschungzentrum Jülich.

c: Then at RWTH Aachen University and Forschungszentrum Jülich GmbH.

d: RWTH Aachen University and Forschungszentrum Jülich GmbH.

Just as quantum computers are conceived as the quantum analogon of classical computers with bits and logic operations switched out by quantum counterparts, so can one devise *networks* of such objects, where quantum information generated or processed at a quantum *node* is distributed across long distances through the quantum counterparts of classical information channels [151, 152]. Famously envisioned by Kimble [153], the concept can be extended to the notion of a *quantum internet* [154].

A wide array of ideas has since been put forth that make use of the theoretical capabilities of such quantum networks. Initially, quantum networks were studied in the context of quantum cryptography [155–158]. There, the no-cloning theorem and clever use of entanglement ensure quantum-secured communication between distant parties that cannot be eavesdropped upon or tampered with by adversarial parties without detection.[1] Considerable attention has also been paid to the notion of distributed quantum computation [161]. As quantum computers do not appear to be on the same course of miniaturization as classical computers have been, there might turn out to be a limit to the physical size of quantum computers and, hence, a limit to the processing power of a monolithic node.

1: As ever in cryptography, new protocols keep getting hacked and loop-holes are discovered [159, 160]. It will be interesting to see, therefore, if the security will faithfully transfer from theory to experiment.

Distributed quantum computation resolves this bottleneck by allowing computations to be executed across separate nodes much like classical supercomputer clusters. Although a comprehensive resource assessment of the feasibility of such approaches is still outstanding, initial results are promising [162], and experimental demonstrations of small computations have recently been shown [163]. A concept combining distributed quantum computation with quantum cryptography is blind quantum computation [164, 165], which promises a form of cloud-based quantum computation. Also here, first experimental demonstrations have recently been achieved [166].

Next, quantum networks have garnered interest in the field of quantum sensing [167]. This term refers to the branch of quantum technology in which individual quantum systems are employed as highly sensitive sensors, for example of magnetic fields [168, 169], or, more generally, to perform measurements of physical quantities [59, 170]. For example, there exist proposals to employ quantum networks for long-baseline telescopes that use optical interferometry to enhance the resolution of astronomical imaging [171, 172], akin to the techniques used to produce the first image of a supermassive black hole [173], or global networks of quantum clocks [174]. Going beyond technological applications, the capability to coherently transmit quantum states across large distances opens the pathway to tests of quantum theory itself, and where it might fail [175]. At least since the publication of the Einstein-Podolsky-Rosen

(EPR)–paradox [176], tests of the non-locality of quantum mechanics have been proposed [177, 178] and performed [179, 180]. More recently, for example, a small quantum network was used to rule out a – certain [181] – description of quantum theory by real numbers [182–185] and we may expect more such experiments to come [186]. Indeed, as the first small-scale experimental demonstrators arrive [187–190], research into complex quantum networks and their properties and possible applications is still in its beginnings [191].[2]

2: Take for example the recent interest in quantum causal order [192–194], originally envisioned to study gravitational effects on quantum coherence.

How does a quantum network work? In the canonical quantization picture we already adopted previously, we might simply replace classical optical links by quantum versions thereof and similarly transmit *flying* qubits instead of bits through those channels. However, even optical fibers, the backbone of the modern internet, are lossy, and since the photon loss scales exponentially with distance, there would be little hope to build networks larger than a few to a few tens of kilometers.[3] In classical networks, this problem is remedied by repeater stations that simply produce copies of incoming photons and thus amplify the signal. In quantum mechanics, however, this is forbidden by the no-cloning theorem, which states that one cannot achieve a perfect copy of a qubit prepared in an arbitrary and unknown quantum state [195, 196]. To the rescue comes, then, entanglement. In order to transmit quantum information between two parties, Alice and Bob, one places a repeater station, Charlie, halfway between Alice and Bob. Charlie shares bipartite maximally entangled states (often referred to as a EPR or Bell pair) with both Alice and Bob and performs a Bell measurement on the two halves of the pairs in his possession. This projects Alice and Bob's halves of the Bell pairs into a state that is maximally entangled between the two of them. This technique of entangling two states that have never interacted with each other is known as entanglement swapping [197, 198]. Briegel et al. [199] and Dür et al. [200] then proposed a quantum repeater protocol that uses entanglement swapping, enhanced by entanglement distillation,[4] to successively entangle neighboring pairs of entangled states whose resource requirements scale logarithmically with the length of the quantum channel between the ends of which entanglement needs to be established. What is more, the protocol tolerates error and loss rates on the percent level and is thus much more benign than quantum error correction (QEC) [201]. Following the initial proposal, more improved schemes were developed that tolerate higher errors [202] or employ entirely different techniques [203], see Reference 67 for a review. Muralidharan et al. [204] analyzed the resource requirements of three quantum repeater generations and identified optimal architectures for different parameter regimes. Recently, small-scale experimental realizations have demonstrated establishment of two-photon-entanglement across a 50 km link by a quantum repeater node in the middle [205].

3: We can expect a survival probability of 1 % over a distance of 100 km [67]. There is therefore arguably no feasible alternative to optical transmission over long distances.

4: Also known as entanglement purification.

A crucial detail of the quantum repeater protocol is that it requires storing Bell states in a quantum memory until the heralding of successful entanglement.[5] In practice, this means that quantum repeaters require a coherent light-matter interface between photonic flying qubits and stationary quantum memory since storing photons is not feasible. Such an interface has been the subject of intense research and there exist a large number of competing approaches that differ in choice of material platform for the memory and choice of encoding for the photon [66, 207]. Among the most advanced are atomic and defect-based systems. Atomic (and ionic) systems are a natural choice as the energy scales of atomic transitions are compatible with photons in the telecom range ($\sim$ 1550 nm) [188, 189, 205, 208, 209], while nuclear spins coupled to defects in crystal lattices

5: Note that there exist also protocols for memoryless, all-optical quantum repeaters [67, 206].

have long spin lifetimes at the same time as optical transitions [187, 190, 210–212].

Optically interfacing semiconductor spin[6] or superconducting qubits, on the other hand, is arguably more challenging because of a separation of energy scales. While qubits in these systems have energy splittings in the GHz regime, telecom photons have energies of hundreds of THz, and so bridging this gap requires some sort of intermediary or *transducer*. For superconducting qubits, approaches based on mechanical resonators [213] and electro-optic nonlinear materials [214] have been pursued among others. For spin qubits, excitons[7] (or complexes thereof) confined in optically active quantum dots (OAQDs) such as self-assembled QDs [215–218] appear to be most promising owing to their excellent optical properties. Due to the fast recombination speeds of excitons, their suitability as a qubit is limited. Single charge carriers confined to self-assembled quantum dots (SAQDs) have been explored instead [103], but still face the problem that the growth of SAQDs is random and as such coupling two or more qubits is extremely challenging. By contrast, spin qubits confined in gate-defined quantum dots (GDQDs) have reached a high level of maturity [219, 220], and the promise of scalability by leveraging the highly advanced industrial semiconductor fabrication technology still holds true.

6: By semiconductor spin qubits, I refer to qubits encoded in the spin of one or more electrons or holes confined in quantum dots (QDs), in contrast to spins attached to charged defect centers or nuclear spins.

7: Excitons are bound electron-hole pairs. See Chapter 11.

Jöcker et al. [221] thus proposed a protocol for transferring the quantum state of a photonic, polarization-encoded qubit to that of a spin qubit confined in a GDQD by means of a OAQD serving as intermediary in order to benefit from the benign optical properties of the latter and the long coherence times and processing capabilities of the former. In the protocol, an incident photon generates an exciton in the OAQD, transferring the quantum state to the exciton. By application of a strong in-plane magnetic field, bright and dark states of the exciton are mixed and electron and hole remain in a product state with all information encoded in the electron spin. This allows discarding the hole and subsequently transferring the electron state to the GDQD, the details of which depend on the encoding chosen, either single-spin (Loss-DiVincenzo (LD)) or two-spin (singlet-triplet) qubit. The protocol is agnostic to the precise realization of the OAQD, although it adopts parameters from SAQDs. A crucial ingredient of the protocol is that the OAQD be in tunnel-coupling distance to the GDQD in order to enable exchange coupling or the adiabatic transfer of the photo-electron. For SAQDs this is a challenging endeavor since, as mentioned above, their growth is random and therefore requires one to first locate the dots and then align the gates during fabrication of the GDQD. Because of the exponential dependence of the tunnel coupling strength on the distance, this alignment needs to be very precise indeed.

An alternative to a SAQD was proposed by Descamps [70] in the form of an electrostatic exciton trap. Here, excitons are designed to be confined not by local modifications of the band structure during growth but by application of out-of-plane electric fields through gate electrodes. This tilts the band structure and lowers the exciton energy by the quantum-confined Stark effect (QCSE), where exciton dissociation is prevented by charge carrier confinement in a quantum well (QW). To avoid dissociation by lateral electric fields, Descamps [70] developed a fabrication process to etch down the semiconductor heterostructure to a 200 nm thin membrane symmetric about the QW, allowing lithographic patterning of laterally aligned, nanometer-scale gate electrodes on both sides of the membrane [110]. Application of voltages of the same magnitude but opposite polarity to gates on the top and bottom side of the membrane then produces – to leading order – a local electric field without changing the chemical potential in the QW itself, while voltages of the same polarity have just the opposite effect. Together with the out-of-plane confinement by the QW, this should in theory provide zero-dimensional confinement of excitons purely by electrostatic means and allow for the top-down, scalable fabrication of OAQDs in close proximity to other, conventional GDQDs. What is more, the confinement strength and potential relative to the neighboring GDQD can be finely tuned by well-established techniques.

In Reference 110, Descamps et al. demonstrated the implementation of the membrane fabrication process as well as initial progress towards forming a GDQD in transport and optical measurement of the lowering of the local exciton potential by the QCSE. Yet unobserved was the signature of QD-behavior in an exciton trap as manifested by, for example, the resolution of orbital splittings or single-photon emission. In this part of the present thesis, I present optical measurements of electrostatic exciton traps towards this goal. It is outlined as follows. In Chapter 11, I give an introduction to the physics of photoluminescence (PL) in semiconductors as well as the influence of electric fields. Following that, I introduce the Python measurement framework I wrote to control the experiment in Chapter 12. Then, in Chapter 13, I present measurements of doped membranes. I show data of an exciton trap and discuss the voltage, power, and position dependence of the PL emission as well as photoluminescence excitation (PLE) measurements. Finally, I perform simulations of the membrane structure to explain the quenching of PL intensity observed when focusing a gate and propose slight modifications to the heterostructure design to mitigate this effect. I conclude with an outlook in Chapter 14.

11 Photoluminescence and excitons in semiconductors

Semiconductors are the foundation of the modern age of technology. In large parts, this is due to their favorable optical properties, determined chiefly by their band gap on the order of 1 eV, that enable wide-ranging applications from semiconductor lasers over photo-detectors to optical communication technology. Semiconductors can be classified into *direct* and *indirect* categories, where the terminology refers to the location of the band gap, that is, the smallest energy difference between valence and conduction band, in the Brillouin zone. A direct semiconductor's band gap features a valence band maximum at the same point in k-space as a conduction band minimum. Such materials, a prime example of which is the compound GaAs with a band gap of $E_\mathrm{g} = 1.519\,\mathrm{eV}$ at $T = 0\,\mathrm{K}$ in the near-infrared (NIR) spectrum [222], emit electromagnetic radiation with energy $h\nu = E_\mathrm{g}$ when electrons excited from the valence into the conduction band recombine with a hole. When the excitation occurs through optical absorption, this effect is known as photoluminescence.

Of course, besides their optical applications, semiconductors are also the backbone of the modern electronics industry. Their band gap allows tuning the conductivity by means of the field effect wherein the potential and thereby the band structure is locally modified by means of a voltage applied to a gate electrode. This facilitates the construction of voltage-controlled transistors. The development of remote-doping techniques then enabled the growth of extremely high-mobility two-dimensional electron gases (2DEGs) in GaAs/$\mathrm{Al}_x\mathrm{Ga}_{1-x}\mathrm{As}$ heterostructures [75]. In these structures, dopants are introduced into the AlGaAs barrier layer some 50 nm away from the GaAs layer and thus spatially separated by a AlGaAs spacer layer. This reduces Coulomb scattering in the 2DEG induced by the doping and produces very high-quality samples as evidenced for example by the observation of the fractional quantum Hall effect [223].

It is hence not at all surprising that early quantum dot experiments also took place in this well-understood material system. While undoped approaches to electron and hole spin qubits in GaAs exist [224–227], they introduce added complications because of the need to electrostatically induce a 2DEG and reliably contacting it [228]. What is more, these issues are exacerbated when the structures need to be thinned down to a membrane [70, 229]. Therefore, to accommodate gate-defined quantum dots (GDQDs) next to the optically active quantum dot (OAQD) in the membrane, the devices measured in this part of the present thesis are fabricated on doped heterostructures hosting a 2DEG. In the following, I discuss the optical behavior of these structures under illumination to develop an intuition for the measurements presented later on.[1] I take a simple analytical approach in order to capture the most relevant physics and obtain a qualitative understanding. For quantitative results, a more detailed method incorporating electrostatics would be required; I refer the reader to Reference 70 for such simulations.

1: The *electrical* behavior is another matter; there, illumination can lead to the creation, trapping, and de-trapping of free charge carriers that alter the transport properties of the device, leading to electrical hysteresis and instability [135–138, 229].

11.1 Photoluminescence in doped GaAs/Al$_x$Ga$_{1-x}$As heterostructures

Consider an intrinsic, direct-gap, Zincblende semiconductor such as GaAs with a band gap of $E_g = 1.519\,\mathrm{eV}$ at zero temperature [222]. At the Γ-point, the conduction band is well approximated by a parabolic band with effective mass $m_c^*/m_e = 0.067$ and is offset by $E_g/2$ above the Fermi level μ. Offset by the same absolute amount in the opposite direction is the valence band, which is fourfold degenerate close to Γ with heavy and light holes with effective masses $m_{hh}^*/m_e = 0.34$ and $m_{lh}^*/m_e = 0.09$, respectively [230]. The third valence band[2] is split off by the spin-orbit interaction and lies $0.34\,\mathrm{eV}$ below the other valence bands [231]. Introducing confinement in one direction, for example by doping a GaAs/Al$_x$Ga$_{1-x}$As heterojunction or growing a GaAs quantum well (QW) sandwiched between two layers of AlGaAs, lifts the fourfold degeneracy of light and heavy holes and leaves – under typical conditions – the latter as the valence band ground state.

2: The valence bands are p-like and hence contain contributions from three twofold degenerate atomic orbitals.

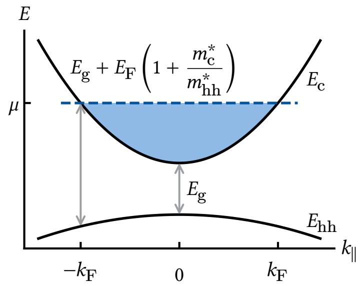

Figure 11.1: Band structure diagram of a doped heterostructure (after Reference 232). Due to the n-type doping, the conduction band is filled up to the Fermi level μ. Photonic excitation of an electron-hole pair can only occur at $|k| > k_F$ into the free states above μ due to the small photon momentum. Recombination can occur within a bandwidth of $E_F(1 + m_c^*/m_{hh}^*)$.

Doping in sufficiently high concentration then raises the Fermi level from mid-gap to inside the conduction band in the plane of confinement, resulting in a band structure as sketched in Figure 11.1. The bands remain parabolic as function of the in-plane wave vector $k_\parallel$ close to Γ. The conduction band is filled up to μ where, measured from the conduction band edge, $E_c(k_\parallel = k_F) = E_F = \hbar^2 k_F^2/2m_c^*$ with the Fermi wave vector $k_F \sim 10^8\,\mathrm{m}^{-1}$. Now, absorption of a photon incident on the semiconductor demands conservation of energy and momentum. The latter condition implies that hole and electron have close to equal momentum because $k_\gamma = 2\pi/\lambda \approx 8 \times 10^6\,\mathrm{m}^{-1}$ is much smaller than k_F. Thus, excitation of an electron-hole pair from the valence into the conduction band occurs only for $k \geq k_F$. By contrast, recombination can take place for any k in principle. In practice, however, photo-electrons quickly relax down to the Fermi level, and recombination takes place between electrons inside the Fermi sea and photo-holes somewhere in the valence band at $k \leq k_F$.[3] The former condition then implies $E_\gamma \geq E_g + E_F\left(1 + m_c^*/m_{hh}^*\right)$ because no free electron states are available in the conduction band below μ due to the Pauli exclusion principle and where the term in parentheses is due to the valence band dispersion. Many-body effects due to localized photo-holes scattering with electrons of the Fermi sea can lead to a strong enhancement of luminescence intensity at the Fermi edge, the so-called Fermi-edge singularity (FES) [233–235]. The intensity of the PL at the Fermi edge relative to recombination at the band edge ($k_\parallel = 0$) is thus an indicator of the QW interface quality and degree of alloy fluctuations, both of which lead to hole localization [236, 237].

3: For sufficiently localized states in real space, a wide range of k states in the valence band is available for recombination as states are extended in k-space. We can estimate the localization length required for states up to k_F to participate by $\Delta x \geq 1/2\Delta k = 1/2k_F \sim 5\,\mathrm{nm}$ for a typical 2DEG.

Free electron-hole pairs created by photo-excitation can form hydrogenic bound states due to the Coulomb attraction of their opposite electric charges, *excitons*. In two dimensions, this effect is strongly enhanced due to the increased overlap of electron and hole wave functions. Furthermore, the reduced dimensionality also enhances the binding energy from $\sim 4\,\mathrm{meV}$ in bulk GaAs to up to four times that in GaAs QWs [238, 239]. Rather than the continuum of the joint density of states of valence and conduction band discussed previously, excitons are discrete states whose energy is lower than the band gap energy by the binding energy, $E_X = E_g - E_b$. In doped QWs hosting a 2DEG, the free carriers can screen the exciton binding energy and lead to an ionized electron-hole plasma [240]. This is known as the insulator-to-metal (Mott) transition in semiconductors [241]. Competing with this are so-called Mahan excitons [233, 234] that give rise to the FES [242].

Next, I discuss the influence of an electric field on excitons. For this, we return to undoped structures for simplicity.

11.2 The quantum-confined Stark effect

Consider the electron-hole pair in bulk GaAs in the co-moving frame of the hole. The hole generates the Coulomb potential

$$V(r) = \frac{e}{4\pi\epsilon_0\epsilon_r|r|}, \tag{11.1}$$

where $\epsilon_r = 13.3$ is the relative permittivity of GaAs at low temperatures, which attracts the electron by the Coulomb energy $E(r) = qV(r) = -eV(r)$. Figure 11.2 depicts the Coulomb potential in magenta and the bound electron's wave function sketched in black. The electron-hole separation r is shown in units of the exciton Bohr radius in two dimensions [243]

$$a_B^{(2D)} = \frac{2\pi\epsilon_0\epsilon_r\hbar^2}{\mu e^2} = 6\,\mathrm{nm} \tag{11.2}$$

with μ the reduced effective mass of conduction and heavy-hole valence bands.

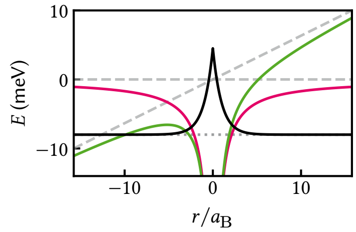

Figure 11.2: Effect of an in-plane electric field on an exciton wave function. In the hole's reference frame, the electron sees a static attractive Coulomb potential (magenta), resulting in a bound state (dotted gray line, wave function sketched in black). Applying an electric field ($F = 100\,\mathrm{mV/\mu m}$, dashed gray lines) tilts the Coulomb potential (green) and leads to a transparent barrier through which the electron can tunnel out.

We now apply an electric field F. This modifies the potential seen by the electron by erF as sketched in green Figure 11.2. Ignoring changes to the electron wave function, we can see that the electron can tunnel out of the Coulomb potential, leading to *field-induced ionization* of the exciton. Now place the exciton in a QW instead of in bulk with the electric field pointed such that it is out-of-plane of the QW. The field still tilts the potential, but because electrons and holes are confined into a quasi-two-dimensional plane, they cannot escape and hence do not dissociate. This is the quantum-confined Stark effect (QCSE) [244].

In order to obtain a qualitative understanding of the QCSE, consider an undoped GaAs/$Al_{0.33}Ga_{0.66}As$ QW of width $L = 20\,\mathrm{nm}$. We take a 57:43 ratio for the band offsets [230], resulting in discontinuities of height $\Delta E_c = 0.24\,\mathrm{eV}$ and $\Delta E_{hh} = 0.18\,\mathrm{eV}$ at the interfaces for the conduction and the heavy-hole valence band, respectively, and $m_c^*/m_e = 0.067$ and $m_{hh}^*/m_e = 0.34$ for the effective masses.[4] Assuming an infinitely deep well for simplicity, the eigenenergies are

4: I note that the literature knows many different values for the hole effective mass in the plane of a quantum well, suggesting that one should actually measure it to be confident in the actual value.

$$E_n = \frac{1}{2m^*}\left[\frac{\pi\hbar n}{L}\right]^2 \tag{11.3}$$

and the eigenstates are

$$\psi_n(z) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi z}{L}\right). \tag{11.4}$$

The ground state energy is then 14 meV (3 meV) above (below) the conduction (valence) band edge, corresponding to 6 % (2 %) of the respective band offsets and implying that the infinite-well approximation is acceptable,[5] while the first excited state lies 42 meV higher than the ground state. The upper panel of Figure 11.3 depicts the first two wave functions of electrons and holes in a band structure diagram. Due to the symmetry

5: In a finite well, the wave functions decay exponentially into the barriers and result in slightly lower eigenenergies. However, the qualitative behavior remains the same.

of the confining potential, the wave functions are symmetric around the center of the well.

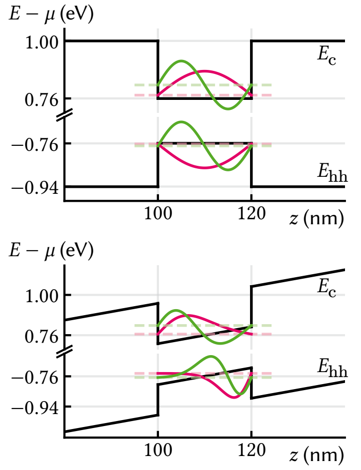

Figure 11.3: QCSE in an undoped QW. Top: conduction and heavy-hole valence band profiles along the growth direction. The wave functions of the first two eigenstates in the well are drawn in magenta and green, respectively. The ground state transition is larger by $\Delta E = 17$ meV than the gap E_g due to the confinement. Bottom: same structure as above with an out-of-plane electric field applied across the structure ($F = 5\,\mathrm{V}/\mu\mathrm{m}$). Analytical wave functions in the infinite-well approximation are shown in magenta and green again. The wave functions get pushed to opposite interfaces of the QW, lowering the ground state transition energy by $\Delta E = -10$ meV. Excitonic effects are not included.

Now, applying an out-of-plane electric field tilts the bands and pulls electrons and holes to opposite interfaces of the QW. The Hamiltonian for the electrons in this case reads [231, 245, 246]

$$H = -\frac{\hbar^2}{2m^*}\frac{\mathrm{d}^2}{\mathrm{d}z^2} + eFz \tag{11.5}$$

if we take z to be zero at an interface and choose $F \geq 0$. Introducing the length and energy scales [231]

$$\tilde{\epsilon} = \left[\frac{(\hbar eF)^2}{2m^*}\right]^{\frac{1}{3}}, \tag{11.6}$$

$$\tilde{z} = \left[\frac{\hbar^2}{2m^* eF}\right]^{\frac{1}{3}} = \frac{\tilde{\epsilon}}{eF}, \tag{11.7}$$

and defining

$$Z_n(z) = z/\tilde{z} - \epsilon_n/\tilde{\epsilon} \tag{11.8}$$

with ϵ_n the eigenvalues of H, the Schrödinger equation becomes [245]

$$\frac{\mathrm{d}^2}{\mathrm{d}Z_n^2}\psi_n(Z_n) = Z_n\psi_n(Z_n). \tag{11.9}$$

Equation 11.9 is known as the Stokes or Airy equation and has the general solution

$$\psi_n(Z_n) = \alpha_n \,\mathrm{Ai}(Z_n) + \beta_n \,\mathrm{Bi}(Z_n), \tag{11.10}$$

where $\mathrm{Ai}(z)$ and $\mathrm{Bi}(z)$ are the Airy functions. $\mathrm{Ai}(z)$ and $\mathrm{Bi}(z)$ oscillate for $z < 0$ and decay (grow) exponentially for $z > 0$, respectively. As we assumed infinitely high barriers at $z = 0$ and $z = L$, the boundary conditions impose

$$\psi_n(Z_n(0)) = \psi_n(Z_n(L)) = 0, \tag{11.11}$$

which completely determines the eigenstates and -energies. For large well widths or fields ($eFL/\epsilon_n \gg 1$), the second term is exponentially suppressed and the eigenenergies are given by the zeros of $\mathrm{Ai}(Z_n)$. For zero field, one recovers the square well solution (Equations 11.3 and 11.4).

The finite-field case is shown in the lower panel of Figure 11.3 for $F = 5\,\mathrm{V}/\mu\mathrm{m}$. Owing to the larger effective mass of the heavy holes, the characteristic length scale $\tilde{z}$ is shorter and hence the corresponding wave functions are narrower (more localized) than their electronic counterparts. The ground state transition energy at this field is 10 meV below the gap or 27 meV lower than in the zero-field case.

For a full quantitative accounting of the transition energies, the exciton binding energy as well as finite barrier heights would need to be included. The former is on the order of 6 meV to 9 meV in GaAs and becomes smaller as the overlap of the electron and hole wave functions is reduced when applying an electric field, pulling the wave functions to opposite interfaces [244]. For the latter, Miller et al. [246] found that finite-well properties could be reproduced by using effective well widths with infinite-well models.

As we tilt the bands, energy levels below the QW become available in the barrier once $\Delta E_{\mathrm{c(hh)}} - \epsilon_n < eFd$ with d the width of the barrier and $\Delta E_{\mathrm{c(hh)}}$ is the height of the conduction (valence) band discontinuity. This allows confined carriers to escape the QW with a finite probability by tunneling through the barrier, an effect known as Fowler-Nordheim tunneling. If this probability becomes too large, trapped excitons can dissociate by electron or hole tunneling through the barriers before recombining, and thereby lower the radiative recombination efficiency (internal quantum efficiency, see Subsection 7.1.2), introducing additional errors in the spin-photon interface. Following Reference 231, we can estimate the tunneling probability as[6]

6: The same result up to a slightly different numerical prefactor in the exponent is obtained more formally from the WKB approximation [70].

$$\mathcal{T}_n(F) \approx \exp\left\{-\frac{\sqrt{4m^*[\Delta E_{\mathrm{c(hh)}} - \epsilon_n]^3}}{eF\hbar}\right\}. \tag{11.12}$$

The tunneling rate Γ_n can then be estimated by the frequency that a confined charge carrier "visits" the edge of the QW multiplied by the probability that it tunnels, $\mathcal{T}_n$ [247]. That is, we take ϵ_n to be a kinetic energy and calculate the velocity as

$$v_n = \frac{\hbar k_n}{m^*_{\mathrm{c(hh)}}} = \sqrt{\frac{2\epsilon_n}{m^*_{\mathrm{c(hh)}}}}. \tag{11.13}$$

Then the frequency of one round trip inside the QW is

$$\tau_n^{-1} = \frac{v_n}{2L} = \frac{1}{L}\sqrt{\frac{\epsilon_n}{2m^*_{\mathrm{c(hh)}}}} \tag{11.14}$$

and

$$\Gamma_n = \frac{\mathcal{T}_n}{\tau_n}. \tag{11.15}$$

What is the order of magnitude of τ_n? For an energy of $\epsilon_n = 10\,\mathrm{meV}$, $\tau_n \approx 200\,\mathrm{fs}$ for electrons and $400\,\mathrm{fs}$ for holes. The tunneling probability $\mathcal{T}_n$ therefore needs to be quite small indeed for the rate Γ_n to remain small and retain carriers inside the QW sufficiently long.

11.2.1 In-plane confinement

So far, we have considered the QCSE in a single dimension, as if we were to apply a global electric field. However, as we saw before, the field lowers the exciton energy below the QW confinement and hence, if applied locally, results in an effective confinement potential in the plane of the QW. In order to understand the energy level structure and optical properties of excitons confined in such an electrostatic trap, we now go beyond the 1D case and add a lateral confinement potential.

Descamps [70] performed numerical simulations for a geometry with circular gate electrodes with 200 nm diameter on both sides of a membrane, finding a confinement depth of $V_0 = -20\,\mathrm{meV}$ at $F = 5\,\mathrm{V/\mu m}$ that is well approximated by a single-particle harmonic potential for the center-of-mass wave function of the exciton, $V(\rho) = M\omega^2\rho^2/2 + V_0$, with mass $M = m^*_{\mathrm{c}} + m^*_{\mathrm{hh}}$ and confinement strength $\omega/2\pi = 738\,\mathrm{GHz}$ corresponding to an oscillator length $\xi = \sqrt{\hbar/M\omega} = 20\,\mathrm{nm}$. To obtain a qualitative picture, let us interpolate this result for different fields. The depth of confinement corresponds to the energy shift induced by the QCSE and is quadratic in F, $V_0 = -\alpha F^2$. At vanishing field, there should be no in-plane potential, $\omega = 0$. Hence, we can assume that $\omega \propto F$ to leading order and thus $V(\rho) = \beta M F^2 \rho^2/2 - \alpha F^2$.

How does this in-plane confinement modify the wave function? Because the exciton Bohr radius in the plane, $a_{\mathrm{B}}^{(2D)}$ (Equation 11.2), is much smaller than the oscillator length ξ, we separate the exciton's center-of-mass motion from the relative electron-hole motion in the plane [248]. Since furthermore the QW well width L is not much larger than the bulk Bohr radius $a_{\mathrm{B}}^{(3D)} = 2a_{\mathrm{B}}^{(2D)}$, we initially also separate electron and hole motions normal to the plane from the in-plane motion, allowing us to make the product ansatz

$$\Psi_{nmp\ell}(z_{\mathrm{e}}, z_{\mathrm{h}}, \rho, \phi, \varrho) = \psi_n(z_{\mathrm{e}})\psi_m(z_{\mathrm{h}})\zeta_{p\ell}(\rho, \phi)F(\varrho), \tag{11.16}$$

where $\psi_n(z)$ are the out-of-plane wave functions derived above, $F(\varrho)$ is the wave function of the relative electron-hole motion, and

$$\zeta_{p\ell}(\rho, \phi) = \chi_{p\ell}(\rho)\exp(i\ell\phi) \tag{11.17}$$

with[7] [249]

7: Note that Karimi et al. [249] miss a factor $1/\sqrt{2\pi}$ in the normalization.

$$\chi_{p\ell}(\tilde{\rho}) = \sqrt{\frac{2p!}{2\pi\xi^2(p+|\ell|)!}}\exp\left(-\tilde{\rho}^2/2\right)\tilde{\rho}^{|\ell|}L_p^{|\ell|}(\tilde{\rho}), \tag{11.18}$$

is the center-of-mass wave function of the exciton in a harmonic potential in cylindrical coordinates. Here, $L_p^{|\ell|}(x)$ is the associated Laguerre polynomial and we used the shorthand $\tilde{\rho} = \rho/\xi$. The numbers $p \in \mathbb{N}$ and $\ell \in \mathbb{Z}$ denote the principal and orbital momentum quantum numbers. The eigenenergies of the harmonic oscillator solution are given by

$$\epsilon_{p\ell} = \hbar\omega\,(2p + |\ell| + 1), \tag{11.19}$$

and so (excluding excitations of the relative degree of freedom as well as the exciton binding energy) the energy of a trapped exciton is given by

$$\Delta E_{nmp\ell} = E_{\mathrm{g}} + \epsilon_n + \epsilon_m + \epsilon_{p\ell} \tag{11.20}$$

with $\epsilon_{n(m)}$ the out-of-plane eigenenergies of an electron (hole) in state n (m).

Now, in order for the exciton to recombine efficiently, electron and hole must be in the same location and hence $F(\varrho) = F(0)$ [248, 250]. In this case, we can also slightly relax the assumption that in- and out-of-plane motion separate and to a first approximation perform the replacement $\rho \to r = \sqrt{\rho^2 + z^2}$ in Equation 11.18 so that $\Psi_{nmp\ell}(z_{\mathrm{e}}, z_{\mathrm{h}}, \rho, \phi, \varrho) = \Psi_{nmp\ell}(z, z, r, \phi, 0)$. Furthermore, we consider only transitions with $n = m$ since those with unequal z quantum numbers are much weaker [231] and write in this case as a shorthand $\Psi_{nmp\ell}(z, z, r, \phi, 0) = \Psi_{np\ell}(z, r, \phi)$ as well as $\Delta E_{np\ell}$. The wave function for vanishing electron-hole separation is shown in Figure 11.4 for $n = \ell = 0$ (which makes it independent of ϕ). For $\ell = 0$, $\Psi_{np\ell}$ has n nodes along z and p nodes along ρ.

We can now use the exciton wave function $\Psi_{np\ell}$ to calculate the *oscillator strength*, a quantity often quoted in optical spectroscopy of semiconductors. The oscillator strength puts in relation the quantum mechanical transition rate with the emission rate of a classical oscillator with frequency $\omega = \Delta E/\hbar$ matching the transition energy [251]. For a dipole transition from state $|i\rangle$ to state $|j\rangle$, it may be written as [231]

$$f_{ji} = \frac{2m\Delta E_{ji}}{\hbar^2}|\langle j|\mathbf{r}|i\rangle|^2. \tag{11.21}$$

The states $|i\rangle$ and $|j\rangle$ are the full Bloch wave functions of valence and con-

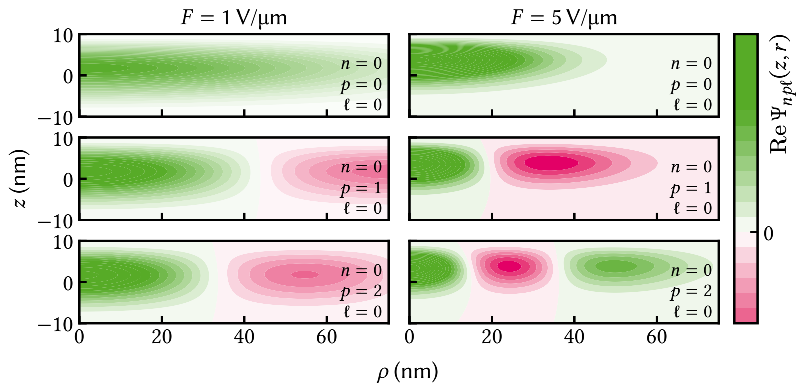

Figure 11.4: Exciton wave function at zero electron-hole separation in a harmonic trap under an electric field. Left column shows the low-field and right column the high-field case. Top row is the ground state, middle and bottom row the first and second excited state in the plane, respectively. The out-of-plane wave function is the ground state in all cases ($n = 0$) and the trap confinement strength $\omega/2\pi = 738\,\mathrm{GHz}$ for $F = 5\,\mathrm{V/\mu m}$ [70, Section 2.2.2] and linearly interpolated for $1\,\mathrm{V/\mu m}$ (see text).

duction band. Adapting the expression given by Andreani and Bassani [250] to account for the in-plane confinement, we posit $\Psi_{np\ell}$ as the envelope function overlap of the Bloch states $|i\rangle$ and $|j\rangle$ so that the matrix element in Equation 11.21 factorizes into a matrix element of the lattice-periodic part and an integral over the envelope function overlap,

$$\begin{aligned} f_{np\ell} &= \frac{2m\Delta E_{np\ell}}{\hbar^2}\left|\iint \mathrm{d}\mathbf{r}\,\Psi_{np\ell}(\mathbf{r})\right|^2 |\langle u_\mathrm{c}|\mathbf{r}|u_\mathrm{hh}\rangle|^2 \\ &= \frac{2m\Delta E_{np\ell}}{\hbar^2}\left|J_r J_\phi F(0)\right|^2 |\langle u_\mathrm{c}|\mathbf{r}|u_\mathrm{hh}\rangle|^2 \end{aligned} \tag{11.22}$$

with [248, 250]

$$J_r = \int_0^L \mathrm{d}z \int_0^\infty \mathrm{d}\rho\,\rho\psi_n^{(\mathrm{e})}(z)\psi_n^{(\mathrm{h})}(z)\chi_{p\ell}(\sqrt{\rho^2+z^2}), \tag{11.22a}$$

$$J_\phi = \int_0^{2\pi} \mathrm{d}\phi\,\exp(i\ell\phi), \tag{11.22b}$$

$$F(0) = \frac{1}{a_\mathrm{B}^{(2D)}}\sqrt{\frac{2}{\pi}}, \tag{11.22c}$$

and where m is the free electron mass. Equation 11.22c holds exactly only for strictly two-dimensional excitons [250] and $|u_{\mathrm{c(hh)}}\rangle$ are the lattice-periodic parts of the Bloch functions of the conduction and valence bands, respectively. Since these are not readily available, we will use the reduced oscillator strengths in the following, assuming that the lattice-periodic matrix element is independent of the electric field,

$$\tilde{f}_{np\ell} = \frac{f_{np\ell}}{|\langle u_\mathrm{c}|\mathbf{r}|u_\mathrm{hh}\rangle|^2}. \tag{11.23}$$

From Equation 11.22b, we immediately see that the oscillator strength of states with nonzero orbital momentum ($\ell \neq 0$) vanishes, $f_{np0} = 0$, meaning that states with $\ell \neq 0$ are not optically active! This in turn implies from Equation 11.19 that the optically resolvable exciton level spacing in a radially symmetric trap is given by $\Delta E = 2\hbar\omega = 1\,\mathrm{meV}$, a factor of two larger than assumed by Descamps [70]. Of course, the radial symmetry will most likely be broken in a realistic experimental scenario, and it would be interesting to investigate if indeed those states are optically inactive.

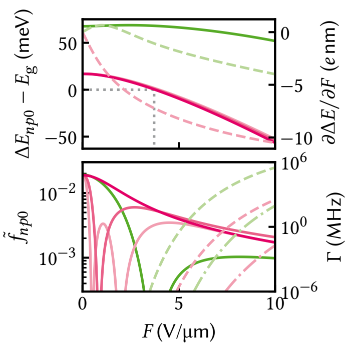

Figure 11.5: Electric field dependence of the QCSE for the first two energy levels in the QW. Top: the ground state energy (magenta) shows the expected quadratic dependence; the confinement energy is compensated at around $F = 3.7\,\mathrm{V/\mu m}$. Higher in-plane orbital levels (p) are shown for the QW ground state in decreasing saturation. For zero field, the splitting is zero as there is no in-plane confinement. At maximum field, the splitting is $2\hbar\omega = 2\,\mathrm{meV}$. Dashed lines (right axis) show the derivative, revealing that the QW excited state is actually raised in energy at low fields. Bottom: oscillator strengths (same color code as above). Dashed lines (right axis) show the electron tunneling rate through the barrier, dash-dotted the holes.

Figure 11.5 summarizes the QCSE in a QW additionally confined in the plane by local electric fields. The top panel shows the exciton transition energy ΔE_{np0} for the optically active $\ell = 0$ states in solid lines.[8] The

8: Note again that we neglected the exciton binding energy, which decreases with increasing field and hence slightly reduces the red shift [244].

first three in-plane sublevels due to the harmonic potential are drawn in less saturated colors but hard to see because the level spacing is much smaller than the out-of-plane QW subband spacing, $\sim 1\,\mathrm{meV}/50\,\mathrm{meV}$. The ground state (magenta) shows the expected quadratic dependence on the electric field also obtained, for example, from perturbation theory. Drawn as a dashed line is the induced dipole moment,

$$\mathbf{p}_{np}(F) = \frac{\partial \Delta E_{np}}{\partial F}\hat{\mathbf{e}}_z, \tag{11.24}$$

which is monotonously decreasing for the ground state as function of electric field F, consistent with a continuous lowering of energy. For the first excited state (green), the induced dipole moment is actually positive below $2\,\mathrm{V}/\mu\mathrm{m}$, leading to a repulsive interaction and hence a raising of the transition energy by up to $1\,\mathrm{meV}$. The lower panel shows the reduced oscillator strength, Equation 11.23. For the ground state it decays approximately exponentially with the electric field as the overlap between electron and hole wave functions, which decay exponentially into the QW themselves, is reduced. At $F = 10\,\mathrm{V}/\mu\mathrm{m}$, $\tilde{f}_{000}$ is reduced by around an order of magnitude, which indicates that the brightness of an exciton trap should not be diminished too much by the applied field. f_{0p0} for higher in-plane orbital states ($p > 0$, magenta, decreasing saturation) has an envelope following the ground state's exponential decay but vanishes at p points in F due to the fact that the wave function has p nodes. By contrast, f_{100} for the first excited QW state (green) does not appear to decay to zero at large fields but also has a zero at an intermediate field because of the wave function's node along z.

Finally, the right axis shows the estimated tunneling rates (Equation 11.15) of the electron (hole) ground (excited) states in dashed (dash-dotted) magenta (green) lines, respectively. Rates of electrons are at least four orders of magnitude larger than those of holes owing to their smaller effective mass despite the larger band offset. At $F = 5\,\mathrm{V}/\mu\mathrm{m}$ corresponding to a voltage of $1\,\mathrm{V}$ across a membrane of $200\,\mathrm{nm}$ thickness, the electron tunnel rate is on the order of $1\,\mathrm{Hz}$, but rises very quickly above that. Considering once again a QW hosting a 2DEG and neglecting associated band bending effects, the rate corresponds to a current through a circular area of $1\,\mu\mathrm{m}$ in diameter of $1\,\mathrm{fA}$ at an electron density of $n = 5\times10^{11}\,\mathrm{cm}^{-2}$, but already $500\,\mathrm{pA}$ ($800\,\mathrm{kHz}$) at $7.5\,\mathrm{V}/\mu\mathrm{m}$. For single excitons, however, which have a lifetime on the order of $1\,\mathrm{ns}$, even a tunnel rate of $1\,\mathrm{MHz}$ results in a sufficiently low tunnel probability $\sim 1\,‰$.

11.3 Excitonic complexes

If we liken the exciton to a Hydrogen atom[9] with modified binding energy E_b and Bohr radius a_B, it is natural to expect molecules and ions of these states to exist. The simplest molecule, or excitonic complex, is the biexciton, the analog of the H_2 molecule. Following a similar argument as for the binding energy of the exciton, we expect that of the biexciton to also increase as the dimensionality of the system is reduced. Indeed, in QWs it has been measured to be on the order of $1\,\mathrm{meV}$, 3 to 4 times larger than in 3D [252], although also negative binding energies, corresponding to an anti-bonding state, have been reported [253–255]. A signature of biexcitons is the power dependence of their PL. The steady-state density of electron-hole pairs, which is proportional to the exciton

9: Although positronium is a better analogon.

density at low enough powers, is proportional to the irradiance, *i. e.*, the excitation power. Conversely, the probability to form a two-body bound state is proportional to the square of the density of those bodies and we hence expect the power under the PL peak of a biexciton emission to be proportional to the square of the laser power used to excite the sample.

Besides the neutral biexciton, there exist also charged states consisting of three or more bodies. Here, additional electrons or holes bind to an exciton, forming negative or positively charged trions similar to H^- or He^+ ions. Naturally, this process is favored when a large number of surplus charge carriers is present in the sample such as is the case in doped QWs with a 2DEG (or two-dimensional hole gas (2DHG)) [256]. In these structures, the trion binding energy has been found to also be on the order of 1 meV [257, 258].

The mjolnir measurement framework 12

To facilitate the optical experiments in this part of the present thesis, I wrote a software package that controls the measurement workflow and all other interactions with the setup described in Part II, `mjolnir`[1] [259]. The package is written in Python, is open source, and has a documentation as well as some rudimentary tests.[2] In this chapter, I lay out the goals and non-goals behind its development and briefly describe the design, features, as well as the typical workflow.

1: https://git.rwth-aachen.de/qutech/mjolnir

2: Testing code interacting with hardware is notoriously difficult to achieve in an automated manner.

12.1 Rationale

There exist various software solutions for measurement control, ranging from commercial (*e.g.*, Keysight Labber[3]) over open source (`quantify`,[4] `labcore`[5]) to home-built (`elicit`,[6] `QuMADA`[7]). While most of these frameworks attempt to achieve some degree of universality and not be tailored towards a specific type of experiment or setup, some bias always exists and most are geared towards fully electrical experiments. On the other hand, software for optical experiments naturally exists as well, but it often suffers the same shortcomings coming from a different direction, and there is fairly little on offer for experiments at the interface of quantum optics and transport. An added complication is the availability of drivers. Striving for full automation of the setup, a number of different drivers are required given the complexity of the setup and the mix of electrical, optical, and mechanical instruments (Part II). Thus, `QCoDeS` forms a good basis for our specific needs due to its large driver coverage. However, the wide array of physical instruments demands some level of abstraction to promote reproducibility and ease-of-use. Furthermore, while `QCoDeS` provides measurement functionality, their definition involves a large amount of repetitive setup code that needs to be duplicated for each measurement, invites errors, and degrades legibility. The `doNd`[8] functionality abstracts some of this away for multi-dimensional loops but is not flexible enough for our purposes.

3: https://www.keysight.com/de/de/products/software/application-sw/labber-software.html

4: https://quantify-os.org/

5: https://github.com/toolsforexperiments/labcore

6: https://git-ce.rwth-aachen.de/qutech/frameworks/qool-tool

7: https://github.com/qutech/qumada

8: https://microsoft.github.io/Qcodes/api/dataset/index.html#qcodes.dataset.dond

By contrast, the experiments conducted in the present thesis do not place high demands on timing accuracy and measurement speed as typical PL integration times are relatively long. The software thus does not need to prioritize sophisticated triggering and pulsing logic required by qubit experiments. Because the requirements are rather specific, `mjolnir` is therefore single-minded. It does not aspire to be suitable for applications other than the ones it is designed for. The implementation is extremely biased towards the particular set of instruments in use in the lab and does not attempt to generalize to allow for different instruments to be used. At the same time, it is designed with modularity in mind, allowing individual components to be readily swapped out for alternative solutions.

12.2 Instrument abstraction

Central to the `mjolnir` package is the abstraction of physical instruments into logical ones, thereby grouping logical functionality provided by different physical devices. Take for instance the tunable continuous-wave (cw) M Squared Solstis laser. It is cooled by a Thermotek T225p chiller

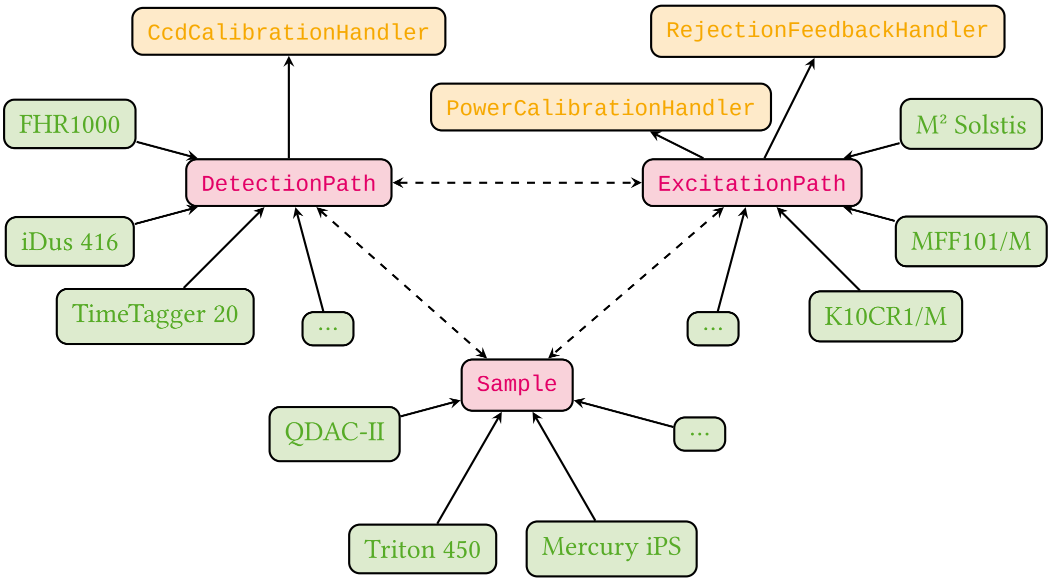

Figure 12.1: Abstraction of physical instruments. `mjolnir` defines three logical instrument classes (magenta) that represent different aspects of the experiment and group various physical instruments (green). Logical instruments expose various logical parameters that include communication with multiple physical instruments. `Sample` can be subclassed to suit the particular type of device under investigation. Calibration handlers (orange) are controlled by the logical instruments governing the optical path. Logical instruments are part of a `QCoDeS Station` and can interact with each other as indicated by the dashed arrows.

and pumped by a Lighthouse Photonics Sprout-G diode-pumped solid state (DPSS) laser. Behind its exit aperture, a Thorlabs MFF101/M acts as a shutter, a neutral-density (ND) filter mounted on a Thorlabs K10CR1/M controls the output power, while a Thorlabs PM100D monitors the power at the optical head and an Attocube AMC100 controls the polarization state of the optical head. Thus, seven different instruments from five different manufacturers are required to control the illumination state of the sample. As a user, however, one would simply like to en- or disable the laser and set wavelength, power, and polarization configuration of the optical head.

To simplify and abstract away the particularities of the physical instruments providing control over parts of a user-facing parameter such as the excitation wavelength,[9] three logical instruments implemented as subclasses of the `QCoDeS Instrument` class govern the experimental apparatus:

9: For example, the power meter needs to be informed when changing the wavelength to keep the internal calibration up to date.

1. `ExcitationPath`. Controls all physical instruments related to illumination of the sample, including the white light source.
2. `DetectionPath`. Controls instruments related to the detection of radiation emitted by the sample, *i. e.*, the spectrometer, CCD, and photon counting card.
3. `Sample`. Controls the QDevil QDAC-II voltage source, cryostat, and magnet, and implements a software representation of the device under test (DUT).

Figure 12.1 shows a graph outlining the relationship between physical and logical instruments. Figure 12.2 shows the repository structure. In the following, I give a brief overview of the functionalities comprised by these logical instruments.

12.2.1 Excitation path

As already outlined above, the `ExcitationPath` object controls everything related to the illumination of the sample. The `active_light_source` parameter switches between laser, white light, and no illumination. Turning on the laser comprises enabling the chiller, switching on the pump laser, waiting for it to ramp up, and asserting the wavelength lock is acquired. For safety reasons, confirmation by the user that they are physically in the lab is required. Furthermore, the `ExcitationPath` instrument implements setting the excitation power by means of a calibration of the ND filter, see Section 12.3. Since the output power of the laser varies depending on wavelength, the object furthermore provides a `wavelength_constant_power` parameter that automatically recalibrates the power once a new wavelength is set. Finally, it manages the state of the automatic excitation rejection control, see Section 12.3.

12.2.2 Detection path

The `DetectionPath` object is responsible for the optical analysis. Chiefly, it manages the Horiba FHR1000 spectrometer and provides convenient shorthands for selecting grating (`active_grating`) and exit port (`active_detection_path`, either `"ccd"` or `"apd"`). It handles initialization of the spectrometer and the charge-coupled device (CCD) including the CCD cooler and oversees calibration of the CCD pixel-to-wavelength relation, see Section 12.3. From the CCD calibration and its pixel size, the dispersion at the imaging plane of the currently selected grating can be obtained, allowing the user to set the monochromator bandwidth of the spectrometer in terms of a wavelength window rather than the more abstract exit slit width.

12.2.3 Sample

The `Sample` class initializes the QDevil QDAC-II digital-to-analog converter (DAC) and magnet power supply. Mainly, though, it serves as an abstraction of the actual DUT and its "control knobs" such as gates. Users implement subclasses for different sample designs. As the samples we are concerned with in the present thesis are all of the same type, I discuss the main implementation, `TrapSample`, that represents membrane samples with exciton traps. On a single sample, there are any number of traps consisting of one or both sets of top and bottom "central" and "guard" gates. Each trap is implemented as a `QCoDeS` `InstrumentModule`, `Trap`, that is part of a `ChannelList`. The currently active trap (*i.e.*, the one currently in focus by the microscope) is selected by the `active_trap` parameter and accessible through the `trap` property of the `TrapSample` object. A `Trap` exposes the virtual gate parameters[10] `difference_mode` and `common_mode` (see Subsection 13.1.1) as the difference and sum of top and bottom gate voltages, respectively, as well as the unmodified top and bottom gate voltage and corresponding leakage current parameters. DAC channels are mapped to their corresponding `Trap` parameters using `yaml` configuration files specific to each sample and hosted in a separate repository.

```
├── doc
├── src
│   └── mjolnir
│       ├── config
│       │   ├── valhalla
│       │   └── examples
│       ├── instruments
│       │   ├── __init__.py
│       │   ├── logical_[…].py
│       │   └── physical_[…].py
│       ├── measurements
│       │   ├── __init__.py
│       │   ├── handler.py
│       │   ├── measures.py
│       │   └── sweeps.py
│       ├── parameters
│       ├── plotting
│       │   ├── __init__.py
│       │   ├── live_view.py
│       │   └── plot_nd.py
│       ├── __init__.py
│       ├── calibration.py
│       └── …
├── main.py
├── pyproject.toml
└── …
```

Figure 12.2: Source tree structure of the `mjolnir` package. Logical `QCoDeS` instruments and parameters are defined in the `instruments` and `parameters` modules, respectively. Instruments are configured using `yaml` files located in a `config` subdirectory. The `measurements` module provides classes for the abstraction of measurements using `QCoDeS` underneath. Live plots of instrument data as well as a plot function for multidimensional measurement data are defined in the `plotting` module. `calibration.py` contains routines for power, CCD, and excitation rejection calibrations. The `main.py` file is a code cell-based script that serves as the entrypoint for measurements.

10: Using the QDevil QDAC-II's built-in `Arrangement_Context` virtual gate functionality.

12.3 Calibrations

mjolnir implements three automatic calibration procedures in the classes CcdCalibrationHandler, PowerCalibrationHandler, and RejectionFeedbackHandler (see Figure 12.1). As their names suggest, these handle calibration of the CCD pixel-to-wavelength conversion, the power transmission through the rotatable ND filter, and the waveplate and polarizer angles for optimal laser rejection, respectively.

12.3.1 CCD calibration

Because the dispersion of the spectrometer gratings depends to a small degree on wavelength, the calibration function converting horizontal CCD screen pixels into wavelengths needs to be updated for every central wavelength selected using the grating. In principle, this requires measuring the position of several reference wavelengths on the CCD screen and fitting a polynomial of second or third degree. As the tunable cw laser is coupled to a wavelength meter with built-in calibration lamp (HighFinesse WS6), it lends itself naturally as a source of the reference wavelength. In practice, the deviation from an established calibration function is usually small enough that simply shifting the center pixel (that is, the zeroth-order polynomial term) suffices for reasonably small wavelength ranges of tens of nanometers. The CcdCalibrationHandler implements three different calibration schemes specified as the ccd_calibration_update_mode parameter of the DetectionPath instrument:

1. "full". The pixel position of the maximum intensity of nine equidistant wavelengths centered around the grating central wavelength and spread out over the entire range of the CCD screen (45 nm for the 600 gr/mm grating and 9 nm for the 1800 gr/mm grating) is measured sequentially and fitted to a third-degree polynomial.
2. "fast". The laser is set to the grating central wavelength and the zeroth-order coefficient of the previous calibration polynomial is shifted by the pixel distance between the peak and the horizontal center of the CCD screen.
3. "dirty". The zeroth-order coefficient of the calibration polynomial is shifted by the difference in nominal grating and laser wavelengths.

By default, the "dirty" mode is enabled as it does not require any grating moves or wavelength changes and is therefore the fastest. Rather than recalibrating, old calibration data can also be loaded from disk if it is not older than a user-specified date. Finally, note that the spectrometer grating motor also requires occasional recalibration. Both the coefficient of linearity (converting motor steps to selected central wavelength) and the offset in motor steps can change over time. Currently, this calibration is not automated but is instead compensated for by shifting the central wavelength on the CCD screen. In principle, the driver for the Horiba FHR1000 spectrometer implements all necessary functionality for automating the calibration procedure already.

12.3.2 Power calibration

11: Thorlabs NDC-25C-4 [260].

The ND filter[11] mounted on the Thorlabs K10CR1/M rotation stage has a continuously variable optical density (OD) (Equation 7.26) of 0.04 to 4.0, which depends quite sensitively on the wavelength. In order to allow reliably setting a specified excitation power at any excitation wavelength, the `mjolnir` package automates the calibration of the filter angle against transmitted power. Two modes are implemented, `"full"` and `"fast"`. In the former mode, the transmitted power is sampled at a given angular interval and the resulting data fitted using a quadratic smoothing spline. Figure 12.3 shows a typical measurement and spline fit at 795 nm. The specified power is then set by inverting the spline and evaluating at the given power.

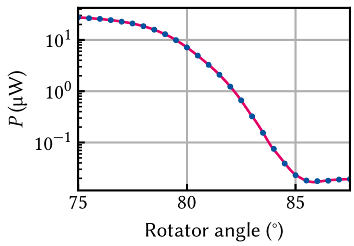

Figure 12.3: Power calibration using the continuously variable ND filter. Error bars from statistics are too small to see. Magenta line is a quadratic smoothing spline.

As performing the full calibration is quite time-consuming and furthermore the inversion not fully reliable, the `"fast"` calibration update mode sets the rotator angle by performing a noisy optimization. Here, first the last valid full calibration is loaded and its spline scaled to the current power level. Then, the residual sum of squares between the target power and the power meter reading is minimized using the `noisyopt` package [261] with the starting value given by the angle predicted by the rescaled calibration spline.

12.3.3 Rejection feedback

The excitation rejection by means of cross-polarization extinction has already been discussed in Subsection 7.1.4. As demonstrated there, the OD achieved by cross-polarizing the excitation beam with the analyzer mounted in the detection path reaches a maximum close to 6 and is extremely sensitive to small variations in the angles of the $\lambda/4$- and $\lambda/2$-plates controlling polarization state. Close to the optimum, the OD is well-described by a paraboloid as function of those angles, with a second-order polynomial coefficient on the order of $-10^{-2}/\mathrm{deg}^2$. Since the optics including the waveplates display chromatic dispersion, the excitation rejection needs to be re-optimized whenever the laser wavelength is changed, and the exponential angular dependence of the transmitted intensity places high demands on the dynamic range of the photodetector used for measurement. Therefore, the reflected laser intensity is measured with the avalanche photodiode (APD) signal digitized by the Swabian Instruments Time Tagger 20 on the side exit port of the spectrometer.[12] The Time Tagger can reliably cover a count rate range from 10^2 cps to 10^6 cps, compared to the Andor iDus 416 CCD covering only 10^3 cps to 10^5 cps.

12: The signals from both APDs are combined to achieve the best signal-to-noise ratio (SNR), see Part II.

In the `mjolnir` package, the calibration is then carried out as a bivariate noisy minimization using `noisyopt` [261]. Upon initialization the power is set to a small value to avoid overexposing the detectors. Then, the spectrometer is set to select the laser wavelength, the side exit port is selected, and the optimization over the $\lambda/4$- and $\lambda/2$-angles is run within conservatively set bounds of $[-2.5°, 2.5°]$, which has been found to be a good compromise of robustness and convergence speed. A caching mechanism stores the optimal angles together with the current wavelength, which serves as a starting guess upon subsequent calibration runs. The calibration is quite tedious and time-consuming for several reasons. Chiefly, moving the waveplate angles is slow simply because it is a mechanical movement, and optimizers typically do not take into account the distance between subsequent sample points, potentially resulting in unnecessarily inefficient exploration of the parameter space. Incorporating a penalty on the sample distance into the optimization algorithm would hence improve the convergence speed. Additionally, due to vibrations in the cryostat (Chapter 8), the root mean square (RMS) intensity of the back-scattered laser is large – and the larger the closer the excitation rejection is to its optimal value[13] – and thus requires long averaging times for a robust measurement. Finally, the algorithm outlined above fails when the starting guess is too far away from the optimum, such as when the new wavelength is on the order of tens of nm away from the previously optimized one. A good – albeit time-consuming – strategy is then to incrementally update the wavelength.

13: The count rate can vary by up to a factor of two with the frequency of the pulse tube refrigerator (PTR) pulses.

12.4 Measurement routines

Measurements in `mjolnir` use the `QCoDeS` infrastructure for data acquisition and storage. To abstract away code for repetitive setup and teardown tasks, the user interacts with a `MeasurementHandler` object through the `measure()` method. The class can be subclassed to customize the aforementioned tasks for different types of experiments (such as those that involve illumination with the laser and detection with the CCD, `LaserCcdMeasurementHandler`), or add default parameters to every measurement (such as leakage currents or the laser power). A measurement's independent parameters are passed to `measure()` as instances of a `Sweep` class which completely determines the measurement structure. `Sweep` objects support syntactic sugar for concatenation (`@`), nesting (`|`), and parallelization (`&`). Dependent parameters (those that should be measured) can be simple parameters or instances of a `Measure` class.[14] The state of external parameters, that is, parameters that are not varied or measured during a measurement, can be defined for the duration of the measurement using the `parameter_contexts` argument, a mapping from `QCoDeS` parameters to valid values thereof. This uses the `Parameter.set_to()` context manager to set the parameters to the given values before the measurement starts and restore their original values upon exit in a controlled manner, allowing users to ensure their device is in a well-defined state. Custom setup and teardown tasks can furthermore be specified through the `add_before_run` and `add_after_run` arguments, respectively.

14: Besides some optimization for measurements of parameters delegating from the same physical parameter, the latter exists mostly for (primitive) live plotting.

A typical workflow for a PL measurement using laser and CCD is sketched in Listing 12.1. The `MeasurementHandler` object automatically takes care of, among other things, arming the CCD, acquiring a background image in the dark, and opening the laser shutter for the duration of the measurement. Acquired data is saved to the `QCoDeS` database, but a helper function exporting to the `xarray` `Dataset` format is available. Saved together with the data are a snapshot of the `QCoDeS` `Station` as well as arbitrary custom metadata. More details and examples on the measurement functionality can be found in the documentation.[15] Lastly, I note that the measurement functionality in `mjolnir` is independent of the instrument abstraction and calibration logic and as such could be replaced or augmented with minimal effort by that of other `QCoDeS`-based frameworks such as `quantify` or `labcore` to make use of their infrastructure. I leave this option for future work.

15: `https://qutech.pages.rwth-aachen.de/mjolnir/`

```python
from mjolnir.measurement.handler import LaserCcdMeasurementHandler
from mjolnir.measurement.sweeps import GridSweep
from mjolnir.measurement.measures import Measure

handler = LaserCcdMeasurementHandler(station)

# excitation_path, sample are instances of the ExcitationPath and
# TrapSample classes, respectively
power_sweep = GridSweep(excitation_path.power_at_sample,
                        rng=(5e-9, 5e-7), num_points=11,
                        spacing='geom')
gate_sweep = GridSweep(sample.trap.central_difference_mode,
                       rng=(-2, -1), num_points=51))

# two-dimensional sweep over power and gate voltage
sweeps = power_sweep | gate_sweep
# keep track of the MXC temperature
measures = sample.fridge.T8

dataset = handler.measure(
    sweeps,
    measures,
    parameter_contexts={
        sample.trap.central_common_mode: -1,
        detection_path.central_wavelength: 825,
    },
    exposure_time=2
)
```

Listing 12.1: Setup and measurement workflow using the `mjolnir` package. `station` is a `QCoDeS` `Station` object managing the instruments. The `sweeps` object describes a nested loop on whose inner iteration the difference mode parameter of the trap's central gate is swept over a linear grid and on whose outer iteration the laser power, adjusted for the beam splitter ratio, is swept over a logarithmically spaced grid. No dependent parameters (`Measure` objects) need to be explicitly specified as the `LaserCcdMeasurementHandler` measures the CCD spectrum as well as laser power and leakage currents of the swept gates by default. The `parameter_contexts` argument is used to set the spectrometer wavelength to 825 nm and the common mode voltage of the active trap to −1 V. The `exposure_time` argument is passed through to the `initialize` method, where it is used to set up the CCD for acquisition.

12.5 Plotting

Optical measurements using `mjolnir` often produce multi-dimensional datasets by virtue of the default parameter measured, the CCD spectrum, being vector-valued or *batched*. To visualize such datasets with more than two dimensions, `mjolnir` provides the `plot_nd()` function that plots 2D slices through the data as a false-color image and generates interactive slider widgets that allow users to modify the slice coordinates. This facilitates interactively exploring large datasets and recognizing trends present in the data beyond two dimensions. Figure 12.4 shows an exemplary plot window for a four-dimensional data set obtained from a CCD measurement sweeping laser wavelength, power, and a gate voltage, with Listing 12.2 showing the code necessary to produce the plot.

```python
import matplotlib as mpl
from mjolnir.plotting import plot_nd

fig, axes, sliders = plot_nd(
    dataset_or_run_id=140, vertical_target='power_at_sample',
    yscale='log', norm=mpl.colors.AsinhNorm(vmin=0),
    fig_kw=dict(figsize=(6.33585, 4.6))
)
```

Listing 12.2: Code to produce the plot shown in Figure 12.4. If `dataset_or_run_id` is an integer, the currently connected `QCoDeS` database is queried for this run identifier. Otherwise, it should be an `xarray` `Dataset`.

The plotting functionality is implemented using `matplotlib`. While not the most performant library, its flexibility in decorating and customizing plots as well as scaling data, such as the asinh-scale normalization used in Figure 12.4, made it the preferred choice. To nonetheless improve the responsivity of the interactive elements, the plotted artists are *blitted*[16] using the `qutil.plotting.BlitManager` class from the `qutil` library [35].

16: Blitting refers to storing constant backgrounds of raster graphics and only redrawing changing elements. See the `matplotlib` documentation.

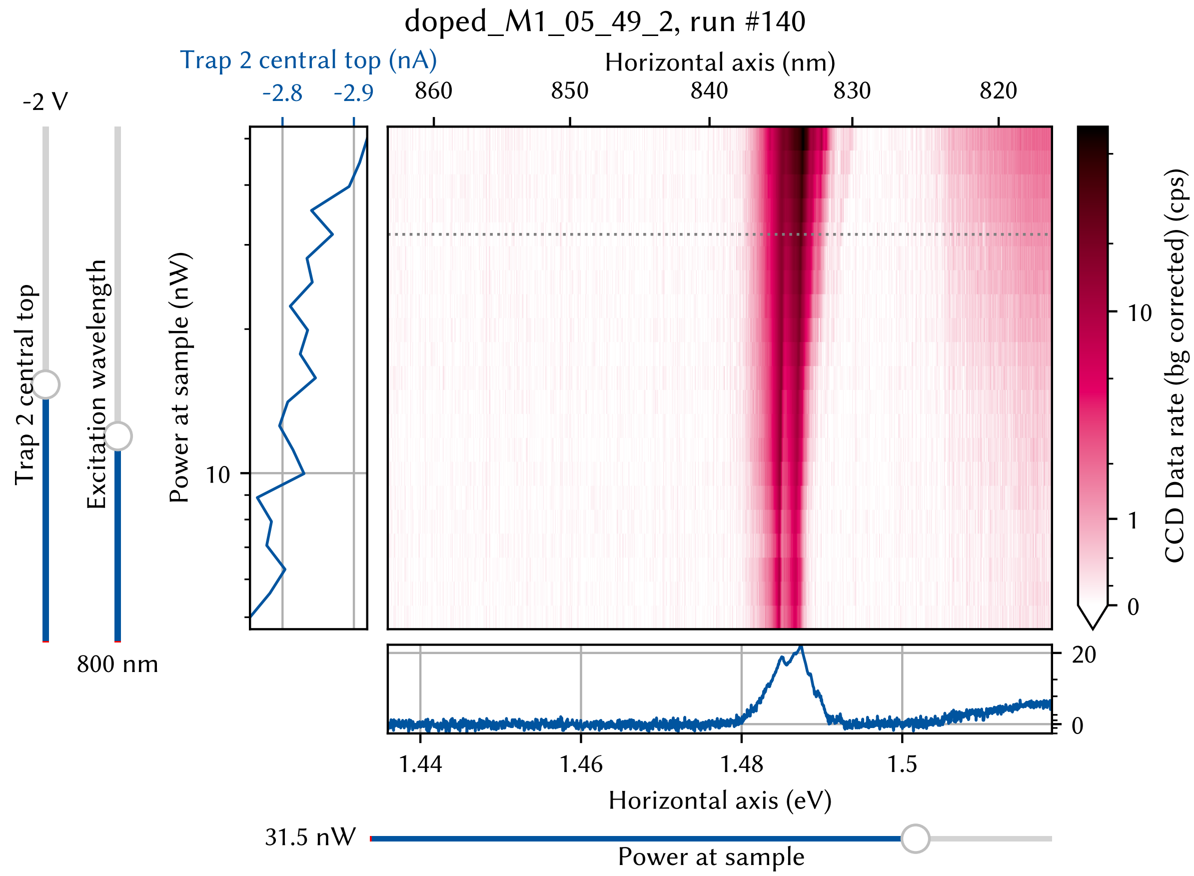

Figure 12.4: The `plot_nd()` plot window. The bottom panel, enabled by default, shows a horizontal line cut through the 2D data of the main panel whose y-value can be set with the adjacent slider widget. The left panel shows the leakage currents of gates that participate in the sweep and, if measured, the laser power. A vertical slider widget is added for each data dimension not displayed on the x or y axis of the main plot. The figure title shows the sample as well as the measurement identifiers assigned by `QCoDeS`.

The data variable to be shown in the main panel can be selected with the `array_target` argument, which accepts a string and chooses the closest match in the dataset since the full parameter names are often a bit unwieldy. Similarly, the coordinates to be plotted on the x- and y-axes are selected with the `horizontal_target` and `vertical_target` arguments, respectively.

In addition to plotting measurement data, `mjolnir` implements live plotting of instrument data leveraging the `qutil.plotting.live_view` module. Since most instruments do not support multiple concurrent connections, switching control between the Python interpreter and the graphical user interface (GUI) program provided by the manufacturer can often be tedious.[17] Thus, making use of their continuous data acquisition functionality is prohibitively time-consuming, but at the same time optical experiments frequently call for live observation of instrument data, for example when aligning the optics. Live plotting is currently implemented for three different instruments and corresponding parameters; the Thorlabs PM100D power meter (power reading), the Andor iDus 416 CCD (spectrum or image), and the Swabian Instruments Time Tagger 20 counting card (count rate(s)). Data acquisition is executed in a background thread while the plot, implemented in `matplotlib`, currently runs in the main thread, and hence does not update while the interpreter is blocked. In principle, though, the `live_view` module also supports plotting in a separate process, making a multiprocessing implementa-

17: For instance, the CCD GUI, Andor Solis, by default disables the thermoelectric cooler when exiting the program.

tion straightforward. Interactive GUI elements in the `matplotlib` figure allow for pausing and automatically rescaling the data.[18] Since for the CCD data acquisition runs using the *run till abort* mode of the device and as such continuously, the instrument is blocked for the duration of the live plot and it therefore needs to be closed before starting a measurement. By contrast, the power meter and Time Tagger parameters are simply queried regularly in sequential mode and can hence in principle remain open and running while performing measurements.

18: These two features can also be activated with the SPACE and R keys.

Photoluminescence measurements of doped semiconductor membranes

13

In this chapter, I present and discuss optical measurements of gated semiconductor membranes. All of the samples under investigation had the almost same nominal heterostructure layout; a 20 nm GaAs QW sandwiched between two modulation-doped 90 nm $Al_{0.33}Ga_{0.67}As$ barriers of which 50 nm are an undoped spacer layer. Together with either a 5 nm or 10 nm GaAs cap to on both sides to protect the aluminium from oxidation, the membranes had a thickness of 210 nm or 220 nm after thinning.

Gate electrodes were patterned on the top and bottom side of the membrane using electron-beam lithography, where here and elsewhere in the present thesis I refer by "top" to gates on the etched and by "bottom" to those on the as-grown surface. That is, "top" gates are on the air side facing the microscope objective when the sample is mounted in the fridge, whereas "bottom" gates are in contact with the epoxy gluing them to the Silicon host chip. Typical designs have gates on the top and bottom sides of the membrane that are contacted at different positions on the edge of the mesa and converge towards the exciton trap site, only close to where they overlap laterally. Figure 13.1 shows SEM images taken of such a trap. For details about the fabrication process, refer to Refences 70 and 229.

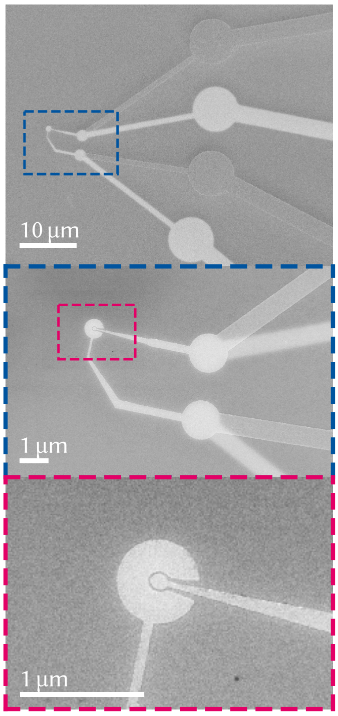

Figure 13.1: SEM image of an exciton trap with both central and guard gates. Gates on the bottom side are visible through the membrane. Images taken by Thomas Descamps.

I installed and cooled the samples to Millikelvin temperatures in the dilution refrigerator (DR) introduced in Part II, typically using the bottom-loading technique. As detailed there, PL measurements can be performed by illuminating the sample with a cw laser above the band gap through an objective lens in front of the sample. The emitted PL radiation is picked up by the same lens and coupled into a single-mode fiber (SMF) outside the fridge, from where it is sent to a diffraction grating spectrometer with a CCD for spectral analysis.

In total, I show measurements from three different samples. Unfortunately, none of the samples tested had a fully functional exciton trap with four working gates. Several old samples fabricated during the work of Descamps [70] appeared to have aged, resulting in broken gates or otherwise unconventional behavior. Newly fabricated samples on old heterostructure pieces frequently showed contact problems, either of optical gates to the mesa, between different lithography resolution steps, or Ohmic contacts, while other gates had leakage to ground or the QW.

I pursued two different approaches to positioning the laser on the samples. The first was using the white-light imaging arm of the confocal microscope (see Part II). The larger, thicker gate structures on the samples typically show good contrast in the CMOS camera image owing to the large reflectance of Au compared to GaAs (at normal incidence and a single interface $\mathscr{R} = 98\,\%$ and $32\,\%$, respectively, see Section 13.3), allowing orientating oneself following the sample design. However, the simplicity of the optics[1] results in a comparably poor contrast for the smallest features on the scale below 1 µm. The magnification factor of the microscope is 30, resulting in a feature size of 160 nm per pixel on the camera. In the best-case scenario, the cryostat vibration noise is on the order of 100 nm RMS (Section 8.3), or very roughly one pixel. Resolving features only a few pixels in size is thus clearly on the edge of the microscope's capabilities. Positioning the sample in this way is usually

1: The objective is just a singlet lens, compared to sophisticated commerical objectives containing a large number of optics inside to correct optical abberations. See for example the Attocube LT-APO/LWD.

fairly reliable with some experience if one takes visual identification of the exciton trap gates as a target. In practice, it is still necessary to fine-tune the position once the light source is switched to the laser because first, the focal distance of the objective lens shifts slightly when switching from a broadband to a monochromatic light source, and second, the focal spot of the laser is much smaller than that of the white light.[2]

2: The white light is launched from a 400 μm diameter multi-mode fiber (MMF), compared to a 5 μm diameter SMF for the laser, resulting in a spot 80 times larger.

The second approach is to roughly align the position on a large gate feature using the camera image, switch to laser illumination, and monitor the PL signal when biasing the gate. One can then move along the gate towards the exciton trap by following the PL features expected for a gate: a reduced QW emission due to absorption (see Section 13.3) and enhanced reflection of the laser line, as well as a Stark-shifted PL. This has the advantage that the functionality of the gate can be monitored. Biasing also the other gates of the trap under investigation, one can then look for additional Stark shifting of the PL. If the effect of each gate voltage can be observed, the position of the laser will be close to the trap center, and can then be optimized further. While this is in a sense flying in the dark, it is a quite reliable method for an experienced experimentalist *if* the gates are fully functional.

13.1 Photoluminescence spectroscopy

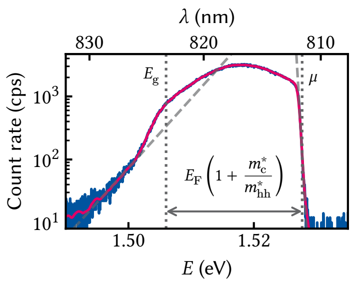

Figure 13.2: PL of the bare 2DEG. Magenta line is a smoothing spline fit to the data. Indicated by dotted gray lines are the Fermi edge at high and the band edge at low energy. The Fermi edge has a Fermi distribution (exponential indicated by a dashed gray line) whose temperature is typically much higher than the lattice temperature (~ 1 K). Below the band edge there is an exponential tail (dashed gray line) due to impurities that permeates far into the gap.

Figure 13.2 shows a typical PL spectrum obtained on the bare, unbiased QW of a doped membrane sample. This measurement corresponds to the configuration already discussed in Section 11.1. Due to the Pauli exclusion principle, electrons require an energy of at least $E_F\left(1+m_c^*/m_{hh}^*\right)$ above the band gap and, because of the vanishingly small photon momentum, a momentum of at least k_F to be excited into a free state above the Fermi level μ (dotted gray line).[3] Once excited, they quickly relax down to the Fermi edge at μ from where they can recombine emitting a photon. As the Fermi sea is at a finite temperature, the high-energy shoulder of the PL spectrum is hence thermally broadened according to the Fermi distribution function of the electron gas (dashed gray line). The associated temperature is around 1.5 K and hence orders of magnitude higher than the lattice temperature of 10 mK. This effect has already been observed by Pinczuk et al. [264]. Like in those experiments, the temperature of the Fermi edge does not vary significantly with excitation power, making local heating of the lattice due to high excitation power an unlikely cause [265]. I show a power-dependence measurement of the Fermi edge supporting this statement in Section B.2. Bockelmann and Bastard [266] showed that in low-dimensional electron gases electron cooling by phonon emission is limited by a finite scattering rate. Thus, under continuous excitation a quasi-equilibrium is established as hot electrons cannot loose heat quickly enough. Although their simulations were performed for higher temperatures and smaller confinement widths, their results of electron temperatures in the range of 2 K for lattice temperatures of 300 mK are comparable to ours.

3: This is known as the Burstein-Moss shift [262, 263].

PL emission is possible also at lower energies as electrons inside the Fermi sea recombine with the free photo-hole that scatters towards the valence band ege. The band gap then defines the low-energy shoulder of the PL spectrum, below which there are – ideally – no states available (dotted gray line). However, the PL reveals there are indeed free states decaying exponentially into the gap, originating most likely from impurities (dashed gray line). Compared to the results of Kamburov et

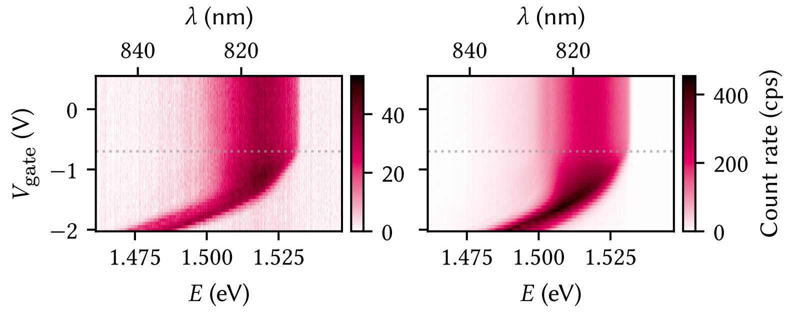

Figure 13.3: PL as function of gate voltage on a single fan-out gate on the bottom (left) and top (right) side of the membrane. The behavior is qualitatively similar but the overall end-to-end efficiency lower by an order of magnitude for gates on the bottom (as-grown buried) side. Dotted gray lines are a guide to the eye demonstrating that the changes to the PL spectrum set in at the same voltage for both types of gates (around −0.7 V).

al. [232], the PL spectra obtained here are much flatter over energy, with the PL peak typically close to the middle between gap and Fermi edge. Conversely, Kamburov et al. [232] observed a strong peak at the band gap,[4] indicating that in our samples holes are more strongly localized and therefore have a wider spread in k-space, enabling transitions in a wider range of wave vectors [235]. This would in turn imply increased alloy disorder or interface roughness [237], an observation we shall come back to in Section 13.2.

4: *Cf.* also Gabbay et al. [237], who observe all but no FES in samples nominally comparable to ours.

From the width of the 2DEG emission, we can calculate the charge carrier density by relating it to the Fermi energy in two dimensions [75, 264],

$$n = \frac{m_{\mathrm{c}}^* E_{\mathrm{F}}}{\pi \hbar^2} = \frac{\mu \Delta E}{\pi \hbar^2}, \tag{13.1}$$

where ΔE is the bandwidth of the emission (dashed gray lines in Figure 13.2) and $\mu = m_{\mathrm{c}}^* m_{\mathrm{hh}}^* / (m_{\mathrm{c}}^* + m_{\mathrm{hh}}^*)$ is the reduced mass of conduction and valence band. For this particular sample, Equation 13.1 yields $n = 5 \times 10^{11}\,\mathrm{cm}^{-2}$ or, equivalently, $E_{\mathrm{F}} = 18\,\mathrm{meV}$ and $k_{\mathrm{F}} = 1.8 \times 10^{8}\,\mathrm{m}^{-1}$. Comparing this value to that obtained from a simulation of the heterostructure with nominal doping concentration $N_{\mathrm{d}} = 6.5 \times 10^{17}\,\mathrm{cm}^{-3}$ using a self-consistent Poisson-Schrödinger solver [267], $n = 1.9 \times 10^{11}\,\mathrm{cm}^{-2}$ (refer to Section B.2 for simulation parameters), shows a significant discrepancy, indicating either a severe mismatch between nominal and actual doping concentrations or a Fermi level pinning very different from the one assumed in the simulation.[5] Finally, we observe that the gap according to the preceding analysis is red-shifted from the undoped bulk gap of 1.519 eV [222] by 13 meV.[6] Descamps [70] hypothesized that the removal of the GaAs substrate and the associated change in strain leads to this lowering of the band gap. However, this effect was also already observed by Pinczuk et al. [264] in "bulk" modulation-doped GaAs QWs. There, the authors put forth a renormalization of the band gap due to many-body interactions as an explanation, an interpretation since well-agreed upon [268]. At high doping concentrations, and hence high carrier densities, electron-electron interactions push the conduction band down while electron-hole interactions lift the valence band up [269]. Another effect leading to an effective narrowing of the band gap is the random distribution of ionized impurities that causes local fluctuations in the band edges and results in tail states decaying exponentially into the gap [269].

5: Note that the carrier density obtained this way does not vary significantly with excitation power, ruling out photo-doping as a source of the discrepancy. See Section B.2 for a measurement of the 2DEG PL as function of power.

6: The red shift is in fact larger still due to the confinement energy of the QW, estimated to be 17 meV in Section 11.2.

13.1.1 Quantum-confined Stark shift

Let us now address the behavior of the PL under electric fields. To this end, the laser is positioned on a gate written by electron-beam lithography and a negative voltage is applied to the gate. Figure 13.3 depicts two measurements on a well-behaved sample. The left (right) panel shows the PL as function of voltage with the laser positioned on a bottom (top) gate. In contrast to the intuition obtained in Section 11.2, there is no immediate effect to be observed once the voltage is switched on. This is most likely due to the presence of the 2DEG screening the external electric field. At $V_{\mathrm{gate}} = -0.7\,\mathrm{V}$, the high-energy shoulder of the emission starts to shift towards lower energies while the low-energy shoulder stays invariant. Per the previous section, we can interpret this as the charge carrier density and thereby the Fermi energy being lowered as the 2DEG is depleted. As the electric field tilts the bands, the band edges of both conduction and valence band shift by the same amount and therefore the band gap is not modified in this regime.[7] As the 2DEG is gradually depleted, a broad exciton peak emerges that shifts approximately quadratically with the applied voltage as ionization from the interaction with the Fermi sea of electrons becomes less pronounced (*cf.* Section 11.1). Beyond $-1.5\,\mathrm{V}$, the 2DEG is completely depleted. The voltage difference of $-0.7\,\mathrm{V}$ between onset and completion of the depletion matches roughly the value expected from a 1D-band structure simulation [267] with the nominal device parameters (see Tables B.1 and B.2).

7: We would in fact expect a slight increase in confinement energy as the carrier density is lowered because the band bending due to the surplus electric charge of the 2DEG is lifted.

Besides the voltage dependence, another feature stands out from Figure 13.3: the PL intensity is lower by an order of magnitude when on top of a back gate compared to a top gate. This is puzzling at first, as in the latter case there is an additional semi-transparent gate[8] absorbing and reflecting both laser and PL radiation, whereas in the former there is only the bare heterostructure, so we would expect the exact opposite! I elucidate this issue in Section 13.3.

8: 7 nm Au with a 2 nm Ti adhesion layer.

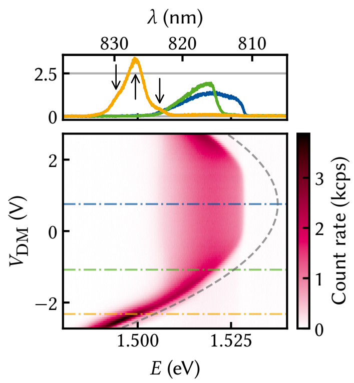

Figure 13.4: PL as function of difference-mode voltage on a large exciton trap. The observed Stark shift follows roughly the expected quadratic dispersion, but is offset by 0.75 V with respect to zero bias. Dashed gray line is a guide to the eye of a parabola with curvature $-3.5\,\mathrm{meV/V^2}$. Line cuts in the upper panel are taken at the voltages indicated by dash-dotted lines in the lower, and arrows indicate positions of individual peaks making up the yellow line shape.

Moving to a large exciton trap with a single set of top and bottom gates and 5 µm diameter, we can measure the behavior of the Stark shift in the intended setting of local confining gates on either side of the membrane. We define the virtual gates

$$V_{\mathrm{DM}} = V_{\mathrm{T}} - V_{\mathrm{B}}, \tag{13.2}$$

$$V_{\mathrm{CM}} = V_{\mathrm{T}} + V_{\mathrm{B}}, \tag{13.3}$$

where DM (CM) stands for difference (common) mode and T (B) for top and bottom, respectively. Clearly, V_{DM} should ideally be proportional to the out-of-plane electric field across the membrane, $V_{\mathrm{DM}} = Ft$ with t the membrane thickness, whereas V_{CM} should tune the band edge offset from the Fermi level μ. As we observed previously, the presence of the 2DEG screens the electric field generated by V_{DM}. A good operating point is therefore at a negative V_{CM} which should deplete the 2DEG (or at least reduce the charge carrier density). Figure 13.4 shows a PL map as function of the difference mode voltage V_{DM} at $V_{\mathrm{CM}} = -1.3\,\mathrm{V}$. From Figure 13.3, where the optical measurement of the charge carrier density results in a similar value as for the sample in Figure 13.4, $n \sim 5{\times}10^{11}\,\mathrm{cm^{-2}}$, we would expect this common mode voltage to suffice in at least overcoming the screening and reducing the carrier density in the QW. However, there is clearly still 2DEG emission present for a large range of V_{DM}, and the Fermi edge is at the same energy as without a gate bias. Overall, the Stark shift pattern is symmetric but offset by $V_{\mathrm{DM}} = 0.75\,\mathrm{V}$. This behavior is also observed in the response of the PL to a single gate in this and several other samples, where the onset of an effect by bottom gate is significantly later than that for the top gate, suggesting that the voltage is screened by some mechanism and the bottom gate has a much smaller lever arm than the top gate. Perhaps surprisingly, the gates on the *bottom* side of the membrane display this behavior, *i. e.*, the gates on the as-grown surface.[9] This makes surface states an unlikely candidate for the screening as we would expect the quality of the grown surface to be better than that of the etched surface [70]. A possible cause might be oxygen segregation during growth[10] or other impurities [270].

The fact that the 2DEG is depleted by V_{DM} in the range $-0.2\,\mathrm{V}$ to $-1.4\,\mathrm{V}$ and above $1.5\,\mathrm{V}$ at all is unexpected. Even with a difference in lever arms as just discussed, the symmetric dependence of the PL emission on V_{DM} implies that $V_{\mathrm{DM}} - 0.75\,\mathrm{V}$ *should* correspond to the out-of-plane electric field such that the energy is unchanged in the middle of the quantum well, $V_{\mathrm{DM}} - 0.75\,\mathrm{V} \propto F(z - t/2)$. Staying in the virtual gate picture, a difference in relative lever arms[11] $\Delta\alpha = \alpha_{\mathrm{T}} - \alpha_{\mathrm{B}}$ between V_{T} and V_{B} should modify the virtual gates as

$$\begin{pmatrix} \tilde{V}_{\mathrm{DM}} \\ \tilde{V}_{\mathrm{CM}} \end{pmatrix} = \frac{1}{\sqrt{1+\Delta\alpha^2}} \begin{pmatrix} 1 & \Delta\alpha \\ -\Delta\alpha & 1 \end{pmatrix} \begin{pmatrix} V_{\mathrm{DM}} \\ V_{\mathrm{CM}} \end{pmatrix} \tag{13.4}$$

$$= \frac{1}{\sqrt{1+\Delta\alpha^2}} \begin{pmatrix} 1+\Delta\alpha & -1+\Delta\alpha \\ 1-\Delta\alpha & 1+\Delta\alpha \end{pmatrix} \begin{pmatrix} V_{\mathrm{T}} \\ V_{\mathrm{B}} \end{pmatrix}, \tag{13.5}$$

where the virtual gate matrix is orthogonal, acting as a rotation in voltage space, and as such not modify the qualitative behavior of V_{DM} besides the offset.[12] That is, in and of itself, V_{DM} should not reduce the charge carrier density in an isolated QW to first order.[13]

One possible explanation for the observed behavior is the depletion of the 2DEG by electrons tunneling out of the well and recombining with the donor ions, rendering one of the doped barriers electrically neutral. With the doping pulling down the conduction band between 50 nm and 90 nm away from the QW, the tunneling rate will be more pronounced than estimated for the undoped case in Section 11.2. Taking the large trap diameter in this case into account, there are $\sim 10^5$ electrons in the unbiased QW on the area of the trap gates. Thus, at a tunneling rate of 1 MHz, all electrons would tunnel out of the QW within 100 ms on average, putting a response of the system into the steady state within the fairly long time scales of a PL measurement[14] well within reasonable bounds. In the literature, this is known as carrier sweep-out and has been studied in the context of solar cells and other electroabsorptive devices [247, 272, 273].

The dashed gray line in Figure 13.4 shows a parabola, $\Delta E \propto \alpha V_{\mathrm{DM}}^2$, with curvature $\alpha = -3.5\,\mathrm{meV/V^2}$ as a guide to the eye. For difference-mode voltages below $-1.8\,\mathrm{V}$, the exciton Stark shift follows this quite closely, establishing that the exciton energy can be tuned by over 20 meV in the depleted regime, which simulations suggest is sufficient for confinement of single excitons in a smaller trap geometry [70]. The upper panel depicts three line cuts at the positions indicated by the dash-dotted lines in the main panel. As can be seen from the cut at $-2.3\,\mathrm{V}$ (orange), the line shape appears to consist of three individual emission lines assigned by Descamps [70] as the neutral, singly, and doubly negatively charged excitons. I return to the question of line assignment below.

13.1.2 Power dependence

As outlined in Section 11.3, the dependence of emitted PL power on excitation power of individual emission lines can help inferring the excitonic species responsible. Moreover, as the density of excitons depends on the excitation power, interaction effects between them influence the

9: I note that the sample of Figure 13.3 was fabricated on the same heterostructure as the device in Figure 4(d) of Reference 110, which showed little to no electrical hysteresis unlike most other samples investigated by Descamps [70].

10: A. Ludwig, private communication.

11: Relative gate lever arms are unitless parameters used to quantify the degree of cross-coupling between gates, in contrast to the physical lever arms used to quantify the change in electron energy versus gate voltage employed in Section 6.2. See for example Reference 271.

12: Setting $\Delta\alpha = 0.6$ yields a $\tilde{V}_{\mathrm{DM}}$ for which the Stark shift in Figure 13.4 is symmetric about zero.

13: We might instead expect a small change in the apparent gap energy as the well is tilted.

14: Say 0.1 s to 10 s.

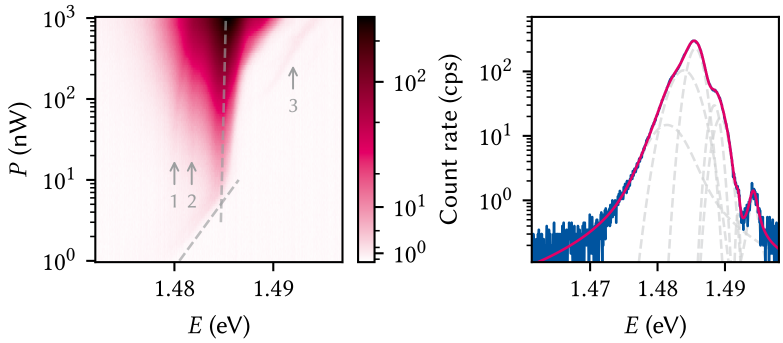

Figure 13.5: PL as function of excitation power P. In the left panel two qualitatively different regimes are indicated by dashed gray lines as guides to the eye; below 10 nW, the main peak displays a blue shift logarithmic in excitation power, $E = 1.485\,\mathrm{eV} + 5\,\mathrm{meV}\log_{10} P$. Above, the blue shift diminishes significantly. Three additional lines, indicated by arrows, with varying power dispersion are visible. Right panel shows a fit to data at $P = 1\,\mu\mathrm{W}$. A sum of seven individual lines (dashed, gray) is required to fit the data. The dashed gray lines are the individual contributions.

energy of the emission. Such a measurement is shown in the left panel of Figure 13.5 for the same common-mode voltage as in Figure 13.4 and $V_{\mathrm{DM}} = -2.7\,\mathrm{V}$ (*i. e.*, corresponding to the lowest line in that plot) where the 2DEG is completely depleted. Two qualitatively different regimes can be observed as indicated by the dashed gray lines as guides to the eye. Below 10 nW excitation power, corresponding to approximately $0.75\,\mathrm{W/cm^2}$ at the beam diameter measured in Part II, the main line displays a blue shift logarithmic in excitation power. Above this value, the blue shift is much less pronounced. It is not understood though why the blue shift abruptly changes in quality at 10 nW excitation power. At powers above 10 nW, two additional lines on the low-energy side of the main peak become visible that appear to converge towards higher powers, while above 100 nW a line appears on the high-energy side that has a similar blue shift as the main peak at low powers. In principle, the blue shift as function of power is to be expected. At high excitation power, the density of free charge carriers is increased, either because they are spatially localized or because of a finite diffusion speed. This screens the Coulomb interaction between electron and hole and thus leads to a reduction in binding energy corresponding to a blue shift of the exciton [274]. At low excitation power, exciton-exciton interaction leads to a renormalization of the self-energy linear in the exciton density that results in an increase in exciton energy by the same amount [275–279]. This mechanism underpins the use of exciton traps such as SAQDs as single-photon sources; upon spectrally filtering on the emission wavelength of a single exciton, only a single photon can be emitted from the trap at a time since the presence of another exciton in the trap would shift the emission energy of both.

The right panel shows a line cut taken at $1\,\mu\mathrm{W}$. A weighted sum of seven individual Voigt profile line shapes [280],

$$V(E;\sigma,\gamma) = G(E;\sigma) * L(E;\gamma), \tag{13.6}$$

that is, a convolution of Gaussian and Lorentzian line shapes given by

$$G(x;\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right), \tag{13.6a}$$

$$L(x;\gamma) = \frac{1}{\pi}\frac{\gamma}{\gamma^2 + x^2}, \tag{13.6b}$$

is required to obtain an adequate fit. The Voigt profile arises from a combination of two separate line broadening mechanisms that manifest as a Gaussian (Equation 13.6a) and Lorentzian (Equation 13.6b) line shape. The former describes inhomogeneous broadening due to noise faster than the data acquisition time, whereas the latter is the homoge-

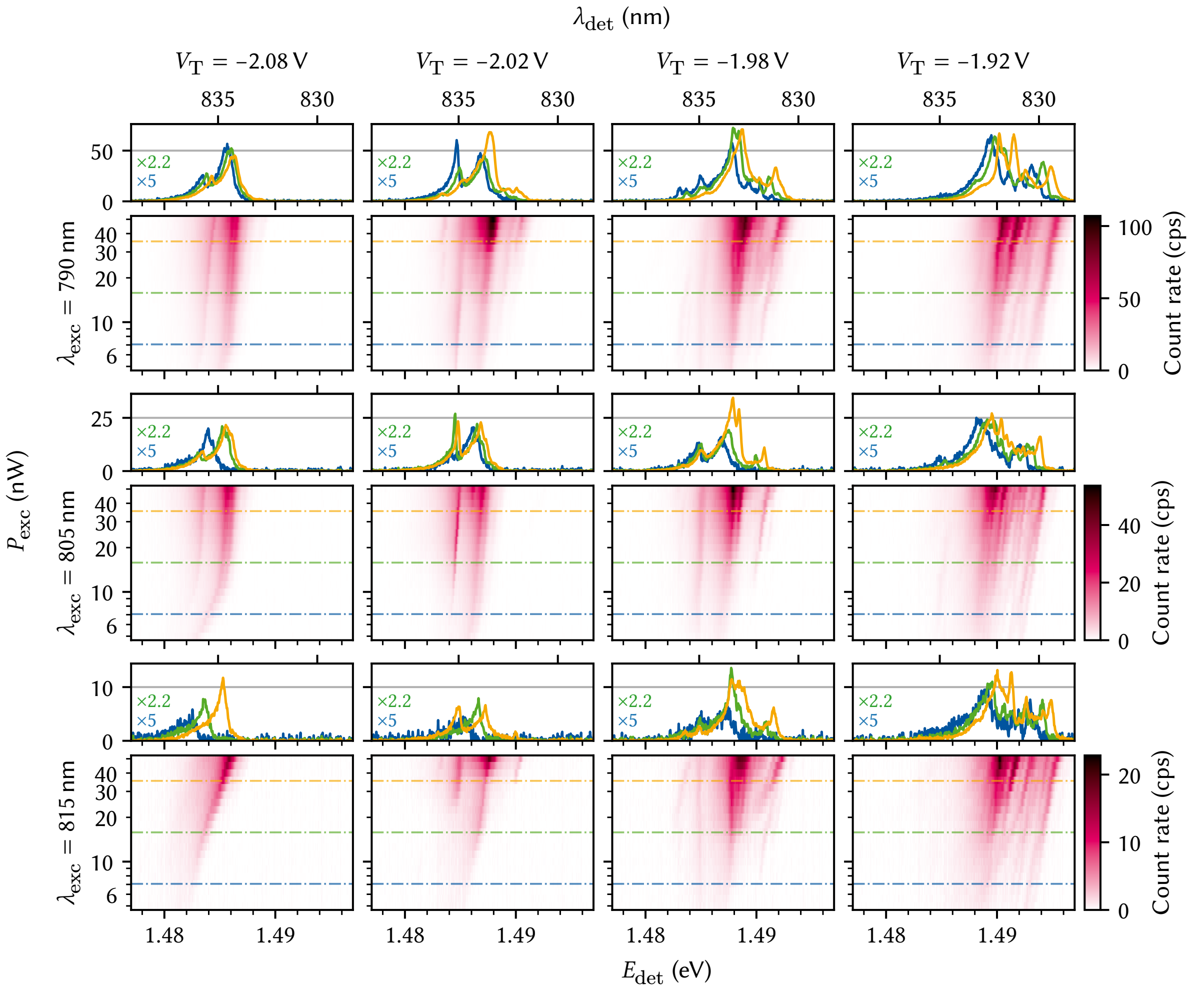

Figure 13.6: Wide-range PL parameter sweep on a large exciton trap plotted as function of excitation power and detection energy. Rows are data for three different excitation wavelengths, columns for four different top gate voltages V_T and share color and line cut scales. Line cuts are taken at the indicated positions of P_{det} = 7 nW, 16 nW, and 35 nW and drawn scaled by the fraction of excitation power with respect to 35 nW.

neous broadening due to for example the finite lifetime of the emitting state[15] or power broadening [281]. All but the two outermost lines in the best fit are dominated by the inhomogeneous, Gaussian contribution to Equation 13.6 with widths σ in the range of 0.1 meV to 1 meV and a peak separation on the order of 2 meV.[16]

According to Reference 70, the lifetime of a Stark-shifted exciton is on the order of 1 ns due to the reduced wave function overlap. This corresponds to a homogeneous linewidth of $2\gamma = \hbar/\tau \sim 660\,\mathrm{neV}$, several orders of magnitude below the observed linewidth, and it is thus consistent with the fact that most peaks are best fit by inhomogeneously broadened line shapes. This large a number of lines is certainly unexpected and cannot be explained by different excitonic species. Where it is possible to track individual peaks as function of excitation power, their power dependence is linear, $\int \mathrm{d}E\, V(E) \propto P$, suggesting neutral excitons or band-to-band recombination as origins of the emission.

A possible explanation for the multitude of lines is thus that they originate from different spatial locations, a hypothesis we return to shortly. First, let let us conclude this section with a large parameter sweep of this trap, shown in Figure 13.6. Each of the three rows (with separate panels

15: This is also known as the linewidth's transform limit; energy and (life)time are Fourier pairs by Heisenberg's uncertainty principle.

16: The large number of Voigt profiles used to fit the data does tend to overfit the data.

for line cuts each) show data for different excitation wavelengths, λ_{exc}, each of the columns show data for different top gate voltage, V_{T}, and the PL is plotted as function of detection energy, E_{det}, and excitation power, P_{exc}. The line cuts taken at lower powers (blue and green) are scaled to match the one at the highest power (orange) assuming a linear power dependence. For all data $V_{\mathrm{B}} = 0\,\mathrm{V}$ so that $V_{\mathrm{T}} = V_{\mathrm{DM}} = V_{\mathrm{CM}}$. Despite the comparatively small changes in voltage, the behavior of the sample changes significantly. Whereas for $V_{\mathrm{T}} = -2.08\,\mathrm{V}$ the observed PL features are similar to those in Figure 13.5, at $V_{\mathrm{T}} = -1.92\,\mathrm{V}$ there is a very large number of lines in the spectrum, most but not all of which share the same blue shift as function of excitation power. The effect of a different excitation wavelength appears to mostly be a shift of the features along the excitation power axis and thus simply a change in absorption efficiency, although some features also change qualitatively. I discuss the wavelength dependence in more detail in Section 13.2.

So what is the origin of the substructure of the PL emission? The peak distance is on the order of 1 meV. This value is on the order of magnitude of the change in ground state energy, 470 μeV, we would expect from fluctuations in the width of the QW by one atomic layer of GaAs with lattice constant $a = 5.65\,\text{Å}$[17] for the design well width $L = 20\,\mathrm{nm}$ (Section 11.1). Indeed, there exists a large body of literature that has examined the influence of interface roughness and QW thickness fluctuations on the exciton PL [282–284]. Not only do atomic steps and islands lead to increased PL linewidth and energy shifts, but in fact also to the possible localization of single excitons [285–287]. That is to say, QDs have been found to form at such interface fluctuations. Comparing the spectra shown in Figure 13.6 to those shown by Brunner et al. [285, Figure 1], our main PL line is more reminiscent of what the authors there label as the 2D exciton; our spectra do not show well-separated and sharp lines on the low-energy side of the spectrum. By contrast, the PL spectrum shown by Zrenner et al. [287, Figure 1] is very similar to the one observed here, although they assign this peak to indirect transitions from the AlAs X valley to the GaAs Γ valence band valley in their AlAs/GaAs coupled QW structure, a situation quite unlike ours.

17: Meaning the thickness of a single atomic layer is $a/2$.

An intricate substructure of the PL emission at lower excitation powers such as this was also observed in other samples. If the individual lines originated from 0D-confined QDs, we would expect their PL emission to display signatures of a single-photon source as outlined above. I therefore performed a $g^{(2)}$ measurement (see Section 7.2) on a line that was narrow (~ 1 meV) and fairly well-separated such that at its full width at half maximum (FWHM) was resolvable. I excited the sample above the gap and spectrally filtered the emitted PL by using the spectrometer as a monochromator with bandwidth given by the line FWHM. Due to the low brightness, a compromise between high excitation power (leading to power broadening) and measurement duration (to gather statistics) was chosen at 100 nW. However, after 48 h no deviation from a flat and uncorrelated $g^{(2)}$ could be observed.[18]

18: I also attempted a $g^{(2)}$ measurement to no avail in a different parameter regime where a single, isolated emission line appears.

13.1.3 Spatial dependence

As mentioned several times already, the nanopositioners on which the sample is mounted show hysteresis and are therefore not suited for reproducible spatial maps of the sample. The hysteresis is due to the non-adiabaticity of the method of movement in the so-called slip-stick mode (*cf.* Chapter 6). What is more, the resistive position readout is also fairly

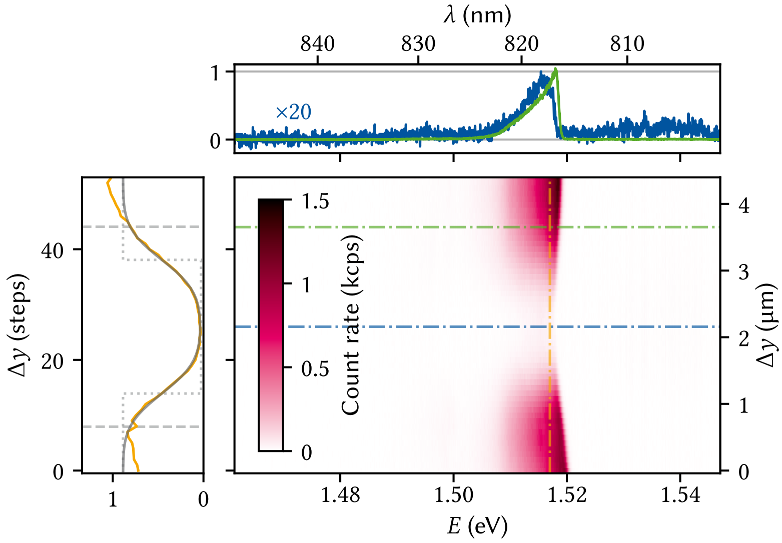

Figure 13.7: PL of the unbiased QW as the laser is stepped across a bottom gate of nominal width $L = 3\,\mu\text{m}$. The Fermi edge shows a slight red shift when on top of the gate in this sample. The line traces in the upper panel are taken at the positions indicated by horizontal dash-dotted lines. Positioner steps are converted to distance using a linear fit of the positioner readout after the initial hysteresis has worn off (about 10 steps). The left panel shows a vertical line cut together with a fit to a Gaussian beam spot convolved with a step profile, indicated by a dotted gray line. From the fit, the width of the gate is $L = 2\,\mu\text{m}$ (beam waist radius $0.75\,\mu\text{m}$, see Equation 7.2), indicating the position readout is off by 50 %. The dashed gray lines in the left panel indicate the expected gate edges for the nominal width.

unreliable below, say, $5\,\mu\text{m}$ resolution. Nonetheless, we can at the very least perform simple one-dimensional sweeps after manually positioning the sample at a given starting position and assume that the displacement per step has some sufficiently narrow distribution around a fixed mean. A more sophisticated algorithmic approach using feature detection may allow also two-dimensional maps with a reasonable accuracy.

Figure 13.7 shows the PL collected from the sample as the positioner is stepped perpendicularly across an unbiased gate with a nominal width of $L = 3\,\mu\text{m}$. The vertical axis also gives a position coordinate which is computed from the steps by fitting the position readout with a linear function once hysteresis has worn off. The blue and green dash-dotted line correspond to the center of and beside the gate, respectively, with line cuts drawn in the upper panel. Clearly, the PL intensity is quenched significantly by the gate, beyond what one could expect from simple reflection and absorption of laser and PL radiation. I perform transfer-matrix method (TMM) simulations to explain this behavior in Section 13.3 (see also Figure 13.3). The left panel shows a line cut taken at 1.517 eV together with a fit to a convolution of a Gaussian beam spot (Equation 8.7) and a step function,

$$\Phi(y) = I(y) * \left[\theta\left(y - y_0 + \frac{L}{2}\right) - \theta\left(y - y_0 - \frac{L}{2}\right)\right] + \Phi_0, \tag{13.7}$$

where $I(y)$ is the one-dimensional Gaussian intensity distribution, $\theta(y)$ the Heaviside step function, and y_0 the center of the gate. Close to the gate, Figure 13.7 shows excellent agreement with Equation 13.7, but yields a gate width of $L = 2\,\mu\text{m}$ at a beam waist radius of $w_0 = 0.75\,\mu\text{m}$, some 50 % off from the nominal design width whose extents are indicated by dashed gray lines in the left panel of the figure. This implies that the position calibration based on the nanopositioner readout in Figure 13.7 is quite crude and inaccurate. Curiously, the map also shows a shifting of the Fermi edge close to the gate. This behavior was observed consistently on this sample next to an unusual PL line shape (*cf.* Figure 13.2). The former effect could potentially be caused by band-deformation due to strain from the metallic gates.

Figure 13.8 shows two sweeps along orthogonal axes across the biased exciton trap discussed before in Figures 13.4 to 13.6. While this is not

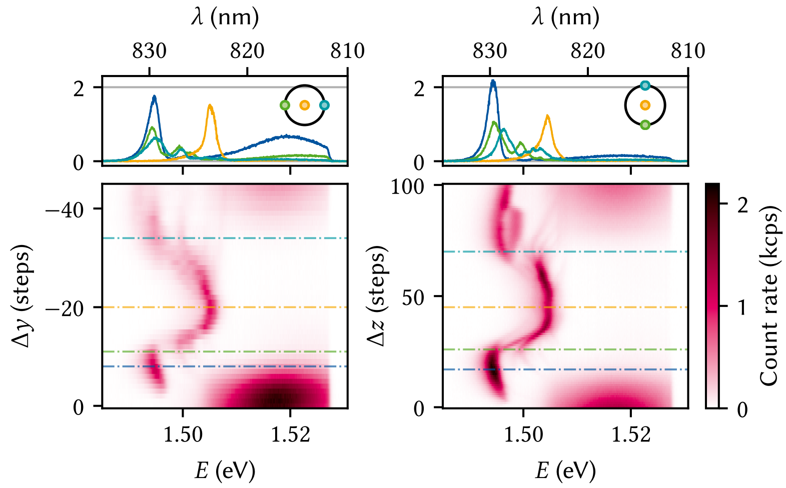

Figure 13.8: PL of a large exciton trap as function of position perpendicular (left) and parallel (right) to gravity. The trap is biased so that the emission is Stark-shifted towards lower energy. Upper panels show line cuts taken along the dash-dotted lines in the lower panels, demonstrating fairly identical behavior along both axes. The cuts in orange, green, and teal are taken at the positions indicated on the circle in the upper panel representing the trap gate.

unequivocal proof of zero-dimensional confinement, it does suggest that the effective exciton potential is laterally lowered. The left panel of Figure 13.8 shows a scan along the in-plane axis perpendicular to gravity, while the right shows the in-plane axis parallel and against gravity. Regrettably, the resistive position readout did not yield anything but noise in these measurements, prohibiting a conversion of positioner steps into relative position as before. Both scans were acquired with the sample voltage applied to the positioners (30 V), which is the same used in Figure 13.7 as well. We can hence roughly expect 10 steps to correspond to 1 μm in y direction. Naturally, a single step against gravity displaces the sample by a smaller amount compared to the perpendicular direction, but given the circular shape of the trap and the similarity of the PL features, the displayed ranges should be roughly the same. The upper panels show line cuts taken at the positions indicated by dash-dotted lines in the lower panels, with the orange, green, and teal lines corresponding to the positions indicated by markers on the black circle in the upper panel. Both sweeps display very similar features. In the center of the trap (yellow), the Stark-shifted emission line consists of a single peak. Towards the edges (teal and green), two surprising effects take place: first, the Stark shift of the dominant peak increases in magnitude, and second, a large number of additional, faint lines appear. The increase in effective electric field towards the edge of an exciton trap is reminiscent of the simulation of the same type of experiment for an undoped QW by Descamps [70, Figure 6.4], although it was not explained there. Indeed, it also appears to be in conflict with Figure 2.16 *ibid.*, where a monotonic increase in effective exciton potential as function of distance from the trap center was predicted. In the device under study here, there is of course a 2DEG whose presence towards the edge of the trap screens the voltages, in theory contributing to an attenuation of the Stark shift. Why we observe to opposite is thus quite puzzling.

The second surprising feature seen in Figure 13.8 is the addition of faint lines that appear to branch out from the main exciton line as the laser spot is moved away from the center of the trap. These could be related to the lines observed in Figure 13.6 as well as the large number of lines required to fit the peak in Figure 13.5. If the origin of these are indeed atomic steps or islands in the QW, it makes sense that we observe them in these measurements, although their appearance strictly at the edges of the trap is suspicious. The Stark shift of the different lines appears depend on the position as well, indicating different coupling to the electric field and further strengthening the hypothesis that they originate from different spatial locations.

13.2 Photoluminescence excitation spectroscopy

The technique of photoluminescence excitation (PLE) can be used to gain insight into the energy level structure of optically active states [239]. While in a PL experiment we only observe the emission, PLE experiments are sensitive to absorption as well. Thus, excited states that quickly decay non-radiatively before recombining can be probed as absorption at that energy will be enhanced even though emission might not. For well-resolved PL lines, a common practice is to spectrally filter the PL radiation while sweeping the excitation laser wavelength, thus obtaining the luminescence flux as function of excitation energy, $\Phi_{\mathrm{PLE}}(E_{\mathrm{exc}}; E_{\mathrm{det}})$, for a single detection energy E_{det}. By contrast, a PL measurement is performed at a fixed excitation energy and yields $\Phi_{\mathrm{PL}}(E_{\mathrm{det}}; E_{\mathrm{exc}})$. Alternatively, the entire PL spectrum can be integrated up to the excitation energy rather than filtered at specific energy,

$$I(E_{\mathrm{exc}}) = \int_0^{E_{\mathrm{exc}}} \mathrm{d}E_{\mathrm{det}}\, \Phi(E_{\mathrm{det}}, E_{\mathrm{exc}}), \tag{13.8}$$

which yields a better SNR at the cost of potentially washing out selective transitions.

What do we expect for the PLE of a QW hosting a 2DEG? By Fermi's exclusion principle, no free states are available below the Fermi level μ and thus excitation of an electron-hole pair from the valence band into the conduction band is only possible for photon energies above $E_{\mathrm{F}}(1 + m_{\mathrm{c}}^*/m_{\mathrm{hh}}^*)$ (Section 11.1). Accordingly, the PLE spectrum should show an onset at that energy.[19] Above it, there exists a continuum of states in the QW plane, and the PLE spectrum should remain approximately constant until higher-lying states can be reached. In a single QW, these can be either higher subbands of heavy-hole valence or conduction bands or indeed light-hole transitions due to heavy-hole–light-hole mixing [288–292]. For the relatively wide QW widths in our samples, we would expect the latter to be on the order of a few meV.

19: In fact, the FES should lead to a peak at the onset of the spectrum.

Applying an external electric field should further enhance the light-hole transitions as tilting the QW breaks the symmetry forbidding light-hole transitions [293, 294]. Furthermore, the electric field depletes the 2DEG as we have seen in Figure 13.4 and should result in a laterally local potential well for excitons with discrete and approximately equidistant energy levels as laid out in Subsection 11.2.1. If such were the case, we would expect resonances in the PLE at intervals corresponding to the level spacing $\hbar\omega$ of the potential well that would confirm 0D-confinement.

Figure 13.9 shows PLE data obtained for the large exciton trap discussed previously obtained at $1\,\mu\mathrm{W}$ excitation power and $V_{\mathrm{CM}} = -1.3\,\mathrm{V}$, *i.e.*, similar conditions as for Figures 13.4 and 13.5. Due to the high excitation power, the measurement does not resolve the multiplet substructure observed in Figure 13.6. The top left panel (a) shows a regular PL map as function of V_{DM} at $\lambda_{\mathrm{exc}} = 795\,\mathrm{nm}$ and essentially reproduces the measurement of Figure 13.4. Panel (b) shows the integrated PLE following Equation 13.8. That is, the rightmost column corresponds to panel (a) integrated along E_{det}. Several interesting features are noteworthy. In line with our considerations above, the onset of PLE absorption coincides with the high-energy shoulder of the PL emission over the entire range of voltages investigated. For $V_{DM} > -1.6$ V, the absorption edge is replicated some 10 meV higher in energy, although the splitting reduces as the carrier density is decreased.

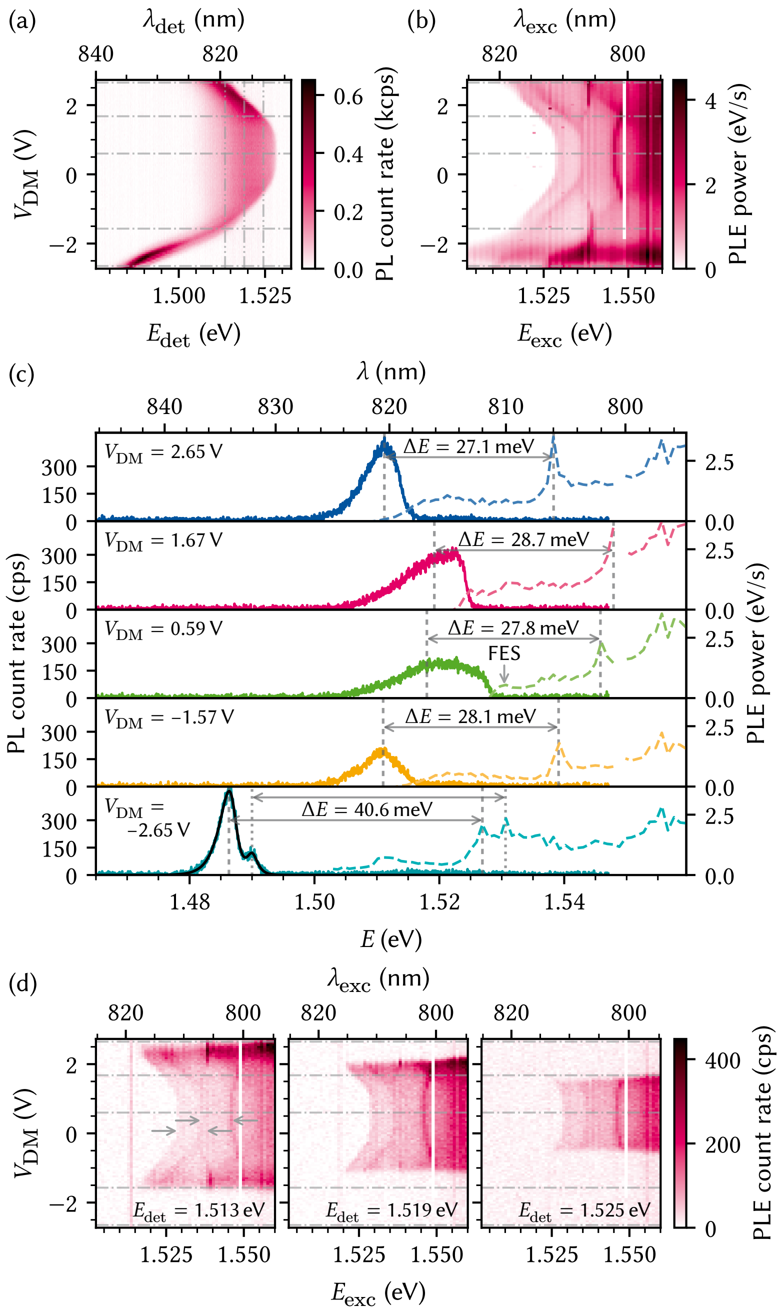

Figure 13.9: Photoluminescence excitation measurement. (a) PL map as function of difference-mode voltage for λ_{exc} = 795 nm (*cf.* Figure 13.4). (b) PLE map. Each column corresponds to a map like in (a) integrated up to the laser line at $E_{det} = E_{exc}$. While a 2DEG exists ($V_{DM} \in [-1.6\,\mathrm{V}, 2.6\,\mathrm{V}]$), absorption in PLE sets on just as emission stops in PL at the Fermi edge μ (see also Figure B.1). The Fermi edge feature is replicated at least once at approximately 10 meV above μ, while a feature with positive Stark shift $\alpha > 0$ duplicated at least once as well appears with apexes at E_{exc} = 1.536 eV and 1.546 eV. A sharp feature appears as the 2DEG is close to depletion at 1.538 eV. The vertical white bar is missing data. (c) Line cuts of the PL (a, solid lines) and PLE maps (b, dashed) taken at the horizontal dash-dotted gray lines there. At $V_{DM} = -2.65$ V, the PL line is split by 3.7 meV which is replicated in the PLE spectrum 40.6 meV higher. The black line shows a fit to a trion and exciton following Reference 295. The features at around 1.555 eV do not shift with the voltage and are thus likely unrelated to the trap. (d) Line cuts of the entire data set at single detection energies indicated by vertical dash-dot-dotted lines in (a). Horizontal dash-dotted lines are again the positions of the line cuts shown in (c). Arrows indicate a splitting of approximately 10 meV.

A few possible explanations for this feature come to mind. First, the two lines could correspond to the negative trion and the neutral exciton. However, the energy difference of ~ 10 meV seems too large for the difference in binding energies, usually on the order of a few meV. Although similar features have been observed and assigned as such [296–298], the splittings in those studies were smaller while at the same time the QWs were thinner and should thus lead to larger binding energies [257, 295].

Similarly, excitons bound to impurities such as ionized donors typically show much lower energy separation to the neutral exciton [299]. A better match with the observed energy difference is the excitonic binding energy of ~ 8 meV, which would imply that the feature at higher energy would correspond to free electron-hole pairs. Alternatively, the lines could correspond to heavy- and light-hole excitons, whose splittings have been found to be on the order of 7 meV for undoped QWs of this width [292] although also smaller values have been reported [300]. The difference in electric-field dependence of the respective binding energies is hard to predict, but the Stark shift should be smaller for light-hole excitons due to their smaller effective mass and hence the narrowing of the energy splitting between the lines for higher voltages is counter-intuitive. Again, though, the presence of the 2DEG in this regime may crucially alter the physics. Finally, we can look to excited QW states as candidates. From Section 11.2, we expect subband splittings to be on the order of 50 meV for the $n_\mathrm{c} = 1$ and $n_\mathrm{hh} = 1$ transition and 8 meV for the (optically suppressed) $n_\mathrm{c} = 0$ and $n_\mathrm{hh} = 1$ transition at zero field. While the last matches fairly closely what we observe at zero field, it too should increase with increasing electric field rather than decrease. Moreover, from the calculation of the oscillator strength (Figure 11.5), we would expect the optical coupling to vanish at a certain electric field value whenever excited states are involved, which is also not observed.

So far, we have exclusively discussed the duplicated feature following the trend of the PL emission edge, but a few more observations can be made by inspecting Figure 13.9(b). Most strikingly, also in the regime of an undepleted 2DEG there is again a duplicated feature with a similar energy splitting as before, but which displays an inverted Stark shift, that is, whose energy is raised by the application of an electric field, or, more phenomenologically, a reduction in charge carrier density. We have seen such a behavior in the theoretical discussion for the first $\Delta n = 0$ excited state of the QW at low fields (Figure 11.5), but similar arguments as above can be made against such an identification. Both of the above features vanish as the 2DEG is depleted at −1.6 V and 2.6 V. In their place, a very sharp feature appears at 1.538 eV that is independent of the electric field. Going to even more negative voltage and fully depleting the 2DEG, the overall luminosity increases (as can also be seen in the PL spectrum) while broad features start to appear as function of excitation power.

Panel (c) of the figure shows line cuts taken at the voltages indicated by dash-dotted lines in panels (a) and (b) with solid lines corresponding the emission (PL) and dashed lines to absorption (PLE). The energy shift of the two most prominent peaks in both spectra is labeled. Although not a perfect match, the shift of around 28 meV is not far off the energy of TO phonons in GaAs, known in bulk to be 33 meV [301]. Similarly, the shift of 40 meV at $V_\mathrm{DM} = -2.65\,\mathrm{V}$ is off by approximately the same amount from the second, AlAs-like TO phonon mode in $\mathrm{Al_{0.32}Ga_{0.68}As}$, 44 meV [283, 302, 303]. It is conceivable that the thinning of the heterostructure to a membrane modifies the phonon energies by 5 meV and it would be interesting to perform Raman spectroscopy on the membranes to investigate this further. Additionally, temperature-dependent measurements could reveal information on the possible existence of a phonon bottleneck [304], which would show as a change in decay rate at a certain temperature corresponding to a crossover in acoustic-phonon–dominated and optical-phonon–dominated cooling of hot electrons.

The data for $V_\mathrm{DM} = 0.59\,\mathrm{V}$ show that the Stokes shift between Fermi edge of the PL emission and the FES in the PLE absorption is small,

$E_S - E_F \sim 2.5$ meV. At the largest voltage, $V_{DM} = -2.65$ V, the PL emission splits up into a doublet peak with a splitting of 3.7 meV. The same splitting is observed between two peaks in the PLE spectrum, but shifted by $\Delta E = 40.6$ meV towards higher energy. These two peaks might correspond to the negative trion and neutral excitons. In fact, their PL emission is well fitted with the lineshape proposed by Esser, Zimmermann, and Runge [295], who describe the trion peak as a convolution of the exciton peak, given by a hyperbolic secant, with a thermally broadened exponential edge. The fit is drawn in black in the same panel, showing good agreement with the data, and yielding $E_T = 1.487$ eV, $E_X = 1.490$ eV, a linewidth of $\Gamma = 770$ µeV, and a temperature of $T = 1.2$ K. The temperature, while hot compared to the lattice temperature, matches fairly well with the temperature obtained from the Fermi edge (Section 13.1). The fact that the peak splits into a doublet structure here whereas it did not in the measurement of Figure 13.4 might be explained by slightly different focal spot positions. Lastly, panel (d) shows line cuts of the data taken at a single detection energy, $I_{PLE}(E_{exc}; E_{det})$, corresponding to the positions indicated by vertical dash-dot-dotted lines in panel (a). The parabolic features discussed previously are clearly not an artefact of integrating over the entire PL spectrum.

To summarize the PLE measurement, several features that systematically shift with the electric field generated by the difference-mode voltage V_{DM} have been observed. However, the data show no indication of an in-plane level splitting arising from confinement, although the resolution in E_{exc} is also likely too low for this large a trap size. Interpolating the harmonic potential found by Descamps [70] for a smaller trap to this trap geometry,[20] a level splitting of $\Delta E = 2\hbar\omega = 40$ µeV would be expected (see Subsection 11.2.1), which is far below the linewidth of the exciton of 770 µeV. I put forth several alternative hypotheses for the origin of the various features, although none match quite precisely with values expected from the literature.

20: For a diameter of 200 nm, $\omega = 738$ GHz, and therefore we can interpolate $\omega = 18.5$ GHz for a diameter of 5 µm because both ω and ρ enter the potential quadratically.

13.3 Transfer-matrix method simulations of the heterostructure membrane

In the previous sections, we have seen that there is a profound difference in the PL intensity emitted from the sample depending on if there is a semi-transparent metallic gate electrode fabricated on either side of the membrane or not. What is more, Figure 13.3 showed that this reduction is in fact, counterintuitively, more pronounced for gates on the backside of the membrane. In the following, I explain this behavior as a consequence of thin-film interference effects. To this end, I perform TMM simulations of the heterostructure membranes investigated in this part of the present thesis using the `PyMoosh` package [305] to elucidate the observed quenching of PL when illuminating gate electrodes as well as the overall optical efficiency.[21] The TMM is a computationally efficient method of computing the amplitude of in- and outgoing electric fields in layered structures. I will first briefly recap the simulation method following Reference 305. For more details, refer to *ibid.* and references therein.

21: Strictly speaking, the term TMM only refers to one of the several formalisms implemented in the `PyMoosh` package. While fast, it is not the most numerically stable, and other methods may be preferred if wall time is not a limiting issue.

13.3.1 Electric fields in layered thin films

Consider a layered structure along z with interfaces at $z_i, i \in \{0, 1, \ldots, N+1\}$ that is translationally invariant along x and y. Each layer i may consist of a different dielectric material characterized by a (complex) relative permittivity $\epsilon_{r,i}$.[22] The electric field component along y of an electromagnetic wave transverse electric (TE) mode originating in some far away point satisfies the Helmholtz equation

22: We disregard magnetic materials with relative permeability $\mu_r \neq 1$ for simplicity.

$$\frac{\partial^2 E_y}{\partial z^2} + \gamma_i^2 E_y = 0, \tag{13.9}$$

where $\gamma_i = \sqrt{\epsilon_{r,i} k_0^2 - k_x^2}$ with $k_0 = \omega/c$ the wave vector in vacuum and k_x the component along x. Because k_x is constant, E_y depends trivially on it and we can without loss of generality set $x = 0$, resulting in the x-dependence $\exp(ik_x x) = 1$. In layer i of the structure, the solution to Equation 13.9 may then be written as a superposition of plane waves incident and reflected on the lower and upper interfaces [305],

$$\begin{cases} E_{y,i}^{+}(z) = A_i^{+} \exp\{i\gamma_i[z - z_i]\} + B_i^{+} \exp\{-i\gamma_i[z - z_i]\}, \\ E_{y,i}^{-}(z) = A_i^{-} \exp\{i\gamma_i[z - z_{i+1}]\} + B_i^{-} \exp\{-i\gamma_i[z - z_{i+1}]\}, \end{cases} \tag{13.10}$$

where the coefficients with superscript + (−) are referenced to the phase at the upper (lower) interface, respectively. Matching these solutions at $z = z_i$ for all i to satisfy the interface conditions imposed by Maxwell's equations gives rise to a linear system of equations, the solution to which can be obtained through several different methods.

A particularly simple method is the transfer-matrix method (T-matrix formalism), which corresponds to writing the interface conditions at $z = z_i$ as the matrix equation

$$\begin{pmatrix} A_{i+1}^{+} \\ B_{i+1}^{+} \end{pmatrix} = T_{i,i+1} \begin{pmatrix} A_i^{-} \\ B_i^{-} \end{pmatrix} \tag{13.11}$$

with

$$T_{i,i+1} = \frac{1}{2\gamma_{i+1}} \begin{pmatrix} \gamma_i + \gamma_{i+1} & \gamma_i - \gamma_{i+1} \\ \gamma_i - \gamma_{i+1} & \gamma_i + \gamma_{i+1} \end{pmatrix} \tag{13.12}$$

the transfer matrix for interface i. Connecting the coefficients for adjacent interfaces within a layer of height $h_i = z_{i+1} - z_i$ requires propagating the phase,

$$\begin{pmatrix} A_i^{-} \\ B_i^{-} \end{pmatrix} = C_i \begin{pmatrix} A_i^{+} \\ B_i^{+} \end{pmatrix}, \tag{13.13}$$

with

$$C_i = \exp\{\mathrm{diag}(-i\gamma_i h_i, i\gamma_i h_i)\}. \tag{13.14}$$

Iterating Equations 13.12 and 13.14, the total transfer matrix $T = T_{0,N+1}$ then reduces to the matrix product

$$T = T_{N,N+1} \prod_{i=0}^{N-1} T_{i,i+1} C_i. \tag{13.15}$$

From T, the reflection and transmission coefficients can be obtained as $r = A_0^{-} = -T_{01}/T_{00}$ and $t = B_{N+1}^{+} = rT_{10} + T_{11}$. Taking the absolute value square of reflection and transmission coefficients then yields the reflectance $\mathscr{R}$ and the transmittance $\mathscr{T}$, which correspond to the fraction of total incident power being reflected and transmitted, respectively. To obtain the absorptance $\mathscr{A}$, the fraction of power being absorbed, in layer i, one can compute the difference of the z-components of the Poynting vectors (*cf.* Equation 7.18) at the top of layers i and $i + 1$. In the TE case considered here, Equation 7.18 reduces to [305]

$$S_i = \mathrm{Re}\left[\frac{\gamma_i^*}{\gamma_0}\left(A_i^+ - B_i^+\right)^*\left(A_i^+ + B_i^+\right)\right] \tag{13.16}$$

and is hence straightforward to extract from the calculation of either the S or T matrices.

Equation 13.15 is simple to evaluate on a computer, making this method attractive for numerical applications. However, the opposite signs in the argument of the exponentials in Equation 13.14 can lead to instabilities for evanescent waves ($\gamma_i \in \mathbb{C}$) due to finite-precision floating point arithmetic [306]. Rewriting Equation 13.12 to have incoming and outgoing fields on opposite sides of the equality alleviates this issue while sacrificing the simple matrix-multiplication composition rule in what is known as the scattering matrix (S-matrix) formalism. Finally, note that for a thorough accounting of in- and out-going field amplitudes, excitonic effects should be included, for example using the approach by D'Andrea and Del Sole [307].

Beyond the calculation of the aforementioned coefficients, the TMM formalism also allows to compute the full spatial dependence of the fields. Two cases are implemented in `PyMoosh`: irradiation of the layered structured with a Gaussian beam rather than plane waves of infinite extent, and a current line source inside the structure. In the first case, the translational invariance along x leading to a plane-wave spatial dependence[23] that was assumed previously is replaced by a superposition of plane waves weighted with a normally distributed amplitude,[24]

23: Recall that $\gamma_i = \sqrt{\epsilon_{r,i}k_0^2 - k_x^2}$.

24: *I. e.*, the inverse Fourier transform of $\mathcal{E}_0(k_x)E_{y,i}(k_x, z)$.

$$E_{y,i}(x, z) = \exp(ik_x x) \rightarrow \int \frac{\mathrm{d}k_x}{2\pi}\mathcal{E}_0(k_x)E_{y,i}(k_x, z)\exp(ik_x x), \tag{13.17}$$

with (*cf.* Equation 8.7)

$$\mathcal{E}_0(k_x) = \frac{w_0}{2\sqrt{\pi}}\exp\left\{-ik_x x_0 - \left[\frac{w_0 k_x}{2}\right]^2\right\} \tag{13.18}$$

and

$$E_{y,i}(k_x, z) = A_i^- \exp\{i\gamma_i(k_x)[z - z_{i+1}]\} + B_i^+ \exp\{-i\gamma_i(k_x)[z - z_i]\}, \tag{13.19}$$

and where we considered only normal incidence for simplicity.

In the second case, Langevin et al. [305] consider an AC current I flowing through a translationally invariant, one-dimensional wire along y at $x = x_\mathrm{s}$. This introduces a source term into the Helmholtz equation, Equation 13.9, which, upon Fourier transforming in x direction, leads to

$$\frac{\partial^2 \hat{E}_y}{\partial z^2} + \gamma_i^2\hat{E}_y = -i\omega\mu_0 I\delta(z)\exp(ik_x x_\mathrm{s}). \tag{13.20}$$

The electric field $\hat{E}_{y,i}(k_x, z)$ is thus proportional to the Green's function of Equation 13.20 and can be obtained using a similar procedure as in the case of a distant source incident on the structure by matching the interface conditions. Performing the inverse Fourier transform by means of Equation 13.17 with constant weights, $\mathcal{E}_0(k_x) \equiv 1$, then yields the two-dimensional spatial distribution of the electric field, $E_{y,i}(x, z)$.

13.3.2 Quantum well absorptance

We now turn to applying the methods laid out above as implemented in `PyMoosh` to the membranes under investigation in the present thesis. For the simulations, I considered a membrane of 10/90/20/90/10 nm thick layers of GaAs/$Al_{0.34}Ga_{0.66}As$/GaAs/$Al_{0.34}Ga_{0.66}As$/GaAs at zero temperature, $\lambda_0 = 825$ nm, and normal incidence. The material parameters were obtained from References 308, 309, and 310 through `PyMoosh`'s interface to the `refractiveindex.info` database [311]. In the main text I take the half-space below the membrane to be the epoxy resin used to glue the sample to the Silicon host chip, which I assume to have a refractive index of $n_{\mathrm{epo}} = 1.55 + 0i$,[25] corresponding to there being no coherent back-scattering at the interface between epoxy and Silicon host chip. I discuss the influence of a finite epoxy thickness in Section B.3. I then consider four different scenarios,

25: Epotek 353ND

1. The bare heterostructure,
2. The heterostructure with only a top gate consisting of 7/2 nm of Au/Ti,
3. The heterostructure with only a bottom gate consisting of 25/5 nm of Au/Ti,
4. The heterostructure with both bottom and top gate with the aforementioned thicknesses,

and compute the absorptance $\mathscr{A}$ in the QW layer and reflectance $\mathscr{R}$ at the respective topmost layer. Both of these parameters will influence the amount of PL collected for a given excitation power. Obviously, the more of the incident laser light is reflected, the less can be absorbed, and the less is absorbed the less can be radiatively emitted.

Table 13.1 presents the results of the simulation, which confirms the behavior observed in the experiment: the presence of a gate on the backside of the membrane is the most significant factor in the reduction of incident power being absorbed in the QW despite the fact there is no additional obstructions in the light's path before the QW, highlighting the importance of multilayer effects. The reason for this lies in the relatively high reflectivity ($\mathscr{R} \sim 75\,\%$ for GaAs/Ti/Au) of the bottom gate stack. Since furthermore the effective wavelength in GaAs is $\lambda = \lambda_0/n \sim 230$ nm, the structure actually resembles a resonant cavity for which the electric field mode has a node in the center of the cavity, *i. e.*, the QW, and thus the absorption is strongly reduced. In turn, this means that changing the thickness of the membrane we can change the resonance condition of the cavity such that the electric field has a peak or trough at the QW position, maximizing the coupling matrix element leading to absorption. As the width of the QW determines the optical properties, the AlGaAs barrier thickness remains as the only parameter we can reasonably vary. *Reducing* the thickness, however, is ill-advised as in doped structures it decreases the mobility in the QW due to enhanced scattering at remote impurities and further increases the probability of tunneling through the barrier (*cf.* Section 11.2). I therefore used `PyMoosh` to optimize QW absorptance over the barrier thickness of a double-gated structure at $\lambda_0 = 825$ nm with the unoptimized case of 90 nm as a lower bound.

Table 13.1: Absorptance $\mathscr{A}$ and reflectance $\mathscr{R}$ in the QW for different configurations of the heterostructure. "Bare" is the standard structure without gate electrodes. "TG" and "BG" are with a gate on either the top or bottom side. "TG+BG" is with gates on both sides as on a trap site.

	$\mathscr{A}$ (%)	$\mathscr{R}$ (%)
Bare	2.9	22.4
TG	1.8	42.0
BG	0.5	82.7
TG+BG	0.4	84.8

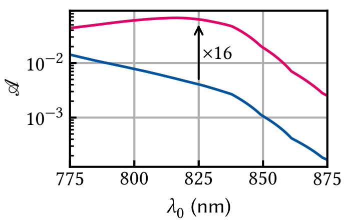

Figure 13.10: QW absorptance $\mathscr{A}$ in a heterostructure with default (blue) and optimized (magenta) barrier thickness and top and bottom gates as function of wavelength. Optimization was performed at 825 nm using the differential evolution algorithm implemented in `PyMoosh`, resulting in a barrier thickness of 122 nm and an absorptance better by a factor of 16 at 6.3 %.

Figure 13.10 depicts the absorptance at the optimal barrier thickness of 122 nm as function of incident wavelength λ_0. $\mathscr{A}$ is enhanced by a factor of up to 16 over a broad wavelength range compared to the unoptimized case. The resulting barrier thickness is only moderately larger than before and should still be thin enough to allow for sufficiently strong

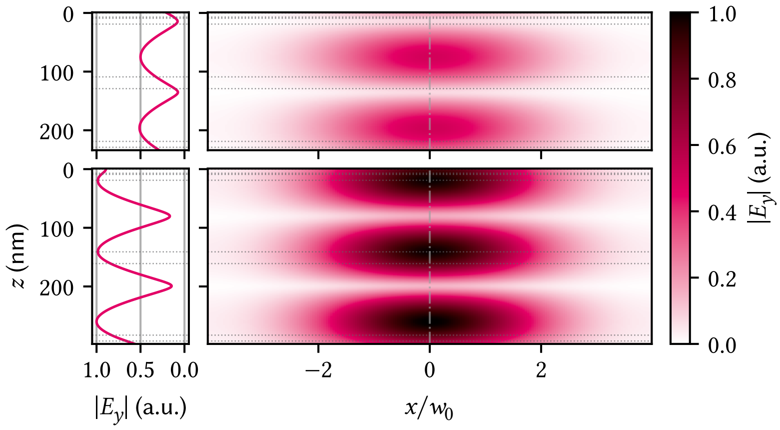

Figure 13.11: Absolute value of the electric field inside the double-gated heterostructure under illumination with a Gaussian beam at $\lambda_0 = 825\,\mathrm{nm}$ from the top. Top (bottom) panels show the structure with the default (optimized) barrier thickness of 90 nm (122 nm), respectively. Dotted horizontal lines indicate interfaces between different materials while the vertical dash-dotted line indicates the position of the line cuts shown in the left column. Increasing the thickness of the barrier has two beneficial effects; first, the overall field intensity inside the structure is higher by a factor of two, and second, there is a peak rather than a knot in the QW at a depth of $\sim$ 120 nm ($\sim$ 150 nm), leading to enhanced absorption.

electric fields at moderate voltages. The electric field distribution inside the membrane for illumination by a Gaussian beam with waist radius $w_0 = 624\,\mathrm{nm}$ (Part II) computed according to Equation 13.17 is shown in Figure 13.11 for the (un)optimized case in the (top) bottom panels. The left panels show a line cut taken at $x = 0$. As argued qualitatively above, in the case of a thinner barrier, the field displays a node at the QW position indicated by dashed lines at a depth of $z = 100$ and 120 nm. Conversely, the situation is reversed for the 30 nm thicker barrier, which hosts a field maximum at the depth of the QW.

13.3.3 Field emission

So far, we have considered the absorption of the incident laser radiation in the QW, a situation akin to placing the light source outside above the simulated structure. To see how multilayer interference affects the extraction of light emitted *inside* the structure, we turn to the Green's function calculation of Equation 13.20. That is, we place an infinitely long and thin current line in-plane (along y) in the center of the QW to model the radiation emitted from a dipole. Evidently, this is an approximation as the solution will be translationally invariant along the current line axis y, whereas a more realistic model would contain a point-dipole at $y = 0$ producing a field distribution as described in Subsection 7.1.2. Nonetheless, in the plane of the point-dipole (*i. e.*, at $y = 0$), its field should be well approximated by that of the infinitely long current line considered here. We once again use `PyMoosh` to perform the simulation.

Figure 13.12 shows the real part of the electric field for two different geometries: the bare heterostructure (left column) and the double-gated structure (right column). The source is indicated by a black dot and interfaces between different materials by dotted lines. The top row shows the results for the unoptimized barrier thickness, where clearly the presence of the gates attenuates the outcoupling from the membrane into the air halfspace above ($z < 0$). For the unoptimized, bare structure the emission pattern is close to cylindrical as would be the case in a homogeneous medium. Increasing the barrier thickness (bottom left panel) in this case results in a complicated standing-wave pattern inside the membrane. When adding the gates on both sides (bottom right panel), an emission pattern close to that of the unoptimized, bare structure is recovered, albeit with a higher extraction efficiency from the membrane.

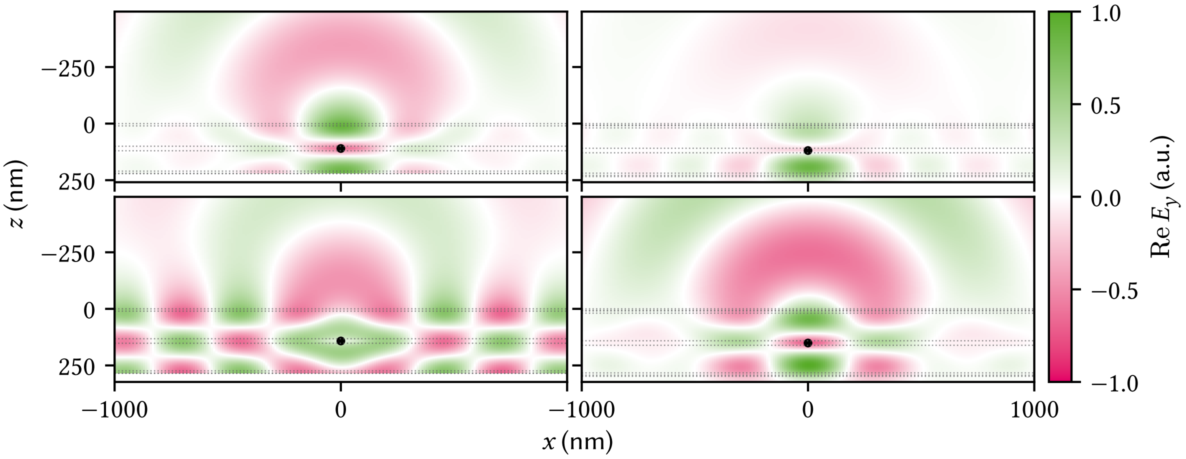

Figure 13.12: Real part of the electric field emitted by a current line located in the QW (black point) for different cases of the unoptimized (top row) and optimized (bottom row) structures. Left column shows the bare structures, right the double-gated ones. The half space $z < 0$ is the air above the membrane in the direction of the objective lens and the dotted lines indicate interfaces between materials. Evidently, the gates reduce the amplitude in the upper half of the membrane and thereby the outcoupling efficiency compared to the bare structure, consistent with what is observed in the experiment. Optimizing the barrier thickness for absorption in the QW drastically improves the outcoupling efficiency into the halfspace $z < 0$ for both the bare and gated structures, although actually the latter (bottom right panel) radiates more power into the surface normal direction when compared with the ungated structure (bottom left).

These results demonstrate that on top of increasing the absorption efficiency, the extraction efficiency of PL radiation from inside the membrane is further enhanced when increasing the AlGaAs barrier thickness by a modest 32 nm, leading us to expect an improvement larger than 16-fold in the end-to-end optical efficiency of PL experiments. While the simulations presented here suggest that this is true for an exciton trap in an unstructured membrane, Wu et al. [139, 312] found that when embedding the trap into a photonic crystal cavity to enhance the single-mode fiber coupling, the overall efficiency quickly falls off for barrier thicknesses above 95 nm (membrane thickness above 230 nm). The discrepancy between the two results indicates that the thicker barriers suggested here result in worse coupling to the Gaussian target mode optimized for by Wu et al. [312]. It would be interesting therefore to also analyze the absorption efficiency of the cavity design, and perform systematic experimental studies of the absorption and emission efficiencies as function of barrier thickness.

Conclusion and outlook 14

In this part of the present thesis, I presented and discussed optical measurements of doped semiconductor membranes under perpendicular electric fields. I first gave an introduction to the physics of photoluminescence (PL) in semiconductor QWs in Chapter 11. Starting with the PL of an unbiased 2DEG in a QW, which is defined by the band gap and the Fermi edge at low and high energies, respectively, I then performed calculations of the electron and hole energies in an undoped QW under the influence of an external electric field. This gives rise to the QCSE and quadratically lowers the energy difference between electron and hole ground states. The QW confinement prevents the field ionization of excitons and instead leads to a continuously decreased wave function overlap resulting in enhanced lifetime and reduced oscillator strength. I showed that, when taking into account lateral confinement by application of a local electric field as envisaged by Descamps [70], the oscillator strength vanishes for finite orbital angular momentum quantum numbers under the assumption of rotational symmetry, resulting in an optically resolvable level splitting of $2\hbar\omega$. Furthermore, the model predicts that the oscillator strength of excited states vanishes at certain electric fields because their wave functions have nodes in the QW. Using simple approximations, I finally estimated the Fowler-Nordheim tunneling rate of charge carriers escaping the QW confinement as function of the electric field, finding that below 5 V/µm rates are below 1 MHz. In the analytical calculations, I neglected excitonic effects, which for quantitative results need to be considered. Furthermore, the samples investigated later on in this part of the present thesis all contained a doped QW hosting a 2DEG whose presence certainly alters the picture. Many-body theory is required to fully capture the arising effects such as the FES and Mahan excitons [234].

In Chapter 12, I introduced the `mjolnir` measurement framework that facilitates optical experiments in the Millikelvin confocal microscope presented in Part II. I made the case for its existence and outlined the design goals before sketching the package's implementation and features. Central to these is the abstraction of physical instruments into logical ones, combining the parameters of different devices into logical functionality. Because of the relatively large range of different devices used to control the optical setup, this significantly reduces the level of complexity exposed to the user during the measurement workflow and enables the concise definition of measurements without large amounts of repetitive setup and teardown code. A very simple example is the `ExcitationPath` instrument's `wavelength` parameter, which not only sets the wavelength of the cw laser but also adjusts the `wavelength` parameter of the power meter, thus ensuring the sensor reading is correctly converted. Furthermore, the package implements multiple calibration routines, for instance to automatically update the energy axis of the CCD mounted after the spectrometer. Measurements are implemented in terms of multidimensional loops defined by composable `Sweep` objects. They are executed through a central `MeasurementHandler` object that can be customized to execute default setup and teardown tasks as well as perform standard modifications of measurement definitions. Finally, `mjolnir` provides live plotting of power meter readings, CCD data, or APD count rate, as well as an interactive data plotter based on 2D-slices through multidimensional data. Since the rest of the package is independent of the measurement functionality, other `QCoDeS`-based measurement frameworks such as `quantify` can readily replace or augment the latter, for instance to benefit from more sophisticated measurement control workflows that include timing control flows and broader support for buffered sweeps, or leverage the data analysis capabilities.

Employing the `mjolnir` software framework, I conducted optical measurements of doped QW membranes that I presented in Chapter 13. With the aim of demonstrating the spatial 0D-confinement of excitons by electrostatic means, I characterized the PL as function of position and electric field applied by means of local gate electrodes on the top and bottom sides of the membrane. The 2DEG PL of the unbiased QW was found to be in good agreement with the intuition obtained in Chapter 11, although it also revealed a significant mismatch between nominal and actual charge carrier densities for multiple samples. Defining the difference-mode (V_{DM}) and common-mode (V_{CM}) voltages as the difference and sum of top and bottom gate voltages, respectively, the electric field across the membrane is expected to be proportional to V_{DM}. In most samples investigated, the QCSE was not symmetric in V_{DM}, however, but showed an offset of around 700 mV, corresponding to an admixture of V_{CM}, that can be explained by built-in screening on one side of the membrane, the origin of which is unknown. Updating the virtual gate matrix to compensate for this effect (Equation 13.4) in future measurements would simplify their interpretation.

For small V_{DM}, the field inside the QW is entirely screened by the 2DEG. Surprisingly, though, at about ±1 V from the symmetry point, the charge carrier density starts to reduce until the 2DEG is fully depleted at about the same amount higher voltages again. I proposed that this is due to carrier sweep-out by Fowler-Nordheim tunneling that establishes an equilibrium between ionized dopants in the AlGaAs barrier and the charge carriers in the well. As the electric field is increased, the tunnel coupling increases and more ionized donors neutralize with electrons from the QW. This reduces the band bending, in turn broadens the tunnel barrier, and hence counteracts the increased transparency due to the tilting of the bands by the electric field. Overall, the emission energy could be tuned by 20 meV in the fully depleted regime.

Measuring the power dependence of the Stark-shifted emission line revealed no biexciton peaks. Instead, a substructure of a considerable number of individual lines appeared. At very small excitation powers, the main emission line displayed a logarithmic blue shift whose strength drastically changes above 10 nW. Several exciton traps showed an intricate substructure of the main emission line, whose constituents coupled differently to the electric field. As one-dimensional position sweeps across a trap also showed, the most likely explanation are a number of different emitters at different spatial locations within the area covered by the microscope focus. Possible candidates are impurities, defects, or interface steps or islands. While the literature has shown these to also be capable of binding excitons, no signatures of confinement were observed in $g^{(2)}$ measurements.

Such could also be revealed by photoluminescence excitation (PLE), a measurement of which I presented in Section 13.2. While the trap diameter was large and the expected orbital splitting of an excitonic quantum dot quite small and not observed, several interesting features appeared nonetheless. I put forth several hypotheses for the origin of the duplicated absorption edge as well as a second duplicated feature with positive Stark shift. Among them were light-hole excitons, excited QW states, and optical phonons, although none yield a quantitative agreement with the literature. For the largest electric field, where the 2DEG is fully depleted, both the PL and PLE displayed a peak splitting of the same magnitude, ~ 3 meV but 40 meV apart. The PL line was well fitted by a trion and exciton line shape. In the regime of finite electron densities it might be interesting to investigate if trion-polaritons occur in this system [313–315].

Finally, I addressed the observed quenching of PL intensity when the microscope focus is moved on top of gates on the backside of the membrane. I appealed to multilayer interference effects that lead to a reduction in both absorption of irradiant photons in the QW as well as in outcoupling efficiency. Using the transfer-matrix method (TMM), I simulated the membrane heterostructure for different configurations of gates on either sides and found good qualitative agreement with the behavior observed in experiments. I then optimized the AlGaAs barrier thickness for higher absorptance and, based on this, proposed increasing the current thickness of 90 nm by a modest 30 nm to achieve a 16-fold improved absorptance and better outcoupling efficiency in unstructured membranes. For exciton traps incorporating photonic crystal cavities for mode-matching to the Gaussian guiding mode of a SMF, however, the results of Wu et al. [312] indicate that the current thickness of 90 nm is already optimal. Systematic studies of the experimentally achievable efficiency should therefore be conducted in order to converge on an optimal heterostructure design.

Ultimately, though, I did not observe signatures of exciton confinement by local electrostatic potentials, which would be a significant step towards the realization of a spin-photon interface to semiconductor spin qubits by top-down fabrication. Where does this leave the concept introduced by Descamps [70]? Firstly, none of the samples that I investigated were fully functional, and as such we cannot make any definitive statements about the feasibility of the concept. Indeed, we can take the opposite point of view: none of the measurements presented in the present thesis provide any negative evidence that would *prohibit* electrostatic exciton trapping as envisioned in Reference 70. All observed deviations from the expected behavior should be possible to resolve by improved sample growth in close feedback with experimental characterization. Access to the full parameter space of virtual difference-mode and common-mode voltages of a sufficiently small central gate pair surrounded by a larger guard gate can reasonably be expected to provide enough tunability to confine single excitons. Given the above, the guard common mode should be used to overcome the screening and deplete the 2DEG. Then, the central difference mode should have a large enough window of operation before the Schottky barrier breaks down ($V_{\mathrm{DM}} \approx \pm 1.4\,\mathrm{V}$) to provide a localized electric field that, going by the calculations in Chapter 11, lowers the exciton energy by up to 25 meV, and results in a quantum dot orbital level splitting on the order of 0.5 meV, large enough to be resolved by PLE. The optimization of the heterostructure design I proposed should furthermore provide higher radiative efficiency and thus allow measurements over a wider range of excitation powers while only modestly lowering the achievable electric field for a given voltage. To enable resonance-fluorescence measurements, the coherence of excitation and subsequent emission under the current polarization configuration of the confocal microscope should be investigated (see Chapter 9).

Moving beyond GaAs, Reznikov [138] proposed an implementation of the device concept in another material system, $\mathrm{Ge/Si_{1-x}Ge_x}$. Hole spin qubits in this system have quickly matured in recent years. What is more, by carefully engineering the alloy composition and strain of the QW, a semi-direct[1] bandgap as well as type-I confinement[2] is projected to be achievable. Favorably, the optical gap can be tuned to the telecom O-band and thus alleviate the need for quantum frequency conversion to match quantum network operation wavelengths. However, the optical characterization of this system will be challenging as the quasi-direct band gap leads to reduced PL efficiency due to valley coupling.

1: *I. e.*, the band structure has conduction and valence band minima and maxima at the Γ-point, but the Γ-valley is not the lowest conduction band valley.

2: *I. e.*, the band alignment of the valence band is inverted compared to the conduction band and both electrons and holes are confined in the QW.

Part IV

A Filter-Function Formalism For Unital Quantum Operations

Introduction 15

Author contributions — *This part of the present thesis is based to large extent on Reference [316], an early draft of which in turn was my Master's thesis [317], and as such contains text contributions from all three authors. Chapter 19 and Section C.4 additionally contain results published in Reference [318], an early draft of which appears in Reference [80]. The first-order concatenation rule was originally derived by Pascal Cerfontaine.*[a] *He also conceived and performed initial calculations of the linearized quantum process expressed in terms of filter functions. Hendrik Bluhm*[b] *initiated the project and wrote the initial draft of the introduction. I recast the approach in the quantum operations formalism based on stochastic Liouville equations and the cumulant expansion, and derived expressions for the second-order terms, as well as developed an optimized expression for periodic Hamiltonians, derived quantities, discussed operator bases, and analyzed the computational efficiency. I also wrote the software package, performed all simulations, and developed all examples. Chapter 17 and Section C.3 contain new results that I derived.*

[316]: Hangleiter et al. (2021), *Filter-Function Formalism and Software Package to Compute Quantum Processes of Gate Sequences for Classical Non-Markovian Noise*

[317]: Hangleiter (2019), *Filter Function Formalism for Unitary Quantum Operations*

[318]: Cerfontaine et al. (2021), *Filter Functions for Quantum Processes under Correlated Noise*

[80]: Cerfontaine (2019), *High-Fidelity Single- and Two-Qubit Gates for Two-Electron Spin Qubits*

a: Then at RWTH Aachen University and Forschungszentrum Jülich GmbH.

b: RWTH Aachen University and Forschungszentrum Jülich GmbH.

Note — *In Reference [316], extensive use was made of Kubo's cumulant expansion [319]. Due to an error in Kubo's paper which was only pointed out several years later by Fox [320] and we were not aware of, those results turned out to be not exact as claimed but approximate [321].*[1] *We address this discrepancy in Chapter 17.*

[321]: Hangleiter et al. (2024), *Erratum: Filter-function Formalism and Software Package to Compute Quantum Processes of Gate Sequences for Classical Non-Markovian Noise [Phys. Rev. Research 3, 043047 (2021)]*

1: Unfortunately, the error has proliferated through the literature and proves quite pervasive despite the significant amount of time that has passed since it was first discovered. Recent examples include Reference 33. One may only speculate if this is because of the 14 years that passed between the original publication and the correction, the lack of an erratum once the mistake had been discovered, or simply because of Kubo's fame. In any case, it might serve as a cautionary tale and possibly an impetus to a less static publication system that allows, for example, cross-linking commentaries and critiques.

In the circuit model of quantum computing, computations are driven by applying time-local quantum gates. Any algorithm can be compiled using sequences of one- and two-qubit gates [322]. Ideal, error-free gates are represented by unitary transformations, so that simulating the action of an algorithm on an initial state of a quantum computer amounts to simple matrix multiplication. Real implementations are subject to noise that causes decoherence resulting in gate errors. If the noise is uncorrelated between gates, its effect can be described by quantum operations acting as linear maps on density matrices, even when several gates are concatenated. A closely related approach is the use of a master equation in Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) form [323, 324], which governs the dynamics of density matrices under the influence of Markovian noise with a flat power spectral density.

Yet many physical systems used as hosts for qubits do not satisfy the condition of uncorrelated noise. One example frequently encountered in solid-state systems is that of $1/f$ noise, which in principle contains arbitrarily long correlation times. It emerges for instance as flux noise in superconducting qubits and electrical noise in quantum dot qubits [101, 325–327]. Whereas simple approaches exist to treat for example quasistatic noise, which corresponds to perfectly correlated noise (*i.e.*, a spectrum with weight only at zero frequency), they cannot be applied to $1/f$ noise because of the wide distribution of correlation times it contains [12]. Thus, there is a gap in the mathematical descriptions of gate operations for noises with arbitrary power spectra that exist between the extremal cases of perfectly flat (white) and sharply peaked (quasistatic) spectra. To capture experimentally relevant effects important to understand the capabilities of quantum computing systems, a universally applicable formalism is hence desirable. For example, one may expect the fidelity requirements for quantum error correction to be more stringent for correlated noise as errors of different gates can interfere constructively [328]. On the other hand, it might also be possible to use correlation effects to one's benefit, attenuating decoherence by cleverly constructing the gate sequences in algorithms.

As experimental platforms begin to approach fidelity limits set by employing primitive pulse schemes [326, 329, 330] and detailed knowledge about noise sources and spectra in solid-state systems becomes available [11, 331, 332], control pulse optimizations tailored towards specific systems will be required to further push fidelities beyond the error correction threshold [333, 334]. This calls for flexible and generically applicable tools as a basis for the numerical optimization of pulses as well as the detailed analysis of the quantum processes they effect. In order to obtain a useful description also for gate operations that decouple from leading orders of noise, such as dynamically corrected gates (DCGs) [335], beyond leading order results are required.

In Reference 318 we presented a formalism based on filter functions and the Magnus expansion (ME) that addresses these needs and limitations of the canonical master equation approach for correlated noise. Specifically, we showed how process descriptions can be obtained perturbatively for arbitrary classical noise spectra and derived a concatenation rule to obtain the filter function of a sequence of gates from those of the individual gates. This work generalizes and extends these results.

Filter functions (FFs) were originally introduced to describe the decay of phase coherence under dynamical decoupling (DD) sequences [336–339] consisting of wait times and perfect π-pulses. The formalism facilitated recognizing these sequences as band-pass filters that allow for probing the environmental noise characteristics of a quantum system through noise spectroscopy [7, 9, 332, 340] or optimizing sequences to suppress specific noise bands [341–344]. It can also be extended to fidelities of gate operations for single [345, 346] or multiple [347, 348] qubits using the ME [349, 350] as well as more general DD protocols [351]. The works by Green et al. [346] and Clausen, Bensky, and Kurizki [352] also introduced the notion of the control matrix as a quantity closely related to the canonical filter function that is convenient for calculations. In this context, the formalism's capability to predict fidelities of gate implementations has been identified and experimentally tested [343, 346, 353, 354]. More recently, it has also proved useful in assessing the performance requirements for classical control electronics [355].

While analytical approaches allow for the calculation of filter functions of arbitrary quantum control protocols in principle, it is in practice often a tedious task to determine analytical solutions to the integrals involved if the complexity of the applied wave forms goes beyond simple square pulses or extends to multiple qubits. Moreover, one does not always have a closed-form expression of the control at hand, such as is the case for numerically optimized control pulses. This calls for a numerical approach which, while giving up some of the insights an analytical form offers, is universally applicable and eliminates the need for laborious analytical calculations.

Here, we build and extend upon our work of Reference 318 and that of Reference 346 to show that the formalism can be recast within the framework of stochastic Liouville equations by means of the cumulant expansion [319, 320, 356, 357]. For Gaussian noise commuting with the control, this entails exact results for the quantum process of an arbitrary control operation using only first and second order terms of the ME [349]. If the noise is Gaussian but in general non-commuting, the cumulant expansion does not terminate [320], but we show that the truncation still yields highly accurate results except in the ultralow-frequency regime by computing the exact filter function from random sampling. Moreover, due to the fact that the ME retains the algebraic structure of the expanded quantity [350] we are able to separate incoherent and coherent contributions to the quantum process. We give explicit methods to evaluate these terms for piecewise-constant control pulses. Moreover, we show that the formalism naturally lends itself as a tool for numerical calculations and present the `filter_functions` Python software package that enables calculating the filter function of arbitrary, piecewise constant defined pulses [358]. On top of providing methods to handle individual quantum gates, the package also implements the concatenation operation as well as parallelized execution of pulses on different groups of qubits, allowing for a highly modular and hence computationally powerful treatment of quantum algorithms in the presence of correlated noise. Given an arbitrary, classical noise spectral density, it can be used to calculate a matrix representation of the error process. From this matrix one can extract average gate fidelities, transition probabilities, and leakage rates as we derive below. To simplify adaptation the software's application programming interface (API) is strongly inspired by and compatible with `QuTiP` [359] as well as `qopt` [360]. This allows users to use these packages in conjunction. Assessing the computational performance, we show that our method outperforms Monte Carlo (MC) simulations for single gates. New analytical results applicable to periodic Hamiltonians and employing the concatenation property make this advantage even more pronounced for sequences of gates. To highlight the main software features, we show example applications below.

We provide this package in the expectation that it will be a useful tool for the community. Besides recasting and expanding on our earlier introduction of the formalism in Reference 318, the present work is intended to provide an overview of the software and its capabilities. It is structured as follows: In Chapter 16 we derive a closed-form expression for unital quantum operations in the presence of non-Markovian Gaussian noise and lay out how it may be evaluated using the filter-function formalism. We review the concatenation of quantum operations shown in Reference 318 and furthermore adapt the method by Green et al. [346] to calculate the filter function of an arbitrary control sequence numerically. We will specifically focus on computational aspects of the formalism and lay out how to compute various quantities of interest. Moreover, we classify its computational complexity for calculating average gate fidelities and remark on simplifications that allow for drastic improvements in performance in certain applications. In Chapter 17, we validate the truncation of the cumulant expansion after the second order using a random sampling approach to compute the exact filter functions of the noisy quantum process for Gaussian noise. In Chapter 18, we then introduce the `filter_functions` software package by outlining the programmatic structure and giving a brief overview over the API. Lastly, in Chapter 19, we show the application of the software by means of four examples that highlight various features of the formalism and its implementation. Therein, we first demonstrate that the formalism can predict average gate fidelities for complex two-qubit quantum gates in agreement with computationally much more costly MC calculations. Next, we show how it can be applied to periodically driven systems to efficiently analyze Rabi oscillations. We finally establish the formalism's ability to predict deviations from the simple concatenation of unitary gates for sequences and algorithms in the presence of correlated noise by simulating a randomized benchmarking (RB) experiment as well as assembling a quantum Fourier transform (QFT) circuit from numerically optimized gates. We conclude by briefly remarking on possible future application and extension of our method in Chapter 20.

Throughout this part we will denote Hilbert-space operators by Roman font, *e.g.* U, and quantum operations and their representations as transfer matrices in Liouville space by calligraphic font, *e.g.* $\mathcal{U}$, which we also use for the control matrix $\tilde{\mathcal{B}}$ to emphasize its innate connection to a transfer matrix. For consistency, a unitary quantum operation will share the same character as the corresponding unitary operator. An operator in the interaction picture will furthermore be designated by an overset tilde, *e.g.* $\tilde{H} = U^\dagger H U$ with U the unitary operator defining the co-moving frame. Definitions of new quantities on the left and right side of an equality are denoted by $\coloneqq$ and $\eqqcolon$, respectively. We use a central dot ($\bullet$) as a placeholder in some definitions of abstract operators such as the Liouvillian, denoted by $\mathcal{L} \coloneqq [H, \bullet]$, which is to be understood as the commutator of the corresponding Hamiltonian H and the operator that $\mathcal{L}$ acts on. The identity matrix is denoted by $\mathbb{1}$ and its dimension always inferred from context. Furthermore, we will use Greek letters for indices that correspond to noise operators in order to distinguish them clearly from those that correspond to basis or matrix elements. Lastly, we work in units where $\hbar = 1$.

Filter-function formalism for unital quantum operations 16

We begin the mathematical part by showing how a superoperator matrix representation of the error process, the *error transfer matrix*, of a unital quantum operation can be computed from the control matrix of the pulse implementing the operation. The control matrix relates the operators through which noise couples into the system to a set of basis operators in the interaction picture and we detail how it can be calculated in a relatively efficient manner for two different situations. First, we consider a sequence of gates whose control matrices have been precomputed. Second, we lay out how the control matrix can be obtained from scratch under the assumption of piecewise-constant control, which is often convenient for approximating continuous pulse shapes. Other wave forms can be dealt with analogously by solving the corresponding integrals. We then move on to show how several quantities of interest can be extracted and present optimized strategies for computing the central objects of the formalism.

16.1 Transfer matrix representation of quantum operations

16.1.1 Brief review of quantum operations and superoperators

The quantum operations formalism provides a general framework for the description of open quantum systems [151, 361]. It forms the mathematical basis for quantum process tomography (QPT) [362, 363] as well as gate set tomography (GST) [364, 365] and has also been extensively employed in the context of RB [366, 367]. Several different representations of quantum operations exist. While all of them are equivalent one typically chooses the most convenient for the problem at hand. For an overview of the most commonly used representations see Reference 365 and for matrix representations in particular Reference 368 and the references therein. In this work we employ the Liouville representation, to the best of our knowledge first formalized by Fano [369], to profit from its simple properties under composition. It is also known as the transfer matrix representation and we will use the terms interchangeably below. We now briefly review the concept and refer the reader to the literature for further details. Concretely, the Liouville representation of an operation $\mathcal{E} : \rho \to \mathcal{E}(\rho)$ acting on density operators in a Hilbert space $\mathscr{H}$ of dimension d is given by

$$\mathcal{E}_{ij} \doteq \mathrm{tr}\Big(C_i^\dagger \mathcal{E}(C_j)\Big) \tag{16.1}$$

with an operator basis $\mathsf{C} = \{C_0, C_1, \dots, C_{d^2-1}\}$ for the space of linear operators over $\mathscr{H}$, $\mathsf{L}(\mathscr{H})$, orthonormal with respect to the Hilbert-Schmidt product $\langle A, B\rangle := \mathrm{tr}\big(A^\dagger B\big)$. In the case that the operator basis corresponds to the Pauli matrices Equation 16.1 is known as the Pauli transfer matrix (PTM). The operation $\mathcal{E}$ is thus associated with a $d^2 \times d^2$ matrix in *Liouville space* $\mathscr{L}$ that describes its action as the degree to which the jth basis element is mapped onto the ith. On $\mathscr{L}$ one can identify a set of basis kets $\{\,|C_i\rangle\!\rangle\}_{i=0}^{d^2-1} = \{\,|i\rangle\!\rangle\}_{i=0}^{d^2-1}$ isomorphic to the operators C_i (and correspondingly bras $\langle\!\langle i|$ to the adjoint $C_i^\dagger$) as well as the inner product $\langle\!\langle i|j\rangle\!\rangle = \langle C_i, C_j\rangle$. As the vectors $|i\rangle\!\rangle$ form an orthonormal basis, any operator on $\mathscr{H}$ can be written as a vector on $\mathscr{L}$, $|A\rangle\!\rangle = \sum_i |i\rangle\!\rangle\langle\!\langle i|A\rangle\!\rangle$, whereas a superoperator on $\mathscr{H}$ becomes a matrix on $\mathscr{L}$, see Equation 16.1. It can then be shown that density operators represented by vectors are propagated by transfer matrices so that the action of a quantum operation $\mathcal{E}$ on a density operator ρ is given by $|\mathcal{E}(\rho)\rangle\!\rangle = \mathcal{E}\,|\rho\rangle\!\rangle = \sum_{ij} |i\rangle\!\rangle\langle\!\langle i|\mathcal{E}|j\rangle\!\rangle\langle\!\langle j|\rho\rangle\!\rangle$. Thus, the composition of two operations $\mathcal{E}_1$ and $\mathcal{E}_2$ corresponds to matrix multiplication in Liouville space, $[\mathcal{E}_2 \circ \mathcal{E}_1]_{ik} = \sum_j [\mathcal{E}_2]_{ij}[\mathcal{E}_1]_{jk}$, a property which makes the representation particularly attractive for sequences of operations. Although from a numerical perspective the computational complexity scales unfavorably with the system dimension d (*cf.* Section 16.9), we will employ the Liouville representation for its transparent interpretation and concise behavior under composition in the following analytical considerations. Lastly, we note that for $C_0 \propto \mathbb{1}$, trace-preservation and unitality are encoded in the relations $\mathcal{E}_{0j} = \delta_{0j}$ and $\mathcal{E}_{j0} = \delta_{j0}$, respectively.

16.1.2 Liouville representation of the error channel

We will now derive an expression for the quantum process of a quantum gate in the presence of arbitrary classical noise. As a single realization of a classical noise generates strictly unitary dynamics, we will be interested in the expectation value of the dynamics over many such realizations, which will lead to a quantum process including decoherence. If the noise is additionally Gaussian and commutes with the control, these results are exact and therefore apply without restrictions to arbitrarily large noise strength as well as to gates that partially decouple from noise. For such DCGs or DD sequences [335, 339] higher order terms can become dominant. Non-commuting noise leads to corrections at higher orders that we investigate in Chapter 17 [321]. In the case that the environment is not strictly Gaussian, our approach becomes perturbative and we recover the results presented in Reference 318. As most of our discussion later on in this part will focus on the approximation neglecting coherent errors, readers not interested in the full generality may refer to that publication for a less general but perhaps more accessible derivation and skip ahead to Section 16.2.

[321]: Hangleiter et al. (2024), *Erratum: Filter-function Formalism and Software Package to Compute Quantum Processes of Gate Sequences for Classical Non-Markovian Noise [Phys. Rev. Research 3, 043047 (2021)]*

The difference is that in Reference 318, the Magnus expansion is applied to the solution of the Schrödinger equation, whereas the approach presented here is based on the theory of stochastic Liouville equations and the cumulant expansion [319, 356]. In the filter function context, the cumulant expansion has been used to express the decay of the off-diagonal terms of a single-qubit density matrix in Reference 339. More recently, Paz-Silva and Viola [351] employed it in conjunction with the ME to obtain the matrix elements of the perturbed density operator after a time T of noisy evolution. In Reference 370, the authors made use of the cumulant expansion and stochastic Liouville equations for the purpose of gate optimization. Here, we combine different aspects of these works and make the connection to the quantum operations formalism by determining the noise-averaged error propagator in the Liouville representation. This form completely characterizes the error process and hence allows for detailed insight into the decoherence mechanisms of the operation.

Concretely, we consider a system described by the stochastic Hamiltonian

$$H(t) = H_{\mathrm{c}}(t) + H_{\mathrm{n}}(t), \tag{16.2}$$
$$H_{\mathrm{n}}(t) = \sum_\alpha b_\alpha(t) B_\alpha(t). \tag{16.3}$$

$H_{\mathrm{c}}(t)$ is implemented by the experimentalist to generate the desired control operation during the time $t \in [0, \tau]$ and $H_{\mathrm{n}}(t)$ describes classical fluctuating noise environments $b_\alpha(t) \in \mathbb{R}$ that couple to the quantum system via the Hermitian noise operators $B_\alpha(t) \in \mathsf{L}(\mathscr{H})$. These may carry a general, deterministic time dependence and without loss of generality, we can require them to be traceless since any contributions proportional to the identity do not contribute to noisy evolution in any case.[1] The $b_\alpha(t)$ are random variables drawn from (not necessarily Gaussian) distributions with zero mean that are assumed to be i.i.d. both with respect to repetitions of the experiment. Note that this concept of independence does not preclude correlations between different noise sources $\alpha \neq \beta$ nor between one noise source at different times $t \neq t'$, but only serves to obtain a well-defined ensemble average. Lastly, to be able to later on relate the correlation functions of the $b_\alpha(t)$ to their spectral density, we require the noise fields to be wide-sense stationary, meaning that their correlation function depends only on the time difference.[2]

1: The identity commutes with the control Hamiltonian at all times and hence does not generate any evolution in the interaction picture in which we work later on (*cf.* Equation 16.13).

2: See also Sidenote 8.

For noise operators without explicit time dependence, Equation 16.3 constitutes a universal decomposition as can be seen by choosing the B_α from an orthonormal basis for $\mathsf{L}(\mathscr{H})$. To motivate the time-dependent form of Equation 16.3, assume the true Hamiltonian is a function of a set of noisy parameters $\tilde{\boldsymbol{\lambda}}(t) = \boldsymbol{\lambda}(t) + \boldsymbol{\delta\lambda}(t)$ where $\boldsymbol{\delta\lambda}(t) = \mathrm{vec}(\{b_\alpha(t)\}_\alpha)$ are the stochastic variables. Expanding the Hamiltonian in an orthonormal operator basis yields $H(\tilde{\boldsymbol{\lambda}}(t)) = \sum_\alpha f_\alpha(\boldsymbol{\lambda}(t), \boldsymbol{\delta\lambda}(t)) B_\alpha$. In general, however, the expansion coefficients f_α will be arbitrary functions of both the deterministic parameters $\boldsymbol{\lambda}(t)$ and the stochastic noises $\boldsymbol{\delta\lambda}(t)$, which prohibits a factorized form like Equation 16.3. We can address this problem by first expanding H around $\boldsymbol{\lambda}(t)$ for small fluctuations $\boldsymbol{\delta\lambda}(t)$. Then, the Hamiltonian approximately becomes $H(\tilde{\boldsymbol{\lambda}}(t)) \approx H(\boldsymbol{\lambda}(t)) + \boldsymbol{\delta\lambda}(t) \cdot \nabla_\lambda H(\boldsymbol{\lambda}(t))$, where we can define the control Hamiltonian as $H_{\mathrm{c}}(t) \coloneqq H(\boldsymbol{\lambda}(t))$. Expanding the second term in the operator basis now results in the form (16.3) for the noise Hamiltonian as it is linear in $\boldsymbol{\delta\lambda}(t)$ and the deterministic time dependence is contained in $\nabla_\lambda H(\boldsymbol{\lambda}(t))$ alone.

This permits us to model complex relations between physical noise sources and the noise operators that capture the coupling to the quantum system, arising for example through control hardware or effective Hamiltonians obtained from *e.g.* Schrieffer-Wolff transformations. While the linearization is in most cases an approximation, it does not impose significant constraints since the noise is typically weak compared to the control.[3] As an example, we could capture a dependence of the device sensitivity on external controls (see also Reference 347). In a widely used setting electrons confined in solid-state quantum dots are manipulated using the exchange interaction J that depends non-linearly on the potential difference ϵ between two dots. Since the dominant physical noise source affecting this control is charge noise, one could include the effect on $J(\epsilon)$ to first order with $s_\epsilon(t) = \partial J(\epsilon(t))/\partial \epsilon(t)$ so that $H_{\mathrm{n}}(t) = b_\epsilon(t) B_\epsilon(t) = b_\epsilon(t) s_\epsilon(t) B_\epsilon$ for some operator B_ϵ which represents the exchange coupling.

3: The same argument forms the basis for the perturbative approach for non-Gaussian noise.

We proceed in our derivation by noting that the control Hamiltonian H_{c}

gives rise to the noise-free Liouville–von Neumann equation

$$\frac{\mathrm{d}\rho(t)}{\mathrm{d}t} = -i[H_\mathrm{c}(t), \rho(t)] = -i\mathcal{L}_\mathrm{c}(t)\rho(t) \tag{16.4}$$

on the Hilbert space $\mathscr{H}$ with the Liouvillian superoperator $\mathcal{L}_\mathrm{c}(t)$ representing the control. Analogous to the Schrödinger equation we may also write this differential equation in terms of time evolution superoperators (superpropagators), $\mathrm{d}\mathcal{U}_\mathrm{c}(t)/\mathrm{d}t = -i\mathcal{L}_\mathrm{c}(t)\mathcal{U}_\mathrm{c}(t)$ where the action of $\mathcal{U}_\mathrm{c}$ on a state ρ is to be understood as $\mathcal{U}_\mathrm{c} : \rho \rightarrow U_\mathrm{c}\rho U_\mathrm{c}^\dagger$ with U_c the usual time evolution operator satisfying the corresponding Schrödinger equation. This allows us to write the superpropagator for the total Liouvillian $\mathcal{L} = \mathcal{L}_\mathrm{c} + \mathcal{L}_\mathrm{n}$ as $\mathcal{U}(t) = \mathcal{U}_\mathrm{c}(t)\tilde{\mathcal{U}}(t)$ where the unitary error superpropagator $\tilde{\mathcal{U}}(t)$ contains the effect of a specific noise realization in Equation 16.3. Next, we transform the noise Liouvillian $\mathcal{L}_\mathrm{n}$ to the interaction picture with respect to the control Liouvillian $\mathcal{L}_\mathrm{c}$ so that $\tilde{\mathcal{U}}(t)$ satisfies the modified Liouville–von Neumann equation

$$\frac{\mathrm{d}\tilde{\mathcal{U}}(t)}{\mathrm{d}t} = -i\tilde{\mathcal{L}}_\mathrm{n}(t)\tilde{\mathcal{U}}(t), \tag{16.5}$$

$$\tilde{\mathcal{L}}_\mathrm{n}(t) = \mathcal{U}_\mathrm{c}^\dagger(t)\mathcal{L}_\mathrm{n}(t)\mathcal{U}_\mathrm{c}(t). \tag{16.6}$$

Equation 16.5 may be formally solved using the Magnus expansion [349] so that at time $t = \tau$

$$\tilde{\mathcal{U}}(\tau) = \exp\{-i\tau\mathcal{L}_\mathrm{eff}(\tau)\} \tag{16.7}$$

with $\mathcal{L}_\mathrm{eff}(\tau) = \sum_{n=1}^{\infty}\mathcal{L}_{\mathrm{eff},n}(\tau)$. A sufficient criterion for the convergence of the expansion is given by Moan, Oteo, and Ros [371] as $\int_0^\tau \mathrm{d}t\,\lVert\tilde{\mathcal{L}}_\mathrm{n}(t)\rVert < \pi$ where $\lVert\,\bullet\,\rVert = \sqrt{\langle\,\bullet\,,\,\bullet\,\rangle}$ is the Frobenius (Hilbert-Schmidt) norm. We may tighten this bound if we assume the noise to have some characteristic correlation time τ_c after which its correlation function has decayed by a certain amount by replacing the upper limit of integration by $\min\{\tau, \tau_\mathrm{c}\}$.[4] The first and second terms of the ME are given by [349, 350]

4: For $\tau_\mathrm{c} \rightarrow 0$ the noise becomes white, *i. e.*, uncorrelated and hence the ME always exists and converges. For $\tau_\mathrm{c} \rightarrow \infty$ the noise becomes perfectly (DC) correlated and the convergence is only guaranteed for special cases.

$$\mathcal{L}_{\mathrm{eff},1}(\tau) = \frac{1}{\tau}\int_0^\tau \mathrm{d}t\,\tilde{\mathcal{L}}_\mathrm{n}(t), \tag{16.8a}$$

$$\mathcal{L}_{\mathrm{eff},2}(\tau) = -\frac{i}{2\tau}\int_0^\tau \mathrm{d}t_1\int_0^{t_1}\mathrm{d}t_2\left[\tilde{\mathcal{L}}_\mathrm{n}(t_1), \tilde{\mathcal{L}}_\mathrm{n}(t_2)\right]. \tag{16.8b}$$

The nth term of the expansion contains n factors of the noise variables $b_\alpha(t)$ and scales with n factors of the control duration τ, suggesting that higher-order terms can be neglected if their product is small. In the Bloch sphere picture this corresponds to requiring that the angle by which the Bloch vector is rotated away from its intended trajectory due to the noise be small. Below, we will use the parameter ξ to denote the magnitude of this deviation. It is properly defined in Subsection C.2.1 where also bounds for the convergence of the ME are discussed. Here, we only state that $\mathcal{L}_{\mathrm{eff},n} \sim \xi^n$ (see also Reference 346).

We have suggestively written the ME in terms of an effective Liouvillian $\mathcal{L}_\mathrm{eff} = [H_\mathrm{eff}, \bullet\,]$ to interpret it as the generator of a *time*-averaged evolution of a single noise realization up to time τ. In order to obtain the *ensemble*-averaged evolution of many realizations of the stochastic Hamiltonian in Equation 16.3, we apply the cumulant expansion to $\tilde{\mathcal{U}}$ (see also Refences 29 and 372),

$$\langle\tilde{\mathcal{U}}(\tau)\rangle = \langle\exp\{-i\tau\mathcal{L}_\mathrm{eff}(\tau)\}\rangle =: \exp\mathcal{K}(\tau) \tag{16.9}$$

with $\langle\,\bullet\,\rangle$ denoting the ensemble average[5] and the cumulant function [319]

$$\mathcal{K}(\tau) = \sum_{k=1}^{\infty}\frac{(-i\tau)^k}{k!}\left\langle\mathcal{L}_{\mathrm{eff}}(\tau)^k\right\rangle_{\mathrm{c}} \tag{16.10}$$

$$= \sum_{k=1}^{\infty}\frac{(-i\tau)^k}{k!}\left\langle\left[\sum_{n=1}^{\infty}\mathcal{L}_{\mathrm{eff},n}(\tau)\right]^k\right\rangle_{\mathrm{c}}. \tag{16.11}$$

The notation $\langle\,\bullet\,\rangle_{\mathrm{c}}$ denotes the cumulant average which prescribes a certain averaging operation. The first cumulant of a set of random variables $\{X_i(t)\}_i$ is simply the expectation value, $\langle X_i(t)\rangle_{\mathrm{c}} = \langle X_i(t)\rangle$, whereas the second cumulant corresponds to the covariance, $\langle X_i(t)X_j(t)\rangle_{\mathrm{c}} = \langle X_i(t)X_j(t)\rangle - \langle X_i(t)\rangle\,\langle X_j(t)\rangle$. Remarkably, third and higher-order cumulants vanish for commuting[6] Gaussian processes [10, 356], making Equation 16.11 exact by truncating the sums already at $k = 2$ and $n = 2$. In this case, the convergence radius of the ME becomes infinite. In the following, we assume this truncation to be a good approximation of the exact dynamics. We investigate this assumption in more detail in Chapter 17 for non-commuting Gaussian noise. For non-Gaussian noise, we note that the approximation error depends on the degree of non-Gaussianity. Due to the central-limit theorem, we may expect corrections to become smaller the more two-level fluctuators (TLFs), whose individual statistics are highly non-Gaussian, couple to a qubit, for example. In general, all we can say is that the error is of $\mathcal{O}(\xi^2)$ and higher order terms include both higher orders of the ME and the cumulant expansion.

We continue by pointing out that the terms with $k = n = 2$ involve fourth-order cumulants and are thus of $\mathcal{O}(\xi^4)$. Since furthermore we assume that the noise fields $b_\alpha(t)$ have zero mean, also the terms with $k = n = 1$ vanish and $\langle X_i(t)X_j(t)\rangle_{\mathrm{c}} = \langle X_i(t)X_j(t)\rangle$. We can hence write the cumulant function succinctly as

$$\mathcal{K}(\tau) = -i\tau\left\langle\mathcal{L}_{\mathrm{eff},2}(\tau)\right\rangle - \frac{\tau^2}{2}\left\langle\mathcal{L}_{\mathrm{eff},1}(\tau)^2\right\rangle. \tag{16.12}$$

Equations 16.9 and 16.12 allow us to compute the full quantum process $\langle\tilde{\mathcal{U}}\rangle : \rho \to \langle\tilde{\mathcal{U}}(\rho)\rangle$ for noise with arbitrary spectral density. Inspecting Equation 16.12, we observe that the first term is anti-Hermitian as it is a pure Magnus term[7] and thus generates unitary, coherent time evolution. Conversely, the second term is Hermitian and thus generates decoherence.[8] The former is more difficult to compute than the latter because the second order of the ME, Equation 16.8b, contains nested time integrals. Arguments can be made [318], however, that for single gates in an experimental context the coherent errors captured by this term can be calibrated out to a large degree [373, 374]. Moreover, many of the central quantities of interest that can be extracted from the quantum process, among which are gate fidelities and certain measurement probabilities, are functions of only the diagonal elements of $\mathcal{K}$. By virtue of the antisymmetry of the second order terms, they do not contribute to these quantities to leading order as we show in Section 16.4.

While we will also lay out how to compute the second order, our discussion will therefore focus on contributions from the incoherent term below. As it turns out, this term can be computed using a filter-function formalism based on that by Green et al. [346]. To see this, we insert the explicit forms of the ME given in Equation 16.8 and the noise Hamiltonian given in Equation 16.3 into Equation 16.12. Together with $[\mathcal{L}, \mathcal{L}'] =$

5: The ensemble average represents the expectation value over identical repetitions of an operation in an experiment. It can be taken to be a spatial ensemble of many identical systems, *e.g.* an NMR system, or, for ergodic systems, a time ensemble of a single system under stationary noise as would be the case for a single spin measured repeatedly, for instance.

6: By *commuting* noise we mean that $[H_{\mathrm{c}}(t), H_{\mathrm{n}}(t)] = 0\,\forall t$. The prototypical example of this is the pure-dephasing Hamiltonian $H(t) = [\Omega(t) + \delta\omega(t)]\,\sigma_z/2$.

7: Remember that the ME preserves algebraic structure to every order.

8: In the Liouville representation, the first term is an antisymmetric matrix that generates a rotation and the second a symmetric matrix that generates a deformation of the generalized, d^2-1-dimensional Bloch sphere.

$[[H, H'], \,\bullet\,]$ and $\mathcal{L}\mathcal{L}' = [H, [H', \,\bullet\,]]$, we find that

$$\begin{aligned}\mathcal{K}(\tau) = -\frac{1}{2}\sum_{\alpha\beta}\Bigg(&\int_0^\tau \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \left\langle b_\alpha(t_1) b_\beta(t_2)\right\rangle \left[\left[\tilde{B}_\alpha(t_1), \tilde{B}_\beta(t_2)\right], \,\bullet\,\right] \\ &+ \int_0^\tau \mathrm{d}t_1 \int_0^{\tau} \mathrm{d}t_2 \left\langle b_\alpha(t_1) b_\beta(t_2)\right\rangle \left[\tilde{B}_\alpha(t_1), \left[\tilde{B}_\beta(t_2), \,\bullet\,\right]\right]\Bigg), \quad (16.13)\end{aligned}$$

where $\tilde{B}_\alpha(t) = U_\mathrm{c}^\dagger(t) B_\alpha(t) U_\mathrm{c}(t)$ are the noise operators of Equation 16.3 in the interaction picture. $\left\langle b_\alpha(t_1) b_\beta(t_2)\right\rangle$ is the cross-correlation function of noise sources α and β which we will later relate to the spectral density. For now, we stay in the time domain and introduce an orthonormal and Hermitian operator basis for the Hilbert space $\mathscr{H}$ to define the Liouville representation,

$$\mathsf{C} = \left\{C_k \in \mathsf{L}(\mathscr{H}) \,:\, C_k^\dagger = C_k \text{ and } \operatorname{tr}(C_k C_l) = \delta_{kl}\right\}_{k=0}^{d^2-1}, \tag{16.14}$$

where we choose $C_0 = d^{-1/2}\,\mathbb{1}$ for convenience so that the remaining elements are traceless. In order to separate the commutators from the time-dependence and hence the integral in Equation 16.13, we expand the noise operators in this basis so that

$$\tilde{B}_\alpha(t) =: \sum_k \tilde{\mathcal{B}}_{\alpha k}(t) C_k. \tag{16.15}$$

The expansion coefficients $\tilde{\mathcal{B}}_{\alpha k}(t) \in \mathbb{R}$ are given by the inner product of a noise operator in the interaction picture on the one hand and a basis element on the other:

$$\tilde{\mathcal{B}}_{\alpha k}(t) = \left\langle \tilde{B}_\alpha(t), C_k\right\rangle = \operatorname{tr}\!\left(U_\mathrm{c}^\dagger(t) B_\alpha(t) U_\mathrm{c}(t) C_k\right). \tag{16.16}$$

In line with Green et al. [346], we call these coefficients the control matrix (see also Refences 375 and 352). In the transfer matrix (superoperator) picture we can take up the following interpretation for the control matrix by virtue of the cyclicity of the trace: it describes a mapping of a state, represented by the basis element C_k and subject to the control operation $\mathcal{U}_\mathrm{c}(t) \,:\, C_k \to U_\mathrm{c}(t) C_k U_\mathrm{c}^\dagger(t)$, onto the noise operator $B_\alpha(t)$. That is, we can write the αth row of the control matrix as $\langle\!\langle \tilde{B}_\alpha(t)| = \langle\!\langle B_\alpha(t)|\, \mathcal{U}_\mathrm{c}(t)$. In this connection lies the power of the FF formalism as will become clear shortly; we can first determine the ideal evolution without noise and subsequently evaluate the error process by linking the unitary control operation to the noise operators.

Having expanded the noise operators in the basis C, we can already anticipate that upon substituting them, Equation 16.13 will separate into a time-dependent part involving on one hand the control matrix and cross-correlation functions and on the other a time-independent part involving commutators of basis elements. This will simplify our calculations in the following. To see this, we recall the definition of the Liouville representation in Equation 16.1 and apply it to the cumulant function so that $\mathcal{K}_{ij} = \operatorname{tr}\!\left(C_i \mathcal{K}[C_j]\right) \in \mathscr{L}$, where the notation $\mathcal{K}[C_j]$ means substituting C_j for the placeholder $\bullet$ in the commutators in Equation 16.13 and we suppressed the time argument for legibility. Finally, we insert the expanded noise operators given by Equation 16.15 and obtain the Liouville representation of the cumulant function,

$$\mathcal{K}_{ij}(\tau) =: -\frac{1}{2}\sum_{\alpha\beta}\sum_{kl}\left(f_{ijkl}\Delta_{\alpha\beta,kl} + g_{ijkl}\Gamma_{\alpha\beta,kl}\right). \tag{16.17}$$

Here, we captured the ordering of the noise operators due to the commutators in Equation 16.13 in the coefficients f_{ijkl} and g_{ijkl}. These are trivial functions of the fourth order trace tensor[9]

$$T_{ijkl} = \mathrm{tr}\left(C_i C_j C_k C_l\right) \tag{16.18}$$

given by

$$f_{ijkl} = T_{klji} - T_{lkji} - T_{klij} + T_{lkij} \quad \text{and} \tag{16.19a}$$

$$g_{ijkl} = T_{klji} - T_{kjli} - T_{kilj} + T_{kijl}. \tag{16.19b}$$

Furthermore, we introduced the frequency (Lamb) shifts Δ and decay amplitudes Γ which contain all information on the noise and qubit dynamics as captured by the control matrix $\tilde{\mathcal{B}}(t)$:

$$\Delta_{\alpha\beta,kl} = \int_0^\tau \mathrm{d}t_1 \int_0^{t_1} \mathrm{d}t_2 \left\langle b_\alpha(t_1) b_\beta(t_2)\right\rangle \tilde{\mathcal{B}}_{\alpha k}(t_1)\tilde{\mathcal{B}}_{\beta l}(t_2), \tag{16.20}$$

$$\Gamma_{\alpha\beta,kl} = \int_0^\tau \mathrm{d}t_1 \int_0^{\tau} \mathrm{d}t_2 \left\langle b_\alpha(t_1) b_\beta(t_2)\right\rangle \tilde{\mathcal{B}}_{\alpha k}(t_1)\tilde{\mathcal{B}}_{\beta l}(t_2). \tag{16.21}$$

The frequency shifts Δ correspond to the first term in Equation 16.12, hence incurring coherent errors, *i. e.* generalized axis and overrotation errors. They reflect a perturbative correction to the quantum evolution due to a change of the Hamiltonian at two points in time, and thus time ordering matters. Conversely, the decay amplitudes Γ correspond to the second term and capture the decoherence. These terms are due to an incoherent average that only takes classical correlations into account, so that time ordering does not play a role. Note that Equation 4 from Reference 318 is obtained from Equations 16.9 and 16.17 by expanding the exponential to linear order and neglecting the second order terms Δ.

For a single qubit and C the Pauli basis one can make use of the simple commutation relations so that the cumulant function takes the form (see Subsection C.1.1)

$$\mathcal{K}_{ij}(\tau) = \begin{cases} -\sum\limits_{k\neq i}\Gamma_{kk} & \text{if} \quad i = j, \\ -\Delta_{ij} + \Delta_{ji} + \Gamma_{ij} & \text{if} \quad i \neq j, \end{cases} \tag{16.22}$$

for $i, j > 0$ and any α, β. As mentioned in Section 16.1 the cases $j = 0$ and $i = 0$ encode trace-preservation and unitality, respectively, and as such $\mathcal{K}_{0j} = \mathcal{K}_{i0} = 0$ since our model is both trace-preserving and unital.[10]

9: Note the similarity to the relationship of a transfer matrix with the χ–matrix, $\mathcal{E}_{ij} = \sum_{kl}\chi_{kl}T_{ikjl}$, with χ_{kl} defined by $\mathcal{E}(\rho) = \sum_{kl}\chi_{kl}C_k\rho C_l$ or, in terms of the Kraus operators K_i of the quantum operation, $\chi_{kl} = \sum_i \mathrm{tr}(K_i C_k)\mathrm{tr}(K_i^\dagger C_l) = \left[\sum_i |K_i\rangle\!\rangle\langle\!\langle K_i|\right]_{kl}$ [365].

10: Unital quantum operations map the identity to the identity (*i. e.*, the identity is a fixed point of $\mathcal{E}$). To violate unitality, the quantum system on $\mathscr{H}$ needs to exchange energy with another system $\mathscr{H}'$, *e. g.* T_1-relaxation, a process which cannot be described by unitary dynamics on $\mathscr{H}$ alone. Because we average over strictly unitary operations on $\mathscr{H}$ to obtain $\langle\tilde{\mathcal{U}}\rangle$, our model cannot capture such processes. The same line of reasoning holds for trace preservation.

16.2 Calculating the decay amplitudes

In order to evaluate the cumulant function $\mathcal{K}(\tau)$ given by Equation 16.17 and thus the transfer matrix $\langle\tilde{\mathcal{U}}(\tau)\rangle$ from Equation 16.9 for a given control operation, we solely require the decay amplitudes Γ_{kl} and frequency shifts Δ_{kl} since the trace tensor T_{ijkl} depends only on the choice of basis and is therefore trivial (although quite costly for large dimensions, *cf.* Section 16.8) to calculate. In this section, we describe simple methods for calculating Γ_{kl} using an extension of the filter-function formalism developed by Green et al. [346] that we introduced in Reference 318. The central quantity of interest will be the control matrix that we already introduced above. It relates the interaction picture noise operators to the operator basis and we will compute it in Fourier space in order to identify the cross-correlation functions with the noise spectral density in Equation 16.21. We distinguish between a sequence of quantum gates, as already presented in Reference 318, and a single gate. In the first case the control matrix of the entire sequence can be calculated from those of the individual gates, greatly simplifying the calculation if the latter have been precomputed. This approach gives rise to correlation terms in the expression for Γ_{kl} that capture the effects of sequencing gates. In the second case, as was shown by Green et al. [346], one can calculate the control matrix for arbitrary single pulses under the assumption of piecewise-constant control and we lay out how to adapt the approach for numerical applications.

We start by noting that, because we assumed the noise fields $b_\alpha(t)$ to be wide-sense stationary, that is to say the cross-correlation functions evaluated at two different points in time t_1 and t_2 depend only on their difference t_1-t_2, we can define the two-sided noise power spectral density (PSD) $S_{\alpha\beta}(\omega)$ as the Fourier transform of the cross-correlation functions $\langle b_\alpha(t_1)b_\beta(t_2)\rangle$ (see also Equation 2.11),

$$\left\langle b_\alpha(t_1)b_\beta(t_2)\right\rangle = \int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi}S_{\alpha\beta}(\omega)e^{-i\omega(t_1-t_2)}. \tag{16.23}$$

Note that the spectrum only characterizes the noise fully in the case of Gaussian noise. For non-Gaussian components in the noise, additional polyspectra have in principle to be considered for higher-order correlation functions [33]. However, since we only discuss second-order contributions which involve two-point correlation functions here, we only need to take $S_{\alpha\beta}(\omega)$ into account. Inserting the definition of the spectral density into Equation 16.21, one finds

$$\Gamma_{\alpha\beta,kl} = \int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi}\tilde{\mathcal{B}}^*_{\alpha k}(\omega)S_{\alpha\beta}(\omega)\tilde{\mathcal{B}}_{\beta l}(\omega) \tag{16.24}$$

with $\tilde{\mathcal{B}}(\omega) = \int_0^\tau \mathrm{d}t\,\tilde{\mathcal{B}}(t)e^{i\omega t}$ the frequency-domain control matrix. Note that $\tilde{\mathcal{B}}^*(\omega) = \tilde{\mathcal{B}}(-\omega)$ because $\tilde{\mathcal{B}}(t)$ is real due to our choice of a Hermitian basis C. In the above equation, the fourth-order tensor[11]

11: Of course, we can also interpret $\mathcal{F}_{\alpha\beta}(\omega) \in \mathscr{L}$ as a linear operator (matrix) in Liouville space, and similarly $\Gamma_{\alpha\beta} \in \mathscr{L}$.

$$\mathcal{F}_{\alpha\beta,kl}(\omega) \coloneqq \tilde{\mathcal{B}}^*_{\alpha k}(\omega)\tilde{\mathcal{B}}_{\beta l}(\omega) \tag{16.25}$$

is the generalized filter function that captures the susceptibility of the decay amplitudes to noise at frequency ω. For $\alpha=\beta, k=l$, and by summing over the basis elements,

$$\mathcal{F}_\alpha(\omega) = \sum_k\left|\tilde{\mathcal{B}}_{\alpha k}(\omega)\right|^2 = \mathrm{tr}\left(\tilde{B}^\dagger_\alpha(\omega)\tilde{B}_\alpha(\omega)\right), \tag{16.26}$$

and this tensor reduces to the canonical *fidelity* filter function [345] from which the entanglement fidelity can be obtained, see Subsection 16.4.1. Thus, if the frequency-domain control matrix $\tilde{\mathcal{B}}_{\alpha k}(\omega)$ for noise source α and basis element k is known, the transfer matrix can be evaluated by integrating Equation 16.24. Moreover, one can study the contributions of each pair of noise sources (α,β) both separately or, at virtually no additional cost and to leading order, collectively by summing over them, $\Gamma_{kl} = \sum_{\alpha\beta}\Gamma_{\alpha\beta,kl}$.

We now discuss how to calculate the control matrix $\tilde{\mathcal{B}}(\omega)$ in frequency space for a given control operation. We focus first on sequences of quantum gates, assuming that the control matrices $\tilde{\mathcal{B}}^{(g)}(\omega)$ for each gate g

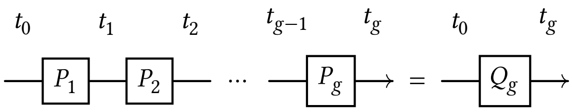

Figure 16.1: Illustration of a sequence of G gates. Individual gates with propagators P_g start at time t_{g-1} and complete at time t_g. The total action from t_0 to t_g is given by Q_g.

have been calculated before.

16.2.1 Control matrix of a gate sequence

For a sequence of gates with precomputed interaction picture noise operators the approach developed by Green et al. [346] based on piecewise-constant control can be adapted to yield an analytical expression for those of the composite gate sequence that is computationally efficient to evaluate [318]. Here we review these results to give a complete picture of the formalism. While our results are general and apply to any superoperator representation, we employ the Liouville representation here for its simple composition operation: matrix multiplication. Computationally, this is not the most efficient choice since transfer matrices have dimension $d^2 \times d^2$ and thus their matrix multiplication scales unfavorably compared to, for example, left-right conjugation by unitaries (*cf.* Section 16.9). However, because the structure of the control matrix $\tilde{\mathcal{B}}$ is similar to that of a transfer matrix,[12] we will obtain a particularly concise expression for the sequence in the following. For a perhaps more intuitive description employing exclusively conjugation by unitaries, we refer the reader to Reference 318.

12: Remember that it corresponds to a basis expansion of the interaction picture noise operators

We consider a sequence of G gates with propagators $P_g = U_\mathrm{c}(t_g, t_{g-1}), g \in \{1, \dots, G\}$ that act during the gth time interval $(t_{g-1}, t_g]$ with $t_0 = 0, t_G = \tau$ as illustrated in Figure 16.1. The cumulative propagator of the sequence up to time t_g is then given by $Q_g = \prod_{g'=g}^{0} P_{g'}$ with $P_0 = \mathbb{1}$ and its Liouville representation denoted by $\mathcal{Q}^{(g)}$. Furthermore, the control matrix of the gth pulse at the time $t - t_{g-1}$ relative to the start of segment g is

$$\tilde{\mathcal{B}}^{(g)}_{\alpha k}(t - t_{g-1}) = \mathrm{tr}\Big(U_\mathrm{c}^\dagger(t, t_{g-1}) B_\alpha(t - t_{g-1}) U_\mathrm{c}(t, t_{g-1}) C_k\Big). \tag{16.27}$$

We can now exploit the fact that in the transfer matrix picture quantum operations compose by matrix multiplication to write the total control matrix at time $t \in (t_{g-1}, t_g]$ as

$$\tilde{\mathcal{B}}(t) = \tilde{\mathcal{B}}^{(g)}(t - t_{g-1}) \mathcal{Q}^{(g-1)} \tag{16.28}$$

since $\mathcal{U}_\mathrm{c}(t, t_0) = \mathcal{U}_\mathrm{c}(t, t_{g-1}) \mathcal{Q}^{(g-1)}$ and we have that $\langle\!\langle \tilde{B}(t) | = \langle\!\langle B(t) | \, \mathcal{U}_\mathrm{c}(t, t_0) = \langle\!\langle B(t - t_{g-1}) | \, \mathcal{Q}^{(g-1)}$. The Fourier transform of Equation 16.28 can then be obtained by evaluating the transform of each gate separately,

$$\tilde{\mathcal{B}}(\omega) = \sum_{g=1}^{G} e^{i\omega t_{g-1}} \tilde{\mathcal{B}}^{(g)}(\omega) \mathcal{Q}^{(g-1)}, \tag{16.29}$$

$$\tilde{\mathcal{B}}^{(g)}(\omega) = \int_0^{\Delta t_g} \mathrm{d}t \, e^{i\omega t} \tilde{\mathcal{B}}^{(g)}(t), \tag{16.30}$$

with $\Delta t_g = t_g - t_{g-1}$ the duration of gate g. Hence, calculating the control matrix of the full sequence requires only the knowledge of the temporal positions, encoded in the phase factors $e^{i\omega t_{g-1}}$, and the total intended action $\mathcal{Q}^{(g-1)}$ of the individual pulses if their control matrices have been precomputed or obtained otherwise.[13] The sequence structure can thus be exploited to one's benefit. If the same gates appear multiple times during the sequence one can reuse control matrices for equal pulses to facilitate calculating filter functions for complex sequences with modest computational effort. Most importantly, Equation 16.29 is independent of the inner structure of the individual pulses and therefore takes the same time to evaluate whether they are highly complex or very simple. In Section 16.9, we will analyze the computational efficiency of capitalizing on this feature in more detail.

13: In particular, one can make use of analytical expressions for the control matrix.

As we have seen, the total control matrix of a composite pulse sequence is given by a weighted sum over the individual control matrices. Since $\tilde{\mathcal{B}}(\omega)$ enters Equation 16.24 twice, this leads to correlation terms between two gates at different positions in the sequence when computing the total decay amplitudes Γ. Inserting Equation 16.29 into Equation 16.24 gives

$$\begin{aligned}\Gamma_{\alpha\beta,kl} &= \sum_{g,g'=1}^{G}\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi}\left[\mathcal{Q}^{(g'-1)\dagger}\tilde{\mathcal{B}}^{(g')\dagger}(\omega)\right]_{k\alpha}S_{\alpha\beta}(\omega)\left[\tilde{\mathcal{B}}^{(g)}(\omega)\mathcal{Q}^{(g-1)}\right]_{\beta l} \\ &\qquad\times e^{i\omega(t_{g-1}-t_{g'-1})} \\ &=: \sum_{g,g'=1}^{G}\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi}S_{\alpha\beta}(\omega)\mathcal{F}_{\alpha\beta,kl}^{(gg')}(\omega) &(16.31) \\ &=: \sum_{g,g'=1}^{G}\Gamma_{\alpha\beta,kl}^{(gg')} &(16.32)\end{aligned}$$

where we defined the pulse correlation filter function $\mathcal{F}^{(gg')}(\omega)$ that captures the temporal correlations between pulses at different positions g and g' in the sequence as well as the corresponding decay amplitudes $\Gamma^{(gg')}$. Unlike regular filter functions, these can be negative for $k=l, g\neq g'$ and therefore reduce the overall noise susceptibility of a sequence given by $\mathcal{F}(\omega)=\sum_{gg'}\mathcal{F}^{(gg')}(\omega)$. By expanding the exponential in Equation 16.9 to linear order, the individual contributions of the pulse correlation FFs can also be traced through to derived quantities such as the infidelity [318, Equation 11]. We have thus gained a concise description of the noise-cancelling properties of gate sequences: in this picture, they arise purely from the concatenation of different pulses, quantifying, for instance, the effectiveness of DD sequences [318].

16.2.2 Control matrix of a single gate

Previous efforts have derived the control matrix analytically for selected pulses such as DD sequences [339], special dynamically corrected gates (DCGs) [347], as well as developed a general analytical framework [345, 346]. However, analytical solutions might not always be accessible, *e.g.* for numerically optimized pulses, and are generally laborious to obtain. Therefore, we now detail a method to obtain the control matrix numerically under the assumption of piecewise-constant control. Our method is similar in spirit to that of Green, Uys, and Biercuk [345] for single qubits with $d=2$, but whereas those authors computed analytical solutions to the relevant integrals during each time step, here we use matrix diagonalization to obtain the propagator of a control operation to make the approach amenable to numerical implementation. This allows carrying out the Fourier transform of the control matrix Equation 16.16 analytically by writing the control propagators in terms of their eigenvalues in diagonal form.

We divide the total duration of the control operation, τ, into G intervals $(t_{g-1}, t_g]$ of duration Δt_g with $g \in \{0, \dots, G\}$ and $t_0 = 0, t_G = \tau$. We then approximate the control Hamiltonian as constant within each interval so that within the g

$$H_{\mathrm{c}}(t) = H_{\mathrm{c}}^{(g)} = \text{const.} \tag{16.33}$$

and similarly the deterministic time dependence of the noise operators as $B_\alpha(t) = s_\alpha(t) B_\alpha = s_\alpha^{(g)} B_\alpha$. Under this approximation we can diagonalize the time-independent Hamiltonians $H_{\mathrm{c}}^{(g)}$ with eigenvalues $\omega_i^{(g)}$ numerically and write the time evolution operator that solves the noise-free Schrödinger equation as $U_{\mathrm{c}}(t, t_{g-1}) = V^{(g)} D^{(g)}(t, t_{g-1}) V^{(g)\dagger}$. Here, $V^{(g)}$ is the unitary matrix of eigenvectors of $H_{\mathrm{c}}^{(g)}$ and the diagonal matrix $D_{ij}^{(g)}(t, t_{g-1}) = \delta_{ij} \exp\{-i\omega_i^{(g)}(t - t_{g-1})\}$ contains the time evolution of the eigenvalues. Using this result together with Q_{g-1}, the cumulative propagator up to time t_{g-1}, we can acquire the total time evolution operator at time t from $U_{\mathrm{c}}(t) = U_{\mathrm{c}}(t, 0) = U_{\mathrm{c}}(t, t_{g-1}) Q_{g-1}$. We then substitute this relation into the definition of the control matrix, Equation 16.16, and obtain

$$\begin{aligned} \tilde{\mathcal{B}}_{\alpha k}(t) &= s_\alpha^{(g)} \operatorname{tr}\Big(Q_{g-1}^\dagger V^{(g)} D^{(g)\dagger}(t, t_{g-1}) V^{(g)\dagger} B_\alpha \\ &\qquad \times V^{(g)} D^{(g)}(t, t_{g-1}) V^{(g)\dagger} Q_{g-1} C_k\Big) && (16.34) \\ &=: s_\alpha^{(g)} \sum_{ij} \bar{B}_{\alpha,ij}^{(g)} \bar{C}_{k,ji}^{(g)} e^{i\Omega_{ij}^{(g)}(t - t_{g-1})}, && (16.35) \end{aligned}$$

where we defined $\Omega_{ij}^{(g)} = \omega_i^{(g)} - \omega_j^{(g)}$, $\bar{C}_k^{(g)} = V^{(g)\dagger} Q_{g-1} C_k Q_{g-1}^\dagger V^{(g)}$, and $\bar{B}_\alpha^{(g)} = V^{(g)\dagger} B_\alpha V^{(g)}$. Carrying out the Fourier transform of Equation 16.35 to get the frequency-domain control matrix of the pulse generated by the Hamiltonian from Equation 16.33 is now straightforward since the integrals involved are over simple exponential functions. We obtain

$$\tilde{\mathcal{B}}_{\alpha k}(\omega) = \sum_{g=1}^{G} s_\alpha^{(g)} e^{i\omega t_{g-1}} \operatorname{tr}\Big(\big[\bar{B}_\alpha^{(g)} \circ I^{(g)}(\omega)\big] \bar{C}_k^{(g)}\Big) \tag{16.36}$$

with $I_{ij}^{(g)}(\omega) = -i(e^{i(\omega + \Omega_{ij}^{(g)})\Delta t_g} - 1)/(\omega + \Omega_{ij}^{(g)})$ and the Hadamard product $(A \circ B)_{ij} := A_{ij} \times B_{ij}$. Equation 16.36 is readily evaluated on a computer and thus enables the calculation of filter functions of arbitrary control sequences, either on its own or in conjunction with Equation 16.29. A similar expression is obtained for representations other than the Liouville representation.

16.3 Calculating the frequency shifts

The frequency shifts $\Delta_{\alpha\beta,kl}$ in Equation 16.17 correspond to the second order of the ME and thus involve a double integral with a nested time dependence. This makes their evaluation more involved than that of the decay amplitudes $\Gamma_{\alpha\beta,kl}$ and, in contrast to the previous section, we cannot single out correlation terms as in Equation 16.31. Moreover, since the integrals do not factorize we also cannot derive a concatenation rule that is entirely independent of the internal structure of the constituent gates as before. However, we can still apply the approximation of piecewise-constant control and follow a similar approach as in Subsection 16.2.2

to compute Δ in Fourier space. In Section C.3, we also lay out a concatenation rule for the second-order terms that contains only a minimal contribution from the internal dynamics of the constituent gates for completeness. Since these terms correspond to a coherent gate error that can in principle be calibrated out to leading order in experiments we will not go into much detail here.

We follow the arguments made above for the decay amplitudes and express the cross-correlation functions $\langle b_\alpha(t) b_\beta(t')\rangle$ by their Fourier transform, the spectral density $S_{\alpha\beta}(\omega)$, using Equation 16.23. Inserting this equation into the definition of the frequency shifts in the time domain, Equation 16.20, yields

$$\Delta_{\alpha\beta,kl} = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S_{\alpha\beta}(\omega) \int_0^\tau \mathrm{d}t\, \tilde{\mathcal{B}}_{\alpha k}(t) e^{-i\omega t} \int_0^t \mathrm{d}t'\, \tilde{\mathcal{B}}_{\beta l}(t') e^{i\omega t'}. \tag{16.37}$$

We again assume piecewise-constant time segments so that the inner time integral can be split up into a sum of integrals over complete constant segments $(t_{g'-1}, t_{g'}]$ as well as a single integral that contains the last, incomplete segment up to time t. That is, taking the time t of the outer integral to be within the interval $(t_{g-1}, t_g]$ we perform the replacement

$$\int_0^t \mathrm{d}t' \rightarrow \sum_{g'=1}^{g-1} \int_{t_{g'-1}}^{t_{g'}} \mathrm{d}t' + \int_{t_{g-1}}^{t} \mathrm{d}t'\,. \tag{16.38}$$

We have thus divided our task into two: The first term allows, as before in Subsections 16.2.1 and 16.2.2, to identify the Fourier transform of the control matrix during time steps g' and g for both the inner and the outer integral according to Equation 16.36. The second term remains a nested double integral, but now the integrand contains only products of complex exponentials because we assume the control to be constant within the limits of integration. As a next step, we also replace the outer time integral by a sum of integrals over single segments, $\int_0^\tau \mathrm{d}t \rightarrow \sum_{g=1}^{G} \int_{t_{g-1}}^{t_g} \mathrm{d}t$, to obtain

$$\begin{aligned}\Delta_{\alpha\beta,kl} = &\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S_{\alpha\beta}(\omega) \sum_{g=1}^{G} \int_{t_{g-1}}^{t_g} \mathrm{d}t \\ &\times e^{-i\omega t} \tilde{\mathcal{B}}_{\alpha k}(t) \left\{ \sum_{g'=1}^{g-1} \int_{t_{g'-1}}^{t_{g'}} \mathrm{d}t' + \int_{t_{g-1}}^{t} \mathrm{d}t' \right\} e^{i\omega t'} \tilde{\mathcal{B}}_{\beta l}(t').\end{aligned} \tag{16.39}$$

Before continuing, we ease notation and define $\tilde{\mathcal{B}}(\omega) =: \sum_g \mathcal{G}^{(g)}(\omega)$ with $\mathcal{G}^{(g)}(\omega)$ obtained from Equation 16.36 and furthermore adopt the Einstein summation convention for the remainder of this section, meaning multiple subscript indices that appear on only one side of an equality are summed over implicitly. We now proceed like in Subsection 16.2.2 and make use of the piecewise-constant approximation to diagonalize the control Hamiltonian during each segment. For the nested integrals, we obtain $\tilde{\mathcal{B}}_{\alpha k}(t)$ from Equation 16.35, whereas the remaining integrals factorize and we can identify the Fourier-transformed quantity $\mathcal{G}^{(g)}(\omega)$.

Equation 16.39 then becomes

$$\Delta_{\alpha\beta,kl} = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S_{\alpha\beta}(\omega) \sum_{g=1}^{G} \Bigg[\mathcal{G}_{\alpha k}^{(g)*}(\omega) \sum_{g'=1}^{g-1} \mathcal{G}_{\beta l}^{(g')}(\omega) + s_\alpha^{(g)} \bar{B}_{\alpha,ij}^{(g)} \bar{C}_{k,ji}^{(g)} I_{ijmn}^{(g)}(\omega) \bar{C}_{l,nm}^{(g)} \bar{B}_{\beta,mn}^{(g)} s_\beta^{(g)} \Bigg] \tag{16.40}$$

with $\bar{B}_{\alpha,ij}^{(g)}, \bar{C}_{k,ij}^{(g)}, \Omega_{ij}^{(g)}$ as defined above in Subsection 16.2.2 and

$$I_{ijmn}^{(g)}(\omega) = \int_{t_{g-1}}^{t_g} \mathrm{d}t\, e^{i\Omega_{ij}^{(g)}(t-t_{g-1})-i\omega t} \int_{t_{g-1}}^{t} \mathrm{d}t'\, e^{i\Omega_{mn}^{(g)}(t'-t_{g-1})+i\omega t'}. \tag{16.41}$$

Explicit results for the integration in Equation 16.41 are given in Subsection C.1.2. To calculate the frequency shifts Δ, we can thus reuse the quantity $\mathcal{G}^{(g)}(\omega)$ also required for the decay amplitudes Γ. The only additional computation, apart from contraction, involves the G integrations $I_{ijmn}^{(g)}(\omega)$. Importantly, Equation 16.40 has the same structure as the corresponding Equation 16.24 for Γ in that the individual entries of Δ are given by an integral over the spectral density of the noise multiplied with a – in this case second-order – filter function that describes the susceptibility to noise at frequency ω:

$$\Delta_{\alpha\beta,kl} = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S_{\alpha\beta}(\omega) \mathcal{F}_{\alpha\beta,kl}^{(2)}(\omega). \tag{16.42}$$

16.4 Computing derived quantities

By means of Equations 16.29, 16.36 and 16.40, one can obtain the cumulant function $\mathcal{K}(\tau)$ from Equation 16.17 and hence the error process $\langle\tilde{\mathcal{U}}(\tau)\rangle$ from Equation 16.9 for an arbitrary sequence of gates. From this, several quantities of interest for the characterization of a given control operation can be extracted. We explicitly review the average gate and state fidelities as well as expressions to quantify leakage here, but emphasize that this is not exhaustive. Because for many applications the noise is weak and hence the parameter $\xi \ll 1$, we will in the following expand the exponential in Equation 16.9 to leading order in ξ. That is, we approximate[14]

$$\langle\tilde{\mathcal{U}}(\tau)\rangle \approx \mathbb{1} + \mathcal{K}(\tau). \tag{16.43}$$

14: Remember that $\|\mathcal{K}(\tau)\| \in \mathcal{O}(\xi^2)$, *i. e.*, second order in the noise strength.

Higher order corrections can be obtained either by explicitly calculating higher powers of $\mathcal{K}$ or by numerically evaluating the exponential of the cumulant function (potentially including higher-order terms from ME and cumulant expansions). The former method often leads to simpler expressions than Equation 16.17 for which the trace tensor T_{ijkl} need not be computed directly. In the weak-noise regime, one can also define specific filter functions for each derived quantity that are given in terms of linear combinations of the generalized filter functions $\mathcal{F}_{\alpha\beta,kl}(\omega)$. The ensemble expectation value of the quantity can then be obtained directly from the overlap with the spectral density, $\int \mathrm{d}\omega/2\pi\, \mathcal{F}(\omega)S(\omega)$. Finally, we will drop the averaging brackets and the argument of the error transfer matrix $\langle\tilde{\mathcal{U}}(\tau)\rangle$ for brevity in the following.

16.4.1 Average gate and entanglement fidelity

The average gate fidelity is a commonly-quoted figure of merit used to characterize physical gate implementations [326, 329, 376–378]. It represents the fidelity between an implementation $\mathcal{U}$ and the ideal gate $\mathcal{Q}$ averaged over the uniform Haar measure. Since $F_{\mathrm{avg}}(\mathcal{U}, \mathcal{Q}) = F_{\mathrm{avg}}(\mathcal{Q}^\dagger \circ \mathcal{U}, \mathbb{1}) = F_{\mathrm{avg}}(\tilde{\mathcal{U}})$, the average gate fidelity can be obtained from the error channel $\tilde{\mathcal{U}}$ as [379, 380]

$$F_{\mathrm{avg}}(\tilde{\mathcal{U}}) = \frac{\operatorname{tr}\tilde{\mathcal{U}} + d}{d(d+1)} \tag{16.44}$$

$$=: \frac{d \times F_{\mathrm{ent}}(\tilde{\mathcal{U}}) + 1}{d+1}, \tag{16.45}$$

where d is the system dimension and $F_{\mathrm{ent}}(\tilde{\mathcal{U}}) = d^{-2}\operatorname{tr}\tilde{\mathcal{U}}$ is the entanglement fidelity. In the low-noise regime where Equation 16.43 holds, we can write the entanglement fidelity in terms of the cumulant function $\mathcal{K}_{\alpha\beta}$ approximately as

$$F_{\mathrm{ent}}(\tilde{\mathcal{U}}) = 1 + \frac{1}{d^2}\sum_{\alpha\beta}\operatorname{tr}\mathcal{K}_{\alpha\beta} \tag{16.46}$$

$$=: 1 - \sum_{\alpha\beta} I_{\alpha\beta}(\tilde{\mathcal{U}}). \tag{16.47}$$

Here, we defined $I_{\alpha\beta}$, the infidelity due to a pair of noise sources (α, β). As we show in Subsection C.1.3, we can simplify the trace of the cumulant function so that the infidelity reads

$$I_{\alpha\beta} = \frac{1}{d}\operatorname{tr}\Gamma_{\alpha\beta}. \tag{16.48}$$

Equation 16.48 reduces to Equation 32 from Reference 345 for a single qubit ($d = 2$) and pure dephasing noise up to a different normalization convention; by pulling the trace through to the generalized filter function $\mathcal{F}_{\alpha\beta,kl}(\omega)$ in Equation 16.24, we recover the relation (setting $\alpha = \beta$ for simplicity)

$$I_\alpha = \frac{1}{d}\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{2\pi} S_\alpha(\omega)\mathcal{F}_\alpha(\omega) \tag{16.49}$$

with the fidelity filter function $\mathcal{F}_\alpha(\omega)$ given by Equation 16.26. Notably, only the decay amplitudes Γ contribute to the fidelity to leading order since the frequency shifts Δ are antisymmetric and therefore vanish under the trace.

16.4.2 State fidelity and measurements

In the context of quantum information processing we are often interested in the probability of measuring the expected state during readout. We can extract this projective readout probability from the transfer matrix in Equation 16.7 by inspecting the transition probability, or state fidelity, between a pure state $\rho = |\psi\rangle\langle\psi|$ and an arbitrary state σ that evolves according to the quantum operation $\mathcal{E} : \sigma \to \mathcal{E}(\sigma)$. Using the double braket notation introduced at the beginning of Chapter 16 we then define the state fidelity as

$$F(|\psi\rangle, \mathcal{E}(\sigma)) = \operatorname{tr}(\rho\mathcal{E}(\sigma)) = \langle\!\langle\rho|\mathcal{E}(\sigma)\rangle\!\rangle = \langle\!\langle\rho|\mathcal{E}|\sigma\rangle\!\rangle\,, \tag{16.50}$$

where we have expressed the density matrices by vectors and $\mathcal{E}$ as a transfer matrix on the Liouville space $\mathscr{L}$. We can thus calculate arbitrary pure state fidelities by simple matrix-vector multiplications of the transfer matrices $\mathcal{E} = \mathcal{Q}\tilde{\mathcal{U}}$ and the vectorized density matrices $|\rho\rangle\!\rangle$ and $|\sigma\rangle\!\rangle$. In Section 19.3 we employ this measure to simulate a RB experiment where return probabilities are of interest so that $F(|\psi\rangle, \mathcal{E}(\rho)) = \langle\!\langle\rho|\, \mathcal{Q}\tilde{\mathcal{U}}\, |\rho\rangle\!\rangle$.

General measurements can be incorporated in the superoperator formalism we have employed here in a straightforward manner using the positive operator-valued measure (POVM) formalism [365, 381]. POVMs constitute a set of Hermitian, positive semidefinite operators $\{E_i\}_i$ (in contrast to the projective measurement $\{|\psi\rangle\langle\psi|, \mathbb{1} - |\psi\rangle\langle\psi|\}$) that fulfill the completeness relation $\sum_i E_i = \mathbb{1}$ and in the double braket notation may be represented as the row vectors $\{\langle\!\langle E_i|\}_i$ in Liouville space. Consequently, the measurement probability for outcome E_i is given by $\langle\!\langle E_i|\mathcal{E}(\sigma)\rangle\!\rangle = \langle\!\langle E_i|\mathcal{E}|\sigma\rangle\!\rangle$ if the system was prepared in the state σ and evolved according to $\mathcal{E}$.

16.4.3 Leakage

In many physical implementations qubits are not encoded in real two-level systems but in two levels of a larger Hilbert space (superconducting transmon [382] or singlet-triplet [383] spin qubits for example) such that population can leak between this computational subspace and other energy levels. Thus, it is often of interest to quantify leakage when assessing gate performance. Recently, Wood and Gambetta [384] have suggested two separate measures for quantifying leakage out of the computational subspace on the one hand and seepage into the subspace on the other. With the filter-function formalism and the transfer matrix of the error process given by Equations 16.7 and 16.17, we can easily extract these quantities.

Using the definitions from Reference 384 and the double braket notation we can write the leakage rate generated by a quantum operation $\mathcal{E}$ as

$$L_c(\mathcal{E}) \coloneqq \frac{1}{d_c} \langle\!\langle\Pi_\ell|\, \mathcal{E}\, |\Pi_c\rangle\!\rangle \tag{16.51a}$$

and the seepage rate as

$$L_\ell(\mathcal{E}) \coloneqq \frac{1}{d_\ell} \langle\!\langle\Pi_c|\, \mathcal{E}\, |\Pi_\ell\rangle\!\rangle\,. \tag{16.51b}$$

Here, $\Pi_{c,\ell}$ are projectors onto the computational and leakage subspaces, respectively, and $d_{c,\ell}$ the corresponding dimensions. For unital channels the leakage and seepage rates are not independent but satisfy $d_c L_c = d_\ell L_\ell$ [384] so that we only need to consider one of the above expressions here (*cf.* Subsection 16.1.2).

Equations 16.51a and 16.51b can be used to determine both coherent and incoherent leakage separately by substituting $\mathcal{Q}$ or $\tilde{\mathcal{U}}$, respectively, for $\mathcal{E}$. While the former is due to systematic errors of the applied pulse and could thus be corrected for by calibration, the latter is induced by noise only. Alternatively, the leakage from both contributions can also be determined collectively by substituting $\mathcal{U}$ for $\mathcal{E}$.

16.5 Performance analysis and efficiency improvements

In this section, we focus on computational aspects of the formalism, remarking first on several mathematical simplifications that make the calculation of control matrices and decay amplitudes more economical. Following this, we investigate the computational complexity of the method and compare it with that of MC techniques, showing that our software implementation surpasses the latter's performance over a broad range of parameter regimes.

16.6 Periodic Hamiltonians

If the control Hamiltonian is periodic, that is $H_\mathrm{c}(t) = H_\mathrm{c}(t+T)$, we can reduce the computational effort of calculating the control matrix by potentially orders of magnitude (see Section 19.2 for an application in Rabi driving). We start by making the following observations: First, the frequency-domain control matrix of every period of the control is the same so that $\tilde{\mathcal{B}}^{(g)}(\omega) = \tilde{\mathcal{B}}^{(1)}(\omega)$. Moreover, $e^{i\omega\Delta t_g} = e^{i\omega T}$ for all g so that $e^{i\omega t_{g-1}} = e^{i\omega T(g-1)}$ and by the composition property of transfer matrices $\mathcal{Q}^{(g-1)} = [\mathcal{Q}^{(1)}]^{g-1}$ where the superscript without parentheses denotes matrix power. We can then simplify Equation 16.29 to read

$$\tilde{\mathcal{B}}(\omega) = \tilde{\mathcal{B}}^{(1)}(\omega) \sum_{g=0}^{G-1} \left[e^{i\omega T} \mathcal{Q}^{(1)}\right]^g . \tag{16.52}$$

Furthermore, if the matrix $\mathbb{1} - e^{i\omega T}\mathcal{Q}^{(1)}$ is invertible, which is typically the case for the vast majority of values of ω, the previous expression can be rewritten as

$$\tilde{\mathcal{B}}(\omega) = \tilde{\mathcal{B}}^{(1)}(\omega)\Big(\mathbb{1} - e^{i\omega T}\mathcal{Q}^{(1)}\Big)^{-1}\Big(\mathbb{1} - \left[e^{i\omega T}\mathcal{Q}^{(1)}\right]^G\Big) \tag{16.53}$$

by evaluating the sum as a finite Neumann series. Equation 16.53 offers a significant performance benefit over regular concatenation in the case of many periods G as we will show in Section 16.9. Beyond numerical advantages, it also provides an analytical method for studying filter functions of periodic driving Hamiltonians.

16.7 Extending Hilbert spaces

Examining Equation 16.16, we can see that the columns of the control matrix and therefore also the filter function are invariant (up to normalization) under an extension of the Hilbert space. This allows parallelizing pulses with precomputed control matrices in a very resource-efficient manner if one chooses a suitable operator basis. Note that the same also applies to other representations of quantum operations.

Suppose we extend the Hilbert space $\mathscr{H}_1$ of a gate for which we have already computed the control matrix by a second Hilbert space $\mathscr{H}_2$ so that $\mathscr{H}_{12} = \mathscr{H}_1 \otimes \mathscr{H}_2$. If we can find an operator basis whose elements separate into tensor products themselves, *i.e.* $\mathsf{C}_{12} = \mathsf{C}_1 \otimes \mathsf{C}_2$ as for the Pauli basis (*cf.* Section 16.8), the control matrix of the composite gate defined on $\mathscr{H}_{12}$ has the same non-trivial columns as that of the original gate on $\mathscr{H}_1$ up to a different normalization factor. The remaining columns are simply zero. This is because the trace over a tensor product factors into traces over the individual subsystems so that $\tilde{\mathcal{B}}_{\alpha k}(t) \propto \operatorname{tr}\bigl([U_1^\dagger \otimes U_2^\dagger][B_\alpha \otimes \mathbb{1}][U_1 \otimes U_2][\mathbb{1} \otimes C_k]\bigr) = \operatorname{tr}\bigl(U_1^\dagger B_\alpha U_1 \mathbb{1}\bigr)\operatorname{tr}\bigl(U_2^\dagger \mathbb{1} U_2 C_k\bigr) = 0$ since we assumed that the noise operators B_α are traceless (*cf.* Subsection 16.1.2).

Generalizing this result to multiple originally disjoint Hilbert spaces we write the composite space as $\mathcal{H} = \bigotimes_i \mathcal{H}_i$ and the corresponding basis as $\mathsf{C} = \bigotimes_i \mathsf{C}_i$. The control matrix of the composite pulse on $\mathcal{H}$ is then a combination of the columns of the control matrices on $\mathcal{H}_i$ for noise operators B_α that are non-trivial, *i. e.* not the identity, only on their original space. For noise operators defined on more than one subspace, *e. g.* $B_{ij} = B_i \otimes B_j, B_i \in \mathcal{H}_i, B_j \in \mathcal{H}_j$, this evidently holds only for those basis elements that are trivial on either of the subspaces and the columns in the composite control matrix corresponding to non-trivial basis elements ($C_k \otimes C_l \,:\, C_k, C_l \neq \mathbb{1}$) need to be computed from scratch.

One can thus reuse precomputed control matrices beyond the concatenation laid out above when studying multi-qubit pulses or algorithms. For concreteness, consider a set of one- and two-qubit pulses whose control matrices have been precomputed. We can then remap those control matrices to any other qubit in a larger register if the entire Hilbert space is defined by the tensor product of the single-qubit Hilbert spaces, and even map the control matrices of two different pulses to the same time slot on different qubits. Thus, we do not need to perform the possibly costly computation of the control matrices again but instead only need to remap the columns of $\tilde{\mathcal{B}}$ to the equivalent basis elements in the basis of the complete Hilbert space, making the assembly of algorithms that consist of a limited set of gates which are used at several points in the algorithm more efficient. In Section 19.4 we simulate a four-qubit QFT algorithm making use of the shortcuts described here.

16.8 Operator bases

Up to this point, we have not explicitly specified the basis that defines the Liouville representation. The only conditions imposed by Equation 16.14 are orthonormality with respect to the Hilbert-Schmidt product and that the basis elements are Hermitian. Yet, the choice of operator basis can have a large impact on the time it takes to compute the control matrix as discussed in the previous section. We therefore give a short overview over two possible choices in the following. As we are mostly interested in the computational properties, we represent linear operators in $\mathsf{L}(\mathcal{H})$ as matrices on $\mathbb{C}^{d\times d}$.

The n-qubit Pauli basis fulfills the requirements set by Equation 16.14 and furthermore allows for the simplifications described before. In our normalization convention where $\langle C_i, C_i\rangle = \mathbb{1}$ it can be written as

$$\mathsf{P}_n := \{\sigma_i\}_{i=0}^{d^2-1} = 2^{-\frac{n}{2}} \times \bigl\{\mathbb{1}, \sigma_x, \sigma_y, \sigma_z\bigr\}^{\otimes n} \tag{16.54}$$

with the Pauli matrices σ_x, σ_y and σ_z . While it is obvious that it is separable, meaning it factors into tensor products of the single-qubit Pauli matrices, the dimension of the Pauli basis is restricted to powers of two, $d = 2^n$. An operator basis without this restriction is the generalized Gell-Mann (GGM) basis [385, 386]. In the following we will discuss optimizations pertaining to this basis that are also implemented in the software (see Chapter 18).

The GGM matrices are a generalization of the Gell-Mann matrices known from particle physics to arbitrary dimensions. In our normalization convention, the basis (excluding the identity element) is given by [387]

$$\Lambda_d := \{\Lambda_i\}_{i=1}^{d^2-1} = \left\{u_{jk}, v_{jk}, w_l\right\}_{j,k,l} \tag{16.55}$$

with

$$u_{jk} = \frac{1}{\sqrt{2}}\left(|j\rangle\langle k| + |k\rangle\langle j|\right), \tag{16.55a}$$

$$v_{jk} = -\frac{i}{\sqrt{2}}\left(|j\rangle\langle k| - |k\rangle\langle j|\right), \tag{16.55b}$$

$$w_l = \frac{1}{\sqrt{l(l+1)}}\left(\sum_{m=1}^{l} |m\rangle\langle m| - l\,|l+1\rangle\langle l+1|\right), \tag{16.55c}$$

for $1 \leq j < k \leq d$, $1 \leq l \leq d-1$, and an orthonormal vector basis $\{|j\rangle\}_{j=1}^{d}$ of the Hilbert space. Expanding an arbitrary matrix $A \in \mathbb{C}^{d\times d}$ in the basis of Equation 16.55 is then simply a matter of adding up the corresponding matrix elements of A according to Equations 16.55a to 16.55c. For instance, the expansion coefficient for the first symmetric basis element can simply be read off to be $\langle u_{12}, A\rangle = 2^{-1/2}(\langle 1|A|2\rangle + \langle 2|A|1\rangle)$. The explicit construction prescription of the GGM basis thus allows calculating inner products of the form $\langle\Lambda_j|A\rangle$ at constant cost instead of the quadratic cost of the trace of a matrix product, speeding up the computation of the transfer matrix from Equation 16.1 (in which case $A = \mathcal{E}(\Lambda_k)$). In numerical experiments, calculating the transfer matrix of a unitary U with dimension d and precomputed matrix products $A_k = U\Lambda_k U^\dagger$ scaled as $\sim d^{4.16}$. This agrees with the expected scaling of $\sim d^4$ (a transfer matrix has $d^2 \times d^2$ elements) and presents a significant improvement over the explicit calculation with trace overlaps $\mathrm{tr}\left(\Lambda_j A_k\right)$ that we observed to scale as $\sim d^{5.93}$ (we expected $\sim d^6$).

Further inspection of the GGM basis additionally reveals an increasing sparsity for large d (the density scales asymptotically as d^{-2}), so that it is well suited for computing the trace tensor Equation 16.18. Since this tensor has d^8 elements, the amount of memory required for a dense representation becomes unreasonably large quite quickly. To overcome this constraint, we can use a GGM basis instead of a comparably dense basis like the Pauli basis. In this case, the resulting tensor is also sparse because the overlap between different basis elements is small.[15] This not only enables storing the tensor in memory but also makes the calculation much faster since one can employ algorithms optimized for operations on sparse arrays (see Chapter 18).

15: The Pauli basis has a density of $1/d$, compared to the GGM's of $\sim 1/d^2$. This results in trace tensors with density $1/d^2$ and $\sim 1/d^4$, respectively.

As an illustration, consider a system of four qubits so that the Hilbert space has dimension $d = 2^4$. An operator basis for this space has $d^2 = 2^8$ elements and consequently the tensor T_{ijkl} has $(2^8)^4 = 2^{32}$ entries. Using 128 bit complex floats to represent the entries the tensor would take up $\approx$ 68 GB of memory in a dense format. Conversely, for a GGM basis stored in a sparse data structure, the resulting trace tensor only takes up $\approx$ 100 MB of memory. Furthermore, calculating T takes $\approx$ 2.89 s on an Intel® Core™ i9-9900K eight-core processor since a GGM basis has a low density. By contrast, the same calculation with a Pauli basis takes $\approx$ 217 s. This is due to the larger density on the one hand and because sparse matrix multiplication algorithms perform poorly with dense matrices on the other.

16.9 Computational complexity

In order to assess the performance of FFs for computing fidelities compared to MC methods, we determine each method's asymptotic scaling behavior as a function of the system dimension d. For the filter functions, we calculate the fidelities using Equations 16.26 and 16.49 in our software implementation, described in more detail in Chapter 18, and hence neglect contributions of $\mathcal{O}(\xi^4)$ from the frequency shifts Δ. Additionally, we distinguish between three different approaches for calculating the control matrix; first, for a single pulse following Equation 16.36, second for an arbitrary sequence of pulses following Equation 16.29, and third for a periodic sequence of pulses following Equation 16.53. For the single pulse, we run benchmarks using exemplary values for the various parameters on a machine with an AMD FX™-6300 processor with six logical cores and 24 GB of memory. We also discuss the filter function method using left-right conjugation by unitaries instead of the Liouville representation. The latter has higher memory requirements and is expected to perform poorly for large system dimensions d since one deals with $d^2 \times d^2$ transfer matrices on a Liouville space $\mathscr{L}$ instead of $d \times d$ unitaries on a Hilbert space $\mathscr{H}$. In the software package, the calculations are currently implemented in Liouville space and calculation by conjugation is only partially supported through the low-level API. However, both representations perform similarly for small dimensions as we show below. Note that for a fair performance comparison the different nature of errors needs to be kept in mind. MC becomes less costly if larger statistical errors can be tolerated, whereas the filter-function formalism is typically limited by higher order errors. For reference, the following considerations are summarized in Table 16.1 for each approach and a representative set of parameters.

The MC algorithm is laid out in detail in Section C.4. To calculate the fidelity using MC, we generate n_{MC} different noise traces that slice every time step Δt of the pulse into $n_{\mathrm{seg}} = f_{\mathrm{UV}}\Delta t$ segments to appropriately sample the spectral density with f_{UV} being the ultraviolet cutoff frequency. In total, there are $n_{\Delta t}n_{\mathrm{MC}}n_{\mathrm{seg}}$ noise samples for each of which the Hamiltonian is diagonalized, exponentiated, and the resulting propagators multiplied to get the final, noisy unitary. The entanglement fidelity is then obtained by averaging the trace overlap $d^{-1}\operatorname{tr}\left(Q^\dagger U\right)$ of ideal and noisy unitary over all noise realizations. Taking the complexity of matrix diagonalization to be $\mathcal{O}(d^3)$ and matrix multiplication to be $\mathcal{O}\left(d^b\right)$ with $b = 3$ for a naive algorithm and $b = 2.376$ for the Coppersmith-Winograd algorithm [388], we expect MC to scale with the dimension d of the problem as $\sim n_{\Delta t}n_{\mathrm{MC}}n_{\mathrm{seg}}(d^b + d^3)$. For simplicity, we use a white noise spectrum for which $S(\omega) = \text{const.}$ but note that sampling arbitrary spectra induces additional overhead for MC, depending on which method is used to generate the noise traces. Typical time-domain methods include the simulation of the underlying physical process (like two-state fluctuators) or the application of an inverse Fourier transform to white noise multiplied by a frequency-domain transfer function.

By contrast, the computational cost of the filter-function formalism as realized by Equation 16.36 is independent of the form of the spectrum. For this approach we find the leading terms to scale as $\sim n_{\Delta t}n_\omega n_\alpha d^4 + n_{\Delta t}d^{b+2}$ with n_α the number of noise operators and n_ω the number of frequency samples. Here, the first term is due to the trace in Equation 16.36 which boils down to the trace of a matrix product, $\sum_{ij} A_{ij}B_{ji}$, that scales with d^2 and is performed once for each of the d^2 basis elements, n_α noise op-

Table 16.1: Complexity scaling of the three approaches for calculating average gate fidelities discussed in the text. "FF (explicit)" stands for calculating filter functions from scratch following Equation 16.36, "FF (concat.)" for sequences following Equation 16.29, and "FF (periodic)" for periodic Hamiltonians following Equation 16.53. $\mathscr{H}$ and $\mathscr{L}$ designate the vector space on which calculations are performed. Example values for the dominant contributions listed in the table are given for matrix multiplication exponent $b = 2.376$, dimension $d = 2$, number of time steps $n_{\Delta t} = 1000$, and number of gates $G = 100$ (corresponding to a sequence of 100 single-qubit gates with 10 time steps each) with the remaining parameters as in Figure 16.2. For increasing d the computational advantage of FF ($\mathscr{L}$) over MC diminishes but is conserved for FF ($\mathscr{H}$).

Method	Reference	Dominating scaling	Example values
MC ($\mathscr{H}$)	Equation C.46	$n_{\Delta t} n_{\mathrm{MC}} n_{\mathrm{seg}}(d^b + d^3)$	1.3×10^8
FF ($\mathscr{L}$, explicit)	Equation 16.36	$n_{\Delta t} n_\omega n_\alpha d^4 + n_{\Delta t} d^{b+2}$	2.4×10^7
FF ($\mathscr{H}$, explicit)	Equation 16.36	$n_{\Delta t} n_\omega n_\alpha (d^2 + d^b)$	1.4×10^7
FF ($\mathscr{L}$, concat.)	Equation 16.29	$G n_\omega n_\alpha d^4 + G d^{2b}$	2.4×10^6
FF ($\mathscr{H}$, concat.)	Equation 16.29	$G n_\omega n_\alpha d^b$	7.8×10^5
FF ($\mathscr{L}$, periodic)	Equation 16.53	$n_\omega (n_\alpha d^4 + d^{2b}[1 + \log G])$	1.0×10^5

erators, $n_{\Delta t}$ time steps, and n_ω frequency points. The second term is due to the transformation $C_k \to \bar{C}_k^{(g)}$ which requires multiplication of $d \times d$ matrices for every time step and basis element. As $n_\alpha n_\omega < n_{\mathrm{MC}} n_{\mathrm{seg}}$ for realistic parameters because the ultraviolet cutoff frequency needs to be chosen sufficiently high and the relative error of the method decreases with $1/\sqrt{n_{\mathrm{MC}}}$, we expect that in the case of a single pulse the filter-function formalism in Liouville representation should outperform Monte Carlo calculations for reasonably small dimensions d. Using left-right conjugation, this advantage should hold also for large d. In this case the Hadamard product ($\sim d^2$) as well as matrix multiplication ($\sim d^b$) are carried out for each frequency, noise operator, and time step to calculate the interaction picture noise operators $\bar{B}_\alpha(\omega)$. We thus find this method to scale with $\sim n_{\Delta t} n_\omega n_\alpha (d^2 + d^b)$.

Figure 16.2 shows exemplary wall times for both methods and $d \in [2, 120]$ that confirm our expectation. Only for about $d \approx 100$ the overhead from the extra time steps and trajectories over which is averaged is compensated for MC. For smaller dimensions the calculation using FFs is faster by almost two orders of magnitude (see the inset showing the same data in a log-log plot). The lines show fits to $t = ad^b$. The data is not quite in the asymptotic regime due to limited memory so that even for large dimension terms of lower power in d contribute significantly to the run time. Even though this causes the fits to underestimate the exponent b, the general trend agrees with our expectation. Note that the crossover does not always occur at the same dimension d. On a different system with an Intel® Core™ i9-9900K eight-core processor the FF method outperformed MC even for $d = 120$ beyond which available memory limited the simulation.

Quantifying the performance gain from using the control matrices' concatenation property to calculate fidelities of gate sequences is more difficult since it strongly depends on the number of gates occurring multiple times in the sequence (enabling reuse of precomputed control matrices) as well as the complexity of the individual gates. The evaluation using the concatenation rule Equation 16.29 performs asymptotically worse than the evaluation for a complete pulse according to Equation 16.36 because of higher powers of d dominating the calculation in the former case. Performing the G matrix multiplications $\tilde{\mathcal{B}}^{(g)}(\omega)\mathcal{Q}^{(g-1)}$ from Equation 16.29 is of order $\sim G n_\omega n_\alpha d^4$, with G the number of pulses in the sequence. Furthermore, calculating the transfer matrix of the total propagators Q_{g-1} involves multiplication of $d \times d$ matrices for all d^4 combi-

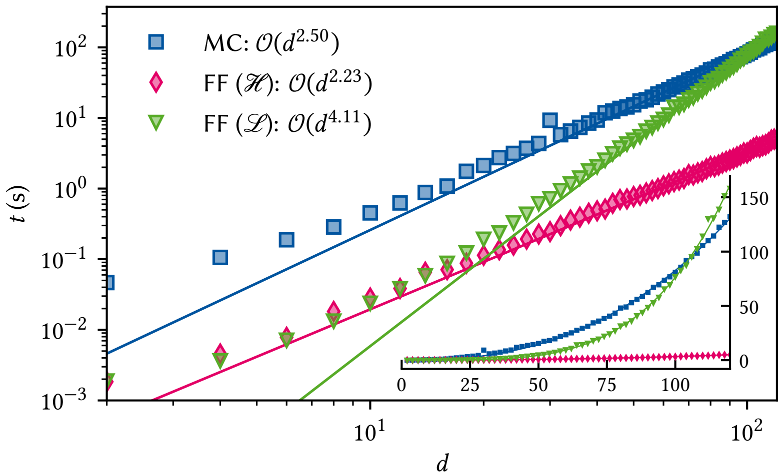

Figure 16.2: Performance of the formalism using Equation 16.36 compared to a Monte Carlo method for a single gate as a function of problem dimension d. Parameters are: $n_{\Delta t} = 1, n_\alpha = 3, n_{\mathrm{MC}} = 100, f_{\mathrm{UV}} = 10^2/\Delta t, n_\omega = 500$ where n_α is the number of noise operators considered, n_{MC} the number of Monte Carlo trajectories over which is averaged, and n_ω the number of frequency samples. The calculation using filter functions clearly outperforms MC for small system sizes. For dimensions larger than $d \approx 100$ (roughly equivalent to 7 qubits) Monte Carlo (blue squares) performs better than the FF calculation with transfer matrices (green triangles) for this set of parameters and processor due to the better scaling behavior. Using conjugation by unitaries (orange diamonds) significantly outperforms MC also for large dimensions. While the fits to $t = ad^b$ (lines) underestimate the leading order exponent due to the data not being in the asymptotic regime, they support the expected relationship of complexity between the approaches. The inset shows the same data on a linear scale, highlighting the different scaling behaviors for large d.

nations of basis elements amounting to $\sim Gd^{b+4}$. In case the Liouville representation of the individual pulses' total propagators P_g, $\mathcal{P}^{(g)}$, have been precomputed, the latter computation can be made more efficient since one can just propagate the transfer matrices $\mathcal{P}^{(g)}$ to obtain the cumulative transfer matrices for the sequence, $\mathcal{Q}^{(g)} = \prod_{g'=g}^{0} \mathcal{Q}^{(g')}$, at cost $\sim Gd^{2b}$. The restriction to small dimensions does not apply for conjugation by unitaries as in this case the matrix multiplications involve $d \times d$ matrices and we do not have to compute the Liouville representation. We thus obtain a more favorable asymptotic scaling of $\sim Gn_\omega n_\alpha d^b$.

Utilizing the concatenation property in the Liouville representation thus corresponds to effectively reducing the number of times the calculations scaling with $\sim n_\omega n_\alpha d^4$ have to be carried out but incurs additional calculations scaling with $\sim d^{2b}$. Accordingly, it provides a performance benefit if a sequence consists of either very complex pulses, in which case single repetitions already make the calculation much more efficient, or of few pulses that occur many times. In the extremal case of G repetitions of a single gate the benefit of employing the concatenation property is most pronounced and can be improved even further utilizing the simplifications laid out in Section 16.6. Since matrix inversion has the same complexity as matrix multiplication and taking a matrix to the Gth power requires $\mathcal{O}(\log G)$ matrix multiplications, Equation 16.53 should scale with $\sim n_\omega(n_\alpha d^4 + d^{2b} + d^{2b}\log G)$ (the first two terms are due to the final matrix multiplications and are independent of G). It hence allows for a vast speedup over Equation 16.29 in that the asymptotic behavior as a function of the number of gates changes from $\sim G$ to $\sim \log G$. An example of this is presented in Section 19.2 for the context of Rabi driving. Note that this closed form is a unique feature of the transfer matrix representation and not applicable to conjugation by unitaries.

Filter functions from random sampling 17

In our derivation of the average error map $\langle \tilde{\mathcal{U}} \rangle$, Equation 16.9, we employed the cumulant expansion due to Kubo [319, 356]. In his papers, Kubo makes the claim that this expansion is exact for Gaussian noise when truncating after the second order, even for *q-numbers*, *i. e.*, linear operators. For *c-numbers* – scalars – this is a well-known result in statistics; the Gaussian probability distribution is fully described by its first two cumulants, the mean μ and variance σ^2. However, Fox [320] showed many years later that Kubo's claim for q-numbers does not hold, and that non-commutativity in the Gaussian variables can cause higher-order cumulants to not vanish. There are two limiting cases in which the expansion remains exact; first, for white noise, whose auto-correlation function is singular at $t = 0$, and thus "decouples" higher-order cumulants, and second, commuting noise, for which $[H_{\mathrm{c}}(t), H_{\mathrm{n}}(t)] = 0 \forall t$. Beyond these two cases, we must therefore assume that the map given by Equation 16.9 is merely approximate when including only up to second-order terms from the Magnus expansion.[1] In this chapter, we investigate the extent of this error. To this end, we compute the *exact* filter function (FF) using a Monte Carlo (MC) approach by randomly sampling monochromatic noise fields.

1: Note that the cluster property still holds in all cases, and hence for Gaussian noise only even-order cumulants do not vanish [320, 357, 389, 390].

17.1 Reconstruction by monochromatic time-domain simulation

For a given interaction-picture quantum operation $\langle \tilde{\mathcal{U}}(\tau) \rangle$ resulting from the quantum system's evolution under the noise fully characterized by its PSD $S(\omega)$, we *define* the filter function $\mathcal{F}(\omega; \tau)$ by

$$\langle \tilde{\mathcal{U}}(\tau) \rangle = \exp \mathcal{K}(\tau) = \exp\left\{ -\int \frac{\mathrm{d}\omega}{2\pi} \mathcal{F}(\omega; \tau) S(\omega) \right\}. \tag{17.1}$$

Now, suppose that

$$S(\omega) = 2\pi\sigma_i^2 \delta(\omega - \omega_i) =: S_i(\omega), \tag{17.2}$$

that is, the PSD of a monochromatic sinusoid of frequency ω_i and root mean square (RMS) σ_i.[2] Equation 17.1 then becomes

2: Equation 17.2 discretizes $S(\omega)$ by sampling it at points ω_i, *i. e.*,

$$S(\omega) \sim \lim_{n\to\infty} \sum_{i=1}^{n} S_i(\omega).$$

See also Sidenote 13.

$$\begin{aligned} \langle \tilde{\mathcal{U}}_i(\tau) \rangle := \langle \tilde{\mathcal{U}}(\tau) \rangle &= \exp\left\{ -2\pi\sigma_i^2 \int \frac{\mathrm{d}\omega}{2\pi} \mathcal{F}(\omega; \tau) \delta(\omega - \omega_i) \right\} && (17.3) \\ &= \exp\left\{ -\sigma_i^2 \mathcal{F}(\omega_i; \tau) \right\}, && (17.4) \end{aligned}$$

where $\langle \tilde{\mathcal{U}}_i(\tau) \rangle$ is the noisy quantum operation generated by monochromatic noise with PSD $S_i(\omega)$ according to Equation 17.2. It is now easy to invert Equation 17.4, and we obtain

$$\mathcal{F}(\omega_i; \tau) = -\sigma_i^{-2} \log \langle \tilde{\mathcal{U}}_i(\tau) \rangle. \tag{17.5}$$

Because we represent quantum operations as matrices in Liouville space, Equation 17.5 is straightforward to implement on a computer; to sample the exact FF at the set of discrete frequencies $\{\omega_i\}_i$ we simply need to compute $\langle\tilde{\mathcal{U}}_i(\tau)\rangle$ using a time-domain simulation method of our choice and take the logarithm![3]

3: A similar approach was pursued by Geck in her PhD thesis to compare gate fidelities [391].

Indeed, we can go a step further and split apart the coherent and incoherent contributions to the noisy evolution. Since (in-)coherent quantum operations are represented by (anti-)symmetric matrices in Liouville space, we may define the incoherent and coherent FFs by

$$\begin{aligned}\mathcal{F}_\Gamma(\omega_i;\tau) &= \frac{1}{2}\left(\mathcal{F}(\omega_i;\tau) + \mathcal{F}(\omega_i;\tau)^\top\right) \\ &= -\frac{1}{2\sigma_i^2}\left(\log\langle\tilde{\mathcal{U}}_i(\tau)\rangle + \log\langle\tilde{\mathcal{U}}_i(\tau)\rangle^\top\right),\end{aligned} \tag{17.6}$$

and

$$\begin{aligned}\mathcal{F}_\Delta(\omega_i;\tau) &= \frac{1}{2}\left(\mathcal{F}(\omega_i;\tau) - \mathcal{F}(\omega_i;\tau)^\top\right) \\ &= -\frac{1}{2\sigma_i^2}\left(\log\langle\tilde{\mathcal{U}}_i(\tau)\rangle - \log\langle\tilde{\mathcal{U}}_i(\tau)\rangle^\top\right),\end{aligned} \tag{17.7}$$

respectively. In the filter-function formalism presented in Chapter 16, Equations 17.6 and 17.7 correspond – up to corrections from non-vanishing higher cumulants – to linear combinations of the generalized filter functions from second- and first-order Magnus expansion, respectively. To be more precise, let us again consider Equation 16.17. Rather than first performing the integrations of Equations 16.20 and 16.21 to obtain the decay amplitudes Γ and frequency shifts Δ, and then the contractions with the trace tensor functions g_{ijkl} and f_{ijkl}, first contract the latter with the control matrices to obtain the time-domain filter functions[4]

4: We consider a single noise operator and drop the index for brevity.

$$\mathcal{F}_{\Gamma,ij}(t_1,t_2) = \frac{1}{2}\sum_{kl} g_{ijkl}\tilde{\mathcal{B}}_k(t_1)\tilde{\mathcal{B}}_l(t_2), \tag{17.8}$$

$$\mathcal{F}_{\Delta,ij}(t_1,t_2) = \frac{1}{2}\sum_{kl} f_{ijkl}\tilde{\mathcal{B}}_k(t_1)\tilde{\mathcal{B}}_l(t_2). \tag{17.9}$$

Again moving to Fourier space and following the same procedure as in Sections 16.2 and 16.3 leads to the coherent and incoherent frequency-domain filter functions $\mathcal{F}_\Gamma(\omega;\tau)$ and $\mathcal{F}_\Delta(\omega;\tau)$ that we must simply integrate over to obtain the average error process $\langle\tilde{\mathcal{U}}(\tau)\rangle$,[5]

5: That is, we have $\mathcal{F}_{\Delta,ij}(\omega;\tau) = 1/2\sum_{kl} f_{ijkl}\mathcal{F}^{(2)}_{kl}(\omega;\tau)$ with the latter defined in Equation 16.42 as well as $\mathcal{F}_{\Gamma,ij}(\omega;\tau) = 1/2\sum_{kl} g_{ijkl}\mathcal{F}_{kl}(\omega;\tau)$ with the latter defined in Equation 16.25.

$$\langle\tilde{\mathcal{U}}(\tau)\rangle = \exp\left\{-\int\frac{\mathrm{d}\omega}{2\pi}S(\omega)\left[\mathcal{F}_\Gamma(\omega;\tau) + \mathcal{F}_\Delta(\omega;\tau)\right]\right\}. \tag{17.10}$$

This allows us to analyze in detail deviations of the closed-form expression obtained by means of the cumulant expansion, Equation 17.10, from the exact filter function sampled at discrete ω_i by Equation 17.5. In the following, we will lay out explicitly how the latter can be computed in the time domain using a MC method.[6]

6: See also Section C.4 for details on MC.

To begin with, observe that

$$\langle\tilde{\mathcal{U}}_i(\tau)\rangle = \mathcal{Q}^\top\langle\mathcal{U}_i(\tau)\rangle, \tag{17.11}$$

where as usual $\mathcal{Q}$ is the complete superpropagator of the noise-free time evolution and $\mathcal{U}_i(\tau)$ that of the noisy time evolution generated by a single realization of the noise $b(t)$, both of which are orthogonal, *i.e.*, the corresponding Hilbert-space operators are unitary. In MC, we generate N realizations of this noise process in the time domain, compute the full time evolution superpropagator $\mathcal{U}_i(\tau)$, and then evaluate the expectation value $\langle\,\bullet\,\rangle$ as the ensemble average to obtain $\langle\mathcal{U}_i(\tau)\rangle$.[7] Inserting into Equa-

7: For N realizations of the stochastic process underlying $b(t)$, the ensemble average of a quantity $A(t)$ that is a function of $b(t)$ is given by

$$\langle A\rangle(t) = \frac{1}{N}\sum_{i=1}^{N} A_i(t)$$

where i enumerates the realizations of the stochastic process. The relative error of this average scales as $N^{-1/2}$.

tion 17.4, we find that

$$\mathcal{F}(\omega_i;\tau) = -\sigma_i^{-2}\log\langle\tilde{\mathcal{U}}_i(\tau)\rangle = -\sigma_i^{-2}\log\mathcal{Q}^\top\langle\mathcal{U}_i(\tau)\rangle\,. \tag{17.12}$$

If we evaluate Equation 17.12 for a set of frequencies $\{\omega_i\}_i$ sampling the true spectrum $S(\omega)$ sufficiently well, we thus obtain the exact filter function $\mathcal{F}(\omega;\tau)$ (within the accuracy of MC), allowing us to compare the accuracy of the formalism developed in Chapter 16.

So what does a single noise realization of Equation 17.2 in the time domain look like? It is a sinusoid with amplitude $A_i \sim \mathrm{Rayleigh}(\sigma_i)$,[8] frequency ω_i, and phase $\phi \sim \mathrm{Unif}(0, 2\pi)$,[9]

$$b(t) = A_i\sin(\omega_i t + \phi). \tag{17.13}$$

To obtain the MC estimate of $\langle\tilde{\mathcal{U}}(\tau)\rangle$, we therefore draw N noise traces according to Equation 17.13, solve the Schrödinger equation, and finally perform the ensemble average from which we can deduce the filter function using Equation 17.12. From the spread of the individual MC samples we obtain confidence intervals on the mean of the error transfer matrix by bootstrapping, which we propagate to the filter function by taking the Fréchet derivative of the matrix logarithm in Equation 17.12 [393].

8: Rayleigh(σ) is the Rayleigh distribution with probability density function [392]

$$\rho(x) = \frac{x}{\sigma^2}\exp\left\{-\frac{x^2}{2\sigma^2}\right\}.$$

It describes the probability distribution of the distance from the origin of a point drawn from a bivariate normal distribution with mean 0 and standard deviation σ.

9: The probability density function of the uniform distribution Unif(a, b) is

$$\rho(y) = \begin{cases}(b-a)^{-1} & \text{if} \quad a \le y < b,\\ 0 & \text{else}\end{cases}$$

17.2 Case studies

Let us now turn to applying the method to a few select case studies. First, we consider a pure-dephasing Hamiltonian with

$$\begin{aligned} H_\mathrm{c}(t) &= \Omega\frac{\sigma_z}{2},\\ H_\mathrm{n}(t) &= b(t)\frac{\sigma_z}{2}. \end{aligned} \tag{17.14}$$

Specifically, we let the system evolve under the Hamiltonian in Equation 17.14 for $\tau = \pi/\Omega$ to undergo a π-rotation around the z-axis of the Bloch sphere. For this model, we expect the cumulant expansion to terminate after the second order because control and noise commute, and hence the interaction-picture noise Hamiltonian, $\tilde{H}_\mathrm{n}(t)$, always commutes with itself. Therefore, the filter-function formalism of Chapter 16 should give the exact result. Equation 17.14 can be solved analytically,

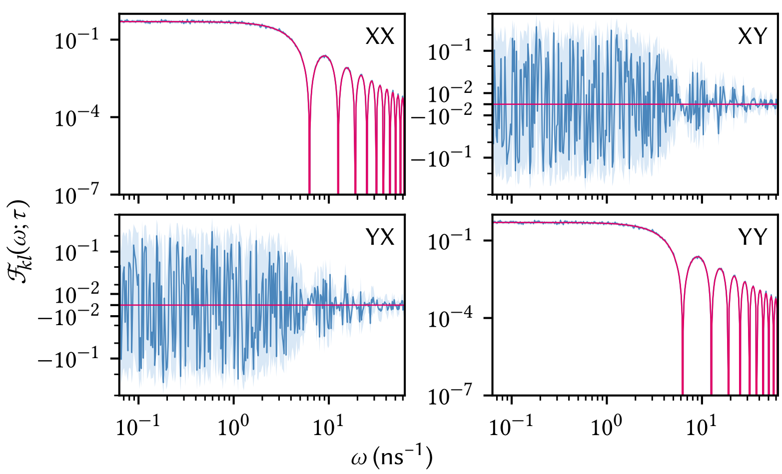

Figure 17.1: Non-zero components of the FF in the Pauli basis P_1 for the pure dephasing Hamiltonian in Equation 17.14. Magenta lines are from the cumulant expansion, Equation 17.10, blue lines the mean of the MC simulation, Equation 17.12, and light blue shaded areas the 95 % confidence interval. The diagonal elements (XX and YY) agree within the MC confidence intervals, with the zeros at $2\pi n/\tau, n \in \{1, 2, \ldots\}$ agreeing down to 10^{-15} where we can expect numerical errors to be limiting. The off-diagonal elements are quite noisy in the MC simulation, but fluctuate about the expected value of zero.

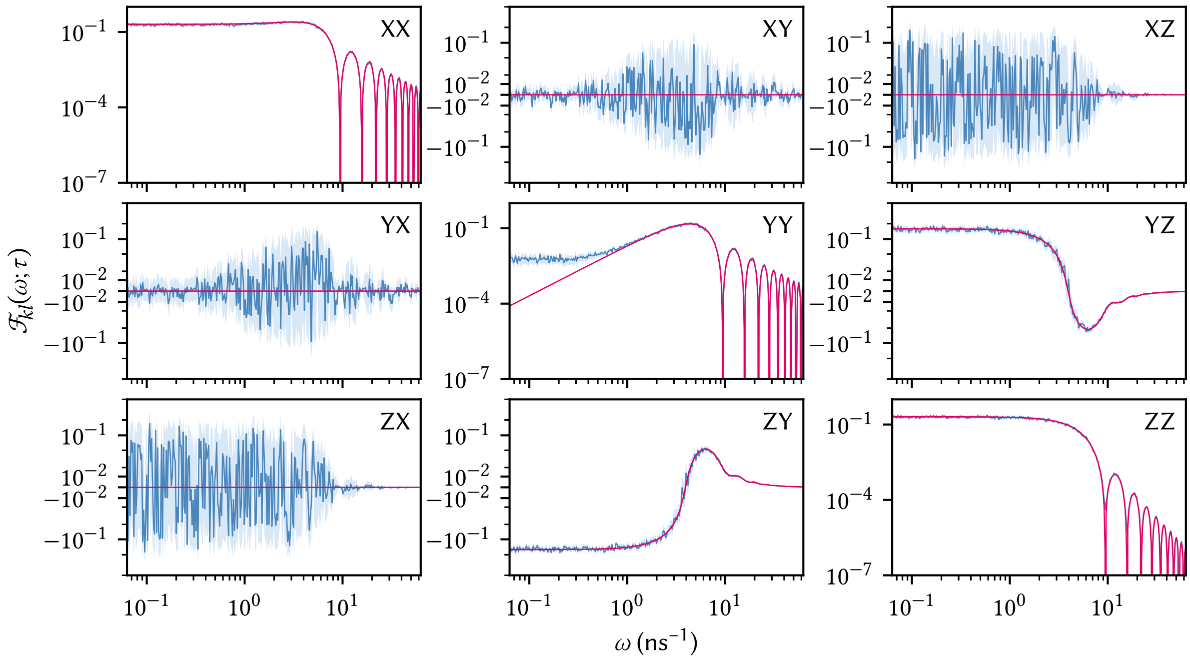

Figure 17.2: FFs for the non-commuting Hamiltonian in Equation 17.15. Cumulant expansion calculation is shown in magenta, MC in blue with light blue shaded areas the confidence intervals. Again, the nominally-zero off-diagonals are susceptible to numerical noise in the MC method. The YY- and ZZ-terms show a deviation of the cumulant expansion calculation, while all others, including the coherent YZ- and ZY-terms appear to be exact up to the numerical uncertainty.

and gives rise to an ideal dephasing channel whose PTM representation is diagonal, $\langle\tilde{\mathcal{U}}(\tau)\rangle = \mathrm{diag}(1, 1-2p, 1-2p, 1)$ with p the dephasing probability whose precise value depends on the form of the PSD $S(\omega)$. Accordingly, the FF $\mathcal{F}_{kl}(\omega;\tau)$ should be zero everywhere but at $k = l \in \{1,2\} \equiv \{x,y\}$ for the Pauli basis P_1. Figure 17.1 shows the total "error transfer matrix filter functions" (Equations 17.10 and 17.12) computed with both methods; magenta for the cumulant expansion and blue for the exact MC method. The diagonals agree perfectly within the confidence intervals, while the off-diagonals are dominated by numerical noise in the MC approach. Since the off-diagonals are zero, only the first-order Magnus terms contribute.[10]

10: Recall that $\mathcal{F}_\Delta(\omega;\tau)$ is an anti-symmetric matrix.

The opposite extreme to the pure-dephasing Hamiltonian is obtained when changing the control to be transverse to the noise,

$$\begin{aligned} H_\mathrm{c}(t) &= \Omega\frac{\sigma_x}{2}, \\ H_\mathrm{n}(t) &= b(t)\frac{\sigma_z}{2}. \end{aligned} \tag{17.15}$$

In this case, control and noise Hamiltonian maximally do not commute at any time ($\tilde{H}_\mathrm{n}(t) = \sigma_z \cos\Omega t + \sigma_y \sin\Omega t$) and we should expect the largest deviations from the exact filter functions in the approximation by truncating the cumulant function after the second order. Figure 17.2 shows the FF for a π-rotation around the x-axis of the Bloch sphere (*i. e.*, $\tau = \pi/\Omega$ once again). This time, all diagonal elements are nonzero, and the YY-term ($\mathcal{F}_{22}(\omega;\tau)$ with $\sigma_2 \equiv \sigma_y$) displays a deviation between cumulant expansion and MC simulation towards low frequencies! This is exactly what we would expect according to Fox [320], namely that the more correlated the noise is, the larger contributions from higher orders will be.[11] Also deviating is the ZZ-term, although not towards DC but where the filter function computed from the approximated cumulant function drops

11: Quasistatic noise is perfectly correlated—it assumes a single value for all times.

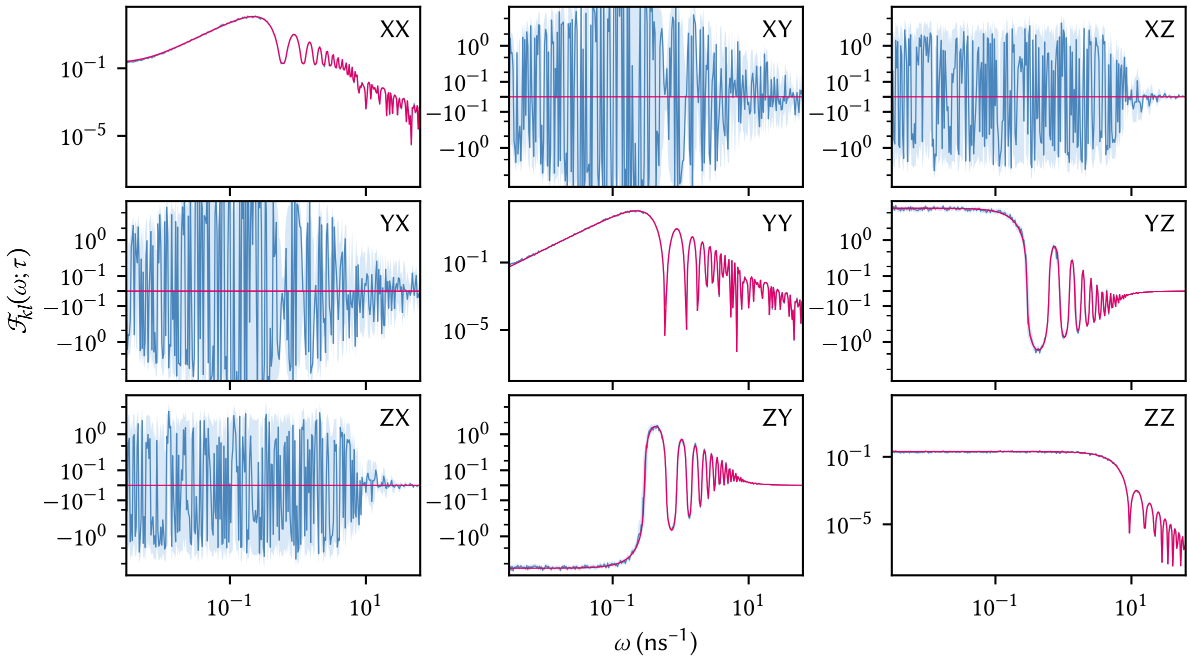

Figure 17.3: FFs for the spin echo (SE) Hamiltonian in Equation 17.16. Cumulant expansion calculation is shown in magenta, MC in blue with light blue shaded areas the confidence intervals. Both coherent and incoherent terms show excellent agreement between both methods.

to zero. By contrast, the XX-term shows good agreement between both methods. Finally, two non-diagonal elements are nonzero, YZ and ZY, anti-symmetric terms which arise from the second-order Magnus and incur coherent errors. For these, the cumulant approximation is excellent, showing no deviation within the confidence intervals.

Finally, as a last example we consider the prototypical filter-function use case: the spin echo (SE). Here, the qubit idles for a time τ_0 both before and after a π rotation around the x-axis of the Bloch sphere, so that the Hamiltonian is given by

$$
\begin{aligned}
H_{\mathrm{c}}(t) &= \Omega(t)\frac{\sigma_x}{2}, \\
H_{\mathrm{n}}(t) &= b(t)\frac{\sigma_z}{2},
\end{aligned}
\tag{17.16}
$$

with $\Omega(t) = \pi/\tau_\pi$ if $t \in [\tau_0, \tau_0 + \tau_\pi]$ and zero else. This pulse sequence is known to cancel quasistatic dephasing noise as the phase picked up on the equator of the Bloch sphere during the first period of free evolution is exactly cancelled by the phase picked up during the second period after the π-pulse. Since the previous example demonstrated that the filter-function formalism based on the second-order cumulant expansion neglects contributions from higher orders in the case of transverse noise, we might expect the SE sequence to show similar behavior. Figure 17.3 shows the results of the simulation. The picture is similar – albeit more complex – to before, with nominally-zero terms suffering from numerical noise. However, in contrast to the simple π-pulse Hamiltonian in Equation 17.15 and Figure 17.2, the YY and ZZ terms show excellent agreement between the exact MC method and the cumulant expansion. We may speculate that the higher-order contributions not captured by the cumulant method in the previous case drop out due to the symmetry of the Hamiltonian.

In conclusion, we have developed a method to numerically compute the exact filter functions of the quantum process description of arbitrary Gaussian noise, and compared the thus obtained filter functions with those from the second-order truncation of the ME and cumulant expansions given in Chapter 16. The approach samples the Liouville-space filter function at discrete frequencies ω_i by simulating the unitary dynamics under monochromatic noise at that frequency and averaging over the outcomes. We found excellent agreement between the nominally exact and approximate methods in the case of commuting noise as well as the SE DD sequence. For non-commuting noise, we observed the expected deviation at very low frequencies that arises from non-vanishing higher order cumulants.

Rather than random sampling using MC, this average could also be performed deterministically as the expectation value over the probability distributions governing A_i and ϕ,

$$\mathrm{E}_{A_i,\phi}[\mathcal{U}] = \int \mathrm{d}x\, \rho_{A_i}(x) \int \mathrm{d}y\, \rho_\phi(y) \mathcal{U}[x, y], \tag{17.17}$$

where we wrote $\mathcal{U}[x, y]$ for the Liouville representation of the propagator for single x and y drawn from their respective distributions, and $\mathrm{E}_X[A]$ denotes the expectation value of an observable A with respect to the random variable X with probability density function ρ_X. This technique allows the expectation value $\mathrm{E}[\mathcal{U}] \equiv \langle\mathcal{U}\rangle$ to be evaluated to arbitrary precision numerically, but is typically less efficient than MC. It would be interesting to see if the numerical noise observed in off-diagonal elements of the filter functions in Figures 17.1 to 17.3 persists with this method.

The `filter_functions` software package

18

IN this chapter, we give a brief overview over the `filter_functions` software package implementing the main features of the formalism derived above.[1] This includes the calculation of the decay amplitudes Γ, frequency shifts Δ, and fidelities as well as the calculation of the control matrices for single gates and both generic and periodic sequences of gates. Moreover, control matrices may be efficiently extended to and merged on larger Hilbert spaces. Calculations using unitary conjugation instead of transfer matrices are implemented but at this point not available in the high-level API.

1: For more details we refer the reader to the documentation.

Our software is written in Python and published under the GPLv3 license [358]. It features a broad coverage through unit tests and extensive API documentation as well as didactic examples (see also Chapter 19). The package relies on the `NumPy` [394] and `SciPy` [395] libraries for vectorized array operations. Data visualization is handled by `matplotlib` [396]. For tensor multiplications with optimized contraction order we use `opt_einsum` [397] for which `sparse` [398], a library aiming to extend the `SciPy` sparse module to multi-dimensional arrays, serves as a backend in the calculation of the trace tensor from Equation 16.18. Lastly, the package is written to interface with `qopt` [360] and `QuTiP` [359], frameworks for the simulation and optimization of open quantum systems, and mirrors the latter's data structure for Hamiltonians ensuring easy interoperability between the two.

18.1 Package overview

In the `filter_functions` package all operations are understood as sequences of pulses that are applied to a quantum system. These pulses are represented by instances of the `PulseSequence` class which holds information about the physical system (control and noise Hamiltonians) as well as the mathematical description (*e.g.* the basis used for the Liouville representation). As indicated above, the Hamiltonians $H_\mathrm{c}(t)$ and $H_\mathrm{n}(t)$ are given in a similar structure as in `QuTiP`. That is, a Hamiltonian is expressed as a sum of Hermitian operators with the time dependence encoded in piecewise constant coefficients so that

$$H_\mathrm{c}(t) = \sum_i a_i^{(g)} A_i = \text{const.} \tag{18.1a}$$

$$H_\mathrm{n}(t) = \sum_\alpha s_\alpha^{(g)} B_\alpha = \text{const.} \tag{18.1b}$$

for $t \in (t_{g-1}, t_g]$, $g \in \{1, \dots, G\}$ and where the $a_i^{(g)}$ are the amplitudes of the ith control field. Note that the noise variables $b_\alpha(t)$ are missing from Equation 18.1b because they are captured by the spectral density $S(\omega)$. In the software, Equations 18.1a and 18.1b are represented as lists whose ith element corresponds to a sublist of two elements: the ith operator and the ith coefficients $[a_i^{(0)}, \dots, a_i^{(G)}]$.

The `PulseSequence` class provides methods to calculate and cache the filter function according to Equation 16.30. Alternatively, filter functions may also be cached manually to permit using the package with analytical solutions for the control matrix. Concatenation of pulses is implemented by the functions `concatenate()` and `concatenate_periodic()` which will attempt to use the cached attributes of the `PulseSequence` instances representing the pulses to efficiently calculate the filter function of the composite pulse following Equation 16.29 and Equation 16.53, respectively.

Operator bases fulfilling Equation 16.14 are implemented by the `Basis` class. There are two predefined types of bases:

1. Pauli bases P_n for $n = 2^d$ qubits from Equation 16.54 and
2. GGM bases Λ_d of arbitrary dimension d from Equation 16.55.

The Pauli basis is both unitary and separable while the GGM basis is sparse for large dimensions but neither unitary nor separable. As mentioned in Section 16.7 (see also Section 19.4), using a separable basis can provide significant performance benefits for calculating the filter functions of algorithms. On the other hand, a sparse basis makes the calculation of the trace tensor T_{ijkl} and therefore also of the error transfer matrix $\tilde{\mathcal{U}}$ much faster (*cf.* Section 16.8). Additionally, the user can define custom bases using the class constructor.

The error transfer matrix $\tilde{\mathcal{U}}$ can be calculated for a given pulse and noise spectrum using the `error_transfer_matrix()` function. Various other quantities can be computed from $\tilde{\mathcal{U}}$ as outlined in Section 16.4. Furthermore, the package includes a plotting module that offers several functions, *e.g.* for the visualization of filter functions or the evolution of the Bloch vector using `QuTiP`.

18.2 Workflow

We now give a short introduction into the workflow of the `filter_functions` package by showing how to calculate the dephasing filter function of a simple Hahn SE sequence [399] as an example. The sequence consists of a single π-pulse of finite duration around the x-axis of the Bloch sphere in between two periods of free evolution. We can hence divide the control fields into three constant segments and write the control Hamiltonian as

$$H_{\mathrm{c}}^{(\mathrm{SE})}(t) = \frac{\sigma_x}{2} \times \begin{cases} \pi/t_\pi\,, & \text{if } \tau \leq t < \tau + t_\pi \\ 0, & \text{otherwise} \end{cases} \tag{18.2}$$

with τ the duration of the free evolution period and t_π that of the π pulse. For the noise Hamiltonian we only need to define the deterministic time dependence $s_\alpha(t)$ and operators B_α since the noise strength is captured by the spectrum $S(\omega)$. Thus we have $s_z(t) = 1$ and $B_z = \sigma_z/2$ for pure dephasing noise that couples linearly to the system.

In the software, we first define a `PulseSequence` object representing the SE sequence. As was already mentioned, the control and noise Hamiltonians are given as a list containing lists for every control or noise operator that is considered. These sublists consist of the respective operator as a `NumPy` array or `QuTiP` `Qobj` and the amplitudes ($a_i^{(g)}$ or $s_\alpha^{(g)}$) in an iterable data structure such as a list. We can hence instantiate the `PulseSequence` with the following code:

```python
import filter_functions as ff
import qutip as qt
from math import pi
tau, t_pi = (1, 1e-3)
# Control Hamiltonian for pi rotation in 2nd time step
H_c = [[qt.sigmax()/2, [0, pi/t_pi, 0]]]
# Pure dephasing noise Hamiltonian with linear coupling
H_n = [[qt.sigmaz()/2, [1, 1, 1]]]
# Durations of piecewise constant segments
dt = [tau, t_pi, tau]
ECHO = ff.PulseSequence(H_c, H_n, dt)
```

where a basis is automatically chosen since we did not specify it in the constructor in the last line. Calculating the filter function of the pulse for a given frequency vector `omega` can then be achieved by calling

```python
F = ECHO.get_filter_function(omega)
```

where `F` is the dephasing filter function $\mathcal{F}_{zz}(\omega)$ as we only defined a single noise operator. Finally, we calculate the error transfer matrix $\tilde{\mathcal{U}}$ for the noise spectral density $S_{zz}(\omega) = \omega^{-2}$,

```python
S = 1/omega**2
U = ff.error_transfer_matrix(ECHO, S, omega)
```

This code uses the control matrix previously cached when the filter function was first computed. Therefore, only the integration in Equation 16.24 and the calculation of the trace tensor in Equation 16.18 are carried out in the last line.

An alternative approach to calculate the spin echo filter function is to employ the concatenation property. For this, we interpret the SE as a sequence consisting of three separate pulses. Each of the pulses has a single time segment during which a constant control is applied and concatenating the separate `PulseSequence` instances yields the `PulseSequence` representing a spin echo. This way analytical control matrices may be used to calculate the control matrix of the composite sequence. Pulses can be concatenated by using either the `concatenate()` function or the overloaded `@` operator:

```python
# Define PulseSequence objects as shown above
FID = ff.PulseSequence(...)
NOT = ff.PulseSequence(...)
# Cache the analytical control matrices at frequency omega
FID.cache_control_matrix(omega, B_FID)
NOT.cache_control_matrix(omega, B_NOT)
# Concatenate the pulses
ECHO = FID @ NOT @ FID
```

Since we have cached the control matrices of the `FID` and `NOT` pulses, that of the `ECHO` object is also automatically calculated and stored. Concatenating `PulseSequence` objects is implemented as an arithmetic operator of the class to reflect the intrinsic composition property of the control matrices.

Further development of the software has focused on making it available in a gate optimization and simulation framework [400]. Besides using it to compute decoherence effects and fidelities, analytical derivatives of the filter functions have been implemented to allow for optimizing pulse parameters in the presence of non-Markovian noise within the framework of quantum optimal control [360, 401].

Additionally, building an interface with `qupulse` [402, 403], a software toolkit for parametrizing and sequencing control pulses and relaying them to control hardware, would implement the capability to compute filter functions of pulses assembled in qupulse, thereby allowing a user in the lab to easily inspect the noise susceptibility characteristics of the pulse they are currently applying to their device.

Example applications 19

We now present example applications of the software package and the formalism. As stated before, we focus on the decay amplitudes Γ and its associated filter functions and assume that the unitary errors generated by the frequency shifts Δ are either small (as is the case for gate fidelities) or calibrated out. All of the examples shown below are part of the software documentation as either interactive `Jupyter` notebooks [404] or Python scripts. In the following, we give angular frequencies and energies in units of inverse times (*e.g.* s^{-1}) while ordinary frequencies are given in Hz and we write $\langle\tilde{\mathcal{U}}(\tau)\rangle = \tilde{\mathcal{U}}$ for legibility.

19.1 Singlet-triplet two-qubit gates

In order to benchmark fidelity predictions of our implementation as well as demonstrate its application to nontrivial pulses, we compute the first-order infidelity of the two-qubit gates presented in Reference 405 and compare the results to the reference's Monte Carlo calculations. There, a numerically optimized gate set consisting of $\{\mathrm{X}_{\pi/2}\otimes\mathrm{I}, \mathrm{Y}_{\pi/2}\otimes\mathrm{I}, \mathrm{CNOT}\}$ for exchange-coupled singlet-triplet spin qubits is introduced, taking into account different noise spectra and realistic control hardware.

For readers unfamiliar with the reference we briefly summarize the physical system and noise model entering the optimization. The authors consider four electrons confined in a linear array of four quantum dots in a semiconductor heterostructure. Each electron $i\in\{1,2,3,4\}$ experiences a different static magnetic field B_i so that there is a gradient $b_{ij} = B_i - B_j$ between two adjacent dots i and j. This gives rise to spin quantization along the magnetic field axis and defines the eigenstates $\{|{\uparrow\downarrow}\rangle, |{\downarrow\uparrow}\rangle\}$ that span the computational subspace of a single qubit so that the accessible Hilbert space of the two-qubit system is spanned by $\{|{\uparrow\downarrow}\rangle, |{\downarrow\uparrow}\rangle\}^{\otimes 2}$. The magnitude of the exchange interaction J_{ij} between two adjacent dots i and j is controlled via gate electrodes located on top of the heterostructure that can be pulsed on a nanosecond timescale with an arbitrary waveform generator (AWG). Changing the gate voltages changes the detuning ϵ_{ij} of the electrochemical potential between dots and in turn leads to a change in exchange coupling according to the phenomenological model $J_{ij}(\epsilon_{ij}) \propto \exp\left(\epsilon_{ij}\right)$.

The pulses are defined by a set of discrete detuning voltages ϵ_{ij} passed to an AWG with a sample rate of 1 GS/s and constant magnetic field gradients b_{ij} are assumed. To reflect the fact that the qubits experience a different pulse than what is programmed into the AWG due to cable dispersion and non-ideal control hardware, the detunings are convoluted with an experimental impulse response [405]. Finally, the signal is discretized as piecewise constant by slicing each segment into five steps, yielding a time increment of $\Delta t = 0.2\,\mathrm{ns}$.

To find optimal detuning pulses, a Levenberg-Marquardt algorithm iteratively minimizes the infidelity, leakage, and trace distance from the target unitary. For the infidelity, contributions from quasistatic magnetic field noise as well as quasistatic and white charge noise are taken into account during each iteration. Because treating colored (correlated) noise using

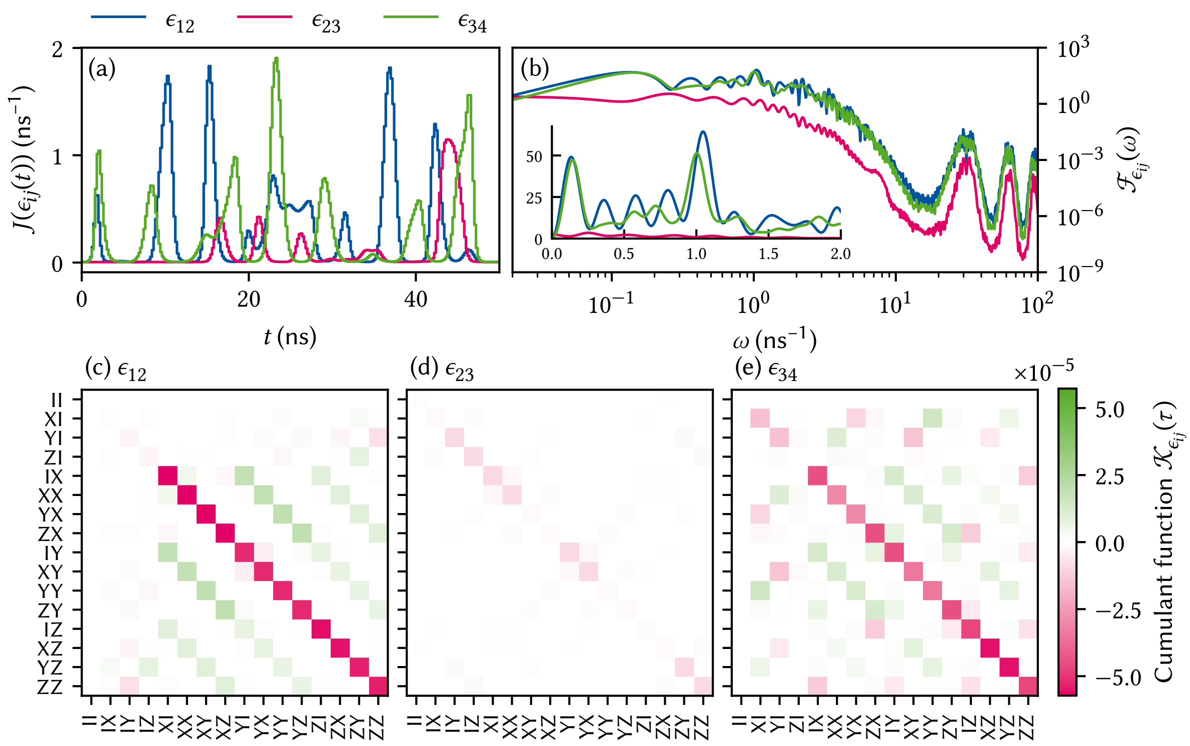

Figure 19.1: (a) Exchange interaction $J(\epsilon_{ij})$ for the CNOT gate presented in Reference 405 as function of time. (b) Filter functions $\mathcal{F}_{\epsilon_{ij}}$ for noise in the detunings evaluated on the computational subspace. The filter functions are modulated by oscillations at high frequencies due to numerical artifacts of the finite step size for the time evolution. The inset shows the filter functions in the DC regime on a linear scale with distinct peaks around $\omega = 2\pi/\tau$ and $\omega = 50/\tau$ ($\tau = 50\,\mathrm{ns}$). (c)–(e) Computational subspace block of the first order approximation of the error transfer matrix, given by the cumulant function $\mathcal{K}_{\alpha\alpha}$ excluding second order contributions, for the CNOT gate and the three detunings $\alpha \in \{\epsilon_{12}, \epsilon_{23}, \epsilon_{34}\}$. Note that in panel (e) the order of the rows and columns was permuted for better comparability.

Monte Carlo methods is computationally expensive (*cf.* Section 16.9), the infidelity due to fast $1/f$-like noise is only computed for the final gate and not used during the optimization.

Two-qubit interactions are mediated via the exchange J_{23} that makes the states $|{\uparrow\uparrow\downarrow\downarrow}\rangle$ and $|{\downarrow\downarrow\uparrow\uparrow}\rangle$ accessible. They constitute levels outside of the computational subspace that ideally should only be occupied during an entangling gate operation. A non-vanishing population of these states after the operation has ended is therefore unwanted and considered leakage, the magnitude of which we could quantify following Subsection 16.4.3. However, here we limit ourselves to determine the infidelity contribution from fast, *viz.* non-quasistatic, charge noise entering the system through ϵ_{ij}. That is, we consider noise sources $\alpha \in \{\epsilon_{12}, \epsilon_{23}, \epsilon_{34}\}$. We take the non-linear dependence of the Hamiltonian on the detunings ϵ_{ij} into account by setting $s_{\epsilon_{ij}}(t) = \partial J_{ij}(\epsilon_{ij}(t)) \big/ \partial\epsilon_{ij}(t) \propto J_{ij}(\epsilon_{ij}(t))$.

Figure 19.1 shows the filtered (convoluted) exchange interaction J_{ij} between each pair of dots during the pulse sequence in panel (a) and filter functions plotted as function of frequency in panel (b) for the three different detunings. For a detailed description on how the filter functions were computed in the presence of additional leakage levels refer to Subsection C.5.1. As one would expect from the fact that the intermediate (inter-qubit) exchange interaction J_{23} (orange dash-dotted lines) is only turned on for short times to entangle the qubits, the filter function for ϵ_{23} is smaller by roughly an order of magnitude than the intra-qubit ex-

	This work		Reference 405	
a	0	0.7	0	0.7
$X_{\pi/2} \otimes I$	1.7×10^{-3}	5.8×10^{-5}	1.9×10^{-3}	5.7×10^{-5}
$Y_{\pi/2} \otimes I$	1.6×10^{-3}	5.7×10^{-5}	1.7×10^{-3}	5.6×10^{-5}
CNOT	1.4×10^{-3}	6.3×10^{-5}	1.6×10^{-3}	6.3×10^{-5}

Table 19.1: Fast charge noise infidelity contributions to the total average gate fidelity of the two-qubit gate set from Reference 405 without capacitive coupling for GaAs S-T_0 qubits compared to the original results. The fidelities are consistent with results from the reference within the uncertainty bounds of 3 % of the Monte Carlo calculation. The infidelities presented here are all average gate infidelities (cf. Equation 16.44, Refences 379 and 380).

change filter functions. Notably, the filter functions for ϵ_{12} and ϵ_{34} show clear characteristics of DCGs, that is they drop to zero as $\omega \to 0$, and decouple from quasistatic noise with an error suppression $\propto \omega^2$. This is not unexpected as the optimization minimizes quasistatic noise contributions to the infidelity. In addition, one can also observe small oscillations with period $5\,\mathrm{ns}^{-1}$ in frequency space that arise as a numerical artifact of the piecewise constant discretization of the control parameters as investigations have shown. If high-frequency spectral components are expected to play a significant role, one needs to be aware of these effects and adjust the simulation parameters appropriately.

The inset of Figure 19.1(b) shows the same filter functions for the DC tail on a linear scale. Most notably, $\mathcal{F}_{\epsilon_{12}}$ and $\mathcal{F}_{\epsilon_{34}}$ have maxima around $\omega = 2\pi/\tau$, *i.e.* exactly the frequency matching the pulse duration, and around $\omega = 50/\tau = 1\,\mathrm{ns}^{-1}$ with $\tau_{\mathrm{CNOT}} = 50\,\mathrm{ns}$. The former is the typical window in which a pulse is most susceptible to noise whereas the latter matches the absolute value of the magnetic field gradients, $b_{12} = -b_{34} = 1\,\mathrm{ns}^{-1}$, indicating that the peak corresponds to the qubit dynamics generated by the magnetic field gradients. Panels (c)–(e) show the cumulant functions $\mathcal{K}_{\epsilon_{ij}}(\tau)$ of the detuning error channels ϵ_{ij} on the computational subspace. $\mathcal{K}_{\epsilon_{12}}$ displays clear characteristics of a Pauli channel with only elements on the diagonal and secondary diagonals deviating from zero significantly whereas $\mathcal{K}_{\epsilon_{34}}$ (the target qubit) possesses a more complicated structure.

We now compute the infidelity contribution originating from fast charge noise using Equation 16.44 but tracing only over the computational subspace to compare to the Monte Carlo calculations of Reference 405 (see Subsection C.5.1 for further details). Like the reference, we use a noise spectrum $S_{\epsilon,a}(f) \propto 1/f^a$ with $S_{\epsilon,a}(1\,\mathrm{MHz}) = 4{\times}10^{-20}\,\mathrm{V}^2/\mathrm{Hz}$ and consider white noise ($a = 0$) and correlated noise with $a = 0.7$ [11] with infrared and ultraviolet cutoffs $1/\tau$ and $100\,\mathrm{ns}^{-1}$, respectively. Table 19.1 compares the results in this work with the reference. The values computed here are consistent with the more elaborate Monte Carlo calculations within a few percent. Notably, the deviation is smaller for the smaller noise levels with $a = 0.7$, in line with the fact that we have only computed the contributions from the decay amplitudes Γ and thus the leading order perturbation.

19.2 Rabi driving

A widely used method for qubit control is Rabi driving [329, 333, 343, 406]. If we restrict ourselves to the resonant case for simplicity, the control Hamiltonian takes on the general form $H_c = \omega_0\sigma_z/2 +$

$A\sin(\omega_0 t + \phi)\sigma_x$. Here, ω_0 is the resonance frequency, A the drive amplitude corresponding to the Rabi frequency in the weak driving limit $A/\omega_0 \ll 1$, $\Omega_\mathrm{R} \approx A$, and ϕ an adjustable phase giving control over the rotation axis in the xy-plane of the Bloch sphere. This Hamiltonian and associated decoherence mechanisms are well-studied in the weak driving regime, where the rotating-wave approximation (RWA) can be applied to remove fast-oscillating terms in the rotating frame [407, 408]. There is a comprehensive understanding of how spectral densities transform to this frame and which frequencies are most relevant to loss of coherence [409].

By contrast, the description of a system in the strong driving regime, where $A/\omega_0 \sim 1$, is more complicated since the RWA cannot be applied without making large errors. Yet, an improved understanding is desirable because strong driving allows for much shorter gate times and thus shifts the window of relevant noise frequencies towards higher energies where the total noise power is typically lower, *e.g.* for $1/f$ noise. Conversely, faster control also requires more accurate timing to prevent rotation errors. It is therefore of interest to have tools that can provide a comprehensive picture for Rabi pulses over a wide range of driving amplitudes. By making use of the concatenation property of the filter functions, our formalism can do just that.

The problem that arises when trying to numerically investigate Rabi pulses in the weak driving regime in the lab frame is that typical control operations have a duration $\tau \gg T$ with $T = 2\pi/\omega_0$. Since the sampling time step Δt should additionally be chosen much smaller than a single drive period in order to sample the time evolution accurately ($\Delta t \ll T$), brute-force simulations are costly.

For $T/\Delta t = 100$ samples per period and assuming Rabi and drive frequencies in typical regimes for SiGe and MOS quantum dots [410, 411] or trapped ions [343], $\Omega_\mathrm{R} = 1\,\mathrm{\mu s}^{-1}$ and $\omega_0 = 20\,\mathrm{ns}^{-1}$, a Monte Carlo simulation of a π-rotation with approximately $3\,\%$ relative error would require 10^9 samples in total. Using the filter-function formalism, we can drastically reduce the simulation time even beyond the improvement gained from concatenating precomputed filter functions of individual drive periods using Equation 16.29. This can be achieved with Equation 16.53, which simplifies the calculation of the control matrix for periodic Hamiltonians.

To benchmark our implementation, we use the parameters from above and calculate the control matrix of a NOT gate generated by a Rabi Hamiltonian with three different methods on an Intel® Core™ i9-9900K eight-core processor. First, we use Equation 16.29 in a brute force approach. Second, we utilize the concatenation property following Equation 16.36. Third, we employ the simplified expression given by Equation 16.53. The brute force approach takes $250\,\mathrm{s}$ of wall time whereas calculating the filter function using the standard concatenation is faster by two orders of magnitude, taking $1.5\,\mathrm{s}$ to run. Lastly, the calculation utilizing the optimized method is faster again by two orders of magnitude and is completed in $0.056\,\mathrm{s}$.

As an example application, we calculate the filter functions for continuous Rabi driving in the weak and strong driving regimes. For weak driving, we use the parameters from the benchmark above for a pulse of duration $\tau_\mathrm{weak} \approx 20\,\mathrm{\mu s}$ that corresponds to 20 identity rotations in total. For the strong driving regime, we use the approximate analytical solution for a flux qubit biased at its symmetry point from Reference 412 with $A = \omega_0/4$ to drive the qubit for $\tau_\mathrm{strong} \approx 4\,\mathrm{ns}$ so that we achieve the same amount of identity rotations as in the weak driving case. In the reference, strong driving in this regime is shown to give rise to non-negligible counterrotating terms that modulate the Rabi oscillations and which are well-described by Floquet theory applied to the Rabi driving Hamiltonian. While for the regime studied here only two additional modes appear, the results extend to the regime where $A > \omega_0$ and up to eight different frequency components were observed.

Figure 19.2 shows the filter functions $\mathcal{F}_x$ and $\mathcal{F}_z$ for the σ_x and σ_z noise operators in the weak (a) and the strong (b) driving regime. Both display sharp peaks at their Rabi frequencies and the resonance frequency for $\mathcal{F}_z$ and $\mathcal{F}_x$, respectively. We expect these features as they correspond to perturbations of the qubit Hamiltonian that are resonant with the qubit dynamics about an axis orthogonal to them. For weak driving, $\mathcal{F}_x$ is constant up to the resonance frequency where it peaks sharply and then aligns with $\mathcal{F}_z$. The latter has a peak at the Rabi frequency before rolling off with ω^{-2} and a DC level that is almost ten orders of magnitude larger than that of the transverse filter function. This behavior is consistent with the results by Yan et al. [409], who show that the noise sources dominating decoherence during driven evolution are $S_x(\omega_0)$ and $S_z(\Omega_\mathrm{R})$. Note that the piecewise-constant control approximation causes the weak driving filter functions to level off towards low frequencies after an initial roll-off (here at $\omega \sim 1\,\mathrm{ms}^{-1}$). By decreasing the discretization time step Δt, one can shift the frequency at which this effect occurs to lower frequencies and thus attribute the feature to a numerical artefact of the approximation. However, the decoupling properties depend quite sensitively on the pulse duration.

In case of strong driving, the two filter functions are closer in amplitude for lower frequencies. In addition, $\mathcal{F}_x$ also peaks at $\omega = \omega_0 \pm \Omega_\mathrm{R}$. These peaks also show up at higher frequencies in the dephasing filter function $\mathcal{F}_z$, reflecting frequency mixing in the strong coupling regime. While both filter functions show characteristics of a DCG in the weak driving regime, that is they drop to zero as $\omega \to 0$, this is not the case in the strong driving regime. Instead, there they approach a constant level for small frequencies. On top of rotation errors from timing inaccuracies, we may thus expect naive strong driving gates to be more susceptible to quasistatic noise than weak driving gates. By shaping the pulse envelope of the strong driving gate the decoupling properties could be recovered.

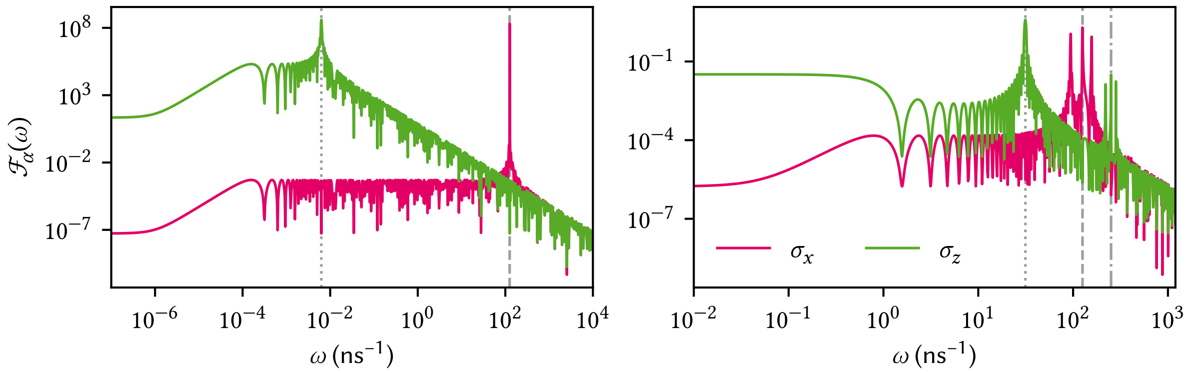

Figure 19.2: Filter functions for weak (a) and strong (b) Rabi driving (20 identity gates in total). gray dashed (dotted) lines indicate the respective drive (Rabi) frequencies ω_0 (Ω_R). (a) Weak driving with $A/\omega_0 \ll 1$. The filter function $\mathcal{F}_x$ for noise operator σ_x is approximately constant up to the resonance frequency where it peaks sharply and then aligns with the filter function $\mathcal{F}_z$ for σ_z. $\mathcal{F}_z$ peaks at the Rabi frequency before rolling off with ω^{-2} and a DC level that is almost ten orders of magnitude larger than the DC level of the transverse filter function $\mathcal{F}_x$. (b) Strong driving with $A/\omega_0 \sim 1$. Again $\mathcal{F}_z$ peaks at Ω_R whereas $\mathcal{F}_x$ has three distinct peaks at ω_0 and $\omega_0 \pm \Omega_\mathrm{R}$. These features also appear at $2\omega_0$ in $\mathcal{F}_z$ due to the strong coupling (dash-dotted line).

19.3 Randomized Benchmarking

RB and related methods, for example interleaved RB, are popular tools to assess the quality of a qubit system and the operations used to control it [366, 413, 414]. The basic protocol consists of constructing K random sequences of varying length m of gates drawn from the Clifford group,[1] and appending a final inversion gate so that the identity operation should be performed in total. Each of these pulse sequences is applied to an initial state $|\psi\rangle$ in order to measure the survival probability $p(|\psi\rangle)$ after the sequence. In reality, the applied operations are subject to noise and experimental imprecisions. This renders them imperfect and results in a survival probability smaller than one. Assuming gate-independent errors, the average gate fidelity F_avg is then obtained by fitting the measured survival probabilities for each sequence length to the zeroth-order exponential model [366]

1: The Clifford group is a subgroup of the special unitary group with the advantage that compositions are easy to compute and that averaging over all unitaries can under reasonable assumptions be replaced by averaging over all Cliffords. This makes the Clifford gates a convenient choice for benchmarking. For a nice, short introduction as well as further references, see [415].

$$p(|\psi\rangle) = A\left(1 - \frac{dr}{d-1}\right)^m + B, \tag{19.1}$$

where $r = 1 - F_\mathrm{avg}$ is the average error per single gate to be extracted from the fit, A and B are parameters capturing SPAM errors, and d is the dimensionality of the system.

One of the main assumptions of the RB protocol is that temporal correlations of the noise are small on timescales longer than the average gate time [366]. If this requirement is not satisfied, *e.g.* if $1/f$ noise plays a dominant role, the decay of the sequence fidelity can have non-exponential components [416–418] and a single exponential fit will not produce the true average gate fidelity [419, 420]. The filter-function formalism suggests itself to numerically probe RB experiments in such systems for two reasons. First, it enables the study of gate performance subject to noise with correlation times longer than individual gate times. This regime, where a simple description in terms of individual, isolated quantum operations fails, is accessible in the filter-function formalism because universal classical noise can be included by the power spectral density $S(\omega)$. Second, the simulation of a RB experiment can be performed efficiently by using the concatenation property. Because RB sequences are compiled from a limited set of gates whose filter functions may be precomputed, one only needs to concatenate m filter functions for a single sequence of length m to gain access to the survival probability.

Since for sufficiently long RB sequences $r \in \mathcal{O}(1)$, and we would need to include the frequency shifts Δ in a full simulation following Equation 16.9 because the low-noise approximation Equation 16.43 does not hold in this regime. Unfortunately, the concatenation property does not hold for Δ. Therefore, we focus on the high-fidelity regime where the exponential decay of the sequence fidelity may be approximated to linear order and only the decay amplitudes Γ need to be considered.

In order to evaluate the survival probability of a RB experiment using filter functions, we employ the state fidelity from Subsection 16.4.2 and focus on the single-qubit case with $d = 2$ and the (normalized) Pauli basis from Equation 16.54. Because the ideal action of a RB sequence is the identity we have $\mathcal{Q} = \mathbb{1}$. Assuming we prepare and measure in the computational basis, $|\psi\rangle \in \{|0\rangle, |1\rangle\}$ so that $\sqrt{2}\,|\rho\rangle\!\rangle = |\sigma_0\rangle\!\rangle \pm |\sigma_3\rangle\!\rangle$, we simplify Equation 16.50 to

$$\begin{aligned} F(|\psi\rangle, \mathcal{U}_{\mathrm{RB}}(|\psi\rangle\!\langle\psi|)) &= \frac{1}{2}\big(\tilde{\mathcal{U}}_{00} + \tilde{\mathcal{U}}_{33} \pm \tilde{\mathcal{U}}_{03} \pm \tilde{\mathcal{U}}_{30}\big) = \frac{1 + \tilde{\mathcal{U}}_{33}}{2} \\ &\approx 1 - \frac{1}{2}\sum_{k\neq 3} \Gamma_{kk}. \end{aligned} \tag{19.2}$$

For the second equality we used that $\tilde{\mathcal{U}}$ is trace-preserving and unital (*cf.* Subsection 16.1.2) while in the last step we approximated the expression using Equations 16.22 and 16.43. For our simulation, we neglect SPAM errors so that $A = B = 0.5$, choose $|\psi\rangle = |0\rangle$, and approximate Equation 19.1 as

$$p(|0\rangle) = F(|0\rangle, \mathcal{U}_{\mathrm{RB}}(|0\rangle\!\langle 0|)) \approx 1 - rm \tag{19.3}$$

for small gate errors $r \ll 1$ since this is the regime which we can efficiently simulate using the concatenation property.

We simulate single-qubit RB experiments using three different gate sets to generate the 24 elements of the Clifford group. For the first gate set we implement the group by naive "single" rotations about the symmetry axes of the cube. Each pulse corresponds to a single time segment during which one rotation is performed so that the jth element is given by $Q_j = \exp\big(-i\phi_j \mathbf{n}_j \cdot \boldsymbol{\sigma}\big)$. We compile the other two gate sets from primitive $\pi/2$ x- and y-rotations so that on average each Clifford gate consists of 3.75 primitive gates (see Reference 421). For the specific implementation of the primitive $\pi/2$-gates we compare "naive" rotations, *i.e.* with a single time segment so that $Q_j = \exp\big(-i\pi\sigma_j/4\big)$ for $j \in \{x, y\}$, and the "optimized" gates from Reference 405. Pulse durations are chosen such that the average duration of all 24 Clifford gates generated from a single gate set is equal for all three gate sets. This is to ensure that the different implementations of the Clifford gates are sensitive to the same noise frequencies.

We investigate white noise and correlated noise with $S(\omega) \propto \omega^{-0.7}$ assuming the same noise spectrum on each Cartesian axis of the Bloch sphere and normalize the noise power for each gate set and noise type (white and correlated) so that the average Clifford infidelity r is the same throughout. We then randomly draw $K = 100$ sequences for 11 different lengths $m \in [1, 101]$ and concatenate the m Clifford gates using Equation 16.29 to compute the control matrix of the entire sequence. For the integral in Equation 16.24 we choose the ultraviolet cutoff frequency two orders of magnitude above the inverse duration of the short-

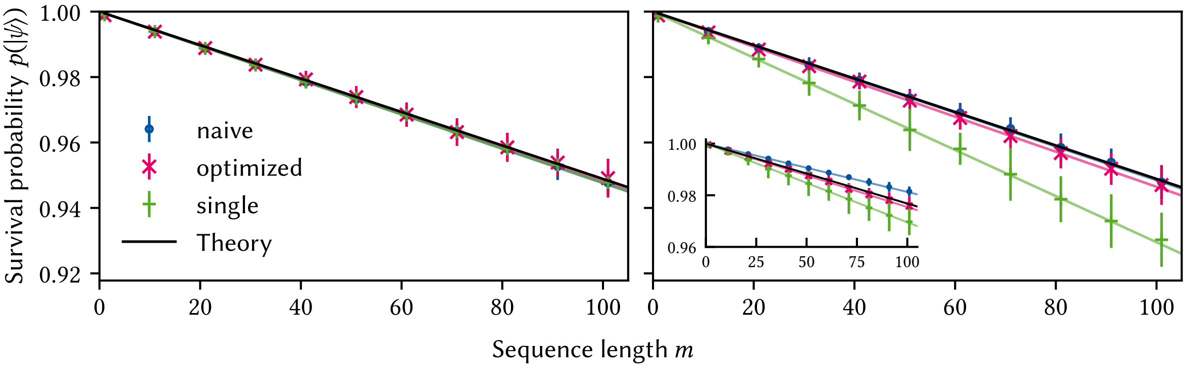

Figure 19.3: Simulation of a RB experiment using 100 random sequences per point for different gate and noise types (see the main text for an explanation of the gate type monikers). Dashed lines are fits of Equation 19.3 to the data while the solid black lines correspond to a zeroth-order RB model with $A = B = 0.5$ and the true average gate infidelity per Clifford r. Errorbars show the standard deviation of the RB sequence fidelities, illustrating that for the "single" gate set noise correlations can lead to amplified destructive and constructive interference of errors. The same noise spectrum is used for all three error channels (σ_x, σ_y, σ_z) and the large plots show the sum of all contributions. (a) Uncorrelated white noise with the noise power adjusted for each gate type so that the average error per gate r is constant over all gate types. No notable deviation is seen between different gate types. (b) Correlated $1/f$-like noise with noise power adjusted to match the average Clifford fidelity in (a). The decay of the "single" gate set differs considerably from that of the other gate sets and the RB decay expected for the given average gate fidelity, whereas "naive" and "optimized" gates match the zeroth order RB model well, indicating that correlations in the noise affect the relation between RB decay and average gate fidelity in a gate-set–dependent way. Inset: contributions from σ_z-noise show that the sequence fidelity can be better than expected for certain gate types and noise channels.

est pulse, $f_{\mathrm{UV}} = 10^2/\tau_{\mathrm{min}}$. Similarly, the infrared cutoff is chosen as $f_{\mathrm{IR}} = 10^{-2}/m_{\mathrm{max}}\tau_{\mathrm{max}}$ with $m_{\mathrm{max}} = 101$ and $\tau_{\mathrm{max}} = 7\tau_{\mathrm{min}}$ (since the longest gate is compiled from seven primitive gates with duration τ_{min}) to guarantee that all nontrivial structure of the filter functions is adequately resolved at small frequencies.[2] Finally, we fit Equation 19.3 to the infidelities computed for the different noise spectra.

2: For a precise fidelity estimate, the infrared cutoff should be extended to $f = 0$. However, we are only interested in a qualitative picture and neglect this part of the spectrum here. At frequencies much smaller than $\approx 1/\tau$ where τ is the duration of the entire control operation, the filter function is constant and we therefore do not disregard any interesting features by setting $f_{\mathrm{IR}} = 10^{-2}/\tau = 10^{-2}/m_{\mathrm{max}}\tau_{\mathrm{max}}$.

The results of the simulation are shown in Figure 19.3(a) and (b) for white and correlated noise, respectively. For white noise, the survival probability agrees well with the RB prediction for all gate types whereas for $1/f$-like noise the "single" gates (green pluses) deviate considerably. Hence, fitting the zeroth-order RB model to such data will not reveal the true average gate fidelity although errors are of order unity. We note that Refences 416 and 354 found similar results using different methods for $1/f$ and perfectly correlated DC noise, respectively. The former observed RB to estimate r within 25 % and the latter found the mean of the RB fidelity distribution to deviate from the mode, thereby giving rise to incorrectly estimated fidelities.

On top of affirming the findings by the references, our results demonstrate that the accuracy of the predictions made by RB theory, *i.e.* that the RB decay rate directly corresponds to the average error rate of the gates, not only depends on the gate implementation but also on which error channels are assumed. This can be seen from the inset of Figure 19.3(b), where only dephasing noise (σ_z) contributions are shown. For this noise channel and the "naive" gates (blue points), one finds a slower RB decay than expected from the actual average gate fidelity, so that the latter would be overestimated by an RB experiment, whereas the "single" gates show the opposite behavior. Depending on the gate set and relevant error channels, non-Markovian noise may thus even lead to improved sequence fidelities due to errors interfering destructively. This behavior is captured by the pulse correlation filter functions whose contributions to the sequence fidelity lead to the deviations from the RB prediction.

Notably, the data for the "optimized" gates (magenta crosses) agree with the prediction for every noise channel individually which implies that correlations between pulses are suppressed. This highlights the formalism's attractiveness for numerical gate optimization as the pulse correlation filter functions $\mathcal{F}^{(gg')}(\omega)$ may be exploited to suppress correlation errors. To be more explicit, the correlation decay amplitudes $\Gamma^{(gg')}$ from Equation 16.31 can be used to construct cost functions for quantum optimal control algorithms like GRAPE [422, 423] or CRAB [424]. By constructing linear combinations of $\Gamma^{(gg')}$ with different pulse indices g and g', correlations between any number of pulses can be specifically targeted and suppressed using numerical pulse optimization.

19.4 Quantum Fourier transform

To demonstrate the flexibility of our software implementation, we calculate filter functions for a four-qubit quantum Fourier transform (QFT) [151, 425] circuit. The QFT routine plays an important role in many quantum algorithms such as Shor's algorithm [426] and quantum phase estimation [151]. For the underlying gate set, we assume a standard Rabi driving model with IQ control and nearest neighbor exchange. That is, we assume full control of the x- and y-axes of the individual qubits as well as the exchange interaction mediating coupling between two neighboring qubits. This system allows for native access to the minimal gate set $\mathbb{G} = \{\mathrm{X}_i(\pi/2), \mathrm{Y}_i(\pi/2), \mathrm{CR}_{ij}(\pi/2^3)\}$ where $\mathrm{CR}_{ij}(\phi)$ denotes a controlled rotation by ϕ about z with control qubit i and target qubit j. Controlled-z rotations by angles $\pi/2^m$ as required for the QFT can thus be obtained by concatenating 2^{3-m} minimal gates $\mathrm{CR}_{ij}(\pi/2^3)$.

Despite native access to all necessary gates, we employ `QuTiP`'s implementation [359] of the GRAPE algorithm [422, 423] to generate the gates in order to highlight our method's suitability for numerically optimized pulses. For the optimization we choose a time step of $\Delta t = 1\,\mathrm{ns}$ and a total gate duration of $\tau = 30\,\mathrm{ns}$. For completeness, see Subsection C.5.2 for details on the optimized gates. We then construct the remaining required gates by sequencing these elementary gates, *i.e.* the Hadamard gate $\mathrm{H}_i = \mathrm{X}_i(\pi/2) \circ \mathrm{X}_i(\pi/2) \circ \mathrm{Y}_i(\pi/2)$, where $\mathrm{B} \circ \mathrm{A}$ denotes the composition of gates A and B such that gate A is executed before gate B. To map the canonical circuit [151] onto our specific qubit layout with only nearest-neighbor coupling, we furthermore introduce SWAP operations to couple distant qubits. These gates can be implemented by three CNOTs, $\mathrm{SWAP}_{ij} = \mathrm{CNOT}_{ij} \circ \mathrm{CNOT}_{ji} \circ \mathrm{CNOT}_{ij}$. The CNOTs in turn are obtained by a Hadamard transform of the controlled phase gate, $\mathrm{CNOT}_{ij} = \mathrm{H}_j \circ \mathrm{CR}_{ij}(\pi) \circ \mathrm{H}_j$. The complete quantum circuit is shown at the top of Figure 19.4; for the canonical circuit with all-to-all connectivity refer to Reference 151. In total, there are 442 elementary pulses, 198 of which are required for the three SWAPs on the first two qubits, so that the entire algorithm would take $\sim 13\,\mu\mathrm{s}$ to run. Note that the circuit could be compressed in time by parallelizing some operations but for simplicity we only execute gates sequentially and do not execute dedicated idling gates.

In order to leverage the extensibility of the filter function approach (see Section 16.7), we use a Pauli basis for the pulses and proceed as follows:

1. Instantiate the `PulseSequence` objects for the elementary gates $\mathbb{G}$ for the first two qubits and cache the control matrices.
2. Compile all required single- and two-qubit pulses by concatenating the `PulseSequences` that implement $\mathbb{G}$.

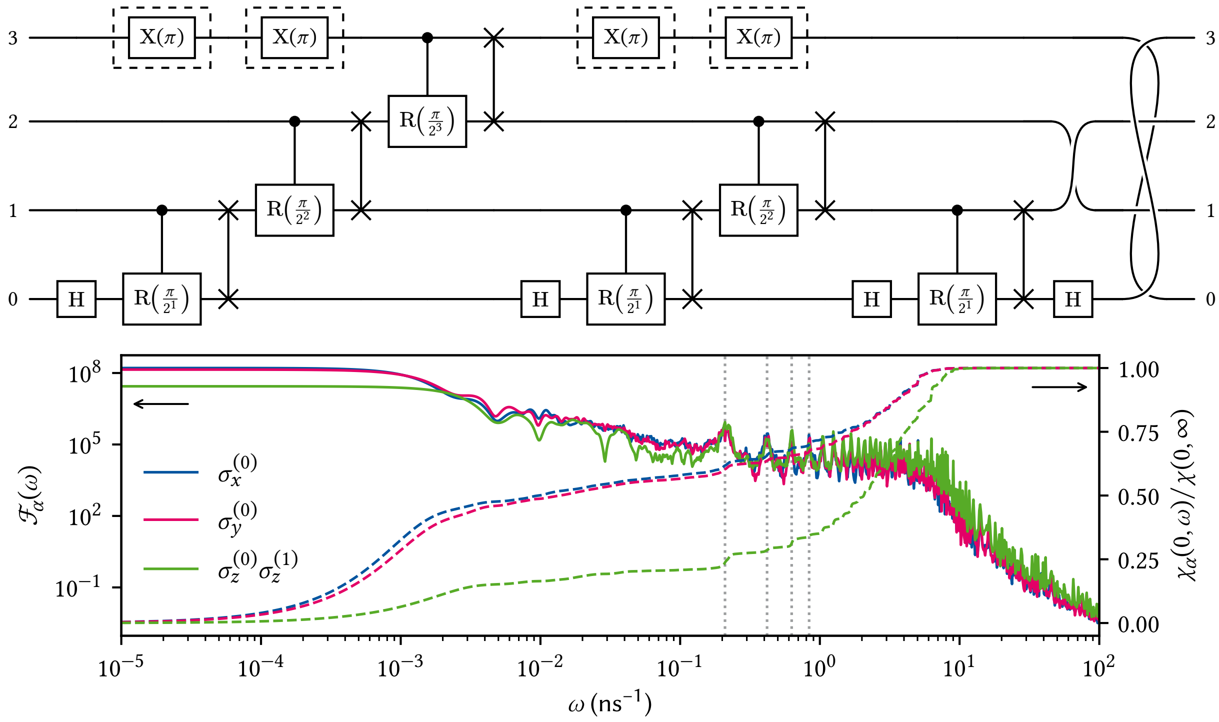

Figure 19.4: Top: Circuit for a QFT on four qubits with nearest-neighbor coupling. Labels next to the wires indicate the qubit index with the permutations at the end of the circuit replacing the final SWAP operations. X-gates in dashed boxes indicate optional echo pulses that leave the action nominally unchanged. Bottom: Filter functions for noise operators on the first qubit ($i = 0$) without echoes. Dotted gray lines indicate the positions of the nth harmonic, $\omega_n = 2\pi n/\tau$ with $\tau = 30\,\mathrm{ns}$ the duration of the gates in $\mathbb{G}$, for $n \in \{1, 2, 3, 4\}$. The filter functions have a baseline of around 10^4 in the range $\omega \in [10^{-1}, 10^1]\,\mathrm{ns}^{-1}$ before they drop down to follow the usual $1/\omega^2$ behavior. The dashed lines show the error sensitivities $\chi_\alpha(\omega_1, \omega_2) := \int_{\omega_1}^{\omega_2} \mathrm{d}\omega\, \mathcal{F}_\alpha(\omega)$ in the frequency band $[0, \omega]$ as a fraction of the total sensitivity $\chi_\alpha(0, \infty)$. These are closely related to the entanglement fidelity (*cf.* Equations 16.26 and 16.49) and suggest that high frequencies up to the knee at $\omega \approx 10\,\mathrm{ns}^{-1}$ cannot be neglected if the cutoff frequency of the noise is sufficiently high or the spectrum does not drop off quickly enough (note the linear scale as opposed to the logarithmic scale for the filter functions).

3. Extend the `PulseSequences` to the full four-qubit Hilbert space.
4. Recursively concatenate recurring gate sequences by concatenating four-qubit `PulseSequences`, *e.g.* $\mathrm{SWAP}_{10} \circ \mathrm{CR}_{10}(\pi/2^1) \circ \mathrm{H}_0$, in order to optimally use the performance benefit offered by Equation 16.29.
5. Concatenate the last `PulseSequences` to get the complete QFT pulse.

For our gate parameters and 400 frequency points, this procedure takes around 5 s on an Intel® Core™ i9-9900K eight-core processor, whereas computing the filter functions naively using Equation 16.36 takes around 4 min. The resulting filter functions are shown in Figure 19.4 for the noise operators affecting the first qubit. Evidently, the fidelity of the algorithm is most susceptible to DC noise; below roughly $1\,\mu\mathrm{s}^{-1}$ the filter functions level off at their maximum value. In the GHz range there is a plateau with sharp peaks corresponding to the n harmonics of the inverse pulse duration $\omega_n = 2\pi n/\tau$, where the leftmost belongs to $n = 1$. The dashed lines show the error sensitivities $\chi_\alpha(\omega_1, \omega_2) := \int_{\omega_1}^{\omega_2} \mathrm{d}\omega\, \mathcal{F}(\omega)$ in the frequency band $[0, \omega]$ relative to the total sensitivity $\chi_\alpha(0, \infty)$. For a white spectrum, *i.e.* $S(\omega) = \mathrm{const.}$, this quantifies the fraction of the total entanglement infidelity that is accumulated up to frequency ω (*cf.* Equations 16.26 and 16.49). Thus, to obtain a precise estimate of the algorithm's fidelity, five frequency decades need to be taken into account.

To obtain insight into the efficacy of running DD sequences on idling

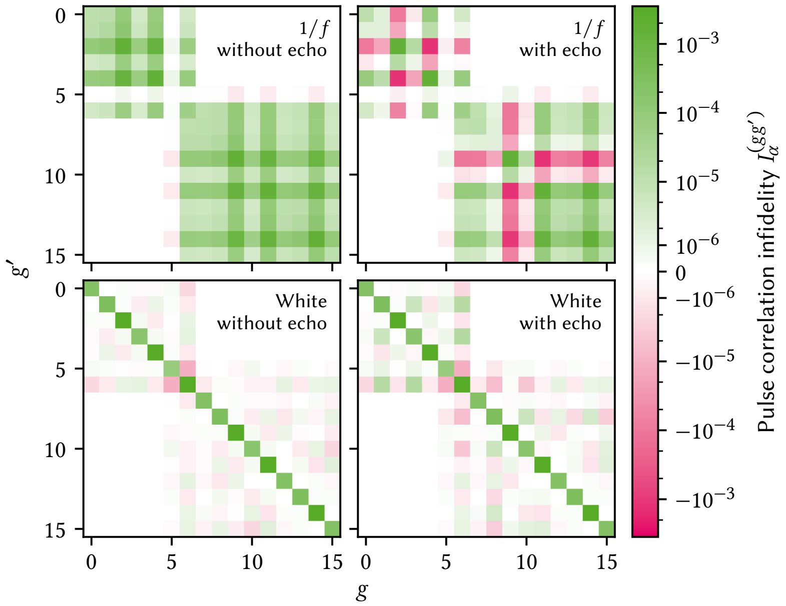

Figure 19.5: Pulse correlation infidelity contributions $I^{(gg')}$ for the noise operator $\sigma_y^{(3)}$ and different noise types (rows) and circuits (columns). The right column corresponds to the circuit with the $\mathrm{X}(\pi)$ pulses indicated with dashed boxes in Figure 19.4 inserted, causing echo effects that reduce the overall infidelity for correlated noise. For uncorrelated noise, their effect is negligible.

qubits in an algorithm, we insert spin echoes (two $\mathrm{X}(\pi)$ gates amounting to an identity gate) on the fourth qubit before and after the gate operations taking place on this qubit. Exploiting the fact that we assemble the final filter function by concatenating the filter functions of each of the 16 clock cycles, we compute the pulse correlation infidelities (*cf.* Equations 16.31 and 16.48 as well as Equation 11 of Reference 318),

$$I_{\alpha\beta}^{(gg')} = \frac{1}{d}\operatorname{tr}\Gamma_{\alpha\beta}^{(gg')}, \qquad g, g' \in \{0, 1, \dots, 15\} \tag{19.4}$$

for $\alpha = \beta \equiv \sigma_y^{(3)}$. The off-diagonal elements of this matrix ($g \neq g'$) capture correlation effects that arise from the concatenation operation alone, while the diagonal elements correspond to the contributions from each single pulse[3] individually. In particular, off-diagonal contributions can be negative, leading to a reduction in infidelity (an enhancement in fidelity) compared to the naive picture of simply adding all individual infidelities,[4] $I = \sum_g I^{(gg)}$. For uncorrelated (white) noise, we expect $I^{(gg')} \approx 0$ for $g \neq g'$ because the noise has no "memory", that is, cannot transport effects from one time step to another. Indeed, the sum rule for fidelity filter functions, which we prove in Subsection C.1.4, states that for white noise the fidelity depends only on the total duration τ and can therefore not be influenced by whichever control operation happened before or after. This stands in contrast to correlated (*e.g.*, $1/f$) noise, where interference can cause random phases picked up at one point in the sequence be reversed at another [318].

3: Or here rather: each single clock cycle of the circuit.

4: Recall that we are approximating the error process to linear order so that $F \sim \prod_g \exp(1 - \epsilon_g) \approx 1 - \sum_g \epsilon_g$.

In Figure 19.5, we show the correlation infidelities $I^{(gg')}$ computed for the QFT algorithm shown in Figure 19.4. The figure shows results for $1/f$ noise in the top row, and white noise in the bottom, while the first column shows the algorithm without the echo pulses and the second with. To facilitate a qualitative comparison, the noise level in the simulation was adjusted so that for white noise the total infidelity without echoes was the same as for $1/f$ noise. The infidelity in the case of white noise is indeed dominated by the diagonal elements as we would expect. Adding decoupling pulses does not change the fidelity significantly as the addition does not change the total duration of the circuit. In the case of $1/f$

noise, however, the picture is quite different. Here, the overall fidelity is reduced in the circuit without echos as there are significant contributions from correlations at $g \neq g'$. Adding the echo pulses causes some of these contributions to change sign and lead to a reduction in total infidelity of this noise channel by a factor of five. In Subsection C.5.2, we perform MC and GKSL master equation simulations of the QFT circuit to validate the fidelity predictions made by the filter-function formalism.

These insights demonstrate that our method represents a useful tool to analyze how and to which degree small algorithms are affected by correlated errors, and how this effect depends on the gate implementation. It could thus also be used to choose or optimize gates in an algorithm-specific way.

Conclusion and outlook 20

Before we conclude, let us address two possible avenues for future work, one for the formalism itself and one for its application. To extend our approach to the filter-function formalism beyond the scope discussed in this work, the most evident path forward is to allow for quantum mechanical baths instead of purely classical ones. Such an extension would facilitate studying for example non-unital T_1-like processes. In fact, the filter-function formalism was originally introduced considering quantum baths such as spin-boson models [337, 338] or more general baths [7, 336, 427], but it remains an open question whether this can be applied to our presentation of the formalism and the numerical implementation in particular. In a fully quantum-mechanical treatment, (sufficiently weak) noise coupling into the quantum system can be modelled via a set of bath operators $\{D_\alpha(t)\}_\alpha$ so that $H_\mathrm{n}(t) = \sum_\alpha B_\alpha(t) \otimes D_\alpha(t)$ (the classical case is recovered by replacing $D_\alpha(t) \rightarrow b_\alpha(t)\mathbb{1}$) [428]. Accordingly, the ensemble average over the stochastic bath variables $\{b_\alpha(t)\}_\alpha$ needs to be replaced by the quantum expectation value $\mathrm{tr}_B(\,\bullet\,\rho_B)$ with respect to the state ρ_B of the bath B. One therefore needs to deal with correlation functions of bath operators instead of stochastic variables. An immediate consequence for numerical applications is hence an increased dimensionality of the system, which could be dealt with by using analytical expressions for the partial trace over the bath.

For future applications of our method, it would be interesting to study the effects of noise correlations on the performance of quantum error correction (QEC) schemes [42, 43, 429–431]. While extensive research has been conducted on QEC [201], noise is usually assumed to be uncorrelated between error correction cycles. In this respect, our formalism may shed light on effects that need to be taken into account for a realistic description of the protocol. As outlined above, we can compute expectation values of (stabilizer) measurements in a straightforward manner from the error transfer matrix. Unfortunately, this implies performing the ensemble average over different noise realizations, therefore removing all correlations between subsequent measurement outcomes for a given noise realization. Hence, the same feature that allows us to calculate the quantum process for correlated noise, namely that we compute only the final map by averaging over all "paths" leading to it, prevents us from studying correlations between consecutive cycles. To overcome this limitation in the context of quantum memory one could invoke the principle of deferred measurement [151] and move all measurements to the end of the circuit, replacing classically controlled operations dependent on the measurement outcomes by conditional quantum operations. Alternatively, to incorporate the probabilistic nature of measurements, one could devise a branching model that implements the classically controlled recovery operation by following both conditional branches of measurement outcomes with weights corresponding to the measurement probabilities as computed from the ensemble-averaged error transfer matrix. An intriguing connection also exists to the quantum Zeno effect, for which quantum systems subject to periodic projective measurements have been identified with a filter function [336, 432, 433] and which recently has been shown to *improve* error correction in the presence of non-Markovian noise [434].

As quantum control schemes become more sophisticated and take into account realistic hardware constraints and sequencing effects, their analytical description becomes cumbersome, making numerical tools invaluable for analyzing pulse performance. In the above, we have shown that the filter-function formalism lends itself naturally to these tasks since the central objects of our formulation, the interaction picture noise operators, obey a simple composition rule which can be utilized to efficiently calculate them for a sequence of quantum gates. Because the nature of the noise is encoded in a power spectral density in the frequency domain, its effects are isolated from the description of the control until they are evaluated by the overlap integral of noise spectrum and filter function. Hence, the noise operators are highly reusable in calculations and can serve as an economic way of simulating pulse sequences.

Building on the results of Reference 318, we have presented a general framework to study decoherence mechanisms and pulse correlations in quantum systems coupled to generic classical noise environments. By combining the quantum-operations and filter-function formalisms, we have shown in Chapter 16 how to compute the Liouville representation of the error channel of an arbitrary control operation in the presence of Gaussian noise. This expression is exact in the case of commuting noise Hamiltonians, and we have investigated the error of the perturbative expansion in all other cases in Chapter 17, finding that the filter functions for single elements of the error transfer matrix deviate from the exact results at very low frequency and close to zeros of the functions. Furthermore, we have introduced the `filter_functions` Python software package that implements the aforementioned method in Chapter 18. We showed both analytically and numerically that our software implementation can outperform Monte Carlo techniques by orders of magnitude. By employing the formalism and software to study several examples in Chapter 19, we demonstrated the wide range of possible applications.

The capacity for applications in quantum optimal control has already been established above. In Reference 401, we presented analytical derivatives for the fidelity filter function, Equation 16.26, and their implementation in the software package. Together with the infidelity, Equation 16.49, they can serve as efficient cost functions for pulse optimization in the presence of realistic, correlated noise [360]. Since our method offers insight into correlations between pulses at different positions in a sequence, the pulse correlation filter function $F^{(gg')}(\omega)$ with $g \neq g'$ can additionally serve as a tool for studying under which conditions pulses decouple from noise with long correlation times. Such insight would be valuable to design pulses for algorithms. Another interesting application could be quantum error correction in the regime of long-time correlated noise as outlined above, where we also briefly touched upon a possible extension of the framework to quantum mechanical baths.

The tools presented here, both analytical and numerical as implemented in the `filter_functions` software package [358], provide an accessible way for computing filter functions in generic control settings across the different material platforms employed in quantum technologies and beyond.

Appendix

Supplementary to Part II: Optical coupling and vibrations

A

In this appendix, I give additional information on the optical coupling efficiency of the confocal microscope described in Chapter 7 as well as the knife-edge measurement used to calibrate the optical vibration spectroscopy technique presented in Chapter 8.

A.1 Optical coupling

A.1.1 Collection efficiency

In this subsection, I derive the expression for the collection efficiency η_c, Equation 7.19, for radiation emitted from a dipole oriented in-plane inside a semiconductor quantum well (QW) with refractive index n and collected with a lens of a given numerical aperture (NA). The dipole fields in spherical coordinates (r, ϑ, φ) in the coordinate system (z, x, y) are given in Equations 7.10 and 7.11.[1] Since we observe the dipole from the side, *i. e.*, perpendicular to its axis, it is useful to rotate the spherical coordinates $(r, \vartheta, \varphi) \to (r, \theta, \phi)$ aligned with the cartesian coordinate system (x, y, z) in Figure 7.3. To this end, observe that with the spherical coordinates

1: That is,

$$
\begin{aligned}
z &= r\sin\vartheta\sin\varphi\\
x &= r\sin\vartheta\cos\varphi\\
y &= r\cos\vartheta.
\end{aligned}
$$

$$x = r\sin\theta\sin\phi \tag{A.1}$$
$$y = r\sin\theta\cos\phi \tag{A.2}$$
$$z = r\cos\theta \tag{A.3}$$

we can express the angles in the dipole coordinate system, (ϑ, φ), in terms of angles of the sample coordinate system, (θ, ϕ), by

$$
\begin{aligned}
\vartheta &= \arctan\left(\sqrt{z^2+y^2}/x\right)\\
&= \arctan\left(\csc\phi\sqrt{\sin^2\phi+\cot^2\theta}\right)
\end{aligned} \tag{A.4}
$$

$$
\begin{aligned}
\varphi &= \arctan(y/z)\\
&= \arctan(\sin\phi\tan\theta).
\end{aligned} \tag{A.5}
$$

The time-averaged Poynting vector, Equation 7.18, is then

$$
\begin{aligned}
\langle \mathrm{S}(r,\theta,\phi)\rangle &= \frac{1}{2}\operatorname{Re}\left[\mathbf{E}(r,\vartheta)\times\mathbf{H}^*(r,\vartheta)\right]\\
&= \frac{|\mathbf{p}|^2ck^4}{32\pi^2\epsilon r^2}\sin^2\vartheta\,\hat{\mathbf{e}}_r\\
&= \frac{|\mathbf{p}|^2ck^4}{32\pi^2\epsilon r^2}\left[\cos^2\theta+\sin^2\theta\sin^2\phi\right]\hat{\mathbf{e}}_r.
\end{aligned} \tag{A.6}
$$

Evaluating the integral in Equation 7.17 for the escape cone angle $\theta_m = \arcsin(\mathrm{NA}/n)$ thus yields

$$
\begin{aligned}
P_m &= \int_0^{\theta_m} d\theta\sin\theta\int_0^{2\pi} d\phi\,\langle \mathrm{S}(r,\theta,\phi)\rangle\\
&= \frac{|\mathbf{p}|^2ck^4}{32\pi^2\epsilon r^2}\left[\frac{4\pi}{3}-\frac{\pi}{3n^3}\left(4n^2-\mathrm{NA}^2\right)\sqrt{n^2-\mathrm{NA}^2}\right],
\end{aligned} \tag{A.7}
$$

whereas the total emitted power is the well-known expression

$$\begin{aligned} P_{\text{tot}} &= \int_0^{\pi} \mathrm{d}\theta \sin\theta \int_0^{2\pi} \mathrm{d}\phi \, \langle \mathrm{S}(r,\theta,\phi) \rangle \\ &= \frac{|\mathbf{p}|^2 c k^4}{12\pi\epsilon r^2}. \end{aligned} \tag{A.8}$$

Together, we therefore find

$$\eta_{\mathrm{c}} = \frac{P_{\mathrm{m}}}{P_{\text{tot}}} = \frac{1}{2} - \frac{1}{8n^3}\left(4n^2 - \mathrm{NA}^2\right)\sqrt{n^2 - \mathrm{NA}^2} \tag{A.9}$$

as given in the main text.

A.1.2 Mode profile

In this subsection, I derive the electric field of a dipole situated inside a dielectric slab and oriented parallel to the surface outside of the dielectric. We use the convention of spherical coordinates from the previous subsection and consider the electric field given in Equation 7.10 in the rotated coordinate system (r, θ, ϕ) by performing the substitutions given in Equations A.4 and A.5, obtaining

$$\mathbf{E}(r,\theta,\phi) = A(r)\left[E_r(r,\theta,\phi)\hat{\mathbf{e}}_r + E_\theta(r,\theta,\phi)\hat{\mathbf{e}}_\theta + E_\phi(r,\theta,\phi)\hat{\mathbf{e}}_\phi\right] \tag{A.10}$$

with

$$E_r(r,\theta,\phi) = f_r(k,r)\sin\theta\cos\phi \tag{A.11}$$

$$E_\theta(r,\theta,\phi) = -f_\vartheta(k,r)\cos\theta\cos\phi \tag{A.12}$$

$$E_\phi(r,\theta,\phi) = f_\vartheta(k,r)\sin\phi \tag{A.13}$$

where we defined

$$f_r(k,r) = -\frac{2i}{kr} + \frac{2}{k^2r^2} \tag{A.14}$$

$$f_\vartheta(k,r) = -1 - \frac{i}{kr} + \frac{1}{k^2r^2} \tag{A.15}$$

for conciseness. Now, when the electric field impinges on the surface of the slab, two things must be taken into account: first, it gets refracted, meaning that $\mathbf{k}$ is transformed according to Snell's law (Equation 7.16), and second, the transmitted field is modified with the Fresnel transmission coefficients [87]

$$t_s = \frac{2\sin\theta\sin\theta'}{\sin(\theta+\theta')} = \frac{2n\cos\theta}{n\cos\theta + \nu(\theta)} \tag{A.16}$$

$$t_p = \frac{t_s}{\cos(\theta-\theta')} = \frac{2n\cos\theta}{\cos\theta + n\nu(\theta)} \tag{A.17}$$

where s (p) stands for the component perpendicular (parallel) to the plane of incidence and we defined

$$\nu(\theta) = \sqrt{1 - n^2\sin^2\theta}. \tag{A.18}$$

In fact, $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ lie in the p plane and $\hat{\mathbf{e}}_\phi$ is parallel to s so that transmission through the interface transforms the electric field components as

$$E_r \to E'_r = t_p E_r = \frac{n^2 f_r(k,r)\,[\sin(\phi + 2\theta) - \sin(\phi - 2\theta)]}{2\,[n\nu(\theta) + \cos\theta]} \tag{A.19}$$

$$E_\theta \to E'_\theta = t_p E_\theta = -\frac{2n f_\vartheta(k,r)\nu(\theta)\cos\theta\cos\phi}{n\nu(\theta) + \cos\theta} \tag{A.20}$$

$$E_\phi \to E'_\phi = t_s E_\phi = -\frac{2n f_\vartheta(k,r)\cos\theta\sin\phi}{\nu(\theta) + n\cos\theta}. \tag{A.21}$$

As it should be, the radial component vanishes in the far field, $kr \gg 1$, and only the transverse field components survive. Furthermore, since $\hat{\mathbf{e}}_\phi$ does not depend on θ, only $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ transform according to Snell's law on refraction, resulting in

$$\hat{\mathbf{e}}_r \to \hat{\mathbf{e}}'_r = \begin{pmatrix} \nu(\theta)\cos\phi \\ \nu(\theta)\sin\phi \\ -n\sin\theta \end{pmatrix} \tag{A.22}$$

$$\hat{\mathbf{e}}_\theta \to \hat{\mathbf{e}}'_\theta = \begin{pmatrix} n\sin\theta\cos\phi \\ n\sin\theta\sin\phi \\ \nu(\theta) \end{pmatrix} \tag{A.23}$$

in the Cartesian basis defined in Figure 7.3. Finally, we can evaluate the electric field amplitude outside the slab in the Cartesian basis,

$$\mathbf{E}'(r,\theta,\phi) = A(r) \sum_{i\in\{x,y,z\}} E'_i(r,\theta,\phi)\hat{\mathbf{e}}_i, \tag{A.24}$$

with

$$E'_x = f_r(k,r)n\sin^2\theta\cos^2\phi - f_\vartheta(k,r)\left[\nu(\theta)\cos^2\phi\cos\theta - \sin^2\phi\right] \tag{A.25}$$

$$E'_y = \left[f_r(k,r)n\sin^2\theta - f_\vartheta(k,r)\{\nu(\theta)\cos\theta + 1\}\right]\sin\phi\cos\phi \tag{A.26}$$

$$E'_z = [f_r(k,r)\nu(\theta) + f_\vartheta(k,r)n\cos\theta]\sin\theta\cos\phi \tag{A.27}$$

where we dropped the arguments for conciseness.

In principle, the angle Θ and radial distance R for the fields outside the slab would need to be modified when keeping the center of the coordinate system at the dipole source. That is, if we denote the vector from the dipole to the point in the surface where a particular ray exits the slab by $\mathbf{r}$ and the vector from that point to the point of observation by $\mathbf{r}'$, then $\theta = \arccos(d/r)$ and $\theta' = \arccos([z-d]/r')$, and the spherical coordinate vector $\mathbf{R} = \mathbf{r} + \mathbf{r}'$ with polar angle Θ. Because of refraction, $\theta \neq \theta'$ and hence $\mathbf{R}$ has norm $R \neq r + r'$. However, outside the slab we are only interested in the far field at the lens where $R, r' \gg d, r$ with d the depth of the dipole inside the slab. Therefore, we can well approximate $\Theta \approx \theta'$ and $R \approx r'$ when dealing with the absolute value of the field, corresponding to placing the center of the coordinate system right below the surface of the slab. For the phase, this approximation likely does not hold well as the radius of the spot on the sample, $\rho_\mathrm{m} = d\tan\theta_\mathrm{m}$, is on the order of the wavelength $\lambda = \lambda_0/n$ and hence the phase is not constant across the radius ρ, even when observed from the far field. A more rigorous, likely numerical, treatment would be needed to fully account for this.

A.1.3 Fraunhofer diffraction

The beam focused into the single-mode fiber (SMF), Equation 7.20, experiences diffraction at the ocular lens aperture. Having approximated the beam as circular, we can write the electric field on the screen, *i. e.*, the fiber end face, as [87]

$$\tilde{E}(q) = \frac{|\mathbf{p}|k^2}{2\epsilon} \exp(ikz) \int_0^a \mathrm{d}\rho\, \rho J_0(k\rho q/f_{\mathrm{oc}}) E_x(\rho) \tag{A.28}$$

with $E_x(\rho)$ given by Equation 7.22 upon substituting $\theta = \arctan(\rho/f_{\mathrm{ob}})$ and $r = \sqrt{\rho^2 + f_{\mathrm{ob}}^2}$, and $J_0(x)$ is the Bessel function of order zero. Equation A.28 is in general only solvable numerically. Furthermore, note that likely the Fraunhofer approximation is only of limited applicability here since the phase of the wave incident on the lens is not constant across ρ as laid out before.

A.2 Vibration spectroscopy

A.2.1 Knife-edge measurement

In Section 8.3, I used a knife-edge measurement to calibrate the readout of the sample position using the count rate of laser radiation reflected off a lateral reflectance gradient. The gradient was determined by the convolution of the finite spatial extent of the laser spot and a step in reflectance from a Au gate with approximately perfect reflectance and the bare GaAs surface. The same measurement can also be used to extract the reflectance $\mathscr{R}_0$ of the bare GaAs surface as well as the spot size radius w_0 of a Gaussian beam by fitting the theoretical dependence of the reflected count rate on the lateral position, Equation 8.8. From the refractive index of GaAs, we would expect

$$\mathscr{R}_0 = \left|\frac{n-1}{n+1}\right|^2 \approx 32\,\% \tag{A.29}$$

at zero temperature [435].

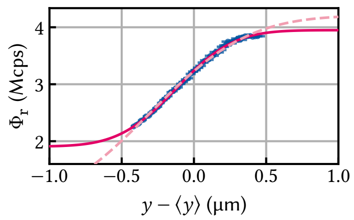

Figure A.1: Reflectance gradient of the edge of a Au gate electrode on the bare GaAs surface. The dashed line is a fit to Equation 8.8 with $\mathscr{R}_0$ fixed while the solid includes this parameter in the fit.

Figure A.1 shows the same data as Figure 8.7 together with fits to Equation 8.8 in magenta. The dashed line is a fit with $\mathscr{R}_0$ fixed, whereas the solid line is a fit including $\mathscr{R}_0$ as a free parameter. Clearly, the latter matches the data better, resulting in

$$\mathscr{R}_0 = 65.1(14)\,\% \tag{A.30}$$
$$w_0 = 0.624(28)\,\mu\mathrm{m}. \tag{A.31}$$

The discrepancy in reflectance might be explained by multilayer and thin-film effects given that the sample is only 220 nm thick and warrants closer investigation. More likely, the assumption that the Au optical gate is perfectly reflecting is to be challenged as its thickness corresponds to only a fifth of the wavelength. In Part III, I carry out transfer-matrix method (TMM) simulations to this end. The Gaussian beam waist radius w_0 resulting from the fit is in quite good agreement with the results obtained in Subsection 7.1.3, where I obtained the value 0.60 μm and 0.84 μm for the y- and z-direction, respectively (see Table 7.2, but note the different coordinate systems).

Supplementary to Part III: Additional measurements and TMM simulations

B

In this appendix, I give additional information on the bandstructure simulations and show further measurements supporting the interpretation of data shown in Chapter 13. Moreover, I investigate the influence of the epoxy thickness on the simulations performed in Section 13.3.

B.1 Self-consistent Poisson-Schrödinger simulation of the membrane band structure

The samples studied in Chapter 13 were all grown using molecular beam epitaxy (MBE) and had similar nominal designs. To compare the design charge carrier density with the measurements obtained from photoluminescence (PL) of the Fermi edge, I simulated the band structure using a self-consistent Poisson-Schrödinger solver [267]. Table B.1 shows the nominal doping concentration N_d as well as the charge carrier density in the QW, n, obtained from a simulation using the parameters given in Table B.2.

Table B.1: Heterostructure parameters used in the present thesis. The doping density N_d is nominal, whereas the charge carrier density in the QW, n, is computed using the nominal doping values with parameters given in Table B.2.

Wafer	N_d (cm^{-3})	n (cm^{-2})
M1_05_49	6.5×10^{17}	1.95×10^{11}
15460 (Honey)	1.8×10^{18}	4.26×10^{11}
15271 (Fig)	8.0×10^{17}	3.91×10^{11}

Table B.2: Simulation parameters used to compute the charge carrier density n in Table B.1.

Parameter	Value	Unit
E_DX [a]	−71.5	meV
ΔE_c [b]	240	meV
m_c	0.067	m_e
Δz	0.5	nm
T	10	mK
V_FP [c]	0.76	V

[a] Energy of the DX-center below the Fermi level.
[b] Conduction band offset at the GaAs/AlGaAs interface.
[c] Fermi level pinning voltage.

B.2 Additional data

B.2.1 Combined plot of PL and PLE data

In Section 13.2, I showed PLE measurements of a large exciton trap. When integrating the PL spectrum over the detection energy, the data showed several distinct features including an onset of the absorption that depends on the electric field applied by the difference-mode voltage V_DM. Figure B.1 shows the same data as panels (a) and (b) of Figure 13.9, but drawn in the same plot. It is clear that the onset of absorption occurs just as the PL emission drops off, indicating a small Stokes shift.

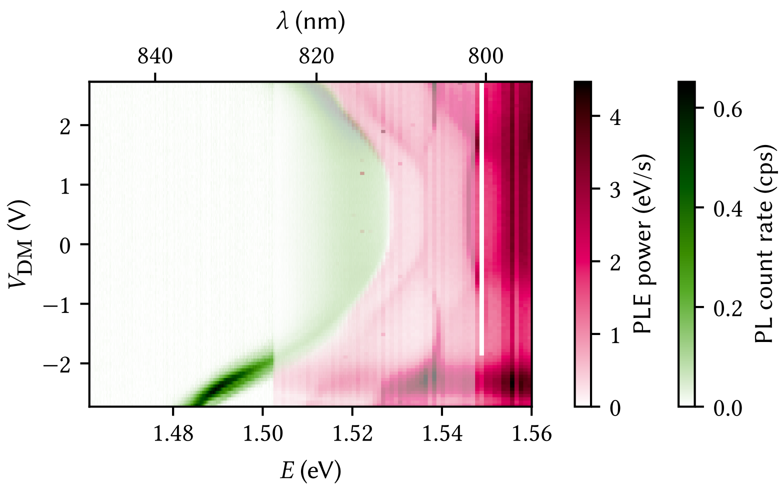

Figure B.1: Combined plot of the PL with excitation at 795 nm (green) and photoluminescence excitation (PLE) (magenta). The data overlap in the range of 801 nm to 825 nm, where they are plotted with 50 % transparency. The onset of the absorption edge in PLE coincides with the high-energy shoulder of the PL emission.

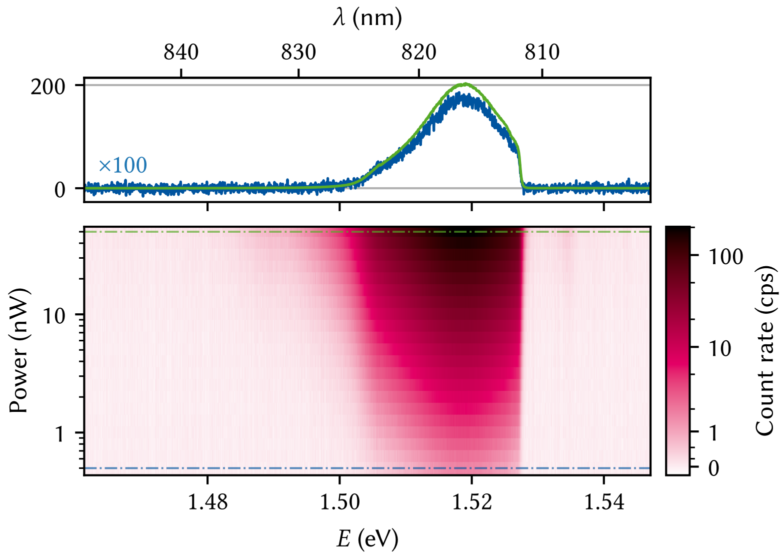

Figure B.2: Two-dimensional electron gas (2DEG) PL of an unbiased QW as function of excitation power. The Fermi edge at 1.5275 eV does not broaden over two orders of magnitude as the line cuts in the upper panels also show.

B.2.2 2DEG PL as function of power

In Section 13.1, it was determined that the electrons recombining from the Fermi edge of the 2DEG were much hotter at $\sim$ 1 K than the cryostat temperature at $\sim$ 10 mK. The fit to the trion-exciton lineshape in Section 13.2 yielded similar results. Figure B.2 demonstrates that this is not a local heating of the lattice due to excess laser power. Shown is the 2DEG PL as function of excitation power from 5 nW to 500 nW, with the upper panel showing line cuts at highest and lowest power. There is no discernible broadening in the Fermi edge on the high-energy side of the spectrum.

B.3 Dependence of TMM simulations on epoxy thickness

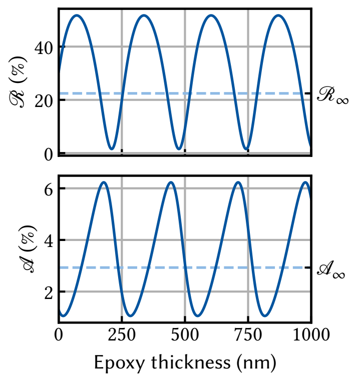

Figure B.3: Reflectance (top) and QW absorptance (bottom) as function of epoxy thickness assuming coherent back-scattering. The period corresponds to the wavelength in epoxy, $\lambda_{\mathrm{epo}} = \lambda_0 / n_{\mathrm{epo}}$.

In Section 13.3, I carried out TMM simulations of the membrane structure to elucidate the quenching of PL intensity when focusing the laser on gates on the top or bottom side of the membrane. There, I assumed incoherent scattering of photons at the epoxy/Si interface below the membrane. This assumption is based on the fact that Descamps [70] found the epoxy thickness, on the order of a few µm, to vary significantly across the chip. If the variation is fast enough, we can expect the phase of back-scattered photons to average out and hence destroy coherence. Nonetheless, we should estimate the influence of coherent back-scattering. Figure B.3 shows the reflectance $\mathscr{R}$ and QW absorptance $\mathscr{A}$ as function of the epoxy thickness. The dashed line indicates the value resulting from the incoherent simulation. Both quantities vary significantly with the epoxy thickness, but the incoherent values $\mathscr{A}_\infty$ and $\mathscr{R}_\infty$ are close to the average. From the fact that neither the PL nor the reflected laser intensity varies spatially by such large amounts in experiments, we can conclude that either the thickness variation is small enough to not play a significant role or that it varies fast enough to validate the assumption of incoherent back-scattering.

Supplementary to Part IV: Filter-function derivations and validation

C

In this appendix, we give additional information on the filter-function formalism presented in Part IV. We first show additional derivations that were omitted from the main text in Section C.1, and derive bounds on the Magnus expansion (ME) convergence in Section C.2. In Section C.3, we augment the filter function (FF) formalism of the main text by deriving a concatenation rule for the second-order terms of the ME. This enables a more efficient calculation of the full quantum process including coherent errors for sequences of quantum gates together with the rule for first-order terms given in Subsection 16.2.1. In Section C.4, we review time-domain simulation methods of noisy quantum dynamics, which we employ in Section C.5 to validate our FF formalism's predicted fidelities for the quantum gates of Reference 421 and discussed in Section 19.1, as well as the quantum Fourier transform (QFT) algorithm presented in Section 19.4. We use the notation set in Section 16.1 except where indicated otherwise.

C.1 Additional derivations

C.1.1 Derivation of the single-qubit cumulant function in the Liouville representation

For a single qubit, the Pauli basis $\mathcal{P}_1 = \{\sigma_i\}_{i=0}^3 = 1/\sqrt{2} \times \left\{\mathbb{1}, \sigma_x, \sigma_y, \sigma_z\right\}$ is a natural choice to define the Liouville representation. In this case, the trace tensor Equation 16.18 can be simplified and thus Equation 16.17 given a more intuitive form which we derive in this subsection. Since the cumulant function is linear in the noise indices α, β we drop them in the following for legibility. Our results hold for both a single pair of noise indices and the total cumulant. We start by observing the relation

$$T_{klij} = \mathrm{tr}\left(\sigma_k\sigma_l\sigma_i\sigma_j\right) = \frac{1}{2}(\delta_{kl}\delta_{ij} - \delta_{ki}\delta_{lj} + \delta_{kj}\delta_{li}) \tag{C.1}$$

for the Pauli basis elements $\sigma_k, k \in \{1, 2, 3\}$. Including the identity element σ_0 in the trace tensor gives additional terms. However, as we show now none of these contribute to $\mathcal{K}$ because they cancel out.

First, since the noise Hamiltonian $H_\mathrm{n}(t)$ is traceless and therefore $\tilde{\mathcal{B}}_{\alpha 0}(t) = 0$, we have $\Gamma_{kl}, \Delta_{kl} \propto (1-\delta_{k0})(1-\delta_{0l})$, *i.e.* the first column and row of both the decay amplitude and frequency shift matrices are zero, and hence terms in the sum of Equation 16.17 with either $k = 0$ or $l = 0$ vanish. Next, for $i = j = 0$ all of the traces cancel out as can be easily seen. The last possible cases are given by $i = 0, j \neq 0$ and vice versa. For these cases we have

$$T_{kl0j} = T_{klj0} = \frac{1}{\sqrt{2}}\,\mathrm{tr}\left(\sigma_k\sigma_l\sigma_j\right) = \frac{i}{2}\varepsilon_{klj} \tag{C.2}$$

with ε_{klj} the completely antisymmetric tensor. Both of the above cases vanish in $\mathcal{K}$ since, taking the case $j = 0$ for example,

$$\frac{1}{2}\left(T_{kl0i} - T_{k0li} - T_{kil0} + T_{ki0l}\right) = \frac{i}{2}\left(\varepsilon_{kli} - \varepsilon_{kli} - \varepsilon_{kil} + \varepsilon_{kil}\right) = 0 \tag{C.3a}$$

for the decay amplitudes Γ and

$$\frac{1}{2}\left(T_{kl0i} - T_{lk0i} - T_{kli0} + T_{lki0}\right) = \frac{i}{2}\left(\varepsilon_{kli} - \varepsilon_{lki} - \varepsilon_{kli} + \varepsilon_{lki}\right) = 0 \tag{C.3b}$$

for the frequency shifts Δ. Hence, only terms with $i, j > 0$ contribute and we can plug the simplified expressions for the trace tensor T_{klij}, Equation C.1, into Equation 16.17 to write the cumulant function for a single qubit and the Pauli basis concisely as

$$\begin{aligned}
\mathcal{K}_{ij}(\tau) &= -\frac{1}{2}\sum_{kl}\Big[\Delta_{kl}\left(T_{klji} - T_{lkji} - T_{klij} + T_{lkij}\right) \\
&\qquad\qquad + \Gamma_{kl}\left(T_{klji} - T_{kjli} - T_{kilj} + T_{kijl}\right)\Big] && \text{(C.4)} \\
&= -\sum_{kl}\left[\Delta_{kl}(\delta_{ki}\delta_{lj} - \delta_{kj}\delta_{li}) + \Gamma_{kl}(\delta_{kl}\delta_{ij} - \delta_{kj}\delta_{li})\right] && \text{(C.5)} \\
&= \Delta_{ji} - \Delta_{ij} + \Gamma_{ij} - \delta_{ij}\operatorname{tr}\Gamma && \text{(C.6)} \\
&= \begin{cases} -\sum_{k\neq i}\Gamma_{kk} & \text{if} \quad i = j, \\ -\Delta_{ij} + \Delta_{ji} + \Gamma_{ij} & \text{if} \quad i \neq j, \end{cases} && \text{(C.7)}
\end{aligned}$$

as given in the main text.

C.1.2 Evaluation of the integrals in Equation 16.40

Here we calculate the integrals appearing in the calculation of the frequency shifts Δ, Equation 16.41, given by

$$I_{ijmn}^{(g)}(\omega) = \int_{t_{g-1}}^{t_g} \mathrm{d}t\, e^{i\Omega_{ij}^{(g)}(t-t_{g-1})-i\omega t}\int_{t_{g-1}}^{t} \mathrm{d}t'\, e^{i\Omega_{mn}^{(g)}(t'-t_{g-1})+i\omega t'}. \tag{C.8}$$

The inner integration is simple to perform and we get

$$\begin{aligned}
I_{ijmn}^{(g)}(\omega) = \int_{t_{g-1}}^{t_g} &\mathrm{d}t\, e^{i\Omega_{ij}^{(g)}(t-t_{g-1})-i\omega(t-t_{g-1})} \\
&\times\begin{cases} \dfrac{e^{i(\omega+\Omega_{mn}^{(g)})(t-t_{g-1})} - 1}{i(\omega+\Omega_{mn}^{(g)})} & \text{if} \quad \omega + \Omega_{mn}^{(g)} \neq 0 \\ t - t_{g-1} & \text{if} \quad \omega + \Omega_{mn}^{(g)} = 0. \end{cases}
\end{aligned} \tag{C.9}$$

Shifting the limits of integration and performing integration by parts in the case $\omega + \Omega_{mn}^{(g)} = 0$ then yields

$$I_{ijmn}^{(g)}(\omega) = \tag{C.10}$$
$$\begin{cases}
\frac{1}{\omega+\Omega_{mn}^{(g)}}\left(\frac{e^{i(\Omega_{ij}^{(g)}-\omega)\Delta t_g}-1}{\Omega_{ij}^{(g)}-\omega} - \frac{e^{i(\Omega_{ij}^{(g)}+\Omega_{mn}^{(g)})\Delta t_g}-1}{\Omega_{ij}^{(g)}+\Omega_{mn}^{(g)}}\right) & \text{if } \omega + \Omega_{mn}^{(g)} \neq 0, \\
\frac{1}{\Omega_{ij}^{(g)}-\omega}\left(\frac{e^{i(\Omega_{ij}^{(g)}-\omega)\Delta t_g}-1}{\Omega_{ij}^{(g)}-\omega} - i\Delta t_g e^{i(\Omega_{ij}^{(g)}-\omega)\Delta t_g}\right) & \text{if } \omega + \Omega_{mn}^{(g)} = 0 \wedge \Omega_{ij}^{(g)} - \omega \neq 0, \\
\frac{\Delta t_g^2}{2} & \text{if } \omega + \Omega_{mn}^{(g)} = 0 \wedge \Omega_{ij}^{(g)} - \omega = 0.
\end{cases}$$

C.1.3 Simplifying the calculation of the entanglement infidelity

In the main text, we claimed that the contribution of noise sources (α, β) to the total entanglement infidelity $I_{\mathrm{ent}}(\tilde{\mathcal{U}}) = \sum_{\alpha\beta} I_{\alpha\beta}$ reduces from the trace of the cumulant function $\mathcal{K}$ to

$$
\begin{aligned}
I_{\alpha\beta} &= -\frac{1}{d^2}\operatorname{tr}\mathcal{K}_{\alpha\beta} && \text{(C.11)}\\
&= \frac{1}{d}\operatorname{tr}\Gamma_{\alpha\beta}. && \text{(C.12)}
\end{aligned}
$$

To show this, we substitute $\mathcal{K}$ by its definition in terms of Δ and Γ according to Equation 16.17. This yields for the trace

$$
\begin{aligned}
\operatorname{tr}\mathcal{K}_{\alpha\beta} &= -\frac{1}{2}\sum_{kl}\delta_{ij}(f_{ijkl}\Delta_{\alpha\beta,kl} + g_{ijkl}\Gamma_{\alpha\beta,kl})\\
&= -\frac{1}{2}\sum_{ikl}\Gamma_{\alpha\beta,kl}\,(T_{klii} + T_{lkii} - 2T_{kili})
\end{aligned} \tag{C.13}
$$

since Δ is antisymmetric. In order to further simplify the trace tensors on the right hand side of Equation C.13, we observe that the orthonormality and completeness of the operator basis C defining the Liouville representation of $\mathcal{K}$ (*cf.* Equation 16.14) is equivalent to requiring that $\mathsf{C}^\dagger\mathsf{C} = \mathbb{1}$ with C reshaped into a $d^2 \times d^2$ matrix by a suitable mapping. This condition may also be written as

$$
\begin{aligned}
\delta_{ac}\delta_{bd} &= \sum_k C^*_{k,ab}C_{k,cd}\\
&= \sum_k C_{k,ba}C_{k,cd}
\end{aligned} \tag{C.14}
$$

because every element C_k is Hermitian. Using this relation in Equation C.13 then yields

$$
\begin{aligned}
\operatorname{tr}\mathcal{K}_{\alpha\beta} &= -\frac{1}{2}\sum_{kl}\Gamma_{\alpha\beta,kl}\,(2d\delta_{kl} - 2\operatorname{tr}(C_k)\operatorname{tr}(C_l))\\
&= -d\operatorname{tr}\Gamma_{\alpha\beta}.
\end{aligned} \tag{C.15}
$$

The last equality only holds true for bases with a single non-traceless element (the identity), such as the bases discussed in Section 16.8. This is because in this case, $\operatorname{tr}(C_k) = 0$ for $k > 0$ whereas $\Gamma_{\alpha\beta,kl} = 0$ for either $k = 0$ or $l = 0$ since Γ is a function of the traceless noise Hamiltonian for which $\operatorname{tr}(C_0 H_{\mathrm{n}}) \propto \operatorname{tr} H_{\mathrm{n}} = 0$ (*i. e.* the first column of the control matrix is zero, see Equations 16.16 and 16.21). Finally, substituting Equation C.15 into Equation C.11 we obtain our result

$$
I_{\alpha\beta} = \frac{1}{d}\operatorname{tr}\Gamma_{\alpha\beta}. \tag{C.16}
$$

C.1.4 Sum rule

For white noise, the infidelity to leading order does not depend on the internal dynamics of the control operation but only on the total duration τ. To see this, take Equation 16.21 and substitute the correlation function of white noise, $\langle b_\alpha(t_1)b_\alpha(t_2)\rangle = S_0\delta(t_1 - t_2)$. The integrals then simplify to

$$\Gamma_{\alpha\alpha,kl} = S_0 \int_0^\tau \mathrm{d}t\, \tilde{\mathcal{B}}_{\alpha k}(t)\tilde{\mathcal{B}}_{\alpha l}(t). \tag{C.17}$$

In order to compute the infidelity, we require the trace over the matrix of decay amplitudes (Equation 16.48), so

$$I_\alpha = \frac{S_0}{d}\operatorname{tr}\Gamma_{\alpha\alpha} = \frac{1}{d}\sum_k \int_0^\tau \mathrm{d}t\, \tilde{\mathcal{B}}^2_{\alpha k}(t). \tag{C.18}$$

Now recall that $\tilde{\mathcal{B}}$ is a vector on Liouville space and we can write $\tilde{\mathcal{B}}_\alpha(t) \doteq |\tilde{B}_\alpha(t)\rangle\!\rangle \equiv \langle\!\langle\tilde{B}_\alpha(t)|$ because it is Hermitian. Thus,

$$I_\alpha = \frac{S_0}{d}\int_0^\tau \mathrm{d}t\, \langle\!\langle\tilde{B}_\alpha(t)|\tilde{B}_\alpha(t)\rangle\!\rangle = \frac{1}{d}\int_0^\tau \mathrm{d}t\, \big\|\tilde{B}_\alpha(t)\big\|^2, \tag{C.19}$$

and if $B_\alpha(t) = B_\alpha$, *i. e.*, is time-independent, the integral evaluates simply to τ,

$$I_\alpha = \frac{S_0\tau}{d}\big\|\tilde{B}_\alpha\big\|^2, \tag{C.20}$$

because the control unitary $U_\mathrm{c}(t)$ conserves the norm. Since also $I_\alpha = \int \mathrm{d}\omega\,/2\pi d\, S(\omega)\mathcal{F}_\alpha(\omega)$ with the fidelity filter function $\mathcal{F}_\alpha(\omega)$ (Equations 16.26 and 16.49), this result also implies the sum rule

$$\int \frac{\mathrm{d}\omega}{2\pi}\mathcal{F}_\alpha(\omega) = \tau\big\|\tilde{B}_\alpha\big\|^2, \tag{C.21}$$

a result previously obtained by Cywiński et al. [339]. This means that pulse shaping and similar techniques never yield a leading-order fidelity enhancement if the noise is white in the relevant regime of frequencies as the fidelity only depends on the total duration of the pulse.

C.2 Convergence bounds

In this section we give bounds for the convergence of the expansions employed in the main text for the case of purely autocorrelated noise, $S_{\alpha\beta}(\omega) = \delta_{\alpha\beta}S_{\alpha\beta}(\omega) = S_\alpha(\omega)$, following the approach by Green et al. [346]. For Gaussian noise, our expansion is exact when including first and second order Magnus expansion (ME) terms in the special cases of commuting and uncorrelated noise (corresponding to $[H_\mathrm{c}, H_\mathrm{n}] = 0$ and $S(\omega) = \text{const.}$, respectively). Hence, the convergence radius of the ME becomes infinite in these cases and the fidelity can be computed exactly by evaluating the matrix exponential Equation 16.11. For the general case the following considerations apply.

C.2.1 Magnus expansion

The ME of the error propagator (Equation 16.8) converges if $\int_0^\tau \mathrm{d}t\, \|\tilde{H}_\mathrm{n}(t)\| < \pi$ with $\|A\|^2 = \langle A, A\rangle = \sum_{ij}|A_{ij}|^2$ the Frobenius norm [371]. If we take the noise to have some characteristic and finite correlation time τ_c after which the autocorrelation function $\langle b_\alpha(t)b_\alpha(s)\rangle$ decays by a fraction (to $1/e$, say), τ_c rather than the total duration of the operation τ becomes the relevant timescale and we can hence replace the upper limit of integration by $\bar{\tau} := \min\{\tau, \tau_\mathrm{c}\}$.

To derive bounds for the convergence of the ME, we assume a time dependence of the noise operators of the form $B_\alpha(t) = s_\alpha(t)B_\alpha$. By the Cauchy-Schwarz inequality we then have

$$\begin{aligned}
\left\|\tilde{H}_{\mathrm{n}}(t)\right\|^2 &= \|H_{\mathrm{n}}(t)\|^2 \\
&= \sum_{\alpha\beta} s_\alpha(t)s_\beta(t)b_\alpha(t)b_\beta(t)\left\langle B_\alpha, B_\beta\right\rangle \\
&\leq \sum_{\alpha\beta} s_\alpha(t)s_\beta(t)b_\alpha(t)b_\beta(t)\|B_\alpha\|\,\|B_\beta\| \\
&\leq \left[\sum_\alpha \sum_{g=1}^{G} \vartheta^{(m)}(t)s_\alpha^{(m)}b_\alpha^{(m)}\|B_\alpha\|\right]^2
\end{aligned} \tag{C.22}$$

where $b_\alpha^{(m)}$ ($s_\alpha^{(m)}$) is the maximum value that the noise (coupling strength) assumes during the pulse, $\vartheta^{(g)}(t) = \theta(t-t_{g-1}) - \theta(t-t_g)$ is one during the gth time interval and zero else, and where we approximated the time evolution as piecewise constant. Then, in order to guarantee convergence of the ME,

$$\begin{aligned}
\int_0^\tau \mathrm{d}t\,\|\tilde{H}_{\mathrm{n}}(t)\| &\leq \int_0^\tau \mathrm{d}t\left|\sum_\alpha \sum_{g=1}^{G} \vartheta^{(m)}(t)s_\alpha^{(m)}b_\alpha^{(m)}\|B_\alpha\|\right| \\
&= \sum_\alpha b_\alpha^{(m)}\|B_\alpha\| \sum_{g=1}^{G} s_\alpha^{(m)} \int_{t_{g-1}}^{t_g} \mathrm{d}t \\
&= \sum_\alpha C_m\delta b_\alpha\|B_\alpha\| \sum_{g=1}^{G} s_\alpha^{(m)}\Delta t_g \\
&=: N
\end{aligned} \tag{C.23}$$

where we have expressed the in principle unknown maximum noise amplitude $b_\alpha^{(m)}$ in terms of the root mean square value δb_α. That is, $b_\alpha^{(m)} = C_m\left\langle b_\alpha(0)^2\right\rangle^{1/2} = C_m\delta b_\alpha$ for a sufficiently large value C_m. Finally, realizing that $\delta b_\alpha^2 = \int \frac{\mathrm{d}\omega}{2\pi}S_\alpha(\omega)$ and by the triangle inequality,

$$\begin{aligned}
N &= C_m \sum_\alpha \|B_\alpha\|\left[\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi}S_\alpha(\omega)\right]^{1/2} \sum_{g=1}^{G} s_\alpha^{(m)}\Delta t_g \\
&\leq C_m\left[\sum_\alpha \|B_\alpha\|^2 \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi}S_\alpha(\omega)\left(\sum_{g=1}^{G} s_\alpha^{(m)}\Delta t_g\right)^2\right]^{1/2} \\
&=: C_m\xi \\
&\overset{!}{<} \pi
\end{aligned} \tag{C.24}$$

where we have introduced the parameter ξ. Thus, the expansion converges if $\xi < \pi/C_m$. However, we note that in practice the rms noise amplitude δb_α will often be infinite, limiting the usefulness of this bound for certain noise spectra.

C.2.2 Infidelity

Again assuming a time dependence $B_\alpha(t) = s_\alpha(t)B_\alpha$ as well as piecewise-constant control, we note that for the infidelity we have (*cf.* Equation 16.46)

$$
\begin{aligned}
|\mathrm{tr}(\Gamma)| &= \left|\sum_\alpha \int_0^{\bar\tau} \mathrm{d}t_2 \int_0^{\bar\tau} \mathrm{d}t_1 \,\langle b_\alpha(t_1) b_\alpha(t_2)\rangle \sum_k \tilde{\mathcal{B}}_{\alpha k}(t_1)\tilde{\mathcal{B}}_{\alpha k}(t_2)\right| \\
&\le \left|\sum_\alpha \int_0^{\bar\tau} \mathrm{d}t_2 \int_0^{\bar\tau} \mathrm{d}t_1 \,\langle b_\alpha(t_1) b_\alpha(t_2)\rangle \sum_{g,g'=1}^{G} \vartheta^{(m)}(t_1)\vartheta^{(g')}(t_2) s_\alpha^{(m)} s_\alpha^{(g')} \lVert B_\alpha\rVert^2\right| \\
&\le \sum_\alpha \lVert B_\alpha\rVert^2 \underbrace{\left\langle b_\alpha^2(0)\right\rangle}_{\int \frac{\mathrm{d}\omega}{2\pi} S_\alpha(\omega)} \sum_{g,g'=1}^{G} s_\alpha^{(m)} s_\alpha^{(g')} \left| \int_{t_{g'-1}}^{t_{g'}} \mathrm{d}t_2 \int_{t_{g-1}}^{t_g} \mathrm{d}t_1 \underbrace{\overline{\langle b_\alpha(t_1) b_\alpha(t_2)\rangle}}_{|\bullet|\le 1} \right| \\
&\le \sum_\alpha \left[\lVert B_\alpha\rVert^2 \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} S_\alpha(\omega) \left(\sum_{g=1}^{G} s_\alpha^{(m)} \Delta t_g \right)^2 \right] \\
&= \xi^2,
\end{aligned}
\tag{C.25}
$$

where, going from the second to the third line, we have factored out the total power of noise source α from the cross-correlation function, $\langle b_\alpha(t_1) b_\alpha(t_2)\rangle = \left\langle b_\alpha^2(0)\right\rangle \left|\overline{\langle b_\alpha(t_1) b_\alpha(t_2)\rangle}\right|$. Thus, the first order infidelity Equation 16.46 is upper-bounded by $d^{-1}\xi^2$, the same parameter also bounding the convergence of the ME, and higher orders can be neglected if $\xi^2 \ll 1$.

Note that similar arguments can be made for the higher orders of the ME [346]. In particular, the nth order ME term containing n-point correlation functions of the noise is of order $\mathcal{O}(\xi^n)$ as stated in the main text.

C.3 Concatenation of second-order filter functions

In this section, we lay out how the second-order filter functions of atomic pulse segments can be reused to compute the filter function of the concatenated sequence. While it is not possible to perform the calculation entirely without concern for the internal structure of the individual segments due to the nested time integral (*cf.* Equation 16.40), it is also not necessary to compute everything from scratch as we show below.

We begin by setting some notation. Wherever possible, we omit indices and thus imply matrix multiplication between objects. We assume a single noise operator and drop the corresponding index; we can easily add them again later since none of the manipulations involve the noise indices. We fully adopt the picture that control matrices can be concatenated by summing over individual time steps, which may either be single piecewise-constant segments or entire sequences, corresponding to Equation 16.36 or Equation 16.29, respectively. To make clear that the control "matrices" are in fact vectors in Liouville space, we write them as bras wherever advantageous,[1]

1: Recall our convention of using Roman font for Hilbert-space operators and calligraphic font for their Liouville-space duals.

$$\tilde{\mathcal{B}}(\omega) \doteq \langle\!\langle \tilde{B}(\omega)| \,. \tag{C.26}$$

In the following, we will consider a sequence of piecewise-constant time steps split up into subsequences ("gates") and will deal on the one hand with quantities that depend exclusively on the internal structure of a subsequence and those that do not on the other. For the former, we will denote their internal time step by a parenthesised superscript, *e.g.* $A^{(i)}$, which means the ith time step of A. The latter will have no superscript as they do not depend on the internal structure. We will furthermore distinguish between *local* quantities, which do not depend on the preceding dynamics (that is, are functions of the subsequence alone), and denote the subsequence index they belong to by a parenthesised subscript, *e.g.* $A_{(i)}$ for some quantity A of sequence i. Quantities which are *non-local* and thus depend on the preceding dynamics, but only on local quantities thereof, will have parenthesised subscripts with arrows indicating the range of the sequence, *e.g.* $A_{(i\to 1)}$ with $A_{(1\to 1)} \equiv A_{(1)}$. By contrast, non-local quantities that depend on other non-local quantities will then be denoted in a "posterior" fashion, *e.g.*, $A_{(i|i-1\to 1)}$ if A is a function of subsequence i and depends on all previous subsequences $i-1,\dots,1$. As an illustrating example, consider the first-order concatenation rule for control matrices, Equation 16.29. With this notation, we can write it as[2]

2: The same definition holds for control matrices computed from Equation 16.36, in which case the subscripts would become superscripts. Similarly, sub- and superscripts can be combined.

$$\begin{aligned}\langle\!\langle \tilde{B}_{(g\to 1)}(\omega)| &= \sum_{g'=1}^{g} \langle\!\langle \tilde{B}_{(g'|g'-1\to 1)}(\omega)| &\quad (\text{C.27})\\ &= \langle\!\langle \tilde{B}_{(1)}(\omega)| + \langle\!\langle \tilde{B}_{(2|1)}(\omega)| + \langle\!\langle \tilde{B}_{(3|2\to 1)}(\omega)| + \cdots\end{aligned}$$

with

$$\langle\!\langle \tilde{B}_{(g|g-1\to 1)}(\omega)| := e^{i\omega t_{g-1}} \langle\!\langle \tilde{B}_{(g)}(\omega)|\, \mathcal{Q}_{(g-1\to 1)}, \qquad (\text{C.28})$$

$$\mathcal{Q}_{(g-1\to 1)} := \prod_{g'=g-1}^{1} \mathcal{Q}_{(g')}, \qquad (\text{C.29})$$

where $\langle\!\langle \tilde{B}_{(g)}(\omega)|$ is the control matrix of the gth gate and $\mathcal{Q}_{(g)}$ the corresponding control superpropagator. For complete sequences, typically denoted by $g = G$, we drop the subscript, $\langle\!\langle \tilde{B}_{(G\to 1)}(\omega)| \equiv \langle\!\langle \tilde{B}(\omega)|$. Finally, we drop the specifier $\mathcal{F}^{(2)}$ distinguishing the second- from the first-order filter function for brevity; in this section, we always mean the former.

We start from Equation 16.40, from which for reasons that will become clear shortly we define the second-order filter function by

$$\mathcal{F}_{\alpha\beta,kl}(\omega) := \mathcal{N}_{\alpha\beta,kl}(\omega) + \sum_{g=1}^{G} \mathcal{I}^{(g|g-1\to 1)}_{\alpha\beta,kl}(\omega) \qquad (\text{C.30})$$

with

$$\mathcal{N}_{\alpha\beta,kl}(\omega) := \sum_{g=1}^{G} \tilde{\mathcal{B}}^{(g|g-1\to 1)\ast}_{\alpha k}(\omega)\tilde{\mathcal{B}}^{(g-1\to 1)}_{\beta l}(\omega), \qquad (\text{C.30a})$$

$$\mathcal{I}^{(g|g-1\to 1)}_{\alpha\beta,kl}(\omega) := s^{(g)}_{\alpha}\bar{B}^{(g)}_{\alpha,ij}\bar{C}^{(g\to 1)}_{k,ji} I^{(g)}_{ijmn}(\omega)\bar{C}^{(g\to 1)}_{l,nm}\bar{B}^{(g)}_{\beta,mn}s^{(g)}_{\beta}, \qquad (\text{C.30b})$$

$\bar{B}$ and $\bar{C}$ defined in Subsection 16.2.2, and where repeated indices are contracted. Comparing to the nested time integral, the first summand in the brackets contains all contributions from complete time segments up to the one containing the inner integration variable t, whereas the second captures the final, incomplete segment with $t_{g-1} < t \leq t_g$. Now imagine the sequence of piecewise-constant time steps, $g \in \{1,\dots,G\}$, being split apart at some index $1 < \gamma < G$ and thereby being divided into two subsequences $g \in \{1,\dots,\gamma\}$ and $h \in \{1,\dots,\eta\} \equiv g \in \{\gamma+1,\dots,G\}$ with $G = \gamma + \eta$. Our goal is to obtain an expression for the second-order filter function $\mathcal{F}(\omega)$ that is – as much as possible – a sum of local terms of these subsequences g and h.

Up to γ, the filter function $\mathcal{F}_{(2\to 1)}(\omega) \equiv \mathcal{F}(\omega)$ is simply that of the first sequence, $\mathcal{F}_{(1)}(\omega)$, and we thus have for $g > \gamma$[3]

$$\begin{aligned}\mathcal{F}_{(2|1)}(\omega) &= \mathcal{F}(\omega) - \mathcal{F}_{(1)}(\omega) \\ &= \sum_{g=\gamma+1}^{G} \left[|\tilde{B}^{(g|g-1\to 1)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}^{(g-1\to 1)}(\omega)| + \mathcal{J}^{(g|g-1\to 1)}(\omega)\right],\end{aligned} \tag{C.31}$$

where we already plugged in Equation C.30a. We must now express the quantities $\tilde{\mathcal{B}}^{(g|g-1\to 1)}(\omega)$, $\tilde{\mathcal{B}}^{(g-1\to 1)}(\omega)$, and $\mathcal{J}^{(g|g-1\to 1)}(\omega)$ locally in terms of the index h. To this end, we first write down the step-wise control matrix $\tilde{\mathcal{B}}^{(g|g-1\to 1)}(\omega)$ in the second sequence and split off phases and propagators from the first sequence,

$$\begin{aligned}\langle\!\langle\tilde{B}^{(g|g-1\to 1)}(\omega)| &= e^{i\omega t_g}\langle\!\langle\tilde{B}^{(g)}(\omega)|\,\mathcal{Q}^{(g-1\to 1)} \\ &= e^{i\omega(t_\gamma+t_h)}\langle\!\langle\tilde{B}^{(h)}_{(2)}(\omega)|\,\mathcal{Q}^{(h-1)}_{(2)}\mathcal{Q}_{(1)} \\ &= e^{i\omega\tau_{(1)}}\langle\!\langle\tilde{B}^{(h|h-1\to 1)}_{(2)}(\omega)|\,\mathcal{Q}_{(1)} \\ &= \langle\!\langle\tilde{B}^{(h|h-1\to 1)}_{(2|1)}(\omega)|,\end{aligned} \tag{C.32}$$

where $\tau_{(1)} = t_\gamma$ is the duration of the first sequence and $h = g - \gamma$. Next, we consider the cumulative control matrix $\tilde{\mathcal{B}}^{(g-1\to 1)}(\omega)$. Because in the total sequence it is given by the sum over all $\tilde{\mathcal{B}}^{(g'|g'\to 1)}(\omega)$ up to $g-1$, we can split off the complete control matrix of the first sequence and express the remainder by summing over $\tilde{\mathcal{B}}^{(h)}_{(2|1)}(\omega)$ from Equation C.32:

$$\begin{aligned}\langle\!\langle\tilde{B}^{(g-1\to 1)}(\omega)| &= \langle\!\langle\tilde{B}_{(1)}(\omega)| + e^{i\omega\tau_{(1)}}\sum_{h'=1}^{g-1-\gamma}\langle\!\langle\tilde{B}^{(h'|h'\to 1)}_{(2)}(\omega)|\,\mathcal{Q}_{(1)} \\ &= \langle\!\langle\tilde{B}_{(1)}(\omega)| + e^{i\omega\tau_{(1)}}\langle\!\langle\tilde{B}^{(h-1\to 1)}_{(2)}(\omega)|\,\mathcal{Q}_{(1)} \\ &= \langle\!\langle\tilde{B}^{(h-1\to 1)}_{(2\to 1)}(\omega)|\end{aligned} \tag{C.33}$$

Finally, we need to unravel $\mathcal{J}^{(g|g-1\to 1)}(\omega)$. We start from Equation 16.36, consider a time step $g \geq \gamma$ in the second sequence with $h = g - \gamma$, and rewrite

$$\begin{aligned}\mathcal{J}^{(g|g-1\to 1)}_{kl}(\omega) =& s^{(g)}\bar{B}^{(g)}_{ij}\bar{C}^{(g\to 1)}_{kji}I^{(g)}_{ijmn}(\omega)\bar{C}^{(g\to 1)}_{lnm}\bar{B}^{(g)}_{mn}s^{(g)} \\ =& s^{(h)}_{(2)}\bar{B}^{(h)}_{(2),ij}\bar{C}^{(h\to 1)}_{(2\to 1),kji}I^{(h)}_{(2),ijmn}(\omega)\bar{C}^{(h\to 1)}_{(2\to 1),lnm}\bar{B}^{(h)}_{(2),mn}s^{(h)}_{(2)} \\ =& \mathcal{J}^{(h|h-1\to 1)}_{(2|1),kl}(\omega)\end{aligned} \tag{C.34}$$

because all quantities except for $\bar{C}^{(g)}_{(2|1)}$ depend on their timestep g alone, and where i, j, m, n index the Hilbert space dimensions of the operators, while k, l are the usual indices for the basis elements and therefore Liouville space dimensions. On that term, we can factor out the propagators of the first complete sequence,[4]

$$\bar{C}^{(h\to 1)}_{(2\to 1),kij} = \left[V^{(h)\dagger}_{(2)}Q^{(h-1\to 1)}_{(2)}Q_{(1)}C_k Q^{\dagger}_{(1)}Q^{(h-1\to 1)\dagger}_{(2)}V^{(h)}_{(2)}\right]_{ij}. \tag{C.35}$$

3: We drop indices for legibility as stated above; $\mathcal{J}^{(g|g-1\to 1)}(\omega)$ is a matrix on Liouville space, whereas $\tilde{\mathcal{B}}^{(g|g-1\to 1)}(\omega)$ and $\tilde{\mathcal{B}}^{(g-1\to 1)}(\omega)$ are Liouville-space row vectors and their product here is an outer product, $|\tilde{B}^{(g|g-1\to 1)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}^{(g-1\to 1)}(\omega)|$, resulting in a matrix on Liouville space.

4: Note that the $Q_{(i)}$ here are Hilbert space propagators, not their Liouville space counter parts $\mathcal{Q}_{(i)}$, and that $Q^{(h-1\to 1)}_{(2)} \equiv Q_{h-1}$ in the notation of Subsection 16.2.1.

We can now finally put all pieces together and, starting from Equation C.31, plug in Equations C.32 to C.34, so that we obtain[5]

$$\begin{aligned}
\mathcal{F}_{(2|1)}(\omega) &= \sum_{h=1}^{\eta}\Big[\,|\tilde{B}^{(h|h-1\to 1)}_{(2|1)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}^{(h-1\to 1)}_{(2\to 1)}(\omega)| + \mathcal{I}^{(h|h-1\to 1)}_{(2|1)}(\omega)\Big] \\
&= \sum_{h=1}^{\eta}\Big\{e^{-i\omega\tau_{(1)}}\mathcal{Q}^{\top}_{(1)}\,|\tilde{B}^{(h|h-1\to 1)}_{(2)}(\omega)\rangle\!\rangle \\
&\qquad\times\Big[\langle\!\langle\tilde{B}_{(1)}(\omega)| + e^{i\omega\tau_{(1)}}\langle\!\langle\tilde{B}^{(h-1\to 1)}_{(2)}(\omega)|\,\mathcal{Q}_{(1)}\Big] + \mathcal{I}^{(h|h-1\to 1)}_{(2|1)}(\omega)\Big\}.
\end{aligned} \tag{C.36}$$

To simplify the unwieldy first summand in the curly braces further, we expand the product,

$$\begin{aligned}
&e^{-i\omega\tau_{(1)}}\mathcal{Q}^{\top}_{(1)}\,|\tilde{B}^{(h|h-1\to 1)}_{(2)}(\omega)\rangle\!\rangle\Big[\langle\!\langle\tilde{B}_{(1)}(\omega)| + e^{i\omega\tau_{(1)}}\langle\!\langle\tilde{B}^{(h-1\to 1)}_{(2)}(\omega)|\,\mathcal{Q}_{(1)}\Big] \\
&= e^{-i\omega\tau_{(1)}}\mathcal{Q}^{\top}_{(1)}\,|\tilde{B}^{(h|h-1\to 1)}_{(2)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}_{(1)}(\omega)| \\
&\quad + \mathcal{Q}^{\top}_{(1)}\,|\tilde{B}^{(h|h-1\to 1)}_{(2)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}^{(h-1\to 1)}_{(2)}|\,\mathcal{Q}_{(1)}.
\end{aligned} \tag{C.37}$$

If we now pull in the sum over the time steps h, we can identify in the first term the control matrix and Equation C.27, and in the second the contribution from complete segments to the second-order filter function, Equation C.30a):[6]

$$\begin{aligned}
&\sum_{h=1}^{\eta}(\text{r.h.s Equation C.37}) \\
&\qquad = e^{-i\omega\tau_{(1)}}\mathcal{Q}^{\top}_{(1)}\,|\tilde{B}_{(2)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}_{(1\to 1)}(\omega)| + \mathcal{Q}^{\top}_{(1)}\mathcal{N}_{(2)}(\omega)\mathcal{Q}_{(1)}.
\end{aligned} \tag{C.38}$$

As a last step, we recognize that the ket in the first term is nothing else but the transpose of Equation C.28 so that we can write the filter function succinctly as

$$\begin{aligned}
\mathcal{F}_{(2|1)}(\omega) &= |\tilde{B}_{(2|1)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}_{(1\to 1)}(\omega)| + \mathcal{Q}^{\top}_{(1)}\mathcal{N}_{(2)}(\omega)\mathcal{Q}_{(1)} + \sum_{h=1}^{\eta}\mathcal{I}^{(h|h-1\to 1)}_{(2|1)}(\omega) \\
&= \mathcal{N}_{(2|1)}(\omega) + \sum_{h=1}^{\eta}\mathcal{I}^{(h|h-1\to 1)}_{(2|1)}(\omega).
\end{aligned} \tag{C.39}$$

In Equation C.39, all terms except the last are known ahead of time if the first- and second-order filter functions of the subsequences as well as the control matrix of the concatenated sequence have been computed. We can extend this result to sequences consisting of an arbitrary number of G subsequences with lengths $\{\eta_g\}_{g=1}^{G}$ by recursively shifting indices in Equation C.39 up by one and subsequently adding $\mathcal{F}_{(2|1)}(\omega)$, allowing us to write the concatenation rule for second-order filter functions as

$$\mathcal{F}(\omega) = \sum_{g=1}^{G}\mathcal{F}_{(g|g-1\to 1)}(\omega) \tag{C.40}$$

$$= \sum_{g=1}^{G}\Big[\mathcal{N}_{(g|g-1\to 1)}(\omega) + \sum_{h_g=1}^{\eta_g}\mathcal{I}^{(h_g|h_g-1\to 1)}_{(g|g-1\to 1)}(\omega)\Big] \tag{C.40a}$$

5: Recall that $\mathcal{Q}$ is the Liouville representation of the unitary operator Q and as such – and because our chosen basis C is Hermitian – is an orthogonal matrix for which $\mathcal{Q}^{\top}\mathcal{Q} = \mathbb{1}$.

6: We write explicitly $\langle\!\langle\tilde{B}_{(1\to 1)}(\omega)|$ to emphasize that this is the *cumulative* control matrix.

with

$$
\begin{aligned}
&\mathcal{N}_{(g|g-1\to 1)}(\omega) \\
&\quad = |\tilde{B}_{(g|g-1\to 1)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}_{(g-1\to 1)}(\omega)| + \mathcal{Q}_{(g-1\to 1)}^{\top}\mathcal{N}_{(g)}(\omega)\mathcal{Q}_{(g-1\to 1)}
\end{aligned}
\tag{C.40b}
$$

and

$$
\begin{aligned}
&\mathcal{J}^{(h|h-1\to 1)}_{(g|g-1\to 1),kl}(\omega) \\
&\qquad = s^{(h)}_{(g)}\bar{B}^{(h)}_{(g),ij}\bar{C}^{(h\to 1)}_{(g\to 1),kji}I^{(h)}_{(g),ijmn}(\omega)\bar{C}^{(h\to 1)}_{(g\to 1),lnm}\bar{B}^{(h)}_{(g),mn}s^{(h)}_{(g)}.
\end{aligned}
\tag{C.40c}
$$

Equation C.40 is our final result. Before we analyze it in more detail, let us first briefly discuss the special case where $G = 1$. Then, $\mathcal{Q}_{(0)} = \mathbb{1}$, $\langle\!\langle\tilde{B}_{(0)}| = 0$, and hence $|\tilde{B}_{(1)}(\omega)\rangle\!\rangle\langle\!\langle\tilde{B}_{(0)}(\omega)| = 0$ so that Equation C.40 reduces to Equation C.30 as it should.

To compute the second-order filter function of a sequence of quantum gates most efficiently, Equation C.40 suggests the following procedure:

1. Compute the control matrix for all gates individually. This requires the quantities $V^{(h)}_{(g)}$, $\bar{B}^{(h)}_{(g)}$, and $Q^{(h)}_{(g)}$ to be computed, allowing them to be reused in Equation C.35.
2. Compute the second-order filter function for all gates individually. This requires the quantities $I^{(h)}_{(g)}(\omega)$ (Equation 16.41) and $\mathcal{N}_{(g)}(\omega)$ to be computed, allowing them to be reused in Equations C.40b and C.40c, respectively.
3. Compute the control matrix of the entire sequence. This requires the quantities $\tilde{\mathcal{B}}_{(g|g-1\to 1)}(\omega)$ and $\tilde{\mathcal{B}}_{(g\to 1)}(\omega)$ to be computed, allowing them to be reused in Equation C.40b.

In this way, the only computations that need to be performed once for each gate g to evaluate Equation C.40 are the outer product and matrix multiplications in Equation C.40b. Additionally, for every time step of the entire sequence, the quantity $\bar{C}^{(h\to 1)}_{(g\to 1)}$ needs to be computed following Equation C.35, the contraction in Equation C.40c performed, and finally $\mathcal{J}^{(h|h-1\to 1)}_{(g|g-1\to 1)}(\omega)$ added to the result. In total, this requires $\mathcal{O}(G + H)$ computations, where $H = \sum_{g=1}^{G}\eta_g$ is the total number of time steps, yielding a favorable linear scaling in H compared to a naive approach that is $\mathcal{O}(H^2)$ due to the nested time integral.

C.4 Monte Carlo and GKSL master equation simulations

In this section, we lay out two common time-domain simulation methods of noisy quantum dynamics for completeness; direct simulation of the Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) master equation as well as stochastic Monte Carlo (MC) simulation of the Schrödinger equation. These can serve as alternative and complementary approaches to the filter-function formalism presented in Part IV. All simulations were performed using the `lindblad_mc_tools` software package written during the course of the present thesis [436].

C.4.1 Simulation methods

To simulate a quantum system under the influence of Markovian (white) noise, a common approach is the GKSL master equation [323, 324]. Here, we give it in superoperator form. We represent linear maps $\mathcal{A} : \rho \to \mathcal{A}(\rho)$ by matrices in the Liouville representation following Equation 16.1 and operators as column vectors (*i.e.*, generalized Bloch vectors) as

$$\rho_i := \mathrm{tr}(\sigma_i \rho), \tag{C.41}$$

allowing us to write the Lindblad equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho(t) = -i[H(t), \rho(t)] + \sum_\alpha \gamma_\alpha \left(L_\alpha \rho(t) L_\alpha^\dagger - \frac{1}{2}\left\{L_\alpha^\dagger L_\alpha, \rho(t)\right\}\right) \tag{C.42}$$

as a linear differential equation in matrix form,

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_i(t) = \sum_j \left(-i\mathcal{H}_{ij}(t) + \sum_\alpha \gamma_\alpha \mathcal{D}_{\alpha,ij}\right)\rho_j(t). \tag{C.43}$$

Here, $\mathcal{H}_{ij}(t) = \mathrm{tr}\left(\sigma_i\left[H(t), \sigma_j\right]\right)$ and $\mathcal{D}_{\alpha,ij} = \mathrm{tr}\left(\sigma_i L_\alpha \sigma_j L_\alpha^\dagger - \frac{1}{2}\sigma_i\left\{L_\alpha^\dagger L_\alpha, \sigma_j\right\}\right)$. The γ_α are coupling constants to the noise bath and can be related to the amplitude of the power spectral density (PSD). For Hermitian L_α, the solution to Equation C.42 is a completely-positive trace preserving (CPTP) as well as unital map. Equation C.43 is readily solved under the approximation of piecewise constant timesteps $\Delta t_g = t_g - t_{g-1}$ and one obtains for the complete superpropagator

$$\mathcal{U}(t_g, t_{g-1}) = \exp\left\{\left(-i\mathcal{H}(t_g) + \sum_\alpha \gamma_\alpha \mathcal{D}_\alpha\right)\Delta t_g\right\} \tag{C.44}$$

with

$$\mathcal{U}(\tau) = \prod_{g=G}^{1} \mathcal{U}(t_g, t_{g-1}). \tag{C.45}$$

The entanglement fidelity can then be computed as $F_{\mathrm{ent}} = d^{-2}\,\mathrm{tr}\left(\mathcal{Q}^\dagger \mathcal{U}\right)$, where $\mathcal{Q}$ is the superpropagator due to the Hamiltonian evolution alone (*i.e.*, the ideal evolution without noise), and F_{avg} obtained using Equation 16.45.

In a MC simulation, we work with single realizations of the noise Hamiltonian in Equation 16.3 and solve the Schrödinger equation governed by it. This results in unitary dynamics. The ensemble-averaged dynamics are then obtained by randomly drawing many realizations, solving the Schrödinger equation and computing the desired quantities for each, before finally averaging over all realizations. To sample the PSD faithfully, the piecewise constant time step needs to be significantly smaller than in a noise-free simulation in order to resolve high frequencies of the noise (*cf.* Chapter 2). In practice, we generate time traces of the noise fields by drawing pseudo-random numbers from a distribution whose PSD is $S(f)$. To do this, we draw complex, normally distributed samples in frequency space (*i.e.* white noise), scale it with the amplitude spectral density (ASD), and finally perform the inverse Fourier transform. We then solve the Schrödinger equation by diagonalizing the full Hamiltonian $H(t) = H_{\mathrm{c}}(t) + H_{\mathrm{n}}(t)$ and computing the propagator for one noise realization as

$$U(t) = \prod_g V^{(g)} \exp\left(-i\Omega^{(g)}\Delta t_{\mathrm{MC}}\right) V^{(g)\dagger}, \tag{C.46}$$

where $V^{(g)}$ is the unitary matrix of eigenvectors of $H(t)$ during time segment g and $\Omega^{(g)}$ the diagonal matrix of eigenvalues. We can then obtain an estimate for the entanglement fidelity F_{ent} as

$$\langle F_{\text{ent}}\rangle = \left\langle \left|\mathrm{tr}\left(Q^\dagger U(\tau)\right)\right|^2\right\rangle, \tag{C.47}$$

and F_{avg} again from Equation 16.45. Here, $Q \equiv U_{\text{c}}(t = \tau)$ is the noise-free propagator at time τ of completion of the circuit and $\langle \bullet \rangle$ denotes the ensemble average over N Monte Carlo realizations of Equation C.46, *i. e.*, $\langle A\rangle = N^{-1}\sum_{i=1}^N A_i$. The standard error of the mean can be obtained as $\sigma_{\langle F_{\text{avg}}\rangle} = \sigma_{F_{\text{avg}}}/\sqrt{N}$ with $\sigma_{F_{\text{avg}}}$ the standard deviation over the Monte Carlo traces.

C.5 Fidelity validation

In this section, we lay out in more detail how the fidelity of the optimized S-T_0 qubit gates from Reference 421 was calculated using filter functions, as well as validate the fidelities computed for the QFT algorithm in Section 19.4 using the methods presented in Section C.4.

C.5.1 Singlet-Triplet Gate Fidelity

In two singlet-triplet qubits, angular momentum conservation suppresses occupancy of states with non-vanishing magnetic spin quantum number m_s so that the total accessible state space of dimension $d = 6$ is spanned by $\{|{\uparrow\downarrow\uparrow\downarrow}\rangle, |{\uparrow\downarrow\downarrow\uparrow}\rangle, |{\downarrow\uparrow\uparrow\downarrow}\rangle, |{\downarrow\uparrow\downarrow\uparrow}\rangle, |{\uparrow\uparrow\downarrow\downarrow}\rangle, |{\downarrow\downarrow\uparrow\uparrow}\rangle\}$. A straightforward method to single out the computational subspace (CS) dynamics from those on the whole space would be to simply project the error transfer matrix $\tilde{\mathcal{U}} \approx \mathbb{1} + \mathcal{K}$ with $\mathcal{K}$ the cumulant function onto the CS as proposed by Wood and Gambetta [384], that is calculate the fidelity as $F_{\text{ent}} = d_c^{-2}\mathrm{tr}\big(\Pi_c\tilde{\mathcal{U}}\big)$ where Π_c is the Liouville representation of the projector onto the CS and $d_c = 4$ the dimension of the CS. However, here we use a more involved procedure in order to gain more insight from the error transfer matrix as well as to obtain a better comparison to the fidelities computed by Cerfontaine et al. [421], who map the final 6×6 propagator to the closest unitary on the 4×4 CS during their Monte Carlo simulation.

To calculate the fidelity of the target unitary on the 4×4 CS, we thus construct an orthonormal operator basis C of the full 6×6 space that is partitioned into elements which are nontrivial only on the CS on the one hand and elements which are nontrivial only on the remaining space on the other such that $\mathsf{C} = \mathsf{C}^c \cup \mathsf{C}^\ell$. Using such a basis, we can then trace only over CS elements of the error transfer matrix $\tilde{\mathcal{U}}$ in Equation 16.46 to obtain the fidelity of the gate on the CS. Moreover, we retain the opportunity to characterize the gates on the basis of the Pauli matrices.

Since there is no obvious way to extend the Pauli basis for two qubits to the complete space we proceed as follows: For the CS, we pad the two-qubit Pauli basis with zeros on the leakage levels, *i. e.*,

$$C_i^c \doteq \begin{array}{c} \\ \langle{\uparrow\uparrow\downarrow\downarrow}| \\ \langle{\downarrow\downarrow\uparrow\uparrow}| \end{array}\!\!\begin{array}{c} \begin{array}{ccc} & |{\uparrow\uparrow\downarrow\downarrow}\rangle & |{\downarrow\downarrow\uparrow\uparrow}\rangle \end{array} \\ \begin{pmatrix} P_i & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{array}, \quad i \in \{0, \dots, 15\}, \tag{C.48}$$

where the P_i are normalized two-qubit Pauli matrices (*cf.* Equation 16.54) in the basis $\{|\uparrow\downarrow\uparrow\downarrow\rangle, |\uparrow\downarrow\downarrow\uparrow\rangle, |\downarrow\uparrow\uparrow\downarrow\rangle, |\downarrow\uparrow\downarrow\uparrow\rangle\}$. To complete the basis we require an additional 20 elements orthogonal to the 16 padded Pauli matrices. We obtain the remaining elements by first expanding the C_i^c in an arbitrary basis $\{\Lambda_i\}_{i=0}^{35}$ of the complete space (we choose a generalized Gell-Mann (GGM) basis, Equation 16.55, for simplicity), yielding a 16×36 matrix of expansion coefficients:

$$M_{ij} = \mathrm{tr}\left(C_i^c \Lambda_j\right). \tag{C.49}$$

We then compute an orthonormal vector basis V (a matrix of size 36×20) for the null space of M using singular value decomposition $M = U\Sigma V^\dagger$ and acquire the corresponding basis matrices as

$$C_i^\ell = \sum_j \Lambda_j V_{ji}, \quad i \in \{0, \dots, 19\}. \tag{C.50}$$

Finally, to account for the fact that Cerfontaine et al. [421] map the total propagator to the closest unitary on the CS, we exclude the identity Pauli element $C_0^c \propto \mathrm{diag}(1,1,1,1,0,0)$ from the trace over the computational subspace part of $\tilde{\mathcal{U}}$ represented in the basis $\mathsf{C} = \mathsf{C}^c \cup \mathsf{C}^\ell$ when calculating the fidelity,

$$F_\mathrm{ent} = \frac{1}{16}\sum_{i=1}^{15} \tilde{\mathcal{U}}_{ii}, \tag{C.51}$$

since for unitary operations on the CS we have $\mathcal{K}_{00} \approx 1 - \tilde{\mathcal{U}}_{00} = 1 - \mathrm{tr}\left(C_0^c \tilde{U} C_0^c \tilde{U}^\dagger\right) = 0$. Hence, excluding $\tilde{\mathcal{U}}_{00}$ from the trace corresponds to partially disregarding non-unitary components of the error channel on the computational subspace. Although not the only element that differs compared to the closest subspace unitary, $\mathcal{K}_{00}$ contains the most obvious contribution, whereas those of other elements are more difficult to disentangle into unitary and non-unitary components.

Similar to the fidelity, we also obtain the canonical filter function shown in panel (b) of Figure 19.1 by summing only over columns one through 15 of the control matrix, $F_{\epsilon_{ij}}(\omega) = \sum_{k=1}^{15}\left|\tilde{\mathcal{B}}_{\epsilon_{ij}k}(\omega)\right|^2$. In fact, including the first column, corresponding to the padded identity matrix C_0^c, in the filter function removes the DCG character of $F_{\epsilon_{12}}(\omega)$ and $F_{\epsilon_{34}}(\omega)$, which instead approach a constant level of around 20 (note that the filter function is dimensionless in our units) at zero frequency. This is consistent with the fact that the gates were optimized using quasistatic and fast white noise contributions to the fidelity after mapping to the closest unitary on the computational subspace. We have performed Monte Carlo resimulations that support this reading. In Figure C.1 we show the filter functions once including and once excluding the contributions from C_0^c.

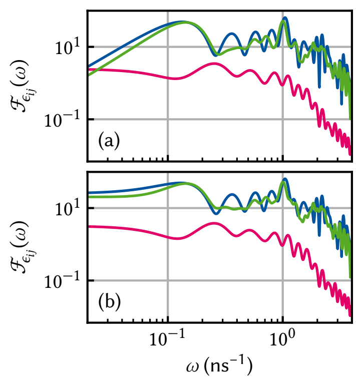

Figure C.1: Filter functions of the voltage detunings ϵ_{ij} excluding (a) and including (b) the zero-padded identity matrix basis element $C_0^c \propto \mathrm{diag}(1,1,1,1,0,0)$ for the computational subspace. Evidently, including C_0^c removes the dynamically corrected gate (DCG) character, namely that $F_{\epsilon_{ij}}(\omega) \to 0$ as $\omega \to 0$, of the gates but has little effect on the high-frequency behavior. As the pulse optimization minimizes, among other figures of merit, the infidelity of the final propagator mapped to the closest unitary on the computational subspace due to quasistatic and fast white noise, this indicates that excluding C_0^c from the filter function corresponds to partially neglecting non-unitary components of the propagator on the computational subspace.

C.5.2 GRAPE-optimized gate set and validation of QFT fidelities

In this section we give details on the GRAPE-optimized pulses for the gate set $\mathbb{G} = \{\mathrm{X}_i(\pi/2), \mathrm{Y}_i(\pi/2), \mathrm{CR}_{ij}(\pi/2^3)\}$ used in Section 19.4 to simulate a QFT algorithm, and validate the fidelities using time-domain simulations. As mentioned in the main text, we consider a toy Rabi driving model with IQ single-qubit control and exchange to mediate inter-qubit coupling. Cast in the language of quantum optimal control theory this translates to a vanishing drift (static) Hamiltonian, $H_\mathrm{d} = 0$, and a control

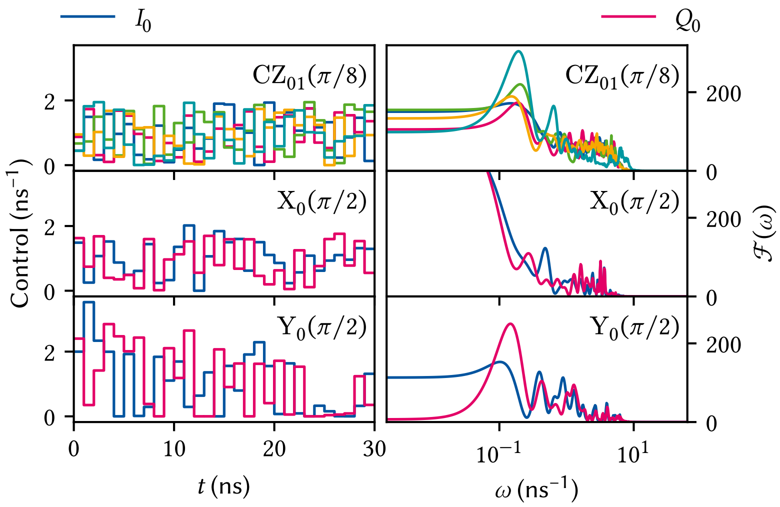

Figure C.2: Control fields (top row) and corresponding filter functions (bottom row) of the GRAPE-optimized pulses in $\mathbb{G}$. (a),(b) $\mathrm{X}_0(\pi/2)$; (c),(d) $\mathrm{Y}_0(\pi/2)$; (e),(f) $\mathrm{CR}_{01}(\pi/2^3)$. Note that the optimization is neither very sophisticated nor realistic as the algorithm only maximizes the systematic (coherent) fidelity $\mathrm{tr}(UQ_{\mathrm{targ}}^\dagger)/d$ and the randomly distributed initial control amplitudes are not subject to any constraints.

Hamiltonian in the rotating frame given by

$$H_\mathrm{c}(t) = H_\mathrm{c}^{(0)}(t) \otimes \mathbb{1} + \mathbb{1} \otimes H_\mathrm{c}^{(1)}(t) + H_\mathrm{c}^{(01)}(t), \tag{C.52}$$

$$H_\mathrm{c}^{(i)}(t) = I_i(t)\sigma_x^{(i)} + Q_i(t)\sigma_y^{(i)}, \tag{C.53}$$

$$H_\mathrm{c}^{(ij)}(t) = J_{ij}(t)\sigma_z^{(i)} \otimes \sigma_z^{(j)}, \tag{C.54}$$

where $I_i(t)$ and $Q_i(t)$ are the in-phase and quadrature pulse envelopes and $\sigma_{x,y}^{(i)}$ are the Pauli matrices acting on the ith and extended trivially to the other qubit. For simplicity, we assume periodic boundary conditions so that qubits 1 and 4 are nearest neighbors as well. As our goal is only of illustrative nature and not to provide a detailed gate optimization, we obtain the controls $\{I_0(t), Q_0(t), I_1(t), Q_1(t), J_{12}(t)\}$ for the gate set $\mathbb{G}$ using the GRAPE algorithm implemented in `QuTiP` [359] initialized with randomly distributed amplitudes. The resulting pulses and the corresponding filter functions for the relevant noise operators are shown in Figure C.2.

We now perform GKSL master equation and MC simulations using the methods laid out in Subsection C.4.1 to verify the fidelities predicted for the QFT circuit in Section 19.4. We focus on noise exclusively on the third qubit, entering through the noise operator $B_\alpha \equiv \sigma_y^{(3)}$. We assemble the QFT circuit discussed in the main text from the gate set $\mathbb{G}$, obtaining the control Hamiltonian

$$H_\mathrm{c}(t) = \sum_{\langle i,j \rangle} I_i(t)\sigma_x^{(i)} + Q_i(t)\sigma_y^{(i)} + J_{ij}(t)\sigma_z^{(i)} \otimes \sigma_z^{(j)}. \tag{C.55}$$

Similarly, we define the noise Hamiltonian as

$$H_\mathrm{n}(t) = \sum_{\langle i,j \rangle} b_I(t)\sigma_x^{(i)} + b_Q(t)\sigma_y^{(i)} + b_J(t)\sigma_z^{(i)} \otimes \sigma_z^{(j)} \tag{C.56}$$

with the noise fields $b_\alpha(t)$ for $\alpha \in \{I, Q, J\}$. For the GKSL master equation, we set $L_\alpha \equiv \sigma_y^{(3)}$ as well as $\gamma_\alpha \equiv S_0/2$ with S_0 the amplitude of the one-sided noise PSD so that $S(\omega) = S_0$. For the MC simulation, we explicitly generate time traces of $b_Q(t)$ (*cf.* Equation C.56). We choose an oversampling factor of 16 so that the time discretization of the simulation is $\Delta t_{\mathrm{MC}} = \Delta t/16 = 62.5\,\mathrm{ps}$ ($\Delta t = 1\,\mathrm{ns}$ is the time step of the pulses used in the FF simulation), leading to a highest resolvable frequency of

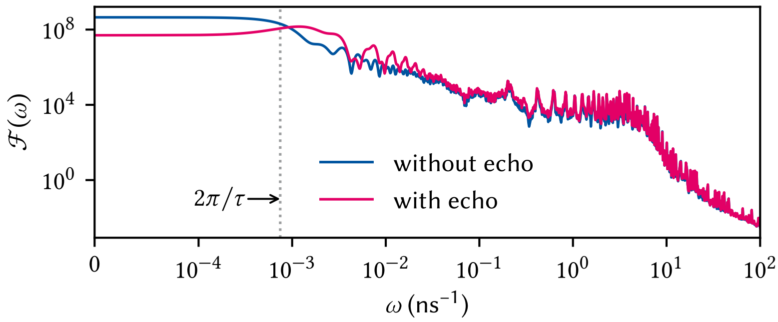

Figure C.3: Filter functions for noise operator $\sigma_y^{(3)}$ for the QFT circuit without (blue) and with (magenta) additional echo pulses. Introducing the echoes shifts spectral weight towards higher frequencies, reducing the DC level of the filter function by two orders of magnitude and thus leading to an improved fidelity for $1/f$ noise.

Table C.1: Infidelities $I_{\text{avg}} = 1 - F_{\text{avg}}$ of the QFT circuit due to noise on $\sigma_y^{(3)}$. MC values are averages over $N = 1000$ random traces and have a relative error of 3 %. We included frequencies in the range of $\omega \in [0, 100]\,\text{ns}^{-1}$ for white noise, and $\omega \in [100\,\text{ms}^{-1}, 100\,\text{ns}^{-1}]$ for pink noise. FF values are computed with $n_\omega = 1000$ samples logarithmically distributed over the same interval. Prefactors in the power law $S(\omega) = A\omega^\alpha$ are $2 \times 10^{-6}\,\text{ns}^{-1}$ and $10^{-9}\,\text{ns}^{-2}$, respectively.

	White noise		1/f noise	
Method	Without echo	With echo	Without echo	With echo
GKSL	8.38×10^{-3}	8.38×10^{-3}	—	—
MC	8.74×10^{-3}	8.00×10^{-3}	2.11×10^{-2}	4.28×10^{-3}
FF	8.38×10^{-3}	8.40×10^{-3}	2.12×10^{-2}	4.46×10^{-3}

$f_{\text{max}} = 16\,\text{GHz}$. Conversely, we increase the frequency resolution by sampling a time trace longer by a given factor, giving frequencies below f_{min} (16 kHz for pink, 0 Hz for white noise) weight zero, and truncating it to the number of time steps in the algorithm times the oversampling factor. This yields a time trace with frequencies $f \in [f_{\text{min}}, f_{\text{max}}]$ and a given resolution (we choose $\Delta f = 160\,\text{Hz}$). For reference, we show the fidelity filter functions for the circuit with and without echo pulses in this frequency band in Figure C.3.

Table C.1 compares the infidelities $I = 1 - F$ from GKSL and MC simulations to the filter function predictions following Equation 16.48. Note that the precise value of the filter function result depends quite sensitively on the frequency sampling due to the sharp peaks in the gigahertz range (Figure C.3). As the table shows, both the GKSL and the MC calculations agree well with the predictions made by our filter-function formalism.

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Glossary

Numbers
2DEG two-dimensional electron gas. 36, 37, 65, 75, 76, 82, 83, 96–100, 104–107, 115–117, 178
2DHG two-dimensional hole gas. 83

A
APD avalanche photodiode. 30, 59, 60, 62, 89, 115
API application programming interface. 17, 30, 123, 143, 153
AR anti-reflection. 34, 65
ASD amplitude spectral density. 17, 189

B
BBAR broadband anti-reflection. 34
BEC Bose-Einstein condensate. 29
BLG bilayer graphene. 67
BOB break-out box. 36
BS beam splitter. 30, 35, 39, 45, 47–50, 66, 91

C
CA clear aperture. 40–45
CCD charge-coupled device. 30, 39, 46, 65, 86–93, 95, 115
CMOS complementary metal-oxide-semiconductor. 39, 46, 47, 59, 95
CPTP completely-positive trace preserving. 189
CRAB chopped random-basis. 165
CS computational subspace. 190, 191
CSD cross power spectral density. 24, 25
cw continuous-wave. 30, 50, 85, 88, 95, 115

D
DAC digital-to-analog converter. 87
DAQ data acquisition. 4, 8, 11, 13–15, 17–19, 23
DCG dynamically corrected gate. 122, 126, 134, 159, 161, 191
DD dynamical decoupling. 5, 122, 126, 134, 152, 166
DMM digital multimeter. 56
DPSS diode-pumped solid state. 86
DR dilution refrigerator. 29, 30, 33–35, 39, 53, 55, 56, 65, 95
DUT device under test. 5, 56, 86, 87

E
ENBW equivalent noise bandwidth. 19
EPR Einstein-Podolsky-Rosen. 71, 72

F
FES Fermi-edge singularity. 76, 77, 97, 105, 107, 115
FF filter function. 122, 130, 134, 143–145, 147–151, 179, 192, 193
FFT fast Fourier transform. 11
FWHM full width at half maximum. 102

G
GDQD gate-defined quantum dot. 36, 37, 66, 73–75
GGM generalized Gell-Mann. 141, 142, 154, 191
GKSL Gorini-Kossakowski-Sudarshan-Lindblad. xiii, 121, 168, 179, 188, 189, 192, 193
GRAPE gradient ascent pulse engineering. 165, 191, 192
GST gate set tomography. 125
GUI graphical user interface. 21, 92, 93

H

HBT Hanbury Brown-Twiss. 30, 49, 60

I

i.i.d. independent and identically distributed. 127
IQ in-phase/quadrature. 165, 191
IRF instrument-response function. 50
ISO International Organization for Standardization. 55, 62

L

LD Loss-DiVincenzo. 73
LIA lock-in amplifier. 5, 12, 17
LOS line-of-sight. 33

M

MBE molecular beam epitaxy. 177
MC Monte Carlo. 123, 140, 143–145, 147–152, 168, 170, 188, 189, 192, 193
ME Magnus expansion. 122, 123, 126, 128, 129, 135, 137, 152, 179, 182–184
MFD mode field diameter. 31, 40–42, 46
MMF multi-mode fiber. 30, 96
MOS metal-oxide-semiconductor. 160
MXC mixing chamber. 30, 33–36

N

NA numerical aperture. 30, 40–44, 173, 174
ND neutral-density. 30, 86–89
NIR near-infrared. 34, 75

O

OAQD optically active quantum dot. 49, 73–75
OD optical density. 48, 49, 89

P

PCB printed circuit board. 36
PDE photon detection efficiency. 46, 48
PL photoluminescence. 45, 50, 74–76, 82, 83, 85, 90, 95–108, 111, 113, 115–118, 177, 178
PLE photoluminescence excitation. 74, 105–108, 116, 117, 177
POVM positive operator-valued measure. 139
PSD power spectral density. xi, 4, 6–8, 10–13, 15, 17, 19, 23–25, 55–57, 60, 61, 66, 132, 147, 150, 189, 192
PT1 first pulse tube stage. 34, 53
PT2 second pulse tube stage. 34, 36, 53
PTM Pauli transfer matrix. 125, 150
PTR pulse tube refrigerator. 31, 53, 55–57, 60, 62, 63, 66, 90

Q

QCSE quantum-confined Stark effect. 73, 74, 77–79, 81, 115, 116
QD quantum dot. 36–38, 65, 73, 74, 102
QE quantum efficiency. 46, 48, 49
QEC quantum error correction. 72, 169
QFT quantum Fourier transform. 124, 141, 165–168, 179, 190–193
QPT quantum process tomography. 125
QW quantum well. 43, 44, 58, 73, 74, 76–83, 95–99, 102–107, 111–113, 115–118, 173, 177, 178

R

RB randomized benchmarking. 124, 125, 139, 162–164
RF radio frequency. 34
RMS root mean square. iii, v, 5, 7, 18–20, 53, 55–57, 59, 60, 62, 63, 66, 90, 95, 147
RWA rotating-wave approximation. 160

S
SAQD self-assembled quantum dot. 31, 49, 50, 73, 100
SE spin echo. 151, 152, 154, 155
SEM Scanning electron microscope. 95
SMA SubMiniature version A. 56
SMF single-mode fiber. 30, 31, 39–47, 51, 57, 62, 65, 95, 96, 113, 117, 175
SNR signal-to-noise ratio. 35, 62, 66, 89, 105
SPAM state preparation and measurement. 162, 163
SPCM single-photon counting module. 30, 46, 48–50

T
TE transverse electric. 109
TEM transverse electromagnetic. 39–41, 45–47, 58, 66
TIA transimpedance amplifier. 15, 37
TLF two-level fluctuator. 11, 129
TMD transition-metal dichalcogenide. 67
TMM transfer-matrix method. xiii, 45, 103, 108–110, 117, 176–178
TO transverse optical. 107

V
VC vibration criterion. 55, 62, 63, 66

W
WD working distance. 41

Figure source files and parameters

Publications

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Software

The following open-source software packages were developed (at least partially) during the work on this thesis.

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Declaration of Authorship

I, Tobias Hangleiter, declare that this thesis and the work presented in it are my own and has been generated by me as the result of my own original research.

I do solemnly swear that:

1. This work was done wholly or mainly while in candidature for the doctoral degree at this faculty and university;
2. Where any part of this thesis has previously been submitted for a degree or any other qualification at this university or any other institution, this has been clearly stated;
3. Where I have consulted the published work of others or myself, this is always clearly attributed;
4. Where I have quoted from the work of others or myself, the source is always given. This thesis is entirely my own work, with the exception of such quotations;
5. I have acknowledged all major sources of assistance;
6. Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself;
7. Parts of this work have been published before as:

[1] P. Cerfontaine, T. Hangleiter, and H. Bluhm. "Filter Functions for Quantum Processes under Correlated Noise." In: *Physical Review Letters* 127.17 (Oct. 18, 2021), p. 170403. DOI: `10.1103/PhysRevLett.127.170403`.

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Aachen, May 1, 2026.